

Emergence and Phase Transitions in Uniform Random Graphs

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1

Introduction

This book is about the structural role of small subgraphs, such as edges and triangles, within large graphs. Given how many edges and triangles there are in a large graph G , we show that a lot can be said about the large-scale structure of G , thinking of that structure as “emerging” from the edges and triangles within it.

To get a feel for how large-scale structure can emerge from small-scale considerations, consider a large number of pennies, say a million, lying without overlap on a square table. We know from Fejes Toth [21] that the area fraction covered by pennies is at most $\rho_{max} = \pi/\sqrt{12} \approx 0.9069$; see Figure 1.1.

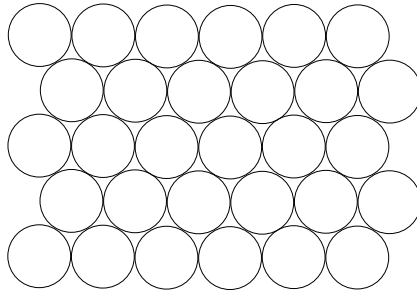


Figure 1.1 The densest packing of equal area disks

Less well known is that extensive computer simulation since about 1950, on the hard disk model of statistical mechanics [74, 37], strongly suggests that there is a partition of the possible range of area fraction $(0, 0.91)$ into three interesting regimes:

1. high area fraction between 0.72 and 0.91;
2. intermediate area fraction between 0.71 and 0.72;

3. low area fraction between 0 and 0.71.

Typical packings with high area fraction, between 0.72 and 0.91, have macroscopic properties that are smooth extensions of those of the densest packing at area fraction 0.91. Typical packings with low area fraction, between 0 and 0.71, have macroscopic properties that are smooth extensions of those with widely separated disks, with area fraction near 0. Finally, packings with area fraction between 0.71 and 0.72 *have no useful description* in terms of macroscopic properties; a large table covered with such area fraction will typically be inhomogeneous, a region filled at area fraction 0.71 and the complement filled at area fraction 0.72.

That is, as the area fraction of the large table covered by pennies increases from 0 to 0.91 one passes from a regime of widely separated disks to a regime of densely packed disks, *the two regimes being sharply distinguishable* by the intermediate regime. The high and low area regimes are called phases.

One macroscopic property that captures the features of the partition is the entropy function: the exponential rate of growth of the number of disk arrangements as the number of disks and table size diverge with limiting area fraction ρ . In the infinite system limit the entropy is expected to be a real analytic function of ρ in the high and low area phases, but from computer simulations ***the restrictions of the entropy to the high and low area phases are not smooth continuations of one another,***

indicating a phase transition. The origin of this singular behavior remains a mystery; its reliability is entirely based on computer simulations and vast evidence from experimental physics.

Replacing these numerical observations with rigorous mathematics is very difficult. By considering such macroscopic properties as joint correlations of centers of disks, the very low density part of regime (3) has been proven to exist in some generality by virial expansions: (see Chapter 4 in [61], and [43]). However, almost everything about the more interesting high-density and mid-density regimes remains unproven.

In this book we consider a different emergent setting, that of constrained graphs, in which we use intuition gained from experience from computer simulations of circle packings [37] to *prove* close relatives of the above picture. We enlarge on this in the following section.

1.1 Graphs constrained by subgraph densities

Among (simple) graphs G on n nodes we consider the subset of those graphs “constrained” by having specified densities of each of a certain finite set of connected subgraphs. Most of the time we constrain two subgraphs, the density e of edges, and the density t of one other subgraph, typically a triangle. We treat those (two) constraints as control parameters, analogous to the (single) area fraction of disk packings, treated above. Our aim is to *prove* the existence of several open sets of constraints (e, t) on which emergent quantities, playing the role that entropy plays for circle packings, are analytic but “essentially different”. Specifically, we aim to prove that restriction of a quantity to one such set is not an analytic extension of its restriction to another; this is how we prove the existence of phase transitions in an emergent sense. Such a quantity is traditionally called an “order parameter” and this important technique will be used at various points of this text, before being discussed abstractly in Section 8.4.

Consider the special case of constraints on the edge and triangles densities. An important first step is determining the range of possible values (e, t) of those densities, and the graphs which attain the extreme possibilities. This is part of the venerable subject of extremal combinatorics, championed by Turán [71], and the solution for edge-triangle graphs was finally completed by Pikhurko and Razborov [50] in 2012 when we started our project; in fact [50] helped motivate our project. See Figure 1.2 and Section 3.2 for a review. We emphasize that although our work was motivated by extremal combinatorics, our work is focused on understanding “emergence” in a graph setting, in the above sense. At first we incorporated this advance in extremal combinatorics to attempt to overcome some problems explored in [14]; see Section 4.7. However this book is a synthesis of the fruits of a sequence of papers since 2013 [54], in which we proved for constrained graphs what cannot yet be proven for the geometric system of circle packings, emergent phase transitions. We prove infinitely many such transitions, and that is the thrust of this book.

We are interested in studying graphs constrained by the density e of edges and the density t of some other subgraph H such as a triangle. To implement the constraints we use uniform families of random variables as follows. Given an achievable pair (e, t) of densities, and a small tolerance δ , we consider all graphs on n vertices with densities within δ of (e, t) , each graph having equal probability. In Chapter 2 we give careful definitions of several elements of our proofs, in particular the graph-limit or graphon formalism, in Section 2.2. That formalism enables us to take limits as $n \rightarrow \infty$ and $\delta \rightarrow 0$ appropriately. For instance this enables us to define, in Section 2.5, the Boltzmann entropy B as

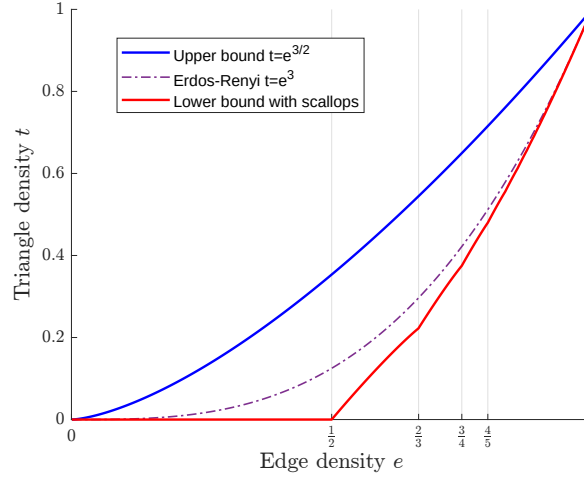


Figure 1.2 The Razborov triangle, distorted to show various features

a function of (e, t) and to do most of our work with graphons with proof that our results will be interpretable for large constrained graphs. We use the term uniform random graphs for models based as above on this limiting process.

As with disk packings a key idea, which we call a *phase* in analogy with physics usage, is a maximal connected open set in the space of the constraint parameters in which global features of a large graph vary smoothly with the parameters. Simple examples of global features are the densities of subgraphs. We will prove that the density of each unconstrained subgraph is an analytic function of the constraints in certain regions (phases), but is nonanalytic and can even jump discontinuously at special values of the constraints.

For instance, consider graphs with edge density $e = 0.6$. It is easy to understand such graphs with triangle density $t = 0.216 = (0.6)^3$; those are Erdős-Rényi graphs, edges chosen independently with probability $p = 0.6$. Understanding graphs with edge density 0.6 and triangle density 0.215 or 0.217 is much harder. (We use the term “Erdős-Rényi graphs” frequently in this book to refer to graphs with independent edges.) In the coming chapters we will prove that triangle densities just above and just below 0.216 lie in different phases. In each phase, the structure of a typical graph is fairly easy to describe, involving a small set of parameters that are analytic functions of (e, t) . However, the two structures are essentially different, yielding a variety of emergent quantities whose behavior changes abruptly at $t = 0.216$.

Our analysis of large graphs with subgraph-density constraints can be viewed

as an extension of extremal graph theory [10], [3], though our focus is on emergence. An important part of our results is the analysis of singular behavior among large graphs with *near-extreme* constraints. For instance, using [50] it is relatively easy to determine the global features of graphs with densities on the *boundary* of the edge-triangle parameter space of Figure 1.2. It is also easy to determine global features of Erdős-Rényi graphs, say with edge density 0.6 and triangle density 0.216. It is much harder, and therefore mathematically more interesting, to understand large graphs constrained very close to the boundary, or very close to the Erdős-Rényi curve $t = e^3$.

Broadly, the mathematics in this book is a natural overlap of the areas of probabilistic combinatorics and extremal combinatorics. The phases that we prove, both near the extreme constraints and near Erdős-Rényi, are of natural interest to combinatorialists once the mathematics is made compatible with existing parts of combinatorics, as we aim to achieve in this book. We simplify the various proofs since [54] and weave them into a coherent formalism for rigorously understanding subtle asymptotic features of multiply-constrained graphs.

Our project evolved from 2013 with the participation of Kui Ren, Rick Kenyon and Joe Neeman. The cumulative results of our project are refined and presented in this book. While many of the proofs presented here are streamlined and contain new features, Ren, Kenyon and Neeman deserve much of the credit for these results.

This is a mathematics book concerned with proofs of multiple phases in emergent systems. The phase transitions of disk (and sphere) packings have not been proven, only supported by computer simulations and many years of experience in experimental physics. But there is a well-known emergent formalism with some proofs of multiple phases, a different toy model of statistical mechanics, known as Markov random fields or lattice spin models [66, 67, 28, 70, 18, 19, 65], of which the best-known examples are the 2-dimensional Ising ferromagnetic and antiferromagnetic models and their variants. There is a wide variety of lattice spin models, but multiple phases have only been proven for interactions of very special type. (It is revealing to note the failure so far to prove anything for a “quasicrystal” model [6, 32].) This book will present a different wide class of emergence models, a different simplified version of statistical mechanics, which we call uniform random graph models. The mathematical relation between the two toy models of statistical mechanics is based on the differing role of convexity in optimization.

In a broader perspective, independent of emergence, we show that constrained graphs are a new setting in which to explore phases and phase transitions associated with changes in boundary-condition constraints in optimization. For

instance consider Plateau’s problem to minimize the surface area of a “soap bubble” bounding a wire frame [40]. Even for a given frame this generates beautiful and deep mathematics. But one can also study how the optimal solutions vary as the wire frame changes shape, perhaps through curves depending on a pair of parameters. Presumably the solution is in some sense smooth in the parameters except at special values, though we do not know of any general attack on this feature. As another example consider Monge’s “optimal transport” problem, to minimize the “cost” to move a pile of sand from one location to another, the cost being the sum of the distances each sand grain is moved [62, 41]. Again this has generated beautiful mathematics. And again imagine Monge’s problem generalized by using a pair of parameters to specify the shape of the sand pile, and then analyzing the dependence on those parameters of the optimal way to move the grains. Although our work is set in large graphs, it is hoped that the tools we develop will be useful in other optimization problems, such as these.

1.2 Preview of coming attractions

To prove delicate qualitative features of emergent systems, such as multiple phases, it is very useful if not necessary to develop an “infinite system” formalism. For statistical mechanics, this was developed around 1970 (see [29, 69, 31]) and involves creating a space X of infinitely many particles, with probability distributions (“Gibbs states”) on X obtained by a complicated process of taking limits of distributions on finite systems.

Limits of graphs as the size goes to infinity are called *graphons*. The Large Deviation Principle (LDP) of Chatterjee and Varadhan [15] is couched in graphons and provides an effective way to connect *optimal* graphons, graphons that optimize the Shannon entropy function, to features of finite graphs. Chapter 2 will provide an introduction, with links to their background, to the powerful formalisms of graphons and the LDP on which our work is based.

To understand the results of this book it is useful to refer to the phase space or parameter space for the edge-triangle model, the so-called “Razborov triangle”. A schematic version, with some features exaggerated for visibility, is shown in Figure 1.2. (See Section 3.2 for some technical details, and [58, 50, 3, 10] for background.

The rest of this section will give a quick overview of the phases which we have proven to exist in this phase space. In 2017 we conjectured [27] the phases which would appear, based on simulation of a relevant optimization problem; see Figure 1.3. In the language of this figure the most important results of

Chapters 3–8 involve proving the existence of (parts of) the bipodal $F(1,1)$, $A(2,0)$ and $B(1,1)$ phases and the multipodal $C(n,2)$ phases and understanding the transitions between them. We also show that Figure 1.3 is incomplete. There is a (very!) small region, with $e < (3 - \sqrt{3})/6 \approx 0.2113$ and t slightly less than e^3 , that we prove *not* to be part of the $A(2,0)$ phase! The nature of the phase (or phases) in this region is not yet understood.

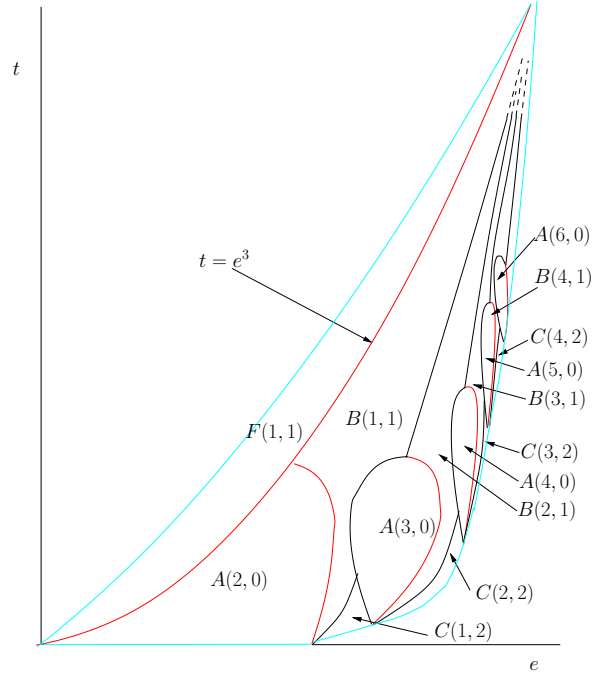


Figure 1.3 Schematic drawing of a conjecture from 2017 [27], indicating the unique optimal graphons associated with the phases of graphs with edge and triangle constraints

In Chapters 2–8 we give a comprehensive presentation of the techniques we have developed to prove phase transitions in uniform random graphs. The subject of phase transitions in random graphs is old and broad, including for instance Markov random fields as noted in the previous section, percolation and K-SAT problems which are well discussed in [42], and exponential random graph models which are well discussed in [14] but also in some depth in Section 4.7. In Appendices A and B we include as side issues some alternative approaches to understanding constrained graphs. These approaches are not

examined in depth; that would require extensive study which we have not performed!

2

Foundations

There are three major themes to this chapter. First, we introduce graphons below starting in Section 2.2: what they are, how they work, and why they are the right tool to think about large graphs. We give a brief introduction; for a compact introduction and history we recommend [13] and for an in-depth presentation and history see [33].

Second, in Section 2.5 we define the Boltzmann entropy $B(e, t)$ which measures the rate of growth in the number of graphs with edge and triangle densities e and t . This notion is taken almost directly from statistical mechanics; for a very clear presentation see [29]. Also in Section 2.5 we give an introduction to the Large Deviation Principle (LDP) for Erdős-Rényi graphs of Chatterjee and Varadhan [15]. We show that $B(e, t)$ is the maximum value of an integral operator $S(g)$, over graphons g satisfying certain integral constraints, (e, t) . When the graphon that maximizes $S(g)$ is unique, typical large graphs are described by the S -optimizing graphon. The problem of understanding large graphs with specified edge and triangle densities then reduces to a constrained optimization problem on the space of graphons, as discussed below.

Finally, we develop methods for solving this constrained optimization problem. We review the method of Lagrange multipliers in infinite dimensions. In addition to the usual functional derivatives used in the calculus of variations, we define a new quantity, called the *worth* of a column of a graphon, and show that all columns of an S -maximizing graphon must maximize worth. Together these techniques make up what we call the “uniform random graph model”, which is the subject of this book.

2.1 Asymptotic notation

Throughout the book we use the following notation to describe limits, both limits as $n \rightarrow \infty$ and limits as our parameters approach a point or curve in the phase plane. Let X be a positive expression such as $\exp(3n^2)$ or $\delta^{2/3}$ or simply 1.

- A quantity is $O(X)$ (“big-O of X ”) if it is bounded by a multiple of X .
- A quantity is $o(X)$ (“little-o of X ”) if it is eventually smaller than any positive multiple of X . For instance, we might write “ $3 + o(1)$ ” for a quantity that approaches 3.
- A quantity is $\Omega(X)$ if its magnitude is eventually bounded below by a positive multiple of X .
- A quantity is $\Theta(X)$ if it is both $O(X)$ and $\Omega(X)$. That is, if its magnitude is eventually bounded both above and below by positive multiples of X .

The most common use of this notation is in proving that certain quantities behave in a certain way for small values of a parameter δ . For instance, we might compute two quantities A and B and show that

$$A = c_1\delta^2 + O(\delta^3), \quad B = c_2\delta^2 + O(\delta^3),$$

where c_1 and c_2 are positive constants with $c_1 > c_2$. Since $A - B$ is within $O(\delta^3)$ of a positive multiple of δ^2 , we conclude that $A - B > 0$ whenever δ is small enough. This argument does not tell us how small δ has to be, yet we know for certain that A dominates B for all sufficiently small values of δ .

2.2 Graphons

The graphon formalism was developed by Lovász and coauthors in an important series of works over several years [11, 12, 34, 35, 36], following the work of Aldous [2], Hoover [23] and Kellenberg [24]; see [8, 16] for more background. Graphons provide a convenient means to analyze graphs as the number of vertices goes to infinity. Used together with the Large Deviation Principle for Erdős-Rényi graphs of Chatterjee and Varadhan [15] the graphon formalism could be compared to the development of equilibrium classical statistical mechanics, which is how we use these two powerful tools.

Definition 2.1 A **graphon** is a symmetric, measurable function $g : [0, 1]^2 \rightarrow [0, 1]$ on the unit square. That is, for all $(x, y) \in [0, 1]^2$,

$$0 \leq g(x, y) = g(y, x) \leq 1. \quad (2.1)$$

We write $\mathcal{W}$ to denote the space of all possible graphons.

There is a (pseudo) norm on the space of bounded symmetric functions on $[0, 1]^2$, called the *cut norm*:

$$\|f\|_{\blacksquare} = \sup_{S, T \subseteq [0, 1]} \left| \int_{S \times T} f(x, y) dx dy \right|.$$

Note that the absolute value goes outside the integral! A function with a large L^1 norm can have a small cut norm if the places where the function is positive and negative are sufficiently scrambled. The cut norm induces a (pseudo) metric on $\mathcal{W}$: if g_1 and g_2 are graphons, then

$$d_{\blacksquare}(g_1, g_2) = \|g_1 - g_2\|_{\blacksquare}.$$

The group Σ of measure-preserving transformations of $[0, 1]$ acts on $\mathcal{W}$ as follows. If g is a graphon and $\sigma : [0, 1] \rightarrow [0, 1]$ is a measure-preserving transformation, let

$$g_{\sigma}(x, y) = g(\sigma(x), \sigma(y)).$$

Modding out by this group action gives a space that is badly behaved. Instead, we mod out by the closure of the group relation as follows. If g_1 and g_2 are graphons, we define the *cut metric*

$$\tilde{d}_{\blacksquare}(g_1, g_2) \equiv \inf_{\sigma, \sigma' \in \Sigma} d_{\blacksquare}(g_1, \sigma, g_2, \sigma'),$$

which measures the separation between the orbits of g_1 and g_2 . Let $\widetilde{\mathcal{W}}$ be the quotient of $\mathcal{W}$ by identification of graphons whose distance in the cut metric is zero. Elements of $\widetilde{\mathcal{W}}$ are called *reduced graphons*.

To each simple graph G on n vertices we associate a *checkerboard graphon*

$$g^G(x, y) = \begin{cases} 1 & \text{if } ([nx], [ny]) \text{ is an edge of } G \\ 0 & \text{otherwise,} \end{cases}$$

where $\lceil y \rceil$ denotes the smallest integer greater than or equal to y . The n intervals of length of $1/n$ along the axes are used to represent the nodes of G . The patterns of 1's and 0's in g^G correspond to the adjacency matrix of G . That is, we have a correspondence

Labeled graphs $\iff$ Adjacency matrices $\iff$ Checkerboard graphons.

As an example, the adjacency matrix

$$\begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix} \quad (2.2)$$

corresponds to the graph and checkerboard graphon shown in Figure 2.1. Note how the checkerboard graphon looks like an *upside-down* version of the adjacency matrix. This is because of the conflicting conventions that the origin is at the *top* left corner of the matrix but is at the *bottom* left corner of $[0, 1]^2$.

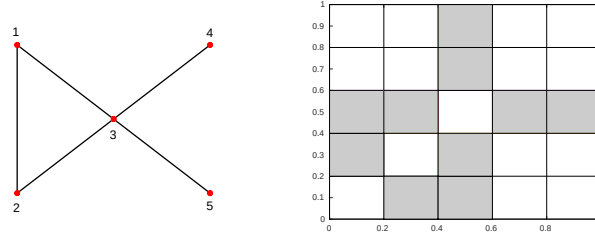


Figure 2.1 A graph and its checkerboard graphon. The graphon takes the value 0 on each white square and 1 on each gray square. Both objects correspond to the adjacency matrix (2.2).

In the topology induced by the cut metric, $\widetilde{\mathcal{W}}$ is compact [35, Theorem 5.1] and the set of reduced checkerboard graphons is dense. That is, every (reduced) graphon is a limit of (reduced) checkerboard graphons, and so can be viewed intuitively as a limit of large graphs. Accordingly, we will say that a graph G is close to a graphon g_0 if its checkerboard graphon g^G is close to g_0 in the cut metric.

In addition to the precise technical definition above, graphons can be viewed heuristically as continuous analogues of adjacency matrices. In graphs with n vertices, the vertices are labeled by integers from 1 to n and the presence or absence of an edge between vertices j and k is given by the (j, k) entry of the adjacency matrix. These integer labels can, of course, be replaced by fractions running from $1/n$ to 1. As $n \rightarrow \infty$, the discrete labels (j, k) turn into continuous parameters $(x, y) \in [0, 1]^2$.

2.3 Graphons as a list of subgraph densities

Let H be a fixed simple graph, such as an edge or a triangle. We wish to count the frequency with which H appears in a large graph G , and by extension define the density of H in a graphon. A useful tool for that is counting the number of *graph homomorphisms* from H to G . That is, of maps from the vertices of H to the vertices of G that send edges to edges. The only difference between graph homomorphisms and ordered embeddings of H in G is that a graph homomorphism might send two or more vertices of H to the same vertex in G . Since H is fixed and we are interested in the limit as the size of G goes to infinity, this distinction will turn out to be unimportant.

Suppose that H has ℓ vertices with edge set $E(H)$ and that G has n vertices. In terms of the adjacency matrix A of G , the number of graph homomorphisms from H to G is

$$\text{hom}(H, G) = \sum_{i_1, \dots, i_\ell=1}^n \prod_{(j,k) \in E(H)} A_{i_j, i_k}.$$

The *density* of H in G is defined to be

$$t_{\text{hom}}(H, G) = \frac{\text{hom}(H, G)}{n^\ell}.$$

This density can also be expressed as an integral involving the checkerboard graphon g^G of G :

$$t(H, G) := \int_{[0,1]^\ell} \prod_{(j,k) \in E(H)} g^G(x_j, x_k) dx_1 \cdots dx_\ell,$$

We then define the density of a graph H in an arbitrary (not necessarily checkerboard) graphon g to be

$$t(H, g) := \int_{[0,1]^\ell} \prod_{(j,k) \in E(H)} g(x_j, x_k) dx_1 \cdots dx_\ell.$$

The function $t(H, g)$ is invariant under measure-preserving transformations of $[0, 1]$ and so can be viewed as a function on $\mathcal{W}$. This function is continuous in the cut metric. If $g_1, g_2, \dots$ represent reduced graphons that converge to g_∞ in the cut metric, then $t(H, g_i)$ converges to $t(H, g_\infty)$.

It turns out that convergence of graphons g_i in the cut metric is equivalent to the convergence of $t(H, g_i)$ for *all* subgraphs H [34, Lemma 4.1]. That is, if we make a list $H_1, H_2, \dots$ of all possible subgraphs, then we can associate

to each reduced graphon g a sequence $\{t(H_i, g)\}$ of subgraph densities. Convergence of graphons in the cut metric is then equivalent to convergence of the corresponding sequences in the product topology.

Another way to phrase this correspondence is that the cut metric on $\widetilde{\mathcal{W}}$ generates the same topology as the metric

$$\delta_{dens}(\tilde{f}, \tilde{g}) \equiv \sum_{j \geq 1} \frac{1}{2^j} |t(H_j, f) - t(H_j, g)|,$$

where $\{H_j\}$ is any fixed sequence of all simple graphs, one from each graph-equivalence class. It provides a homeomorphism between $\widetilde{\mathcal{W}}$ and the set of sequences of achievable subgraph densities $t(H_j, g)$. This use of densities constitutes a change of scale; subgraph densities are at a scale different from individual edges in a finite graph, and we now have a mental picture of reduced graphons only using such large scale quantities. This connects the discussion in Chapter 1 concerning macroscopic versus microscopic properties with constrained graphs.

This change of scale in constrained graphs is similar to that achieved when statistical mechanics, based on probability distributions on finite collections of particles, is used to derive the macroscopic properties of bulk materials as analyzed in thermodynamics. A mathematical treatment of statistical mechanics uses probability distributions on *infinite* configurations of particles, so-called “Gibbs states”, characterized as solutions of the Dobrushin-Lanford-Ruelle (DLR) equations [69].

A key issue is that our results for infinite systems, and in particular phase transitions, should be relevant to large finite graphs. This is discussed in Section 2.6. While the bulk of this book is devoted to studying graphons, we always bear in mind the implications for actual finite graphs.

2.4 Graphons as random processes

Given a graphon g , we can generate a random graph with n vertices as follows:

1. Pick n numbers $x_1, \dots, x_n$ from $[0, 1]$, uniformly and independently.
2. Make $\binom{n}{2}$ independent biased coin flips to determine which of the $\binom{n}{2}$ possible edges are actually present. Vertices i and j are connected with probability $g(x_i, x_j)$.

To understand what this process is doing, we examine a graph G with n vertices generated by a graphon g and estimate the number of edges and triangles

it contains. Let H_e be a connected graph on two vertices and H_t be a triangle graph on three vertices.

Consider two vertices in G with coordinates x and y . The probability that they are connected by an edge is $g(x, y)$. The coordinates x and y are themselves chosen independently and uniformly, so the overall probability that two vertices are connected is

$$\int_0^1 \int_0^1 g(x, y) dx dy.$$

This means that the expected number of edges in G is exactly

$$\binom{n}{2} \iint g(x, y) dx dy.$$

The expected value of $t_{hom}(H_e, G)$, the density of graph homomorphisms from H_e to G , is then

$$\frac{n-1}{n} t(H_e, g).$$

(The factor $\frac{n-1}{n}$ is because a potential graph homomorphism might send both vertices of H_e to the same vertex of G .) As n goes to infinity, this approaches $t(H_e, g)$.

We now consider the variance in $t_{hom}(H_e, G)$. If two potential edges do not share a vertex, then their appearance is independent and they do not contribute anything to the variance of $t_{hom}(H_e, G)$. The fraction of edge pairs that share a vertex is $O(n^{-1})$. These do contribute to the variance. The details of the contribution depend on the graphon g , but without doing any calculations we can conclude that the variance of $t_{hom}(H_e, G)$ is $O(n^{-1})$. As $n \rightarrow \infty$, the random variable $t_{hom}(H_e, G)$ converges, in probability, to the constant $t(H_e, g)$.

The density of triangles is similar. Given three distinct vertices with coordinates x, y , and z , the probability that they form a triangle is $g(x, y)g(y, z)g(z, x)$. The coordinates are independent and uniformly distributed, so the probability that three randomly chosen distinct vertices form a triangle is

$$\iiint g(x, y)g(y, z)g(z, x) dx dy dz = t(H_t, g).$$

The fraction of triples of vertices that are distinct is $(n-1)(n-2)/n^2$, so the expectation of $t_{hom}(H_t, G)$ is

$$\frac{(n-1)(n-2)}{n^2} t(H_t, g) = t(H_t, g) + o(n^{-1}).$$

For the variance, we must consider all ways to pick six of the vertices of G (with replacement) and have the first three and the last three form triangles. When the vertex sets are distinct the two triangles are independent, contributing nothing to the variance of $t_{hom}(H_t, G)$. The fraction of 6-tuples for which the first three and the last three share a vertex is $O(n^{-1})$. We conclude that $t_{hom}(H_t, G)$ is a random variable with mean $t(H_t, g) + O(n^{-1})$ and variance $O(n^{-1})$, which then converges in probability to the constant $t(H_t, g)$ as $n \rightarrow \infty$.

The situation is similar for any fixed subgraph H with k vertices. The probability that k randomly chosen distinct vertices contain edges corresponding to those of H is exactly $t(H, g)$. The situation when two or more vertices are repeated is more complicated, (a complication that did not appear with edges and triangles) but this only happens a fraction $O(n^{-1})$ of the time. As a result, the expectation of $t_{hom}(H, G)$ is $t(H, g) + O(n^{-1})$. The variance of $t_{hom}(H, G)$ involves considering $(2k)$ -tuples of vertices and considering whether the first k and last k have edges corresponding to those of H . If the first and last k vertices are distinct, these are independent events and do not contribute to the variance of $t(H, G)$. Having shared vertices only occurs with frequency $O(n^{-1})$. As with edges and triangles, $t_{hom}(H, G)$ is a random variable with mean $t(H, g) + O(n^{-1})$ and variance $O(n^{-1})$, which then converges in probability to the constant $t(H, g)$ as $n \rightarrow \infty$.

Note that this argument is for a fixed (but arbitrary) H as $n \rightarrow \infty$. We do not consider the question of what happens when H has a size comparable to that of G .

Now we consider the distance between g^G (the checkerboard graphon built from G) and g in the cut metric. Since the graph G is random, this is a random variable. For each subgraph H_j , $t(H_j, g^G) = t_{hom}(H_j, G)$ converges to $t(H_j, g)$ in probability. This implies that $\delta_{dens}(g^G, g)$ converges to zero in probability as $n \rightarrow \infty$, which in turn implies that $\tilde{d}_{\blacksquare}(\tilde{g}^G, \tilde{g})$ converges to zero in probability. In other words,

The random process described by g generates large random graphs G that “look like” g , in the sense that their checkerboard graphons are close to g in the cut metric.

2.4.1 Graphon entropy

We next consider the entropy associated with the random generation of graphs via a graphon g . Once the coordinates $\{x_i\}$ of the n vertices have been chosen,

our process is a collection of coin flips. The Shannon entropy of flipping a coin with the probability p of getting “heads” is

$$-[p \ln(p) + (1 - p) \ln(1 - p)],$$

where this expression is understood to be zero when $p = 0$ or $p = 1$. As a result, the entropy of making $\binom{n}{2}$ coin flips with probability $g(x_i, x_j)$ becomes

$$- \sum_{i < j \leq n} g(x_i, x_j) \ln[g(x_i, x_j)] + [1 - g(x_i, x_j)] \ln[1 - g(x_i, x_j)]. \quad (2.3)$$

As $n \rightarrow \infty$, n^{-2} times the sum (2.3) approaches the integral

$$S(g) := \iint H[g(x, y)] dx dy,$$

where

$$H(p) := -\frac{1}{2}[p \ln(p) + (1 - p) \ln(1 - p)].$$

The factor of $1/2$ in the definition of $H(p)$ is to account for $\binom{n}{2}$ being approximately $n^2/2$, not n^2 . Note that $S(g)$ is not changed by measure-preserving transformations of $[0, 1]$, so S can be viewed as a function on $\widehat{\mathcal{W}}$ as well as a function on $\mathcal{W}$.

The Shannon entropy associated with the $\binom{n}{2}$ coin flips needed to generate a random graph is thus

$$n^2 S(g) + o(n^2).$$

For this reason, we call $S(g)$ the *Shannon entropy* of the graphon g . Heuristically, this measures the number of different outcomes that are reasonably likely. That is, we should expect the number of graphs whose checkerboard graphons are close to g to scale as $\exp(n^2 S(g) + o(n^2))$. If several different graphons can generate graphs with a certain property (say, having a given edge and triangle density, up to a small tolerance), then most such graphs with that property will be generated by the graphon with the greatest value of $S(g)$.

That is, counting and describing large random graphs subject to certain constraints should reduce to maximizing $S(g)$ among all graphons satisfying analogous constraints. In the next section, we will use the Large Deviations Principle of [15] to upgrade the previous sentence from a vague heuristic argument to a precise and rigorous theorem.

2.5 Boltzmann entropy and large deviations

The results of this section apply to all ensembles where we fix the densities $t(H_i, G)$ of a finite set $\{H_1, H_2, \dots, H_k\}$ of subgraphs. In practice, however, we are mostly interested in constraining just two densities, namely those of edges and of one other connected subgraph H . The reader is therefore invited to simplify the following theorems by replacing k with 2, t_1 with the edge density e , and t_2 with the density $t(H, G)$ of H 's.

The large deviations principle (LDP) of Chatterjee and Varadhan [15] is a broadly applicable tool for counting graphs with various properties. For each subset $A \subset \widetilde{\mathcal{W}}$, let $|A|$ denote the number of graphs in $\mathcal{G}_n$ that lie in A , in the sense that their corresponding checkerboard graphons lie in A . (We suppress the dependence of $|A|$ on n .)

Theorem 2.2 (*Chatterjee-Varadhan LDP*) *For any closed set $\tilde{F} \subseteq \widetilde{\mathcal{W}}$,*

$$\limsup_{n \rightarrow \infty} \frac{1}{n^2} \ln(|\tilde{F}|) \leq \sup_{\tilde{g} \in \tilde{F}} S(\tilde{g})$$

and for any open set $\tilde{U} \subseteq \widetilde{\mathcal{W}}$,

$$\liminf_{n \rightarrow \infty} \frac{1}{n^2} \ln(|\tilde{U}|) \geq \sup_{\tilde{g} \in \tilde{U}} S(\tilde{g}).$$

One can also describe the LDP in terms of a “rate function” that is minus the Shannon entropy $S(g)$. We next define the Boltzmann entropy and relate it to the Shannon entropy.

Let $Z_{t_1, \dots, t_k}^{n, \delta}$ be the number of simple graphs G on n nodes such that each density $t(H_i, G)$ is within δ of t_i . Let $B^{n, \delta}(t_1, \dots, t_k) = \ln(Z_{t_1, \dots, t_k}^{n, \delta})/n^2$ and consider

$$B(t_1, \dots, t_k) := \lim_{\delta \rightarrow 0^+} \limsup_{n \rightarrow \infty} B^{n, \delta}(t_1, \dots, t_k). \quad (2.4)$$

The consecutive limits exist by compactness and monotonicity. We call $B(t_1, \dots, t_k)$ the *Boltzmann entropy* function.

For brevity, we sometimes write $B(\mathbf{t})$, where $\mathbf{t} = (t_1, \dots, t_k)$. We also write $\tau(g)$ as shorthand for $(\tau_1(g), \dots, \tau_k(g))$

Note that the Boltzmann entropy is a function of the constraint values $t_1, \dots, t_k$, while the Shannon entropy is a function of a graphon. They are not the same! However, they are closely related. The following is a generalization of results proven in [54, 55], first for edges and triangles and then for edges and one other subgraph.

Theorem 2.3 *For any achievable k -tuple $\mathbf{t} = (t_1, \dots, t_k)$, $B(\mathbf{t}) = \max S(g)$, where the maximum is over all graphons g with $\tau(g) = \mathbf{t}$.*

Proof of Theorem 2.3 We need to define a few sets. Let U_δ be the set of graphons g with each $h_i(g)$ strictly within δ of t_i , i.e. the preimage of an open k -cube of side 2δ in t -space. Let F_δ be the preimage of the closed k -cube. Let $\tilde{U}_\delta^n$ and $\tilde{F}_\delta^n$ be the corresponding sets in $\widehat{\mathcal{W}}$. By Theorem 2.2,

$$\limsup_{n \rightarrow \infty} \frac{\ln |\tilde{F}_\delta^n|}{n^2} \leq \sup_{\tilde{g} \in \tilde{F}_\delta} S(\tilde{g}),$$

which also equals $\sup_{g \in F_\delta} S(g)$, and

$$\liminf_{n \rightarrow \infty} \frac{\ln |\tilde{U}_\delta^n|}{n^2} \geq \sup_{\tilde{g} \in \tilde{U}_\delta} S(\tilde{g}),$$

which also equals $\sup_{g \in U_\delta} S(g)$. This yields a chain of inequalities

$$\begin{aligned} \sup_{U_\delta} S(g) &\leq \liminf_{n \rightarrow \infty} \frac{\ln |\tilde{U}_\delta^n|}{n^2} \leq \limsup_{n \rightarrow \infty} \frac{\ln |\tilde{U}_\delta^n|}{n^2} \\ &\leq \limsup_{n \rightarrow \infty} \frac{\ln |\tilde{F}_\delta^n|}{n^2} \leq \sup_{F_\delta} S(g) \leq \sup_{U_{\delta+\delta^2}} S(g). \end{aligned}$$

As $\delta \rightarrow 0^+$, the limits of $\sup_{U_\delta} S(g)$ and $\sup_{U_{\delta+\delta^2}} S(g)$ are the same, and everything in between is trapped.

So far we have proven that $B(t_1, \dots, t_k)$ equals

$$\lim_{\delta \rightarrow 0^+} \sup_{g \in U_\delta} S(g).$$

This limit is manifestly at least as big as $\max S(g)$, the maximum value of $S(g)$ among graphons with each $\tau_i(g)$ exactly equal to t_i . To see that it cannot be greater, imagine a sequence of graphons g_j with each $\tau_i(g_j)$ converging to t_i , and with $S(g_j)$ greater than $\max S(g) + \epsilon$. By the compactness of $\widehat{\mathcal{W}}$, there is a subsequence whose classes in $\widehat{\mathcal{W}}$ converge to that of a graphon g_∞ . The densities τ_i are continuous in the cut metric, so $\tau_i(g_\infty) = t_i$. The entropy functional is upper-semicontinuous [15], so $S(g_\infty) \geq \max S(g) + \epsilon$, which contradicts the definition of $\max S(g)$.

□

Theorem 2.3 relates the concepts of Boltzmann and Shannon entropy. For each point $\mathbf{t}$ in a phase (defined below), $B(\mathbf{t}) = S(g_{\mathbf{t}})$, where $g_{\mathbf{t}}$ is the unique optimal graphon. Conversely, for any optimal graphon g , $S(g) = B(\mathbf{t}(g))$. However, this correspondence only applies to *optimal* graphons. This may be compared with the well-known difficulty within statistical physics of modeling entropy for nonequilibrium states [38].

The double limit in equation (2.4) specifies precisely the key feature of

“uniform random graph models”. First we consider all graphs with specified subgraph densities $t_1, \dots, t_k$, up to a small tolerance δ . Among such graphs, we consider the uniform distribution, where each graph is equally likely. We take a limit as $n \rightarrow \infty$ to obtain a result about large graphs, and finally we take a limit as the tolerance δ goes to zero.

Our particular construction involved cubes U_δ and F_δ , but we could just as well have used balls or parallelepipeds or ellipsoids. All that was really required for the construction was a set of nested open and closed neighborhoods of a point $t \in [0, 1]^k$, a uniform distribution among graphs whose densities lie in those neighborhoods, a limit as $n \rightarrow \infty$, and finally a limit as those neighborhoods shrink to a point.

Theorem 2.3 allows us to associate a well-defined set of subgraph densities to each point in the space of constraint parameters for which there is a unique optimal graphon. For any subgraph H , most graphs with constrained densities close to $\mathbf{t}$ are close to g_0 in the cut metric, and so have a density of H ’s close to that of $t_H(g_0)$. We can then think of t_H as a function of $\mathbf{t}$. With this in hand, we present the precise definition of a phase.

Definition 2.4 *A phase is a maximal connected open set in the space of constraints (the “phase space”) in which the Boltzmann entropy and each subgraph density t_H is real analytic in $\mathbf{t}$.*

Requiring analyticity for every subgraph density t_H is a very strong condition. *A priori*, it is not obvious that phases even exist! The bulk of this book is devoted to proving that phases really do exist in many models and developing tools for understanding the optimal graphon in each phase.

While the strength of the definition makes proving the existence of phases difficult, it makes proving the existence of phase *transitions* relatively easy. Once we have established the existence of two phases, we merely have to find a polynomial (or more generally an analytic) function of a finite number of subgraph densities whose value in one phase is not the analytic continuation of its value in the other. Such a function is called an *order parameter*, and is discussed in Section 8.4 Any path from one phase to the other must encounter a point where the order parameter fails to be analytic. Such a point cannot be part of any phase.

2.6 Ensembles of large finite graphs

We are studying random graphs in the limit as the number n of nodes goes to infinity. Mathematically, taking the $n \rightarrow \infty$ limit is essential, as the space $\mathcal{G}_n$ of simple graphs on n nodes is finite and discrete, so there would not be mathematically sharp transitions on that space. Phases and phase transitions are only sharply defined in infinite systems. This is why we study (reduced) graphons, which can be viewed as infinite graphs.

However, the underlying combinatorial questions are about large but *finite* graphs. This gets to the heart of what an emergent system is. Our results about graphons become more meaningful when related to properties of typical *large* graphs. The following theorem and discussion do exactly that.

As before, we consider a model with constraints on the densities of k sub-graphs $H_1, \dots, H_k$. Given an achievable ordered k -tuple $t = (t_1, \dots, t_k)$ of densities and a tolerance δ , we let U_δ be the set of (reduced) graphons g with each $\tau_i(g)$ within δ of t_i . Let $\mathcal{G}_{n,\delta}$ be the set of graphs on n vertices that lie in U_δ , in the sense that their checkerboard graphons lie in U_δ .

Theorem 2.5 *Suppose there is a unique (reduced) graphon g_0 that maximizes $S(g)$ subject to the constraints $\tau_i(g) = t_i$. Then, for any $\epsilon > 0$ and sufficiently small $\Delta > 0$ there exist positive constants N and K such that, for all $n > N$ and $0 < \delta < \Delta$, the fraction of graphs in $\mathcal{G}_{\delta,n}$ that are within ϵ of g_0 in the cut metric exceeds $1 - e^{-Kn^2}$.*

Proof Let V_ϵ denote the open set in $\widehat{\mathcal{W}}$ of graphons whose cut distance from g_0 is strictly less than ϵ , and let $\bar{V}_\epsilon$ be those graphons of distance ϵ or less. The complements V_ϵ^c (resp. $\bar{V}_\epsilon^c$) are then closed (resp. open) sets of graphons whose distance from g_0 is greater than or equal to ϵ (resp. strictly greater than ϵ). For each $\delta > 0$, let U_δ be as above. Note that the closure $\bar{U}_\delta$ is the set of graphons for which each $\tau_i(g)$ lies in the closed interval $[t_i - \delta, t_i + \delta]$.

Since $\epsilon > 0$ is fixed, we will suppress the dependence of various quantities on ϵ . By contrast, we will use subscripts to denote how quantities depend on δ and use quantities without such subscripts to denote the limit as $\delta \rightarrow 0$.

If $U_\delta \cap \bar{V}_\epsilon^c$ is empty then *all* graphs in U_δ are close to g_0 and there is nothing left to prove. Otherwise let $S_0 = S(g_0)$, let $S_{3,\delta}$ be the supremum of $S(g)$ on $U_\delta \cap \bar{V}_\epsilon^c$ and let $S_3 = \lim_{\delta \rightarrow 0} S_{3,\delta}$.

The proof proceeds in four steps:

1. Showing that $S_3 < S_0$. We then pick intermediate numbers S_1 and S_2 with $S_3 < S_2 < S_1 < S_0$ and define $K = S_1 - S_2$. (For instance, we could take $K = (S_0 - S_3)/3$, $S_1 = S_0 - K$ and $S_2 = S_0 - 2K$.)

2. Showing that, for any $\delta > 0$ and for all sufficiently large values of n , $|\mathcal{G}_{\delta,n}| > \exp(S_1 n^2)$.
3. Showing that, for δ sufficiently small and for all sufficiently large values of n , $|\mathcal{G}_{\delta,n} \cap V_\epsilon^c| < \exp(S_2 n^2)$.
4. Concluding that if δ is sufficiently small and n is sufficiently large, then $|\mathcal{G}_{\delta,n} \cap U_\epsilon^c|/|\mathcal{G}_{\delta,n}| < \exp(-K n^2)$.

Step 1: Suppose instead that $S_3 \geq S_0$. Then there exists a sequence of graphons $g_1, g_2, \dots$ in V_ϵ^c , with densities approaching $(t_1, \dots, t_k)$, with $\limsup S(g_j) \geq S_0$. As in the proof of Theorem 2.3, the compactness of $\widetilde{\mathcal{W}}$, the continuity of τ_i , and the semi-continuity of S implies the existence of a subsequential limit $g_\infty \in \bar{V}_\epsilon^c$ with densities equal to $(t_1, \dots, t_k)$ and with $S(g_\infty) \geq S_0$. Note that $g_0 \notin \bar{V}_\epsilon^c$, so $g_\infty \neq g_0$, contradicting the assumption that g_0 is a unique entropy maximizer.

Step 2: For any $\delta > 0$, the supremum of $S(g)$ over U_δ is at least S_0 , and so is strictly greater than S_1 . Since U_δ is an open set, the second half of the LDP (Theorem 2.2) tells us that

$$\liminf_{n \rightarrow \infty} \frac{1}{n^2} \ln(|\mathcal{G}_{n,\delta}|) \geq S_0 > S_1,$$

so for all sufficiently large values of n , $|\mathcal{G}_{n,\delta}| > \exp(S_1 n^2)$.

Step 3: To get an upper bound on a number of graphs, we sandwich the closed set $\bar{U}_\delta \cap V_\epsilon^c$ between the sets $U_\delta \cap V_\epsilon^c$ and $U_{\delta+\delta^2} \cap V_\epsilon^c$. Since $\lim_{\delta \rightarrow 0} S_{3,\delta+\delta^2} = \lim_{\delta \rightarrow 0} S_{3,\delta} = S_3$, which is strictly smaller than S_2 , we conclude that for all sufficiently small values of δ , $S_{3,\delta+\delta^2} < S_2$. Since $\bar{U}_\delta \cap V_\epsilon^c \subset U_{\delta+\delta^2} \cap V_\epsilon^c$, the supremum of the entropy on $\bar{U}_\delta \cap V_\epsilon^c$ is bounded by $S_{3,\delta+\delta^2} < S_2$. The second half of the LDP thus tells us that

$$\limsup_{n \rightarrow \infty} \frac{1}{n^2} \ln(|\bar{U}_\delta \cap V_\epsilon^c|) < S_2,$$

so $|U_\delta \cap V_\epsilon^c|$ grows strictly slower than $\exp(n^2 S_2)$, and in particular is eventually bounded by $\exp(n^2 S_2)$.

Step 4: Now we consider the order of operations. Given $\epsilon > 0$, we first compute S_3 and define S_1, S_2 and K . We then pick a $\delta > 0$ sufficiently small, so $|U_\delta \cap V_\epsilon^c|$ is bounded by $\exp(n^2 S_2)$ for all sufficiently large n . That is, there is a number N_1 such that the bound applies for all $n > N_1$. Meanwhile, $|\mathcal{G}_{n,\delta}|$ is at least $\exp(n^2 S_1)$ for all n greater than another constant N_2 . Pick $N = \max(N_1, N_2)$.

The upshot is that for given $\epsilon > 0$ and any sufficiently small $\delta > 0$, and for

all $n > N$:

$$\frac{|\mathcal{G}_{n,\delta} \cap V_\epsilon^c|}{|\mathcal{G}_{n,\delta}|} \leq \frac{\exp(n^2 S_2)}{\exp(n^2 S_1)} = \exp(-Kn^2).$$

□

As a guide, we consider what holds for statistical mechanics models of macroscopic materials. Imagine a scientist who wants to show that the idea of distinct phases, say distinct solid phases for a material such as iron, is a reasonable and useful description of what is seen in physical experiments. While an infinite-particle statistical mechanics model might have phases in which properties such as compressibility are given by analytic functions, an actual physical material consists of a large but finite number of molecules. Furthermore, quantities such as temperature, pressure and compressibility can never be measured exactly; there are *always* uncertainties. Finally, a finite number of measurements, say at finitely many values of the temperature T and pressure p , can never show that compressibility is a smooth function of (T, p) . All the scientist can do is show that one can measure any temperature and pressure, up to a small tolerance, and that the experimental data points of compressibility and other bulk quantities lie on smooth surfaces over (T, p) up to those tolerances.

We take the same approach to understanding constrained finite graphs. Say we prove that a unique optimal graphon varies smoothly in an open set and consider to what extent this implies smoothness in all sufficiently large finite graphs. In practice we can only consider finitely many $\mathbf{t}$ in the open set of parameter space. For graphs on a very large number n of nodes we can only determine the constraint densities $\mathbf{t}$, and nonconstraint densities, say of tetrahedra and finitely many other subgraphs, to within a tolerance δ , for instance by sampling. We prove that for any given δ there is a large enough N so that the above “data points” are consistent with the smooth surfaces over parameter space of the graphon subdensities, for all but exponentially few of the large constrained graphs.

In short, our theorems about phases and phase transitions in the world of graphons apply to large but finite graphs in *exactly* the same way that phases and phase transitions in statistical mechanics apply to large but finite samples of physical materials.

2.7 Multipodality and emergence

A graphon g is said to be n -podal if we can divide the unit interval into pieces $I_1, \dots, I_n$ such that g is constant on each “rectangle” $I_i \times I_j$. After applying a measure-preserving transformation of $[0, 1]$, we can assume that the sets I_i are consecutive intervals. We call these intervals *podes*. 2-podal and 3-podal graphons are said to be *bipodal* and *tripodal*, respectively. n -podal graphons, for any value of n , are said to be *multipodal*. (The term *multipodal* is a recent coinage; many authors refer instead to *step functions*.) An $n + m$ -podal graphon is said to have (n, m) symmetry if the graphon is invariant under permutation of the first n podes, and also under permutation of the last m .

All of the entropy-maximizing graphons that we will encounter in this book, without exception, are multipodal. In the conjectured phase diagram of Figure 1.3, each phase is labeled by two integers, indicating the (n, m) symmetry of that phase. We conjecture that all entropy-maximizing graphons in the edge-triangle model are multipodal, but this has not yet been proved.

Multipodality is important for several reasons. First, it is an example of emergence. If the unique entropy-maximizing graphon is n -podal, then Theorem 2.5 states that all but a small fraction of large graphs with the specified edge and triangle densities are approximately n -podal. That is, the vertices of such a graph can be partitioned into n groups, with the connectivity within each group, and between each pair of groups, described by a single number. In more technical terms, the largest eigenvalues of the adjacency matrix correspond to eigenvectors that are nearly constant on each group.

This partition is not imposed from outside in any way. The only external constraints are on the total number of edges and the total number of triangles. Instead, the vertices of almost all graphs *organize themselves* into podes with a certain characteristic symmetry that varies from phase to phase.

There is also a technical reason why multipodality is important. The space of n -podal graphons (up to equivalence) is finite-dimensional. Each n -podal graphon is described by the sizes of the n podes and by the $\binom{n+1}{2}$ (or fewer) distinct values that the graphon takes. The Shannon entropy $S(g)$ and every subgraph density $t_H(g)$ is an analytic function of these $n + \binom{n+1}{2} - 1$ parameters. Proving that the optimal graphon is described by parameters that are analytic functions of $\mathbf{t}$ is thus sufficient to show that we are in a well-defined phase.

Finally, multipodality presents an intriguing mystery. Many, not all, problems in the calculus of variations have solutions that are smooth solutions of differential equations. By contrast, our optimal graphons are discontinuous functions of (x, y) . Why is this?

2.8 Lagrange multipliers

For a good introduction to Lagrange multipliers and other aspects of variational calculus we recommend [75]. We develop below a variant form of Lagrange multipliers that is useful for our proofs.

Let $x(s)$ be a path on a domain $M \subset \mathbb{R}^n$, with initial position $x(0) = x_0$ and initial velocity v . For any differentiable function $f : M \rightarrow \mathbb{R}$, the *directional derivative along v of f* is the derivative, at $s = 0$, of $f(x(s))$. By the chain rule, this depends only on the vector v and not on the specific choice of path. The *gradient of f at x_0* is a vector ∇f such that the directional derivative of f along v is given by the inner product $\nabla f \cdot v$.

If we want to maximize a function $f(x)$ on M , we look for points in M where $\nabla f = 0$. This is because, if $\nabla f \neq 0$, we can vary x in a way that increases f to first order: just move in the direction of any vector v for which $\nabla f \cdot v > 0$.

Now suppose that we are trying to maximize $f(x)$ while holding some other function $t(x)$ equal to a fixed value a . The constrained maximum of f is usually not at a point where $\nabla f = 0$. Rather, it is at a point where we cannot increase f while holding t fixed. At such a point, assuming $\nabla t \neq 0$, ∇f must be a multiple of ∇t , as otherwise we could find a tangent vector v for which $\nabla f \cdot v > 0$ and $\nabla t \cdot v = 0$. By the implicit function theorem, there would then be a path along which t is constant but f is increasing. Therefore, at the constrained maximum there must be a proportionality constant λ , called a Lagrange multiplier, such that

$$\nabla f = \lambda \nabla t.$$

If there are multiple constraints, say $t_1(x) = a_1, t_2(x) = a_2, \dots, t_k(x) = a_k$, then ∇f must be a linear combination of the gradients $\nabla t_1, \dots, \nabla t_k$. That is, there are coefficients $\lambda_1, \dots, \lambda_k$ such that

$$\nabla f = \lambda_1 \nabla t_1 + \dots + \lambda_k \nabla t_k, \quad (2.5)$$

This assumes that the gradients $\nabla t_1, \dots, \nabla t_k$ are themselves linearly independent. When the gradients ∇t_i are linearly dependent, the implicit function theorem does not apply and turning a violation of (2.5) into an increase in f without changing any of the t_i 's becomes more difficult. Furthermore, even if equation (2.5) applies, the values of $\lambda_1, \dots, \lambda_k$ are not uniquely defined.

Often, the set M of possible x -values has a boundary ∂M . At a point $x_0 \in \partial M$, we must distinguish between vectors tangent to ∂M , for which the previous discussion applies verbatim, and inward-pointing vectors. We call a vector $v \in T_{x_0} \partial M$ *reversible*, as there are paths in our space with derivative v and also paths with derivative $-v$. We call an inward-pointing vector $v \in T_{x_0} M$

irreversible, as there is a path in M starting at x_0 with derivative v but not one with derivative $-v$. Every linear combination of reversible vectors is reversible, but when dealing with irreversible vectors we can only take linear combinations with positive coefficients.

For an irreversible vector w , we cannot have $\nabla f \cdot w > \sum \lambda_i \nabla t_i \cdot w$, as that would imply the existence of a path (with derivative w plus a linear combination of reversible vectors) that increased f while holding the quantities t_i fixed. However, it may be possible to have $\nabla f \cdot w < \sum \lambda_i \nabla t_i \cdot w$, as there are no paths in the $-w$ direction that would yield a contradiction. Applied to irreversible vectors, the equality (2.5) becomes an inequality:

$$\nabla f \cdot w \leq \lambda_1 \nabla t_1 \cdot w + \cdots + \lambda_k \nabla t_k \cdot w. \quad (2.6)$$

Having reviewed the procedure in finite dimensions, we turn to the setting of graphons g with specified values (e, t) of $(\varepsilon(g), \tau(g))$. There is a standard formalism for Lagrange multipliers in infinite dimensions, involving C^1 paths in a Banach manifold. This formalism, while useful, is insufficient for graphons, as many of the paths we wish to consider are not C^1 with respect to the L^2 metrics. We will take a two-step approach, first using C^1 variations to define our Lagrange multipliers and then treating other variations as irreversible. Each such irreversible variation must satisfy an inequality analogous to (2.6), but not necessarily an equality analogous to (2.5).

The simplest changes involve varying the value of our graphon g slowly at each point, as indicated by a bounded symmetric function $g_1 : [0, 1]^2 \rightarrow \mathbb{R}$. We consider a family of graphons g_s of the form

$$g_s(x, y) = g_0(x, y) + s g_1(x, y). \quad (2.7)$$

For this to make sense, we must have $0 \leq g_0(x, y) + s g_1(x, y) \leq 1$ for all sufficiently small positive s . For the change to be reversible, we must have $0 \leq g_0(x, y) + s g_1(x, y) \leq 1$ for all sufficiently small negative s as well. This can always be achieved by choosing g_1 to be supported on a subset of $[0, 1]^2$ on which $g_0(x, y)$ is bounded away from 0 and 1. For now, we do not consider variations that change $g(x, y)$ at points where $g_0(x, y) = 0$ or 1.

The resulting changes to the Shannon entropy, edge density and triangle are:

$$\begin{aligned} \Delta S &:= S(g_s) - S(g_0) = s \iint H'(g_0(x, y)) g_1(x, y) dx dy + o(s), \\ \Delta \varepsilon &:= \varepsilon(g_s) - \varepsilon(g_0) = s \iint g_1(x, y) dx dy + o(s), \\ \Delta \tau &:= \tau(g_s) - \tau(g_0) = s \iint 3G(x, y) g_1(x, y) dx dy + o(s), \end{aligned}$$

where

$$G(x, y) = \int_0^1 g_0(x, z)g_0(y, z) dz.$$

That is, the directional derivatives of S , ε and τ in the g_1 direction are given by the inner product (in L^2) of g_1 with the quantities $H'(g_0(x, y))$, 1 and $3G(x, y)$. These quantities, analogous to gradients, are called the *functional derivatives* of $S(g)$, $\varepsilon(g)$, and $\tau(g)$ with respect to g at $g = g_0$ and are denoted

$$\frac{\delta S(g)}{\delta g(x, y)}, \quad \frac{\delta \varepsilon(g)}{\delta g(x, y)}, \quad \text{and} \quad \frac{\delta \tau(g)}{\delta g(x, y)}.$$

A *column* g_x of a graphon g is the function of y obtained by fixing x :

$$g_x(y) = g(x, y).$$

Similarly, a *row* is the function of x obtained by fixing y . The quantity $G(x, y)$ is the L^2 inner product of the columns of g at x and at y .

Lemma 2.6 *If the function G is constant, then g is a constant graphon.*

Proof Suppose that G is constant. By Cauchy-Schwarz,

$$G(x, y) \leq \sqrt{G(x, x)G(y, y)} \leq \max(G(x, x), G(y, y)).$$

The only way for $G(x, y)$, $G(x, x)$ and $G(y, y)$ to all be equal is if the columns of g at x and y are identical. So the only way for G to be constant is for all of the columns of g to be identical, that is for $g(x, y)$ to be independent of x . Since $g(x, y) = g(y, x)$, $g(x, y)$ must also be independent of y , so g is a constant graphon. $\square$

We call an entropy-maximizing graphon *non-singular* if it is possible to vary $\varepsilon(g)$ and $\tau(g)$ independently via appropriate choices of $g_1(x, y)$. Since g_1 must be supported on points where $0 < g(x, y) < 1$, this involves the values of $G(x, y)$ at such points but not at points where $g(x, y) = 0$ or $g(x, y) = 1$. By Lemma 2.6, the only way that an entropy maximizer g at $t \neq e^3$ can fail to be non-singular is if $g(x, y)$ equals 0 or 1 on a set of positive measure and if $G(x, y)$ is constant on the rest of $[0, 1]^2$. In that case, we say that g is *singular*. Singular entropy maximizers appear at the boundary of the Razborov triangle, where by definition we cannot vary ε and τ independently, and on the Erdős-Rényi curve, where $g(x, y)$ and $G(x, y)$ are constant. It is not known whether they appear anywhere else in the interior of the Razborov triangle.

Theorem 2.7 *Suppose that g is a non-singular constrained entropy maximizer. Then there exist unique constants (α, β) such that*

$$H'(g(x, y)) = \alpha + \beta G(x, y) \tag{2.8}$$

almost everywhere.

Proof By assumption, the functional derivatives of ε and τ are linearly independent functions. We claim that the functional derivatives of S , ε and τ cannot be linearly independent, so $\delta S/\delta g = H'(g(x, y))$ must be a linear combination of $\delta\varepsilon/\delta g = 1$ and $\delta\tau/\delta g = 3G$, which is exactly what equation (2.8) says.

To see that $\delta S/\delta g$, $\delta\varepsilon/\delta g$ and $\delta\tau/\delta g$ cannot be linearly independent, suppose instead that they are. Then they are also linearly independent when restricted to some set $A_\epsilon = \{(x, y) | \epsilon < g(x, y) < 1 - \epsilon\}$. We can then find three functions $g_{1,1}$, $g_{1,2}$ and $g_{1,3}$, supported on A_ϵ , such that the first-order changes to (S, ε, τ) obtained by varying g in these three directions are linearly independent vectors in $\mathbb{R}^3$. For $s = (s_1, s_2, s_3) \in \mathbb{R}^3$, let

$$g_s(x, y) = g(x, y) + s_1 g_{1,1}(x, y) + s_2 g_{1,2}(x, y) + s_3 g_{1,3}(x, y).$$

Since each function $g_{1,i}$ is supported on A_ϵ ,

$$0 \leq g_s(x, y) \leq 1$$

for all sufficiently small $s \in \mathbb{R}^3$. By the inverse function theorem on $\mathbb{R}^3$, we can vary S , ε and τ independently by varying s in a neighborhood of $(0, 0, 0)$, and in particular we can increase S while holding ε and τ fixed, contradicting the assumption that g was a constrained entropy maximizer. (Note that this crucial step does not require the inverse function theorem in infinite dimensions. Just the inverse function theorem in 3 dimensions.)

Note that we have only established equation (2.8) as an identity in L^2 . Viewed pointwise, $H'(g(x, y))$ can differ from $\alpha + \beta G(x, y)$, but only on a set of measure zero.

□

Corollary 2.8 *If g is a non-singular constrained entropy maximizer, then g is bounded away from 0 and 1 almost everywhere.*

Proof Equation (2.8) holds for finite constants α and β . Since $0 \leq G(x, y) \leq 1$, $|H'(g(x, y))|$ can never be greater than $|\alpha| + |\beta|$. But $H'(u)$ diverges as $u \rightarrow 0$ and $u \rightarrow 1$, so $g(x, y)$ must be bounded away from 0 and 1 at all points where equation (2.8) applies, which is to say almost everywhere. □

Graphons that agree almost everywhere represent the same function in L^2 and the same reduced graphon, so by changing g on a set of measure zero we can get our graphon to be bounded away from 0 and 1 literally everywhere.

In addition to pointwise-small variations of the form (2.7), we can change a graphon by a macroscopic amount on a set of small measure. The most important example of this is changing the sizes of the podes in a multipodal

graphon, which requires replacing some of the columns (and rows) associated with one node with the columns (and rows) associated with a different node. Other examples might involve changing $g(x, y)$ from one number to another in a neighborhood of a specific point.

Such variations are not C^1 in $L^2([0, 1]^2)$ and we cannot directly apply the theory of Lagrange multipliers that we have just developed. However, we can still derive Lagrange-like inequalities by treating such variations like irreversible changes at the boundary of a domain.

Proposition 2.9 *Suppose that g_0 is a non-singular entropy maximizer with Lagrange multipliers α and β . Let g_s be a family of graphons obtained by changing g_0 on sets of measure s . Then*

$$\Delta S \leq \alpha \Delta \varepsilon + \frac{\beta}{3} \Delta \tau + o(s), \quad (2.9)$$

where $\Delta S = S(g_s) - S(g_0)$, $\Delta \varepsilon = \varepsilon(g_s) - \varepsilon(g_0)$, and $\Delta \tau = \tau(g_s) - \tau(g_0)$.

Proof Since g_0 is a non-singular entropy maximizer, we can vary its edge and triangle density independently using pointwise small variations, with the resulting change to the entropy being given to leading order by

$$\Delta S = \alpha \Delta \varepsilon + \frac{\beta}{3} \Delta \tau.$$

We apply these changes to cancel the changes in ε and τ caused by replacing g_0 with g_s . By construction, the combination of the two moves leaves ε and τ exactly unchanged. If (2.9) is not satisfied, then the combined move increases S by an amount that is at least of order s , plus the effect of whatever cross terms come from the interactions of the two moves. We will show that those cross terms only contribute $o(s)$, yielding a net gain in entropy, which contradicts the assumption that g_0 was a constrained entropy maximizer.

The cross terms involve changes to the graphon of size $O(s)$ on a set of measure $O(s)$. But that can only change the entropy by $O(s^2 \ln(s))$, the edge density by $O(s^2)$ and the triangle density by $O(s^2)$, resulting in an $o(s)$ change to $S - \alpha \varepsilon - \frac{\beta}{3} \tau$. $\square$

Remark 2.10 *Varying a graphon on a set of measure $s > 0$ is never reversible, as we cannot change the graphon on a set of measure $-s$. However, it sometimes happens that there is an “opposite” variation for which the changes to S , ε and τ are minus those of the original variation, at least to leading order. For instance, if a graphon contains some columns C_1 and some columns C_2 , then changing some of the C_2 ’s to match C_1 has the opposite effect of changing some of the C_1 ’s to match C_2 . When dealing with irreversible changes, we can only*

take positive linear combinations. However, by considering such “opposite” changes, we can effectively reproduce many of the constructions that apply to reversible changes.

In particular, if we have two “opposite” variations, then the inequalities (2.9) applied to both imply that each variation satisfies $\Delta S = \alpha \Delta \varepsilon + \frac{\beta}{3} \Delta \tau + o(s)$.

Of course, varying a graphon can involve both pointwise-small changes on all of $[0, 1]^2$ and large changes on a set of small measure. If a variation involves changing a graphon by up to s on a set of measure $1 - s$ and by arbitrary amounts on a set of measure s , then we can understand the effect of this variation by considering the two kinds of changes separately. The pointwise small changes are governed by Theorem 2.7 and the changes on a set of small measure are governed by Proposition 2.9. As in the proof of Proposition 2.9, the cross terms are $o(s)$. We conclude that such compound variations always satisfy the inequality (2.9).

Remark 2.11 *The situation is subtler when g is a constant (Erdős-Rényi) graphon and $t = e^3$. In that case, the functional derivatives of S , ε and τ are constant functions so equation (2.8) still applies. However, there are two important complications. First, the coefficients α, β are not unique. Each Erdős-Rényi graphon is associated with infinitely many possible pairs (α, β) . Second, the linear dependence of $\delta \varepsilon / \delta g(x, y) = 1$ and $\delta \tau / \delta g(x, y) = 3G(x, y)$ means that the inverse function theorem does not apply, so equation (2.9) does not necessarily hold for irreversible changes. Depending on the choice of (α, β) , there may indeed exist changes on sets of measure s for which $\Delta S - \alpha \Delta \varepsilon - \frac{\beta}{3} \Delta \tau$ is of order s .*

2.8.1 The worth functional

Once we have established the Lagrange multipliers (α, β) for a non-singular constrained entropy maximizer g , it is convenient to work with a quantity

$$F(g) = S(g) - \alpha \varepsilon(g) - \frac{\beta}{3} \tau(g).$$

To first order, pointwise small changes to g cannot change F , while large changes on sets of small measure can decrease F but cannot increase F .

The functionals S and ε are local, in that there is a contribution from each point $(x, y) \in [0, 1]^2$ and we integrate the local contributions to get the global quantity. If the triangle density were also local, then F would be the integral of a local density and our variational equations would come from setting the derivative of this density with respect to $g(x, y)$ equal to zero. That is, finding

a (constrained) stationary point of $S(g)$ would be equivalent to solving (2.8) at every point (x, y) .

However, the triangle density

$$\tau(g) = \iiint g(x, y)g(y, z)g(z, x) dx dy dz$$

is *not* local! It involves interactions between the values of g at the three points (x, y) , (y, z) and (z, x) . To accommodate this complication, we consider the effect of changing entire columns of g .

We define the *worth* of a column, or more generally of any function $C : [0, 1] \rightarrow [0, 1]$, to be

$$W(C) = \int_0^1 2H(C(y)) - 2\alpha C(y) dy - \beta \iint C(y)C(z)g(y, z) dy dz.$$

$W(C)$ measures the contribution per unit width of a column to $F(g)$. Specifically, let $I_s \subset [0, 1]$ be a set of measure s . Let C be a fixed column. We compute the change in F , to first order in s , resulting from changing all of the columns for $x \in I_s$ to C , and likewise changing the corresponding rows. (This leads to an ambiguity for $g(x, y)$ when both x and y are in I_s . The square where the ambiguity occurs only has area s^2 , which does not matter for our first order calculation. We can resolve the ambiguity on this square however we wish, for instance by setting $g(x, y) = 0$ or $g(x, y) = 1$.)

The contribution of $I_s \times [0, 1]$ to $\varepsilon(g)$ changes by

$$\int_{I_s} \int_0^1 (C(y) - g_x(y)) dy dx,$$

as does the contribution from $[0, 1] \times I_s$. The two sets have an overlap of area s^2 , so the total change to ε is

$$\Delta\varepsilon = 2 \int_{I_s} \int_0^1 (C(y) - g_x(y)) dy dx + O(s^2).$$

Likewise, the change to the entropy is

$$\Delta S = 2 \int_{I_s} \int_0^1 (H(C(y)) - H(g_x(y))) dy dx + O(s^2).$$

For the triangle count, we look for triangles with coordinates x , y , and z , at least one of which is in I_s . We do this by counting the triangles with $x \in I_s$, multiplying by 3, and making $O(s^2)$ corrections for the possibility that two or more coordinates are in I_s . The resulting change in τ is

$$\Delta\tau = 3 \int_{I_s} \int_0^1 \int_0^1 g(y, z)(C(y)C(z) - g_x(y)g_x(z)) dy dz dx + O(s^2).$$

The change to F is then

$$\Delta F = \Delta S - \alpha \Delta \varepsilon - \frac{\beta}{3} \Delta \tau = \int_{I_s} (W(C) - W(g_x)) dx + O(s^2).$$

Theorem 2.12 *If g is a non-constant constrained entropy maximizer with Lagrange multipliers (α, β) determined by equation (2.8), then almost every column of g maximizes W . In particular, (almost) all columns of g have the same worth.*

Proof Suppose that a set of columns of positive measure have worths that are not maximal. Then there exists a $\delta > 0$, a hypothetical column C , and a set I_s of positive measure s such that $W(g_x) < W(C) - \delta$ for every $x \in I_s$. For fixed δ and C , we can pick s to be arbitrarily small. The change in F resulting from changing the columns in I_s to C is then

$$\int_{I_s} (W(C) - W(g_x)) dx + O(s^2) \geq s\delta + O(s^2),$$

which contradicts Proposition 2.9. $\square$

In conclusion, the local variational equations (2.8) are extremely important. We will use them throughout the book. However, they are not sufficient. We will see many examples of graphons that satisfy (2.8) at every point $(x, y) \in [0, 1]^2$, yet are not constrained entropy maximizers. They are not even stationary points of the entropy! Finding entropy maximizers requires additional tools such as Theorem 2.12. In particular, as we will see in Chapter 4, Theorem 2.12 is the key tool for maximizing $S(g)$ near the boundary of the Razborov triangle.

2.8.2 Singular entropy maximizers

We now return to singular constrained entropy maximizers. If g is such a graphon, we call $\tau(g)$ a *singular value of t* for the given edge density $e = \varepsilon(g)$. We claim that singular t 's are too sparse to matter. We begin with their measure.

Lemma 2.13 *For each $e \in (0, 1)$, the set of singular t -values has measure zero.*

Proof First suppose that we have a singular t -value with $t < e^3$. Consider the family of graphons

$$g_s(x, y) = se + (1 - s)g_0(x, y),$$

which is defined for all $s \in [0, 1]$. $S(g_s)$ is an increasing function of s . Since $g_0(x, y)$ equals 0 or 1 on a set of positive measure, $S(g_s) - S(g_0)$ scales as

$s \ln(1/s)$ as $s \rightarrow 0$. Meanwhile, we must have $\tau(g_s) > \tau(g_0)$ for small s . (If not, since $\tau(g_1) = e^3 > t$, there must be a positive s with $\tau(g_s) = t$. But then $S(g_s) > S(g_0)$, contradicting the assumption that g_0 was entropy-maximizing.) However, $\tau(g_s)$ is a polynomial in s and cannot grow faster than linearly for small s . Thus

$$\lim_{s \rightarrow 0^+} \frac{S(g_s) - S(g_0)}{\tau(g_s) - \tau(g_0)} = +\infty.$$

Since the Boltzmann entropy $B(e, \tau(g_s))$ is at least $S(g_s)$, the Boltzmann entropy, viewed as a function of triangle density for fixed edge density, must be increasing infinitely fast at t . That is, t is a singular point of the measure dB . Since dB is a finite measure, the set of singular t 's between 0 and e^3 has measure zero.

The same arguments work for $t > e^3$, only with $\tau(g_s)$ being a decreasing function of s and with B decreasing at infinite rate at singular t -values. $\square$

We can treat singular maximizers with $t < e^3$ as having $\beta = +\infty$. There is a critical threshold μ for $G(x, y)$, with $G(x, y) = \mu$ at all points where $0 < g(x, y) < 1$. All points (x, y) with $G(x, y) > \mu$ have $g(x, y) = 0$ and all with $G(x, y) < \mu$ have $g(x, y) = 1$. Otherwise, we could decrease t while increasing S by taking g_1 to be negative where G is larger and positive where G is smaller. Likewise, we can view singular maximizers with $t > e^3$ as having $\beta = -\infty$.

Throughout this book, we use Lagrange multipliers to characterize entropy-maximizing graphons in various regions of the Razborov triangle. In each case, we show that the Boltzmann entropy is an analytic function of (e, t) . However, those arguments assume that α and β exist, which assumes that t is non-singular. The following theorem shows that working with such non-singular t 's is sufficient.

Theorem 2.14 *Fix the edge density e and consider an open interval I of triangle densities. Suppose that there is a differentiable function $f(t)$, defined for all $t \in I$, that equals the Boltzmann entropy $B(e, t)$ whenever t is non-singular. Then every $t \in I$ is non-singular.*

Proof Below the Erdős-Rényi curve, the Boltzmann entropy is a strictly increasing function of t . The set of singular t 's has measure zero, so the Boltzmann entropy equals $f(t)$ on a dense subset of $I \cap (0, e^3)$. Since $f(t)$ is continuous (being differentiable), $B(e, t)$ is continuous in t and equals $f(t)$ on all of $I \cap (0, e^3)$. But then $B(e, t)$ is differentiable at every point in $I \cap (0, e^3)$, so every $t \in I \cap (0, e^3)$ is non-singular. A similar argument applies to $I \cap (e^3, e^{3/2})$.

Finally, the Boltzmann entropy is never differentiable at $t = e^3$ (see Theorem 3.6), so $e^3 \notin I$. $\square$

In practice, we will not explicitly distinguish between singular and non-singular entropy maximizers in this book. Instead, we will do calculations using Lagrange multipliers as if all entropy maximizers were non-singular. When proving that certain phases exist, we will show that the optimal graphons in a given region are unique and of a particular form, with an entropy that depends analytically on (e, t) . This approach is justified, a posteriori, by Theorem 2.14. Since our arguments apply for almost every t (that is, at all non-singular values of t) and yield a differentiable entropy function, they actually apply everywhere.

Singular entropy maximizers may or may not exist at phase boundaries in the Razborov triangle not analyzed in this book, with the Lagrange multipliers associated with a phase diverging as we approach the boundary. This has been ruled out in some cases, as on the hypothesized boundary between the B(1,1) and A(3,0) phases [27], but has not been ruled out in general.

3

Basic results for the edge-triangle model

In this chapter we begin our detailed analysis of the edge-triangle model. That is, for each ordered pair (e, t) of edge and triangle densities we look for the graphons g that maximize the Shannon entropy $S(g)$ subject to the constraints

$$\varepsilon(g) = e; \quad \text{and} \quad \tau(g) = t.$$

We then see how the answer varies with e and t . Here we collect the low-hanging fruit. In later chapters we will reach for the higher branches.

3.1 Methods

There are two essential goals to this chapter. The first is to develop basic results about the edge-triangle model. The second is to develop basic proof strategies. There are three such strategies that we will use over and over again. The first two appear in this chapter. The third will appear in subsequent chapters, as will some more sophisticated methods.

1. **Continuity and compactness.** Let $Q(g)$ be a function (such as edge density, triangle density, or entropy) of the graphon g . Subject to certain constraints, we wish to determine the possible values of Q . If Q and the constraints are continuous (or in some cases semi-continuous) in the cut metric, then the compactness of the space of reduced graphons implies that there are maximum and minimum values $Q_{\max}$ and $Q_{\min}$ of $Q(g)$, and graphons $g_{\max}$ and $g_{\min}$ that achieve these values. If we can construct a suitable path of graphons from $g_{\max}$ to $g_{\min}$ then all values of Q between $Q_{\max}$ and $Q_{\min}$ are possible. That is, the set of possible values of Q is either a closed interval $[Q_{\min}, Q_{\max}]$ (if $Q_{\max}$ is strictly greater than $Q_{\min}$) or a single point.

2. **Saturated inequalities.** We bound the value of $Q(g)$ using a sequence of inequalities. If we can find a graphon g_0 that saturates all of the inequalities, then $Q(g_0)$ is the maximum value of $Q(g)$. If g_0 is the only graphon (up to the usual equivalences) that saturates the inequalities, then g_0 represents the unique reduced graphon that maximizes Q . Uniqueness is especially important when Q is the Shannon entropy, since (as explained in Section 2.6) it implies that all but exponentially few of the large graphs with edge and triangle densities close to (e, t) are close to g_0 .
3. **Competing inequalities.** In many cases we cannot find a graphon that saturates all of the inequalities simultaneously. However, if we find an explicit graphon g_0 that comes *close* to saturating each inequality, and therefore comes close to achieving the upper bound on the quantity Q , then we have an upper bound on the extent to which the true Q -maximizer can fail to satisfy each inequality. This implies that the true maximizer is close, in some integral norm, to g_0 . We can then find the true maximizer using perturbation theory around g_0 .

The details of the proofs vary from case to case, of course, but those are our big guns. The Index to this book contains links to the various proof strategies and methods.

3.2 The Razborov triangle

The most basic question about the edge-triangle model is

What are the possible values of (e, t) ?

Part of the answer is simple. We will show that, for each value of $e \in (0, 1)$, the set of possible values of t is an interval. We will compute the upper bound for all e . We will compute the lower bound for $e \leq 1/2$. What remains is computing the lower bound for $e > 1/2$. This is a hard problem that was only solved by Razborov [58] in 2008. See [3] for background on extremal combinatorics.

Theorem 3.1 *For each $e \in (0, 1)$, the set of possible values of t is a closed interval $[t_{\min}, t_{\max}]$, where $t_{\max}$ is strictly greater than $t_{\min}$.*

Proof This proof uses continuity and compactness. We will spell out the details in this instance, as an illustration of the method.

Let $\mathcal{G}_e$ be the set of graphons with $\varepsilon(g) = e$. This set is non-empty, since it contains the Erdős-Rényi graphon

$$g_{\text{ER}}(x, y) = e.$$

Let $t_{\max}$ and $t_{\min}$ be the supremum and infimum of τ on $\mathcal{G}_e$, and let $g_1, g_2, \dots$, be a sequence of graphons with $\lim \tau(g_n) = t_{\max}$. These represent classes $\tilde{g}_1, \tilde{g}_2, \dots$ of reduced graphons. By the compactness of the space of reduced graphons, there is a subsequence that converges to a reduced graphon $\tilde{g}_{\max}$ which is represented by a graphon $g_{\max}$. By the continuity of τ and ε ,

$$\tau(g_{\max}) = \lim \tau(g_n) = t_{\max} \quad \text{and} \quad \varepsilon(g_{\max}) = \lim \varepsilon(g_n) = \lim e = e.$$

Likewise, there is a graphon $g_{\min}$ with $\tau(g_{\min}) = t_{\min}$.

Next we show that $t_{\max} > t_{\min}$. This follows from the fact that the graphon

$$g_M(x, y) = \begin{cases} 1 & x < \sqrt{e}, y < \sqrt{e}, \\ 0 & \text{otherwise,} \end{cases}$$

has

$$\tau(g_M) = e^{3/2} > e^3 = \tau(g_{\text{ER}}).$$

Finally, we show that all values of τ from $t_{\min}$ to $t_{\max}$ are achievable. For each $s \in [0, 1]$, let

$$g_s = s g_{\max} + (1 - s) g_{\min}.$$

Since ε is a linear functional,

$$\varepsilon(g_s) = s \varepsilon(g_{\max}) + (1 - s) \varepsilon(g_{\min}) = s e + (1 - s) e = e.$$

Meanwhile, $\tau(g_s)$ is a continuous function of s . Since $\tau(g_0) = t_{\min}$ and $\tau(g_1) = t_{\max}$, the intermediate value theorem implies that each value of τ between $t_{\min}$ and $t_{\max}$ is achieved by at least one graphon g_s . $\square$

Theorem 3.2 *For any $e \in [0, 1]$, the maximum value of $\tau(g)$ is $t_{\max} = e^{3/2}$ and is uniquely achieved by the class of g_M .*

Proof For this proof, we use saturated inequalities. Viewed as an integral operator on $L^2([0, 1])$, a graphon g is compact and Hilbert-Schmidt. This implies that it is diagonalizable with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$.

The triangle density is the trace of g^3 :

$$\tau(g) = \iiint_{[0,1]^3} g(x, y) g(y, z) g(z, x) dx dy dz = \sum_{j=1}^{\infty} \lambda_j^3.$$

By Jensen's inequality applied to the function $x^{3/2}$,

$$\sum_{j=1}^{\infty} \lambda_j^3 \leq \left(\sum_{j=1}^{\infty} \lambda_j^2 \right)^{3/2}, \quad (3.1)$$

with equality if and only if g has rank 1 and $\lambda_1 \geq 0$. Meanwhile, the fact that $0 \leq g(x, y) \leq 1$ pointwise implies that $g(x, y)^2 \leq g(x, y)$, so

$$\sum_{j=1}^{\infty} \lambda_j^2 = \iint_{[0,1]^2} g(x, y)^2 dx dy \leq \iint_{[0,1]^2} g(x, y) dx dy = e, \quad (3.2)$$

with equality if and only if $g(x, y)$ is pointwise equal to 0 or 1.

Taken together, inequalities (3.1) and (3.2) imply that $t_{\max} \leq e^{3/2}$. It is easy to see that g_M saturates both inequalities and actually achieves that upper bound.

What remains is showing that g_M is the unique graphon, up to equivalence, that maximizes τ . To saturate inequality (3.1), g must have rank 1 with positive eigenvalue, so $g(x, y) = v(x)v(y)$ for some function $v : [0, 1] \rightarrow [0, 1]$. To saturate inequality (3.2), $g(x, y)$ (and therefore $v(x)$) must be pointwise equal to 0 or 1. That is, $v(x)$ is the indicator function of a subset I of $[0, 1]$ of measure c . The overall edge density is then c^2 , so $c = \sqrt{e}$. By applying a measure-preserving transformation, we can take I to be the interval $[0, \sqrt{e}]$. But then $g = g_M$. $\square$

Theorem 3.3 When $0 \leq e \leq \frac{1}{2}$, $t_{\min} = 0$.

Proof Since the triangle density is *a priori* non-negative, we just have to display a graphon with edge density e and triangle density 0. Let

$$g_m = \begin{cases} 2e & x < \frac{1}{2} < y, \\ 2e & y < \frac{1}{2} < x, \\ 0 & \text{otherwise.} \end{cases}$$

This graphon generates bipartite graphs, where vertices with coordinates less than $1/2$ are only connected to vertices with coordinates greater than $1/2$, and vice-versa. Such graphs have no triangles. $\square$

Note that, for $e \in (0, \frac{1}{2})$, the entropy of g_m is $\frac{1}{2}H(2e)$, which is positive. At the top boundary ($t = e^{3/2}$) the entropy is zero, but on the bottom boundary ($t = 0$, $0 < e < 1/2$), it is not. However, the entropy is only positive on the interior of the interval $(0, \frac{1}{2})$. At $e = 0$ and $e = \frac{1}{2}$, the entropy is once again zero.

When $e > 1/2$, the triangle density is always positive. The graph of $t_{\min}$ as a function of e is a series of “scallop”, taking one form when $\frac{1}{2} \leq e \leq \frac{2}{3}$, another when $\frac{2}{3} \leq e \leq \frac{3}{4}$, another when $\frac{3}{4} \leq e \leq \frac{4}{5}$, etc. The proof is difficult, as is determining the form of the graphons that achieve $\tau(g) = t_{\min}$. We state the results in Theorem 3.4; for the proof see [50].

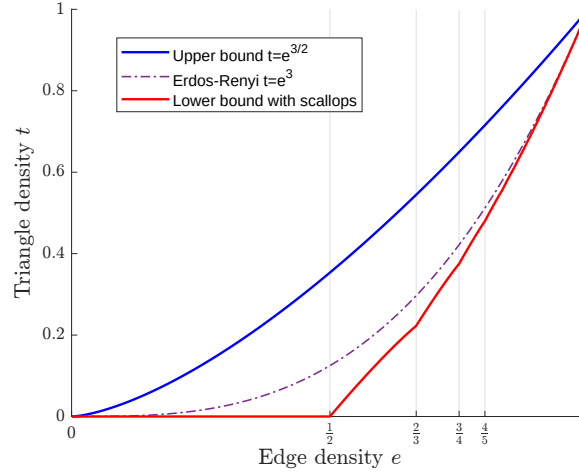


Figure 3.1 The Razborov triangle. The lower boundary has cusps at $e = (k-1)/k$ and $t = 2e^2 - e$. The curve between successive cusps is called a “scallop”.

Theorem 3.4 (Pikhurko and Razborov [50]) *Let $k > 1$ be an integer. When $1 - \frac{1}{k} \leq e \leq 1 - \frac{1}{k+1}$, the minimum triangle density is*

$$t_{\min} = \frac{k(k-1)}{(k+1)^2} \left(1 + \sqrt{1 - e \frac{k+1}{k}} \right)^2 \left(1 - 2\sqrt{1 - e \frac{k+1}{k}} \right).$$

The triangle-minimizing graphons all have the following form. We divide the unit interval into $k+1$ sub-intervals $I_1, \dots, I_k$, the first $k-1$ of which have length

$$c = \frac{1 + \sqrt{1 - e \frac{k+1}{k}}}{k+1}.$$

We set $g(x, y) = 1$ when $x \in I_i$ and $y \in I_j$ and $i \neq j$. We set $g(x, y) = 0$ when x and y are in I_j with $j < k$. The graphon on $I_k \times I_k$ is not constrained, except to have the correct average value, and not to allow any triangles where all three vertices are in I_k .

The set of all possible ordered pairs (e, t) is roughly triangular in shape and is sometimes called the “Razborov triangle”. See Figure 3.1, which reproduces Figure 1.2.

When e is not $1 - \frac{1}{k}$ or $1 - \frac{1}{k+1}$, there are minimizing graphons with positive entropy, thanks to the $I_k \times I_k$ square. Our first entropy-maximization problem

is to determine the graphon(s) that maximize $S(g)$ subject to the constraints $\varepsilon(g) = e$ and $\tau(g) = t_{\min}$.

Theorem 3.5 *We classify the graphons that maximize entropy subject to the constraints $\varepsilon(g) = e$ and $\tau(g) = t_{\min}$, where the dependence of $t_{\min}$ on e is given by Theorems 3.3 and 3.4. When $e \leq 1/2$, the unique entropy-maximizing graphon is g_m , (3.2). When $k > 1$ and $1 - \frac{1}{k} < e < 1 - \frac{1}{k+1}$, the unique entropy-maximizing graphon is modeled as follows on g_m . The interval I_k is split into two pieces $I_k^{(1)}$ and $I_k^{(2)}$ of equal size, with $g(x, y)$ equaling zero on $I_k^{(1)} \times I_k^{(1)}$ and $I_k^{(2)} \times I_k^{(2)}$ and taking a constant value on $(I_k^{(1)} \times I_k^{(2)}) \cup (I_k^{(2)} \times I_k^{(1)})$.*

Proof Let g be a graphon with $\tau(g) = 0$. That is, for almost every $(x, y, z) \in [0, 1]^3$, $g(x, y)g(y, z)g(z, x) = 0$. Let $\bar{g}(x, y)$ be a graphon that is 1 wherever $g(x, y)$ is nonzero and is zero wherever $g(x, y)$ is zero. Then $\tau(\bar{g}) = 0$, which is only possible if $\varepsilon(\bar{g}) \leq 1/2$. We conclude that the support of $g(x, y)$ in $[0, 1]^2$ has measure at most $1/2$ and that if it equals $1/2$ then the support takes the form $I \times I^c \cup I^c \times I$, where I is a subset of $[0, 1]$ of measure $1/2$.

Since $H(u)$ is a convex function of u , we maximize $S(g) = \iint H(g(x, y)) dx dy$ by making the support of g as large as possible and by making $g(x, y)$ constant on that support. That is, we must take $g(x, y) = 2e$ on $I \times I^c \cup I^c \times I$, where I is a subset of measure $1/2$. Without loss of generality we can take $I = [0, 1/2]$, in which case our graphon is precisely g_m .

When $e > 1/2$, the triangle-minimizing graphons are determined by Theorem 3.4, except on the square $I_k \times I_k$. On that square, we are trying to maximize the entropy for a fixed edge density subject to the condition that there are no triangles. But that is precisely the problem that we just solved. To maximize entropy, the graphon on $I_k \times I_k$ must be modeled on g_m . $\square$

When $e = 1 - \frac{1}{k}$ or $e = 1$, there is only a single graphon with $\tau(g) = t_{\min}$ and this graphon has zero Shannon entropy. As a result, the Boltzmann entropy is zero at all of the cusps of the scallops. In between, the entropy function goes up and then down as the value of $g(x, y)$ on $(I_k^{(1)} \times I_k^{(2)}) \cup (I_k^{(2)} \times I_k^{(1)})$ increases from 0 to 1. At the end of the scallop, the intervals $I_1, \dots, I_{k-1}, I_k^{(1)}, I_k^{(2)}$ all have length $1/(k+1)$. At this point, the final interval splits in half and the process begins anew.

3.3 The Erdős-Rényi phase transition

For each (e, t) in the Razborov triangle, let $\mathcal{G}_{e,t}$ be the set of graphons g with $\varepsilon(g) = e$ and $\tau(g) = t$. We are interested in the Boltzmann entropy function

$B(e, t)$, which from Theorem 2.3 satisfies

$$B(e, t) = \max_{g \in \mathcal{G}_{e,t}} S(g).$$

When the maximum is achieved by a unique (reduced) graphon, Theorem 2.5 tells us that this graphon also governs the structure of all but exponentially few large graphs.

As noted earlier, a *phase* is a maximal connected open set on which $B(e, t)$ and every subgraph density t_H are analytic functions of e and t . Where $B(e, t)$ or some t_H changes from one analytic function to another, we say that there is a *phase transition*. While non-analyticity on a set of constraints is a key feature of a phase transition, the real challenge is proving that $B(e, t)$ and t_H are analytic near the set. That is, proving that the set forms the boundary between two genuine phases.

The Erdős-Rényi curve $t = e^3$ is in fact the boundary between phases. In this section we will prove the easy part, namely that $B(e, t)$ is not differentiable when $t = e^3$. In Chapter 7 we will prove that the region just above the Erdős-Rényi curve, where t is slightly greater than e^3 , is part of a single phase that we call F(1,1). In Chapter 8 we will prove that the region just below the Erdős-Rényi curve, with $e > 1/2$, forms another phase that we call B(1,1). Taken together, these results prove that the portion of the Erdős-Rényi curve with $e > 1/2$ is the boundary between two bone-fide phases. (It appears that there are also at least two phases just below Erdős-Rényi when $e < 1/2$, but this has not yet been proven rigorously. See Theorem 8.13 and Section A.3.)

Theorem 3.6 *The function $B(e, t)$ is not differentiable at any point on the Erdős-Rényi curve $t = e^3$.*

Proof The function $H(u)$ is convex. This means that, if we fix e but not t , the maximum possible entropy of a graphon is achieved when the graphon is constant. Such a constant (Erdős-Rényi) graphon has $t = e^3$. Therefore, the maximum value of $B(e, t)$ for fixed e is $H(e)$ and is achieved when $t = e^3$.

For other values of t , let $\Delta t = t - e^3$ and let $\Delta B = B(e, t) - H(e)$. If the function $B(e, t)$ (viewed as a function of t for fixed e) were differentiable at $t = e^3$, then $|\Delta B|$ would be $o(\Delta t)$. Instead, we will show that $|\Delta B|$ is $\Theta(\Delta t)$ when $\Delta t > 0$ and is $\Theta(|\Delta t|^{2/3})$ when $\Delta t < 0$.

For any graphon g with edge density e , let $\Delta S(g) = S(g) - H(e)$. Since $B(e, t)$ is the maximum value of $S(g)$ over graphons with triangle density t , estimates of ΔS are easily converted to estimates of ΔB .

Consider the function

$$f(e, e') = \begin{cases} \frac{H(e') - H(e) - (e' - e)H'(e)}{(e - e')^2} & e' \neq e, \\ \frac{1}{2}H''(e) & e' = e. \end{cases} \quad (3.3)$$

Thanks to the convexity of $H(u)$, $f(e, e')$ is everywhere negative. By compactness, there is a negative number K (which turns out to equal -1) such that $f(e, e') \leq K$ for all $(e, e') \in [0, 1]^2$. See Appendix C.

Let g be a graphon in $\mathcal{G}_{e,t}$ and let

$$g_1(x, y) = g(x, y) - e.$$

Since $\iint_{[0,1]^2} g_1(x, y) dx dy = 0$, we can express the entropy of g as:

$$\begin{aligned} \Delta S(g) &= \iint H(g(x, y)) - H(e) dx dy \\ &= \iint H(g(x, y)) - H(e) - g_1(x, y)H'(e) dx dy \\ &= \iint f(e, g(x, y)) g_1(x, y)^2 dx dy \\ &\leq K \|g_1\|_2^2, \end{aligned} \quad (3.4)$$

where $\|g_1\|_2$ denotes the L^2 norm of g_1 .

Next we consider how Δt depends on g_1 . A direct computation shows that

$$\begin{aligned} \tau(g) &= \iiint (e + g_1(x, y))(e + g_1(y, z))(e + g_1(z, x)) dx dy dz \\ &= e^3 + 3e \int_0^1 \left(\int_0^1 g_1(x, z) dz \right)^2 dx \\ &\quad + \iiint g_1(x, y)g_1(y, z)g_1(z, x) dx dy dz. \end{aligned} \quad (3.5)$$

The second term on the second line is non-negative and is bounded by $3e\|g_1\|_2^2$. The triple integral on the third line can be either positive or negative and is bounded in norm by $\|g_1\|_2^3$ (just as $\tau(g)$ is bounded by $\|g\|_2^3$). There is no term proportional to e^2 , since $\iint g_1(x, y) dx dy = 0$.

When $\Delta t > 0$, we have $\Delta t = O(\|g_1\|_2^2)$, so $\|g_1\|_2$ is bounded below by a multiple of $\sqrt{\Delta t}$. Since the entropy cost is bounded below by a multiple of $\|g_1\|_2^2$, the entropy cost is bounded below by a multiple of Δt .

When $\Delta t < 0$, we have $|\Delta t| = O(\|g_1\|_2^3)$, so $\|g_1\|_2$ is bounded below by a multiple of $|\Delta t|^{1/3}$. The entropy cost is still bounded below by a multiple of $\|g_1\|_{L^2}^2$, which is now bounded below by a multiple of $|\Delta t|^{2/3}$.

□

Our lower bounds for $|\Delta S|$, and thus for $|\Delta B|$, were enough to show that the partial derivative of $B(e, t)$ with respect to t does not exist at $t = e^3$. In addition, we will show that these bounds are sharp, with $\Delta B = \Theta(\Delta t)$ when $\Delta t > 0$ and $\Delta B = \Theta(|\Delta t|^{2/3})$ when $\Delta t < 0$. To that end, we exhibit a graphon g_{sb} with t slightly less than e^3 for which $\Delta S = O(|\Delta t|^{2/3})$ and a graphon g with t slightly greater than e^3 for which $\Delta S = O(\Delta t)$.

For $t < e^3$, let $\delta = (e^3 - t)^{1/3}$ and let

$$g_{sb}(x, y) = \begin{cases} e - \delta & x, y < \frac{1}{2} \text{ or } x, y > \frac{1}{2}, \\ e + \delta & x < \frac{1}{2} < y \text{ or } y < \frac{1}{2} < x. \end{cases} \quad (3.6)$$

The subscript sb stands for “symmetric bipodal”. Graphons of this form will play a large role in the edge-triangle model. It is easy to check that

$$\varepsilon(g_{sb}) = e, \quad \tau(g_{sb}) = e^3 - \delta^3 = t,$$

and

$$\Delta S(g_{sb}) = \frac{1}{2}(H(e + \delta) + H(e - \delta) - 2H(e)) = O(\delta^2) = O(|\Delta t|^{2/3}).$$

For $t > e^3$, let $c > 0$ be a small parameter and consider the graphon

$$g(x, y) = \begin{cases} e + \frac{(2e-1)(2c-c^2)}{(1-c)^2} & x, y > c \\ 1 - e, & \text{otherwise.} \end{cases}$$

Both Δt and ΔS scale as c^1 , showing that $\Delta S = O(\Delta t)$, as claimed.

3.4 The line segment $e = 1/2, t < 1/8$

In section 4.8 of [15], devoted to open problems, Chatterjee and Varadhan asked whether it is possible to explicitly compute an optimal graphon with a “broken symmetry,” meaning one that is not constant. Their Theorem 4.3 provides barriers to finding such nonconstant optimal graphons. This problem was discussed further, and in some depth, by Chatterjee and Diaconis in [14, Theorems 4.1, 4.2 and 6.2]. The following theorem, proved in [55], was the first positive answer to Chatterjee and Varadhan’s question, giving an explicit formula for the unique and nonconstant optimal graphon when $e = 1/2$ and $t < e^3$.

Theorem 3.7 *Suppose that $e = 1/2$ and $0 < t < 1/8$. Let $\delta = \left(\frac{1}{8} - t\right)^{1/3}$. Then $B(e, t) = H(\frac{1}{2} + \delta)$ and the unique entropy-maximizing graphon is the symmetric bipodal graphon g_{sb} defined in (3.6).*

Proof We use the method of saturated inequalities; see Section 3.1. As in the proof of Theorem 3.6, we set $g_1(x, y) = g(x, y) - \frac{1}{2}$ and seek to

1. Minimize $\|g_1\|_2$ for a given $\Delta t < 0$.
2. Maximize $\Delta S(g)$ (or equivalently minimize $|\Delta S|$) for a given $\|g_1\|_2$.

We will show that g_{sb} achieves both goals and is the only graphon (up to equivalence) to do so.

By equation (3.5) and Jensen's inequality,

$$\begin{aligned} \|g_1\|_2^3 &\geq -\text{tr}(g_1^3) \\ &= 3e \int_0^1 \left(\int_0^1 g_1(x, z) dz \right)^2 dx - \Delta t \\ &\geq -\Delta t. \end{aligned}$$

We achieve equality on the first line if (and only if) g_1 has rank 1 with a non-positive eigenvalue. We achieve equality on the third line if (and only if) $\int_0^1 g_1(x, z) dz = 0$ for almost every x . Together, they imply that equality depends on having

$$g_1(x, y) = -\delta v(x)v(y)$$

for a function $v(x)$ with

$$\int_0^1 v(x) dx = 0, \quad \int_0^1 v(x)^2 dx = 1.$$

Note that the symmetric bipodal graphon g_{sb} is of the defined form, with

$$v(x) = \begin{cases} 1 & x < 1/2, \\ -1 & x > 1/2. \end{cases}$$

Next we maximize ΔS for a fixed $\|g_1\|_2$. The Taylor Series

$$H(u) = H(1/2) + \sum_{k=1}^{\infty} \frac{H^{(k)}(1/2)}{k!} \left(u - \frac{1}{2}\right)^k$$

for $H(u)$ around $u = 1/2$ is absolutely convergent for $u \in [0, 1]$. The odd derivatives are all zero. Applying this series to $g(x, y)$ and integrating over the unit square gives

$$\Delta S = \sum_{k=1}^{\infty} \frac{H^{(2k)}(1/2)}{(2k)!} \mu_{2k},$$

where

$$\mu_{2k} = \iint g_1(x, y)^{2k} dx dy.$$

Since the even derivatives $H^{(2k)}(1/2) = -(2k-2)!2^{2k-1}$ are all negative, maximizing ΔS is equivalent to minimizing all of the moments μ_{2k} . The second moment μ_2 is fixed, being $\|g_1\|_2^2$. For given second moment, all of the remaining even moments are minimized when $g_1(x, y)^2$ is pointwise constant. Once again, we see that g_{sb} fits the bill.

There are many graphons that meet the rank-1 condition with $\int_0^1 v(x) dx = 0$. There are many graphons with $g_1(x, y)^2$ constant. However, in order to meet both conditions simultaneously we must have $|v(x)|$ constant, so $v(x)$ must equal $+1$ on half of the unit interval and -1 on the other half. Without loss of generality we can assume that the two halves are $[0, 1/2]$ and $(1/2, 1]$. The resulting graphon is g_{sb} .

In short, g_{sb} maximizes the Shannon entropy $S(g)$ by satisfying both min-max goals and any other graphon that satisfies both goals is equivalent to g_{sb} .

Finally,

$$B(e, t) = S(g_{sb}) = \frac{1}{2} \left(H\left(\frac{1}{2} + \delta\right) + H\left(\frac{1}{2} - \delta\right) \right) = H\left(\frac{1}{2} + \delta\right),$$

since $H(\frac{1}{2} - \delta) = H(\frac{1}{2} + \delta)$. □

4

Phases near the boundary of the parameter space

In Chapter 3, we determined the form of the entropy-maximizing graphon on the boundary of the Razborov triangle, on the Erdős-Rényi curve $t = e^3$, and on the line segment $e = 1/2$, $0 < t < 1/8$. This was enough to see that the edge-triangle model has an interesting phase structure, with a phase transition on the Erdős-Rényi curve and with emergent behavior (namely segregation of vertices into two groups) on the $e = 1/2$ line. However, we did not prove the existence of any phases, in the sense of Definition 2.4! We solved the constrained entropy maximization problem on codimension-1 sets but not on open subsets of the Razborov triangle.

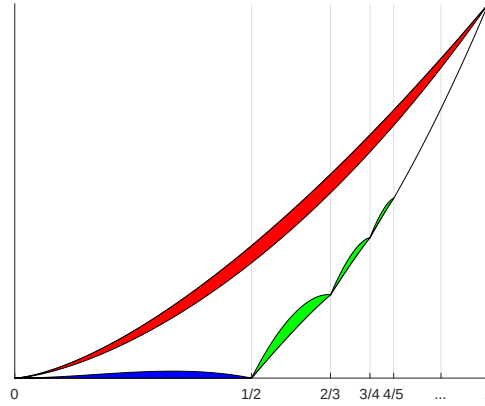


Figure 4.1 Proven phases near the boundary of the Razborov triangle

In this chapter we prove the existence of an infinite number of distinct phases near the boundary of the Razborov triangle. We solve the constrained optimization problem on a neighborhood of the flat part of the bottom boundary

in Theorem 4.1, on a neighborhood of each scallop in Theorem 4.2, and on a neighborhood of the top boundary in Theorem 4.3. See Figure 4.1, where these regions are shaded blue, green, and red, respectively. In the conjectured phase structure of Figure 1.3, these are the $A(2,0)$, $C(n,2)$, and $F(1,1)$ phases. In each case, we show that the optimal graphon is unique and varies analytically with (e, t) , implying that the Boltzmann entropy $B(e, t)$ and the density of every embedded subgraph H are analytic functions of (e, t) within the phase. It is also multipodal, with the same (n, m) symmetry as the optimal graphon on the boundary that we found in Chapter 3. For each pair of phases, we construct an *order parameter* from the densities of embedded subgraphs, with the order parameter being identically zero in one phase and never zero in the other. This shows that the phases are truly different; there cannot be an analytic continuation from one to another.

4.1 Results and Strategy

In this section we present the theorems concerning the existence of phases and sketch the common structure of their proofs. Order parameters will be developed later, in Section 4.5.

Our first result concerns graphons with edge density less than $1/2$ and triangle density close to zero, with (e, t) in the region shaded blue in Figure 4.1.

Theorem 4.1 *For each fixed $e < 1/2$ and all t sufficiently small, the optimal graphon with edge/triangle densities (e, t) is unique and $(2,0)$ -symmetric, with parameters that vary analytically with (e, t) . As $t \rightarrow 0$, the increase $\Delta B = B(e, t) - B(e, 0)$ in the Boltzmann entropy scales as $t \ln(1/t)$ and the Lagrange multiplier β scales as $\ln(1/t)$.*

We then consider graphons with edge density greater than $1/2$ and near-minimal triangle density, with (e, t) just above one of the scallops of the Razborov triangle. These are the regions shaded green in Figure 4.1.

Theorem 4.2 *Let $n \geq 1$ be an integer. For every $e \in \left(\frac{n}{n+1}, \frac{n+1}{n+2}\right)$, with corresponding minimal triangle density t_0 (depending on e), and for all Δt sufficiently small, the optimal graphon with edge/triangle densities $(e, t_0 + \Delta t)$ is unique and $n+2$ -podal, with $(n, 2)$ symmetry. Asymptotically, $\Delta B = B(e, t) - B(e, t_0)$ scales as $\sqrt{\Delta t}$, and the Lagrange multiplier β scales as $1/\sqrt{\Delta t}$. In the optimal graphon, the diagonal entries are all $\exp(-\Theta(\beta))$ and, except for the $(n+1, n+2)$ and $(n+2, n+1)$ entries, the off-diagonal entries are all $1 - \exp(-\Theta(\beta))$.*

Finally, we consider graphons with near-maximal triangle densities. This is the region shaded red in Figure 4.1.

Theorem 4.3 *For each fixed $e \in (0, 1)$ and all t sufficiently close to (but below) $e^{3/2}$, the optimal graphon with edge/triangle densities (e, t) is unique and bipodal. As $t \rightarrow e^{3/2}$, $B(e, t)$ scales as $-(e^{3/2} - t) \ln(e^{3/2} - t)$ and the Lagrange multiplier β scales as $\ln(e^{3/2} - t)$.*

The proofs of Theorems 4.1, 4.2 and 4.3 all follow the same general outline, which we now sketch. We fill in the proofs in the following sections.

We begin with our knowledge, from Theorems 3.2, 3.3, and 3.5, of the unique entropy-maximizing graphon at each point on the boundary of the Razborov triangle. We argue that any entropy-maximizing graphon g near the boundary must be close in L^2 , and not just close in the cut metric, to the unique entropy-maximizer g_0 on the boundary. This implies that all of the columns of g , aside from a set of small measure, are close in L^2 to the corresponding columns of g_0 . Since g_0 is n -podal, this makes g approximately $(n + 1)$ -podal, with n podes that are similar to the corresponding podes of g_0 and a small exceptional pode where we have no a priori estimates.

Knowing the columns outside the exceptional set (to within a small error) gives us control of $G(x, y)$ as long as neither x nor y lies in the exceptional set. The pointwise variational equations (2.8) then determine, to within small pointwise bounds, the behavior of $g(x, y)$ as long as neither x nor y is exceptional.

We then invoke Theorem 2.12. We show that any column that maximizes worth must be L^2 -close to a column of g_0 . That is, the exceptional set is empty and g is approximately n -podal, not $(n + 1)$ -podal. The pointwise bounds of the second step then apply everywhere.

Using the pointwise variational equations (2.8), we develop a bootstrap that shows that the maximum fluctuation of $g(x, y)$ within each rectangle is bounded by a small multiple of itself. This fluctuation must therefore be zero and g must be exactly n -podal.

Finally, we work in the finite-dimensional space of n -podal graphons to show that the optimal graphon g must have the same symmetry as g_0 . This is the step where the three proofs differ the most.

All three proofs take for granted that the Lagrange multipliers α and β are well-defined and finite throughout the region being studied. While this may seem like a leap of faith, it is justified by Lemma 2.13, which states that α and β are well-defined and finite for all optimal graphons for every e and *almost* every t . The proofs of Theorems 4.1, 4.2 and 4.3 then show that the optimal graphon for almost every t takes a certain multipodal form, with an entropy function

that is the restriction of a differentiable function of t . Theorem 2.14 then states that “almost every t ” actually means “every t ”, and that our analysis actually applied to *all* relevant values of t . For more details, see the end of Section 2.8.

4.2 Optimal graphons with almost no triangles

We follow the 5-step strategy outlined above to prove Theorem 4.1.

4.2.1 Defining approximate podes

By Theorem 3.3, the unique entropy maximizing graphon g_0 at $(e, 0)$ is symmetric bipodal, with podes $I_1 = [0, 1/2]$ and $I_2 = (1/2, 1]$, and with $g(x, y)$ equaling 0 on $I_1 \times I_1$ and $I_2 \times I_2$ and $2e$ on $I_1 \times I_2$ and $I_2 \times I_1$.

Now consider a sequence $t_i \rightarrow 0$ of triangle densities and let g_i be an entropy-maximizer at (e, t_i) . By the compactness of the space of reduced graphons, there is a subsequence of g_i 's that converges (after applying measure-preserving transformations) in the cut metric to a graphon g_∞ whose edge and triangle densities are $(e, 0)$. The Shannon entropy S is semi-continuous in the cut metric, so $S(g_\infty) \geq \lim S(g_i)$. However, the limit of $S(t_i)$ is at least the limit of the entropies of symmetric bipodal graphons at (e, t_i) , which is $S(g_0)$. Since $S(g_\infty) \geq S(g_0)$, and since g_0 is the unique entropy maximizer at $(e, 0)$, we must have $g_\infty = g_0$.

What's more, we claim that the entire sequence of graphons g_i (and not just a subsequence) must converge to g_0 . If it didn't, we could pick a subsequence that was bounded away from g_0 . By the argument of the previous paragraph, a subsequence of that subsequence would have to converge to g_0 , which is a contradiction.

So far we have been considering convergence in the cut metric. If the graphons g_i converged to g_0 in the cut metric but not in L^2 , then $S(g_0)$ would be strictly greater than $\lim S(g_i)$. Since we can find graphons at each t whose entropies converge to that of g_0 , this is not the case, implying that the convergence of g_i to g_0 occurs in L^2 as well as in the cut metric. That is, for any $\epsilon > 0$ and all sufficiently small t , we must have $\|g - g_0\|_2 < \epsilon$, where g is any entropy maximizer at (e, t) .

Since

$$\epsilon^2 \geq \int_0^1 dx \int_0^1 dy (g(x, y) - g_0(x, y))^2,$$

the quantity

$$\int_0^1 (g(x, y) - g_0(x, y))^2 dy$$

can only be greater than or equal to ϵ on a set I_3 of x 's of measure ϵ or less. Let I_1 and I_2 be the intersection of I_3^c with $[0, 1/2]$ and $(1/2, 1]$, respectively. That is, we have broken the unit interval into three pieces I_1, I_2, I_3 , such that:

- For all $x \in I_1$, $\|g_x - C_1\|_2 < \sqrt{\epsilon}$,
- For all $x \in I_2$, $\|g_x - C_2\|_2 < \sqrt{\epsilon}$,

where C_1 and C_2 are the columns of g_0 in the first and second nodes. Specifically, C_1 is $2e$ times the indicator functions of $(1/2, 1]$ and C_2 is $2e$ times the indicator function of $[0, 1/2]$. We will refer to the sets I_1, I_2 and I_3 as nodes, but at this point we are **not** assuming that the graphon g is exactly tripodal.

4.2.2 Estimating $G(x, y)$

$G(x, y)$ is the L^2 -inner product of g_x and g_y , which we denote $\langle g_x | g_y \rangle$. That is,

$$G(x, y) = \langle g_x | g_y \rangle = \int_0^1 g(x, z)g(y, z)dz.$$

If x and y are both in I_1 , or both in I_2 , then $G(x, y) = 2e^2 + O(\sqrt{\epsilon})$. If one is in I_1 and the other is in I_2 , then $G(x, y) = O(\sqrt{\epsilon})$. If either or both are in I_3 , then our estimates do not apply.

Now consider the variational equations (2.8), which we repeat here for convenience:

$$H'(g(x, y)) = \alpha + \beta G(x, y).$$

Since some values of g go to zero as $t \rightarrow 0$ and others do not, β must diverge as $t \rightarrow 0$. Since other values of g remain bounded away from 0 as t (and therefore ϵ , and therefore $G(x, y)$) go to zero, α either does not diverge or diverges more slowly than β . For $(x, y) \in I_1 \times I_1$ or $I_2 \times I_2$, we then have

$$H'(g(x, y)) = \alpha + 2\beta e^2(1 + O(\sqrt{\epsilon})) = 2\beta e^2(1 + O(\sqrt{\epsilon})),$$

so

$$g(x, y) = \exp(-4e^2\beta(1 + O(\sqrt{\epsilon}))). \quad (4.1)$$

We would like to apply similar reasoning to points $(x, y) \in I_1 \times I_2$, estimating $G(x, y)$ and therefore $g(x, y)$, but we don't yet have the tools. By equation (4.1), the inner product $\langle g_x | g_y \rangle$ only gets an exponentially small contribution from

I_1 and I_2 , but it might get a contribution, potentially of order $\sqrt{\epsilon}$, from I_3 . So our next step is eliminating I_3 .

4.2.3 Maximizing worth and eliminating I_3

Let $C : [0, 1] \rightarrow [0, 1]$ be a function whose worth function

$$W(C) = 2 \int_0^1 H(C(y)) - \alpha C(y) dy - \beta \iint C(y)C(z)g(y, z) dy dz.$$

we wish to estimate. Let

$$a = 2 \int_0^{1/2} C(y) dy, \quad b = 2 \int_{1/2}^1 C(y) dy.$$

That is, a and b are the average values of C on $[0, 1/2]$ and $[1/2, 1]$.

Consider the three expressions that contribute to $W(C)$:

- The entropy term $2 \int_0^1 H(C(y)) dy$ is bounded above by $H(a) + H(b)$, thanks to H'' being everywhere negative.
- The α term is exactly $-\alpha(a + b)$.
- The β term is approximately $-e\beta ab$.

Up to small deviations due to the differences between g and g_0 , maximizing $W(C)$ amounts to taking $C(y)$ to be constant on $[0, 1/2]$ and constant on $[1/2, 1]$ and choosing a and b to maximize

$$H(a) + H(b) - \alpha(a + b) - e\beta ab. \quad (4.2)$$

Setting the derivatives of (4.2) to zero gives the equations

$$H'(a) = \alpha + \beta eb, \quad H'(b) = \alpha + \beta ea.$$

Since there is a worth-maximizer with a close to 0 and b close to $2e$ (namely any column with $x \in I_1$), and another worth-maximizer with a close to $2e$ and b close to 0, α must be close to $H'(2e)$, while β is large and positive.

If $a > 1/\sqrt{\beta}$, then $e\beta a$ is gigantic and b is extremely close to zero, being $O(\exp(-\sqrt{\beta}))$. This makes βeb tiny so $H'(a) \approx \alpha \approx H'(2e)$ and $a \approx 2e$. Similarly, if $b > 1/\sqrt{\beta}$ then a is tiny and $b \approx 2e$. In both those cases, $W(C) \approx H(2e) - 2eH'(2e) = -\frac{1}{2} \ln(1 - 2e)$. A third possibility is that a and b are both less than $1/\sqrt{\beta}$, but in that case $W(C) \approx 0$, which is strictly less than $-\frac{1}{2} \ln(1 - 2e)$.

The upshot is that there are three stationary points of (4.2) but only two maxima, one that resembles C_1 and one that resembles C_2 . Since every column g_x with $x \in I_3$ must be a worth-maximizer, we can reassign the points of I_3 to I_1 or I_2 depending on the nature of g_x .

4.2.4 Exact bipodality

Our next step is to upgrade our L^2 estimates on the different columns to pointwise estimates. Each column of g is L^2 -close to a column of g_0 , so the function $G(x, y) = \langle g_x | g_y \rangle$ is pointwise close to $2e^2$ on $I_1 \times I_1$ and on $I_2 \times I_2$. By (2.8), this forces $g(x, y)$ to be exponentially small (in β) in these quadrants. This in turn makes $G(x, y)$ exponentially small on $I_1 \times I_2$ and $I_2 \times I_1$, which means that $H'(g)$ is exponentially close to α in these quadrants, so $g(x, y)$ is pointwise close to constant in these rectangles.

We next show that the optimal graphon g is *exactly* constant on each of those rectangles. Let A , B , and D be the average values of $g(x, y)$ on $I_1 \times I_1$, $I_2 \times I_2$ and $I_1 \times I_2$, respectively. Let Δ_A , Δ_B and Δ_D be the difference between the maximum and minimum values of $g(x, y)$ on those rectangles. Let c be the width of I_1 .

On $I_1 \times I_1$, the quantity $G(x, y)$ is bounded below by

$$c(A - \Delta_A)^2 + (1 - c)(D - \Delta_D)^2$$

and bounded above by

$$c(A + \Delta_A)^2 + (1 - c)(D + \Delta_D)^2.$$

The difference between these two expressions is $4cA\Delta_A + 4(1 - c)D\Delta_D$.

All points satisfy the variational equations

$$H'(g(x, y)) = \alpha + \beta G(x, y). \quad (4.3)$$

Subtracting this equation at the smallest value of $G(x, y)$ from that at the largest value, applying the mean value theorem to the left hand side, and applying our bounds on the variation in $G(x, y)$, we obtain

$$-H''(A_0)\Delta_A \leq 4Ac\beta\Delta_A + 4D(1 - c)\beta\Delta_D,$$

where A_0 is some number between $A + \Delta_A$ and $A - \Delta_A$. A little algebra then shows that

$$\Delta_A \leq \frac{4D(1 - c)\beta}{-H''(A_0) - 4Ac\beta}\Delta_D \leq \frac{3D\beta}{-H''(A)}\Delta_D,$$

where in the last step we have used the difference between $4(1 - c) \approx 2$ and 3 to cover for simplifying the denominator and replacing A_0 with A . A similar analysis on $I_2 \times I_2$ shows that

$$\Delta_B \leq \frac{3D\beta}{-H''(B)}\Delta_D.$$

Meanwhile, on $I_1 \times I_2$, $G(x, y)$ is bounded above and below by

$$c(A \pm \Delta_A)(D \pm \Delta_D) + (1 - c)(B \pm \Delta_B)(D \pm \Delta_D),$$

where the plus signs give an upper bound and the minus signs give a lower bound. The difference between the upper and lower bounds is

$$2\beta[(cA + (1 - c)B)\Delta_D + (c\Delta_A + (1 - c)\Delta_B)D].$$

This implies that

$$-H''(D_0)\Delta_D \leq 2\beta(cA + (1 - c)B)\Delta_D + 2\beta D(c\Delta_A + (1 - c)\Delta_B).$$

A little algebra then gives

$$\begin{aligned} \Delta_D &\leq \frac{2\beta D(c\Delta_A + (1 - c)\Delta_B)}{-H''(D_0) - 2\beta(cA + (1 - c)B)} \\ &\leq \frac{3\beta D(\Delta_A + \Delta_B)}{-2H''(D)} \\ &\leq \frac{9\beta^2 D^2}{-2H''(D)} \left(\frac{-1}{H''(A)} + \frac{-1}{H''(B)} \right) \Delta_D. \end{aligned}$$

Now recall that A and B are exponentially small in β and that

$$\frac{-1}{H''(A)} = 2A(1 - A) < 2A, \quad \text{and} \quad \frac{-1}{H''(B)} = 2B(1 - B) < 2B.$$

The coefficient of Δ_D on the right hand side of the last line goes to zero roughly as $\beta^2 \exp(-2e^2\beta)$ as $t \rightarrow 0$ and $\beta \rightarrow \infty$. Once the coefficient is less than one, the only solution is $\Delta_D = 0$, which then implies that $\Delta_A = 0$ and $\Delta_B = 0$. In other words, our optimal graphon is exactly bipodal.

4.2.5 Symmetric bipodality

All that remains is showing that the best bipodal graphon is symmetric, with node sizes $\frac{1}{2}$ and $\frac{1}{2}$ and with $A = B$. This requires extensive calculations but no sophisticated analysis. Ultimately, it is just a (grungy) problem in multivariable calculus.

For each triple (e, t, c) we consider the bipodal graphon that maximizes the entropy, subject to the constraints that the edge and triangle densities are (e, t) and that the first node has width c . Let $S(e, t, c)$ be the entropy of this optimal graphon. We must show that this entropy is maximized at $c = 1/2$. Note that this function is even in $\Delta c := c - \frac{1}{2}$ and analytic in c for fixed (e, t) , insofar as the parameters are determined by analytic Euler-Lagrange equations.

When $t = 0$, the function is easy to compute. The graphon must be zero on $I_1 \times I_1$ and $I_2 \times I_2$ and take on the constant value $\frac{e}{2c(1-c)} = \frac{2e}{1-4\Delta c^2}$ on $I_1 \times I_2$. The entropy is then

$$S(e, 0, c) = \frac{1}{2}(1 - 4\Delta c^2)H\left(\frac{2e}{1 - 4\Delta c^2}\right)$$

$$= S(e, 0, 1/2) + 2 \ln(1 - 2e)\Delta c^2 + O(\Delta c^4).$$

There is an $\Omega(\Delta c^2)$ entropy cost associated with having $\Delta c \neq 0$.

Now consider the effect of having t nonzero. Having the graphon nonzero on $I_1 \times I_1$ and $I_2 \times I_2$ provides additional entropy of order $t \ln(1/t)$. Shifting the value of the graphon on $I_1 \times I_2$ by an $O(t)$ amount changes the entropy by an additional $O(t)$, but since this is small compared to $t \ln(1/t)$, $S(e, t, c) - S(e, 0, c)$ is still $O(t \ln(1/t))$. In order to overcome the $-2 \ln(1 - 2e)\Delta c^2$ cost, we must have $\Delta c = O(\sqrt{t \ln(1/t)})$. Since $t \sim \exp(-2e^2\beta)$, this means that Δc must be exponentially small in β and in particular that $\beta\Delta c$ is a small parameter.

We now compute the quantity $G(x, y)$ in each rectangle and look at the Euler-Lagrange equations for a particular value of β :

$$\begin{aligned} H'(A) &= \alpha + \frac{\beta}{2}(A^2 + D^2) - \beta\Delta c(D^2 - A^2) \\ &\approx \alpha + \frac{\beta}{2}D^2 - \beta\Delta cD^2 \\ H'(B) &= \alpha + \frac{\beta}{2}(B^2 + D^2) + \beta\Delta c(D^2 - B^2) \\ &\approx \alpha + \frac{\beta}{2}D^2 + \beta\Delta cD^2 \\ H'(D) &= \alpha + \frac{\beta D}{2}(A + B + 2\Delta c(A - B)) \\ &\approx \alpha \end{aligned}$$

where in our approximations we use the fact that A and B are exponentially small in β . Since $D \approx 2e$, this makes $\alpha \approx H'(2e)$. The terms proportional to Δc serve to multiply A by a factor of $\exp(-4e^2\beta\Delta c) \approx 1 - 4e^2\beta\Delta c$ and to multiply B by a factor of $\exp(4e^2\beta\Delta c) \approx 1 + 4e^2\beta\Delta c$. These changes in the values of A and B (relative to their values when $\Delta c = 0$) slightly change the triangle density for a given value of β , but only by a fraction $O(\beta\Delta c^2)$. Likewise, the contribution to the entropy of the $I_1 \times I_1$ and $I_2 \times I_2$ squares changes by a fraction $O(\beta\Delta c^2)$. However, that entropy is only $O(t \ln(1/t))$, so we are dealing with an expression that is

$$O(\beta t \ln(1/t)\Delta c^2) = O(t(\ln(1/t))^2\Delta c^2),$$

since $\beta = O(\ln(1/t))$. This possible entropy gain from having $\Delta c \neq 0$ is much smaller than the $-2 \ln(1 - 2e)\Delta c^2$ cost, so the optimal value of Δc is exactly zero. That is, we must have $c = 1/2$.

When $c = 1/2$, two of the Euler-Lagrange equations read:

$$\begin{aligned} H'(A) &= \alpha + \beta(A^2 + D^2)/2 \\ H'(B) &= \alpha + \beta(B^2 + D^2)/2. \end{aligned}$$

If $A > B$, then the right hand side of the first equation is greater than that

of the second, so $H'(A) > H'(B)$. But that is a contradiction, since $H'(u) = \ln(1-u) - \ln(u)$ is a decreasing function of u . Likewise, we cannot have $A < B$. So A and B must be equal, making our optimal graphon symmetric bipodal.

The parameters of a symmetric bipodal graphon are uniquely (and analytically) determined by (e, t) . Specifically,

$$A = B = e - \sqrt[3]{e^3 - t}, \quad D = e + \sqrt[3]{e^3 - t}.$$

We also consider how various quantities scale as $t \rightarrow 0$. After setting $c = 1/2$ and $B = A$, a direct calculation shows that

$$t = \frac{3}{4}AD^2 + \frac{1}{4}A^3,$$

so

$$A = \frac{4t}{3D^2} + O(t^3) = \frac{t}{3e^2} + O(t^2),$$

where we have used the fact that $D = 2e - A$. Since A was exponentially small in β , β must scale as $\ln(1/t)$. The entropy is

$$\frac{1}{2}(H(A) + H(2e - A)) = \frac{1}{2}H(2e) - \frac{1}{2}A \ln(A) + O(A),$$

so $S(g) - \frac{1}{2}H(2e)$ scales as $t \ln(1/t)$.

The Boltzmann entropy $B(e, t)$ is equal to the Shannon entropy $S(g)$ of the optimal graphon at (e, t) , so $\Delta B := B(e, t) - B(e, 0) = S(g) - \frac{1}{2}H(2e)$. Since $A \approx (t/3e^2)$ and $A \approx \exp(-2e^2\beta)$, $\beta \approx \frac{-1}{2e^2} \ln(t/3e^2) \sim \ln(1/t)$. $\square$

4.3 Optimal graphons near the scallops

To prove Theorem 4.2, we follow the same script as in the proof of Theorem 4.1, with one important difference. The additional nodes that appear on the scallops provide an additional way to generate entropy at the expense of added triangles. As a result, ΔB scales as $\sqrt{\Delta t}$ rather than $\Delta t \ln(1/\Delta t)$. Before getting into the details of the proof, we explain how this works.

For simplicity, suppose that we are over the first scallop. On the scallop itself, the optimal graphon takes the form shown in Figure 4.2. The total edge density is

$$e = 2c(1 - c) + \frac{1}{2}p(1 - c)^2,$$

so we must have

$$p = \frac{2(e - 2c(1 - c))}{(1 - c)^2}. \quad (4.4)$$

The triangle density is

$$t = \frac{3}{2}pc(1-c)^2 = 3c(e - 2c(1-c)) = 3ec - 6c^2 + 6c^3.$$

Since we are minimizing t , we must have

$$0 = \frac{dt}{dc} = 3(6c^2 - 4c + e),$$

so

$$c = \frac{1}{3} \left(1 + \sqrt{1 - \frac{3e}{2}} \right).$$

Meanwhile, $d^2t/dc^2 = 36c - 12 > 0$ is always positive.

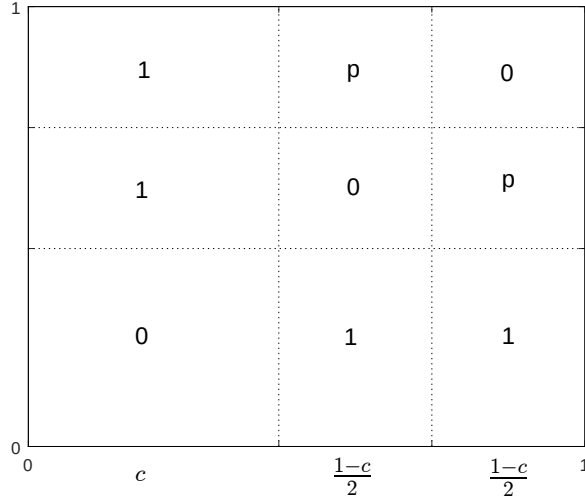


Figure 4.2 A tripodal graphon on the first scallop

Now imagine varying c and p while preserving the structure of Figure 4.2. The entropy of the graphon displayed in Figure 4.2 is

$$S = \frac{1}{2}(1-c)^2 H(p),$$

where p is given by equation (4.4). Setting the edge density equal to e gives

$$p = 4 - 4(1-c)^{-1} + 2e(1-c)^{-2},$$

so

$$\frac{dp}{dc} = 4e(1-c)^{-3} - 4(1-c)^{-2} = \frac{4(e - (1-c))}{(1-c)^3}.$$

We then compute

$$\begin{aligned}
\frac{dS}{dc} &= -(1-c)H(p) + \frac{1}{2}(1-c)^2 H'(p) \frac{dp}{dc} \\
&= -(1-c)H(p) + \frac{2H'(p)(e+c-1)}{1-c} \\
&= -(1-c)H(p) + \frac{H'(p)}{1-c} (p(1-c)^2 + 6c - 4c^2 - 2) \\
&= (1-c)(pH'(p) - H(p)) + (4c-2)H'(p) \\
&= (1-c)\ln(1-p) + 4c-2)(\ln(1-p) - \ln(p)) \\
&= (3c-1)\ln(1-p) + 2(1-2c)\ln(p).
\end{aligned}$$

Since c is between $\frac{1}{2}$ and $\frac{2}{3}$, the coefficients of $\ln(1-p)$ and $\ln(p)$ are both positive, making $\frac{dS}{dc}$ negative.

The key point is that $\frac{dS}{dc}$ is nonzero, so we can increase the entropy to first order by decreasing c . That only increases the triangle count to second order in Δc , so we have achieved an entropy increase that scales as the square root of Δt , thereby proving that $\Delta B = \Omega(\sqrt{t})$. While the optimal graphon at (e, t) isn't exactly of this form, we will see that almost all of the increased entropy comes from this shift in c and p , so $\Delta B = \Theta(\sqrt{t})$.

The situation is similar near the other scallops. Near the n -th scallop, there is a family of graphons parametrized by the size c of each of the n identical nodes. There is a value c_0 that minimizes the triangle density and $\frac{dS}{dc}$ is negative at $c = c_0$. As a result, we can increase S to first order in Δc by decreasing c while only increasing t to second order, so $\Delta S \sim \sqrt{\Delta t}$. This proves that $\Delta B = \Omega(\sqrt{t})$. Proving that the optimal graphon is approximately of this form, and hence that $\Delta B = \Theta(\sqrt{t})$, takes more work.

4.3.1 Defining approximate nodes

Let g_0 be the unique entropy maximizer g_0 at (e, t_0) . This graphon takes the form shown in Figure 4.3, with

$$c = c_0 = \frac{1 + \sqrt{1 - \frac{n+2}{n+1}e}}{n+2},$$

which is the value of c that minimizes

$$t = n(n+1)(n+2)c^3 - 3n(n+1)c^2 + 3nec.$$

As we approach the scallop, any sequence of entropy maximizers must converge to g_0 in L^2 by exactly the same argument as in the proof of Theorem 4.1.

As in the proof of Theorem 4.1, we pick a sufficiently small value of ϵ and consider values of t small enough that $\|g - g_0\|_{L^2} < \epsilon$ for each optimal graphon

1	1	1	p	0
1	1	1	0	p
1	1	0	1	1
1	0	1	1	1
0	1	1	1	1
c	$\dots$	c	$\frac{1-nc}{2}$	$\frac{1-nc}{2}$

Figure 4.3 A multipodal graphon of the form seen on the scallops, in this case with $n = 3$

g . Let $I_1, I_2, \dots, I_{n+2}$ be the subsets of the podes of g_0 for which g_x lies within $\sqrt{\epsilon}$ in L^2 of the corresponding column of g_0 and let I_{n+3} be the exceptional set where g_x is not close to the corresponding column of g_0 . Note that I_{n+3} may contain points x where g_x is close to a different column of g_0 . Those points will soon be reassigned.

4.3.2 Variational equations

We compute $G(x, y)$ in different cases. Let $G_{i,j}$ denote a typical value of $G(x, y)$ when $x \in I_i$ and $y \in I_j$. We can estimate these quantities to within $O(\sqrt{\epsilon})$ using the columns of g_0 . Thanks to the $(n, 2)$ symmetry of g_0 , there are only five different numbers to compute, namely $G_{1,1}$, $G_{1,2}$, $G_{1,n+1}$, $G_{n+1,n+1}$ and $G_{n+1,n+2}$. The results are

$$\begin{aligned}
 G_{1,1} &\approx 1 - c, \\
 G_{1,2} &\approx 1 - 2c, \\
 G_{1,n+1} &\approx (n-1)c + \frac{1-nc}{2}p, \\
 G_{n+1,n+1} &\approx nc + \frac{1-nc}{2}p^2,
 \end{aligned}$$

$$G_{n+1,n+2} \approx nc,$$

where “ $\approx$ ” means “equal to within $O(\sqrt{\epsilon})$ ”. Note that $G_{1,1}$ and $G_{n+1,n+1}$ are greater than $G_{n+1,n+2}$ by amounts that are $\Omega(1)$ as $t \rightarrow 0$ while $G_{1,2}$, and $G_{1,n+1}$ are less than $G_{n+1,n+2}$ by amounts that are $\Omega(1)$. Multiplying by β and adding α , and using the fact that

$$H'(p) = \alpha + \beta G_{n+1,n+2} = (\alpha + nc) + O(\beta\sqrt{\alpha}).$$

we get that $H'(g(x, y))$ is $\Omega(\beta)$ on the diagonal rectangles that do not involve I_{n+3} and is $-\Omega(\beta)$ on the off-diagonal blocks that do not involve I_{n+3} , with the exception of $I_{n+1} \times I_{n+2}$ and $I_{n+2} \times I_{n+1}$. This implies that $g(x, y)$ is exponentially close to 0 or 1 on all of the blocks except the two where $g(x, y) \approx p$ and the ones involving I_{n+3} .

4.3.3 Maximizing worth

Let $C : [0, 1] \rightarrow [0, 1]$ be a function whose worth $W(C)$ we aim to maximize. For each $i = 1, 2, \dots, n+2$, let a_{n+2} be the average values of $C(y)$ for $y \in I_i$. Of the terms contributing to $W(C)$, the entropy term is bounded above by $2c \sum_{i=1}^n H(a_i) + (1-nc)(H(a_{n+1}) + H(a_{n+2}))$, since fluctuations in C within each node can only decrease the entropy. The edge density term is $-2\alpha(c \sum_{i=1}^n a_i + \frac{1-nc}{2}(a_{n+1} + a_{n+2}))$.

The most important term comes from triangles. To within the accuracy of our approximation that $g(y, z)$ is constant on each rectangle, this term is the quadratic function

$$-\beta \sum_{i,j=1}^{n+2} M_{ij} a_i a_j,$$

where M_{ij} is the integral of $g(y, z)$ over $I_i \times I_j$.

If we are at a maximum of W , then the gradient of W must be zero and the Hessian must be negative semi-definite. The Hessian of W with respect to the variables $\{a_i\}$ is precisely $-2\beta M$ plus diagonal terms from the entropy term that are proportional to $H''(a_i)$. The matrix M is (nearly) zero on the diagonal, with all of the off-diagonal terms being close to 1 or p , and so has eigenvalues of both signs. The only way for the Hessian to be negative semi-definite is for all but one of the diagonal entries $H''(a_i)$ to be at least of order β . In other words, all columns that maximize W must have every entry but one (or possibly every entry) approximately equal to 0 or 1. In terms of G , for y in any node but one, $|G(x, y) - nc|$ must exceed $\Theta(1/\beta)$. We then say that a_i is “pegged to 0” or “pegged to 1”.

There are several possibilities for which variables are pegged to 0 or 1 and which (if any) are unpegged.

- If a_1 is unpegged, then $G(x, y) \approx nc$ for $y \in I_1$, so $2c \sum_{i=2}^n a_i + (1 - nc)(a_{n+1} + a_{n+2}) \approx nc$. But that is impossible if all of the other a_i 's are pegged. So a_1 must in fact be pegged. The same argument shows that $a_2, \dots, a_n$ are all pegged.
- If two or more of the variables $a_1, \dots, a_n$ are pegged to 0, then $G(x, y) < nc$ for all y , so $g(x, y) \approx 1$ for all y , which is a contradiction. Thus either one or none of the first n a_i 's is pegged to 0 and the rest are pegged to 1.
- If exactly one of these variables is pegged to 0, then for $y \in I_{n+1}$ or $y \in I_{n+2}$ we have $G(x, y) \leq p \frac{1-c}{2} + (n-1)c < nc$, so a_{n+1} and a_{n+2} are pegged to 1. In other words, C approximates a column of g_0 in one of the first n poles.
- If all of the variables $a_1, \dots, a_n$ are pegged to 1, then we examine a_{n+1} and a_{n+2} . Neither one is pegged to 1, since $G(x, y)$ is at least nc for y in either I_{n+1} or I_{n+2} . They cannot both be pegged to 0, since that would make $G(x, y) = nc$ in both poles, meaning that the values are *not* pegged and the Hessian is not negative-definite. Thus one value must be pegged to 0 while the other is intermediate between 0 and 1. The closeness of all but one a_i to 0 or 1 gives us the same equation for the remaining a_i as satisfied by the actual columns of I_{n+1} or I_{n+2} , implying that the final a_i must be close to p . That is, C approximates a column of g_0 from one of the last two poles.

The upshot is that all worth-maximizers are L^2 -close to columns of g_0 , allowing us to reassign all of the points of I_{n+3} to other poles.

Controlling the columns to within a small L^2 error gives us pointwise control over $G(x, y) = \langle g_x | g_y \rangle$. On all rectangles except for $I_{n+1} \times I_{n+2}$ and $I_{n+2} \times I_{n+1}$, this forces $g(x, y)$ to be exponentially close to 0 or 1. This in turn makes $G(x, y)$ exponentially close to constant on $I_{n+1} \times I_{n+2}$ and $I_{n+2} \times I_{n+1}$ and so makes $g(x, y)$ exponentially close to constant on these last two rectangles.

4.3.4 Exact multipodality

To show that the graphon is exactly $(n+2)$ -podal, we repeat the corresponding step of the proof of Theorem 4.1, only with more terms and slightly different notation. Let g_{ij} be the average value of the graphon on the rectangle $I_i \times I_j$ and let Δg_{ij} be the difference between the greatest and lowest value of the graphon in that rectangle. Let c_i be the width of I_i . We have already determined that all g_{ij} 's except for $g_{n+1, n+2}$ and $g_{n+2, n+1}$ are exponentially close (in β) to 0 or 1, and hence that $1/H''(g_{ij})$ is exponentially small.

If $x \in I_i$ and $y \in I_j$, then the maximum and minimum possible values of $G(x, y)$, and their difference, are

$$\begin{aligned}\max &= \sum_{k=1}^{n+2} c_k (g_{ik} + \Delta g_{ik})(g_{jk} + \Delta g_{jk}), \\ \min &= \sum_{k=1}^{n+2} c_k (g_{ik} - \Delta g_{ik})(g_{jk} - \Delta g_{jk}), \\ \text{difference} &= \sum_{k=1}^{n+2} 2c_k (g_{ik}\Delta g_{jk} + g_{jk}\Delta g_{ik}).\end{aligned}$$

Applying the mean value theorem to the Euler-Lagrange equations, and noting that $H''(u)$ is always negative, we have

$$-H''(g_{ij,0})\Delta g_{ij} \leq 2\beta \sum_{k=1}^{n+2} c_k (g_{ik}\Delta g_{jk} + g_{jk}\Delta g_{ik}),$$

where $g_{ij,0}$ is some number between the values of g corresponding to the maximum and minimum possible values of $G(x, y)$.

The sum on the right contains terms proportional to Δg_{ij} itself, coming from $k = i$ or $k = j$. We bring those terms to the left hand side, noting that coefficients of those terms are much smaller than $H''(g_{ij,0})$. When $\{i, j\} \neq \{n+1, n+2\}$, this is because $g_{ij,0}$ is exponentially close to 0 or 1, so $H''(g_{ij,0})$ is exponentially large, while the coefficients on the right hand side are $O(\beta)$. When $\{i, j\} = \{n+1, n+2\}$, this is because the coefficients of $\Delta g_{n+1, n+2}$ on the right hand side are proportional to $\beta g_{n+1, n+1}$ or $\beta g_{n+2, n+2}$, both of which are exponentially small. By changing the factor of 2 on the right hand side to a 3, we can absorb these small corrections to the coefficient of Δg_{ij} and also replace $H''(g_{ij,0})$ with just $H''(g_{ij})$. We also bound g_{ik} and g_{jk} by 1. The upshot is that

$$\begin{aligned}\Delta g_{ij} &\leq \frac{3\beta}{-H''(g_{ij})} \sum_k c_k (\Delta g_{jk} + \Delta g_{ik}) \\ &\leq \frac{6\beta}{-H''(g_{ij})} \max(\Delta g_{ik} \text{ or } \Delta g_{jk}),\end{aligned}$$

where the sum on the first line and the maximum on the second line skips terms involving Δg_{ij} itself.

Whenever $\{i, j\} \neq \{n+1, n+2\}$, $\frac{6\beta}{-H''(g_{ij})}$ is exponentially small, so Δg_{ij} is bounded by a tiny multiple of a sum of similar errors. In particular, the largest Δg_{ij} of this sort is bounded by a sum of contributions much smaller than itself, possibly plus a contribution from $\Delta g_{n+1, n+2}$. The conclusion is that all Δg_{ij} 's other than $\Delta g_{n+1, n+2}$ are bounded by $\beta \exp(-\Omega(\beta)) \Delta g_{n+1, n+2}$.

Now consider the equation for $\Delta g_{n+1,n+2}$. This equation indicates that $\Delta g_{n+1,n+2}$ is bounded by an $O(1)$ multiple of β times the largest of the remaining Δg_{ij} 's, and so is bounded by a constant times $\beta^2 \exp(-\Omega(\beta)) \Delta g_{n+1,n+2}$. When β is large, $\Delta g_{n+1,n+2}$ is thus bounded by a constant (less than one) times itself, and so must be zero. But then all of the other Δg_{ij} 's must also be zero, so our graphon is multipodal.

4.3.5 Graphons with $(n, 2)$ symmetry

Finally, we show that the optimal graphon is symmetric in the first n nodes and symmetric in the last two. Let $c_1, \dots, c_{n+2}$ be the sizes of the various nodes, let $\bar{c}$ be the average size of the first n nodes, and let Δc_i be $c_i - \bar{c}$ or $c_i - \frac{1-n\bar{c}}{2}$, depending on whether we are talking about the first n nodes or the last two. Let W_i be the worth of columns in the i -th node. There are five kinds of rectangles, namely $I_i \times I_j$ with $i = j \leq n$, with $i < j \leq n$ or $j < i \leq n$, with $i \leq n < j$ or $j \leq n < i$, with $i = j > n$, and finally with $\{i, j\} = \{n+1, n+2\}$. In each class, let $\bar{g}_{ij}$ be the average value of the graphon and let $\Delta g_{ij} = g_{ij} - \bar{g}_{ij}$. We also refer to $g_{n+1,n+2}$ as p .

Recalling our calculations for maximizing worth, all entries g_{ij} except for $g_{n+1,n+2}$ (and $g_{n+2,n+1}$) are pegged to 0 or 1. The Euler-Lagrange equations for $g_{n+1,n+2}$ say that

$$\begin{aligned} H'(p) &\approx \alpha + \beta \sum_{i=1}^n c_i, \\ \alpha &\approx H'(p) - \beta \sum_{i=1}^n c_i, \end{aligned}$$

where “ $\approx$ ” now means “equal up to exponentially small corrections”.

Suppose that $i, j \leq n$. Since each g_{ik} is pegged to 0 or 1, the entropy contributions to W_i are exponentially small. The coefficient of α in W_i is $\sum_{k \neq i} c_k = 1 - c_i$, while the coefficient of $\beta/2$ is the integral of the graphon over everything that doesn't involve the i -th node. The upshot is that

$$\begin{aligned} W_i &\approx -\alpha(1 - c_i) - \frac{\beta}{2}(e - 2c_i(1 - c_i)) \\ &= -\left(\alpha + \frac{\beta e}{2}\right) + c_i(\alpha + \beta) - c_i^2 \beta, \end{aligned}$$

with a similar result for W_j . Taking the difference gives

$$\begin{aligned} 0 &= W_i - W_j \\ &\approx (c_i - c_j)(\alpha + \beta - \beta(c_i + c_j)) \end{aligned}$$

$$\approx (c_i - c_j)(H'(p) + \beta(c_{n+1} + c_{n+2} - c_i - c_j)),$$

where we have used the fact that $\sum_{i=1}^n c_i = 1 - c_{n+1} - c_{n+2}$. However, c_{n+1} and c_{n+2} are close to $\frac{1-n\bar{c}}{2}$, while c_i and c_j are close to $\bar{c}$, so the coefficient of β is bounded away from zero. We conclude that $c_i - c_j$ must be exponentially small. More precisely, $c_i - c_j$ must be exponentially smaller than the largest $|\Delta g_{ik}|$ or $|\Delta g_{jk}|$. A similar argument shows that $c_{n+1} - c_{n+2}$ is also exponentially smaller than the largest $|\Delta g|$.

We now examine the Euler-Lagrange equations for $g_{i\ell}$ and $g_{j\ell}$, where ℓ is different from i or j . The difference between $G(x, y)$ in $I_i \times I_\ell$ and $I_j \times I_\ell$ is

$$\begin{aligned} \sum_{k=1}^{n+2} c_k g_{\ell k} (g_{ik} - g_{jk}) &= \sum_{k=1}^{n+2} c_k g_{\ell k} (\Delta g_{ik} - \Delta g_{jk}) \\ &\quad + (c_i g_{\ell i} - c_j g_{\ell j})(\bar{g}_{ii} - \bar{g}_{ij}). \end{aligned}$$

The first line is of the order of the largest Δg . The second line has a similar contribution from the difference of $\Delta g_{\ell i}$ and $\Delta g_{\ell j}$, plus a contribution of order $c_i - c_j$. But then

$$H'(g_{i\ell}) - H'(g_{j\ell}) = \beta(G_{i\ell} - G_{j\ell}),$$

which is β times a linear combination of Δg 's and $c_i - c_j$. Since H'' is enormous on the interval from $g_{i\ell}$ to $g_{j\ell}$ (both of which are exponentially close to 1), $\Delta g_{i\ell} - \Delta g_{j\ell}$ is bounded by a tiny combination of other Δg 's and Δc 's.

Repeating this argument for $g_{ii} - g_{jj}$ and for $g_{n+1, n+1} - g_{n+2, n+2}$, we get that

- The biggest Δc_i is bounded by a tiny constant times the biggest Δg .
- The biggest Δg is bounded by a tiny constant times the biggest Δc .

We conclude that all of the Δc 's and Δg 's are zero.

We have determined the form of the optimal graphon g at (e, t) . Since the Boltzmann entropy $B(e, t)$ equals the Shannon entropy $S(g)$ of the optimal graphon g , all statements about S are easily converted into statements about B .

Finally, we must show that the values of g on each rectangle, and the sizes of the different nodes, are given by analytic functions of (e, t) . This follows from a general principle in algebraic geometry that is an analytic version of the implicit function theorem. Within the product of the Razborov triangle and the finite-dimensional space of graphons with $(n, 2)$ symmetry, the set of optimal graphons is a 2-dimensional analytic variety, cut out by the analytic Euler-Lagrange equations. As long as the tangent space does not degenerate, and we have just shown that it doesn't, we can write all but two of the variables as analytic functions of the last two, which we can choose to be (e, t) .

4.4 Optimal graphons with near-maximal triangle densities

In proving Theorem 4.3, we follow the same roadmap as in the proofs of Theorems 4.1 and 4.2. Specifically,

1. Using the proximity to the upper boundary, we break $[0, 1]$ into two large podes I_1 and I_2 and a small exceptional set I_3 such that g_x is L^2 -close to the indicator function of I_1 when $x \in I_1$ and is L^2 -close to zero when $x \in I_2$.
2. Equating the worths of g_x when $x \in I_1$ to those of g_x when $x \in I_2$, we determine that $\beta/\alpha \approx -2/e$. The multiplier β is large and negative, while α is large and positive.
3. Maximizing $W(C)$ for an arbitrary $C : [0, 1] \rightarrow [0, 1]$, we show that every column is close to a typical column in the first or second pde. After reassigning points, I_3 is then empty. The control this gives us on $G(x, y)$ shows that $g(x, y)$ is everywhere exponentially close to 0 or 1.
4. Bounding the fluctuations in each rectangle by multiples of the fluctuations in other rectangles, we show that all fluctuations are in fact zero. In other words, our optimal graphon is exactly bipodal with values that are exponentially close to 0 or 1.

Step 1 is exactly as before. The unique graphon at $(e, e^{3/2})$ is bipodal and equals 1 on $I_1 \times I_1$ and zero elsewhere. Every graphon with t close to $e^{3/2}$, and in particular any entropy-maximizing graphon, must be L^2 close to g_0 . This means that for all x 's outside of a set of small measure, g_x is L^2 -close to the corresponding column of g_0 . This also implies that $G(x, y)$ is close to $\sqrt{e}$ when x and y are both in I_1 and is close to zero when either is in I_2 .

The worth of a column that is nearly zero is of course nearly zero. The worth of a column that is nearly 1 on I_1 and nearly zero elsewhere is approximately

$$-2\alpha\sqrt{e} - \beta e^{3/2}.$$

Since all columns must have the same worth, we must have $\beta/\alpha \approx -2/e$. The Lagrange multipliers α and β diverge at the same rate as we approach the boundary, with $\alpha \rightarrow \infty$ and $\beta \rightarrow -\infty$.

Now consider an arbitrary function $C : [0, 1] \rightarrow [0, 1]$. Let a be the average of $C(y)$ on I_1 and let b be the average on I_2 . Since G is L^2 -close to $\sqrt{e}$ times the indicator function of $I_1 \times I_1$,

$$W(C) \lesssim 2\sqrt{e}H(a) + 2(1 - \sqrt{e})H(b) - 2\alpha\sqrt{e}a - 2\alpha(1 - \sqrt{e})b - \beta e^{3/2}a^2,$$

with equality if C is constant on I_1 and constant on I_2 . Since α is large and positive, we must have b exponentially close to 0. Setting $b \approx 0$, our worth is

then approximately

$$2\alpha\sqrt{e}(a^2 - a).$$

This is of course maximized at the endpoints $a = 1$ and $a = 0$, being negative when $a \in (0, 1)$. In other words, any worth-maximizing column must either have $a \approx 1$ and $b \approx 0$, and so must be close to the columns in I_1 , or $a \approx 0$ and $b \approx 0$, and so must be close to the columns in I_2 . Reassigning the points of I_3 to I_1 or I_2 accordingly, we obtain a situation where I_3 is empty.

Showing that $g(x, y)$ is constant in each rectangle follows exactly the same argument as before. So does the analyticity of g as a function of (e, t) .

Finally, we do some calculations in the space of bipodal graphons. To leading order, $e^{3/2} - t$ is a linear function of $1 - g_{11}$, g_{12} , and g_{22} , where we adjust c_1 to keep the edge density fixed. However, $1 - g_{11}$, g_{12} and g_{22} all scale as exponents of β , so β must scale as $\ln(e^{3/2} - t)$. The entropy goes as $-(1 - g_{11}) \ln(1 - g_{11}) - g_{12} \ln(g_{12}) - g_{22} \ln(g_{22})$, which then scales as $-(e^{3/2} - t) \ln(e^{3/2} - t)$.

4.5 Distinct phases and rank

We now show that the phases proven in Theorems 4.1, 4.2 and 4.3 are different in the sense that no path from one of these phases to another, parameterized by a real variable, can be followed analytically.

By Theorem 3.6, there is a singularity in $B(e, t)$ along the entire Erdős-Rényi curve. This separates the phases of Theorems 4.1 and 4.2 from the phase of Theorem 4.3.

To distinguish the phases of Theorems 4.1 and 4.2, we construct a sequence of *order parameters*. These are analytic functions of subgraph densities that are never zero in a particular phase but are identically zero in other phases. Since an analytic function that is zero on a connected open set is zero everywhere, the phase where the function is nonzero cannot be an analytic continuation of any of the phases where it is identically zero.

Before we construct these order parameters, we consider the role of order parameters in general, especially in statistical mechanics, where they were originally developed. An order parameter is meant to “essentially” distinguish the states in one phase, where the parameter is zero, from the states in a different phase, where it is nonzero. For a model to exhibit multiple phases is thus a technical way of saying it exhibits essentially different states, perhaps exhibiting different symmetries. By understanding the order parameters that

distinguish between phases, we can begin to understand why a material exhibits different phases in the first place.

Figure 4.4 shows the phases of a typical material. One sees a large *fluid* phase with a curve poking into it, with a so-called *critical point* at its end. The word liquid appears on the high pressure side and gas on the low pressure side of the curve. Although there is certainly a transition as we cross this curve, there is no *essential* difference between liquid and gas. It is possible to go smoothly from one to the other by going around the critical point. As such, we only speak of a single fluid phase.

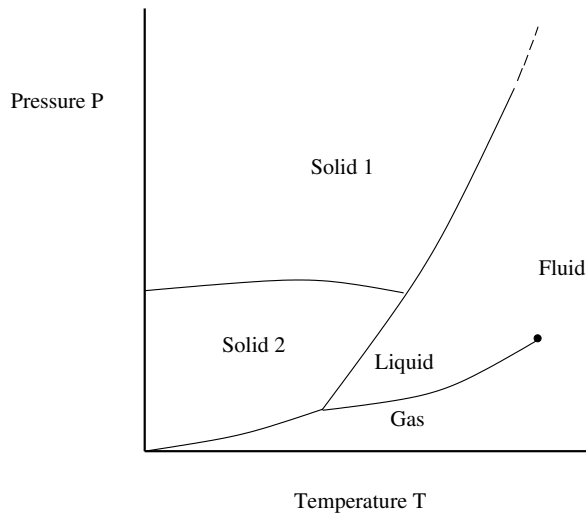


Figure 4.4 A simplified diagram of the phases of matter

By contrast, understanding the essential difference between fluids and solids in matter is a major unsolved problem [72, 64, 76]. Physicists invented order parameters for this purpose. We quote here [5, page 19] an attempt to do this:

It was Landau (1958) who, long ago, first pointed out the vital importance of symmetry in phase transitions. This, the First Theorem of solid-state physics, can be stated very simply: it is impossible to change symmetry gradually. A given symmetry element is either there or it is not; there is no way for it to grow imperceptibly. This means, for instance, that there can be no critical point for the melting curve as there is for the boiling point: it will never be possible to go continuously through some high-pressure phase from liquid to solid.

While appealing, Landau's argument *as applied to the fluid/solid phase transition* is by no means generally accepted; see [49, page 122]. We do not in fact know a convincing order parameter to explain this transition. (For a nonemergent physics approach based on resistance to shear see [73, 7].)

Returning to our graph problem, the optimal graphons described by Theorem 4.1 have rank 2, while the optimal graphons above the n -th scallop described by Theorem 4.2 have rank $n + 2$. We will construct order parameters that detect rank. Specifically, the k -th order parameter will be identically zero whenever the rank of the optimal graphon is $k - 1$ or less, and will never be zero when the rank of the graphon is equal to k .

Newton's identities¹ [39] relate the determinant of a $k \times k$ matrix A to the traces of A^j for $j = 1, 2, \dots, k$. For instance, if we let $t_j = \text{Tr}(A^j)$, then the determinants of small matrices are given by the formulas

$$\det(A) = \begin{cases} (t_1^2 - t_2)/2 & k = 2, \\ (t_1^3 - 3t_1t_2 + 2t_3)/6 & k = 3, \\ (t_1^4 - 6t_1^2t_2 + 8t_1t_3 - 3t_4)/24 & k = 4. \end{cases}$$

Let $p_k(A)$ be the polynomial in the variables $\{t_j\}$ that gives the determinant of a $k \times k$ matrix A .

The same ideas work for arbitrary diagonalizable linear operators, for which the rank equals the number of nonzero eigenvalues, counted with multiplicity. If we evaluate p_k on any diagonalizable trace-class operator, we get zero if the rank of the operator is less than k and the product of the nonzero eigenvalues (counted with multiplicity) if the rank is equal to k .

In particular, we can apply these formulas to graphons, which are symmetric and trace class and therefore diagonalizable. For instance, the expression $(t_1^4 - 6t_1^2t_2 + 8t_1t_3 - 3t_4)/24$, where now $t_j = \text{Tr}(g^j)$, gives zero if the rank of the graphon g is less than 4 and gives a nonzero number if the rank is equal to 4.

When $j > 2$, t_j is the density of j -gons, which is *a priori* an analytic function of (e, t) in each phase. The problem is that t_1 and t_2 are not subgraph densities. In principle, we could get around this by brute force, using Theorems 4.1 and 4.2 to show that t_1 and t_2 are analytic.

There is a simpler solution. We define our k th order parameter to be $p_k(g^3)$ rather than $p_k(g)$. This is still a polynomial in $\{t_j\}$, only now the index j ranges from 3 to $3k$ in steps of 3. We no longer have to worry about t_1 and t_2 . The k -th order parameter is then zero if g^3 has rank less than k and is nonzero if

¹ According to Stigler's Law of Eponymy, attributed by Stigler [68] to Robert Merton, scientific discoveries are never named after their actual originators. Some of Newton's identities were first discovered by Albert Girard, though not published.

g^3 has rank k . But g^3 has the same rank as g , with the same eigenvectors as g and eigenvalues equal to the cubes of the eigenvalues of g , so we are actually testing the rank of g .

In summary: the k th order parameter $p_k(g^3)$ is an analytic function of (e, t) in each phase, being built from subgraph densities. It is identically zero on the regions above the 0th, 1st, . . . , $(k-3)$ rd scallops but is never zero on the region above the $(k-2)$ scallop. Thus the region above the $(k-2)$ scallop is in a different phase from the regions above all the previous scallops.

4.6 Convexity and Legendre Transforms

For background on convexity and Legendre Transforms see [60]. A subset $A \subset \mathbb{R}^n$ is called *convex* if, for every pair of points $x, y \in A$, the entire line segment $\{tx + (1-t)y | t \in [0, 1]\}$ lies in A . Meanwhile, a function $f : A \rightarrow \mathbb{R}$ is called *convex* if

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y)$$

for all triples $(x, y, t) \in A \times A \times (0, 1)$. If we can replace $\leq$ with $<$ whenever $y \neq x$, then we say that f is *strictly convex*. Convex functions are also said to be *concave up*. If $-f$ is convex, we say that f is *concave*, or *concave down*.

The *Legendre transform* of f is a function $f^* : \mathbb{R}^n \rightarrow \mathbb{R}$:²

$$f^*(p) = \inf_x [f(x) - p \cdot x].$$

If A is compact, then the infimum exists and is actually a minimum. It occurs either on the boundary of A , at a point where $\nabla f = p$, or at a point where ∇f is not defined.

Now suppose that f is a differentiable and strictly convex function on all of $\mathbb{R}^n$. Then f^* is a differentiable and strictly concave function on the set of p 's for which $\inf_x [f(x) - p \cdot x] > -\infty$. The function ∇f gives a homeomorphism between $\mathbb{R}^n$ and this set of p 's. The inverse map involves the gradient of f^* .

$$p = \nabla f(x); \quad x = -\nabla f^*(p).$$

Furthermore, f is itself a (slightly modified) Legendre transform of f^* .

$$f(x) = \sup_p [f^*(p) + p \cdot x].$$

If f is convex but not strictly convex, there may be multiple values of x

² Many authors use the opposite sign convention, defining $f^*(p)$ to be the supremum of $p \cdot x - f(x)$ rather than the infimum of $f(x) - p \cdot x$.

corresponding to the same value of p . In that case, f^* may not be differentiable. If f is strictly convex but not everywhere differentiable, then the points where f is not differentiable may correspond to multiple values of p and f^* may not be strictly convex. Even so, there is a smooth and invertible correspondence $x \leftrightarrow p$ between each region where f is analytic and strictly convex and a corresponding region where f^* is analytic and strictly concave.

When f is only defined on a convex set A , rather than on all of $\mathbb{R}^n$, the picture is similar. Points on the boundary of A , or points where ∇f changes discontinuously, may correspond to multiple values of p . However, points in the interior of A , where f is smooth (resp. analytic) and strictly convex, are mapped diffeomorphically (resp. analytically) onto regions in p -space where f^* is strictly concave and smooth (resp. analytic).

The following examples show how this works, and also how crucial the assumptions of convexity of the domain A , and convexity of the function f , really are.

- Let A be the interval $[0, \infty)$ and let $f(x) = \frac{1}{3}x^3$. Both the domain and the function are convex. We then compute

$$f^*(p) = \inf_x \left[\frac{1}{3}x^3 - px \right].$$

When $p \leq 0$, the infimum is a minimum, achieved at $x = 0$, so $f^*(p) = 0$.

When $p \geq 0$, the minimum is achieved at $x = \sqrt{p}$, so

$$f^*(p) = \begin{cases} 0 & p \leq 0, \\ -\frac{2}{3}p^{3/2} & p > 0. \end{cases}$$

The endpoint $x = 0$ is associated with all $p \leq 0$, while the relation $p = x^2$ maps the interior of A , namely the open interval $(0, \infty)$, diffeomorphically onto the interval $(0, \infty)$ in p -space.

- Next let $A = [-2, \infty)$ and let $f(x) = \frac{1}{3}x^3$. The function f is no longer convex, having a negative second derivative when $x < 0$. The minimum of $\frac{1}{3}x^3 - px$ is achieved at $x = -2$ when $p < 1$ and at $x = \sqrt{p}$ when $p > 1$. When $p = 1$, it is achieved both at $x = -2$ and at $x = +1$. The Legendre transform is then

$$f^*(p) = \begin{cases} -\frac{8}{3} + 2p & p \leq 1, \\ -\frac{2}{3}p^{3/2} & p > 1. \end{cases}$$

The minimum of $f(x) - px$ is never achieved at any point strictly between -2 and 1 ! Those points play no role in defining f^* and cannot be recovered from the structure of f^* . They might as well be invisible.

It should come as no surprise that negative values of x , where the function

f is not convex, are not visible. But the interval $[0, 1]$, where f is convex, is also invisible! Non-convexity in one region can mess up the Legendre transform in others.

- Finally, let $A = \mathbb{R}$ and let $f(x) = \frac{1}{3}x^3$. Then $f^*(p) = -\infty$ for all p , since $f(x)$ goes to $-\infty$ faster than linearly as $x \rightarrow -\infty$. In this example, the non-convexity of f in the region $x < 0$ is sufficient to destroy the Legendre transform for all x and all p .
- Sometimes we can see the breakdown in a Legendre transform simply from the shape of the domain and some basic properties of the function f , without having to analyze the convexity of f . For instance, suppose that our region is the portion of the first quadrant of $\mathbb{R}^2$ with $x_1 x_2 \leq 1$, that f is differentiable, that f is bounded on the curve $x_1 x_2 = 1$, and that at each point where $x_1 x_2 = 1$ either $\partial f / \partial x_1 > 0$ or $\partial f / \partial x_2 > 0$. Then there is a neighborhood of the curve $x_1 x_2 = 1$ that is invisible, in the sense that there aren't any values of p for which an x that minimizes $f(x) - p \cdot x$ lies in that neighborhood.

Let x^* be a point sufficiently close to the curve $x_1 x_2 = 1$ that $\nabla f(x^*)$ has at least one positive component. The only candidate value of p is $\nabla f(x^*) = (a, b)$. If $a > 0$, then $-p \cdot x$ goes to $-\infty$ as the curve $x_1 x_2 = 1$ approaches the x_1 axis. If $b > 0$, it goes to $-\infty$ as the curve approaches the x_2 axis. Either way, the function $f(x) - p \cdot x$ is unbounded below, and in particular does not achieve a minimum at x^* .

This example may seem artificial, but we will see something very similar when we consider graphs, the Razborov triangle, and the Boltzmann entropy function $B(e, t)$, in the next section.

4.7 ERGM invisibility and convexity

Exponential Random Graph Models (ERGMs) are parametric models used to investigate the relevance of features in data. As we will see, they are closely related structurally to the uniform random graphs that we analyze in this book. Although they are not closely related *conceptually*, there is value in seeing the connections and differences, in particular because the two subjects use key terms such as “phase” differently. For a history of ERGMs up to 2013, especially relevant for this book, when our work on uniform random graphs began, we know of nothing better than section 2.1 in [14].

In an ERGM one constrains the densities of several subgraphs using an exponential family on the ensemble of all graphs on n vertices. For the edge-triangle model, this means picking constants α and β and giving each graph G

the probability

$$P(G) = \frac{1}{Z} \exp \left(-n^2 \left(\alpha \varepsilon(G) + \frac{\beta}{3} \tau(G) \right) \right), \quad (4.5)$$

where $\varepsilon(G)$ and $\tau(G)$ are the edge and triangle densities of the graph G and Z is a normalization constant called the *partition function*. (ERGMs are usually expressed using parameters $\beta_1 = -\alpha$ and $\beta_2 = -\beta/3$, but working with (α, β) is more convenient for our purposes.)

Chatterjee and Varadhan introduced their LDP for Erdős-Rényi graphs in [15] and illustrated its use in analyzing ERGMs. As we see from Theorem 2.3, for large n typical graphs in the ERGM ensemble are approximated by a graphon that maximizes

$$S(g) - \alpha \varepsilon(g) - \frac{\beta}{3} \tau(g).$$

One therefore defines a *free energy*

$$\Psi(\alpha, \beta) = \sup_g \left(S(g) - \alpha \varepsilon(g) - \frac{\beta}{3} \tau(g) \right). \quad (4.6)$$

Many papers have been written to study Ψ and its singularities; see Chapter 7 in [13] for a clear introduction and numerous references.

The maximization in equation (4.6) can be done in steps.

1. First, for each pair (e, t) , find a graphon g with $\mathbf{t}(g) = (e, t)$ that maximizes $S(g)$. This gives the Boltzmann entropy $B(e, t)$.
2. Next, find the value of t that maximizes $B(e, t) - \frac{\beta}{3}t$. This gives a function that we denote $\psi(e, \beta)$.
3. Finally, find the value of e that maximizes $\psi(e, \beta_2) - \alpha e$. This gives the free energy $\Psi(\alpha, \beta)$.

In other words, the free energy is a Legendre transform of the Boltzmann entropy:

$$\Psi(\alpha, \beta) = \sup_{e, t} \left(B(e, t) - \alpha e - \frac{\beta}{3}t \right).$$

Since this transform uses supremums rather than infimums, invertibility depends on $B(e, t)$ being a concave (rather than convex) function and on the convexity of the Razborov triangle.

Unfortunately, $B(e, t)$ is not concave and the Razborov triangle is not convex. We call a point (e, t) *ERGM-visible* if there exists a value of (α, β) for which the maximum in equation (4.5) occurs at a graphon with $(\varepsilon(g), \tau(g)) = (e, t)$. Otherwise, we say that (e, t) is *ERGM-invisible*. For fixed e , we call a point t

ψ -visible if there exists a value of β for which the maximum of $B(e, t) - \frac{\beta}{3}t$ occurs at that point, and ψ -invisible if no such β exists. Note that ψ -invisibility of t implies ERGM-invisibility of (e, t) , but ψ -visibility does not necessarily imply ERGM-visibility.

In 2011, Chatterjee and Diaconis [14, Theorem 6.2] noticed a problem with ERGMs. Many constraint values are ERGM-invisible. In particular, all points with $t > e^3$ are ERGM-invisible, as are all points with t moderately less than e^3 . The Erdős-Rényi curve itself is ERGM-visible, but each point (e, e^3) corresponds to infinitely many values of (α, β) . The only places where there might be a local bijection between (e, t) and (α, β) are near the bottom boundary of the Razborov triangle. We now show that the nonconvexity of the Razborov triangle rules out most of those points, too.

Theorem 4.4 *If n is a positive integer and $\frac{n}{n+1} < e < \frac{n+1}{n+2}$, and if t is sufficiently close to the minimum triangle density t_0 , then (e, t) is ERGM-invisible.*

Proof Fix a value of e strictly between $\frac{n}{n+1}$ and $\frac{n+1}{n+2}$. The points with edge density e and triangle density just above the minimum must have large values of the Lagrange multipliers α and β . However, the scallop itself is concave down, so for positive values of β , the linear function $-\alpha e - \frac{\beta}{3}t$ is less at one or both of the neighboring cusps (at edge density $\frac{n}{n+1}$ and $\frac{n+1}{n+2}$) than near the interior of the scallop. For large positive values of β (and correspondingly large negative values of α), this difference is greater than the bounded difference in $B(e, t)$ between any two values of (e, t) . Since large values of β correspond to small values of Δt , we conclude that, for all sufficiently small values of Δt , the point $(e, t_0 + \Delta t)$ is ERGM-invisible. $\square$

A similar result applies at the top of the Razborov triangle.

Theorem 4.5 *For each $e \in (0, 1)$ and for all t sufficiently close to (and less than) $e^{3/2}$, (e, t) is ERGM-invisible.*

Proof Near the top boundary, α is large and positive while β is large and negative. However, the boundary curve $t = e^{3/2}$ is concave up, so we can decrease $\alpha e + \beta t/3$ by moving to one endpoint $(0, 0)$ or the other $(1, 1)$. Whenever α and β are big enough in magnitude, in other words whenever we are close enough to the top boundary, these gains swamp any changes in $B(e, t)$. For all (e, t) sufficiently close to the top boundary, either $B(0, 0) - 0\alpha - 0\frac{\beta}{3}$ or $B(1, 1) - 1\alpha - 1\frac{\beta}{3}$ is greater than $B(e, t) - \alpha e - \frac{\beta}{3}t$. $\square$

The statement of Theorem 4.5 is not new; it was already covered by [14, Theorem 6.2]. However, the simplicity of the proof gets to the heart of why points in this region are ERGM-invisible.

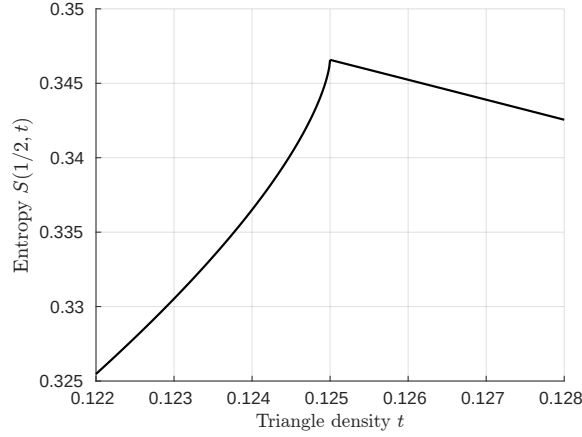


Figure 4.5 $B(e, t)$ as a function of t when $e = 1/2$. The behavior for $t < 0.125$ is given by the exact formula of Theorem 3.7. The behavior for $t > 0.125$ is only approximate, using estimates derived in Chapter 7. However, we have already proven the key feature when $t > 0.125$, namely that ΔB scales as Δt^1 , in Theorem 3.6.

4.7.1 ERGM-invisibility below the Erdős-Rényi curve

Just below the Erdős-Rényi curve, the function $B(e, t)$ is strongly concave up as a function of t , with $\partial^2 B / \partial t^2$ diverging as $(e^3 - t)^{-4/3}$. This casts a shadow on the lower part of the Razborov triangle, rendering regions where $\partial^2 B / \partial t^2 < 0$ ψ -invisible and therefore ERGM-invisible. (Recall how including the interval $[-2, 0]$ in the domain of $\frac{1}{3}x^3$ made $[0, 1]$ invisible when applying a Legendre transform. This phenomenon is very similar.)

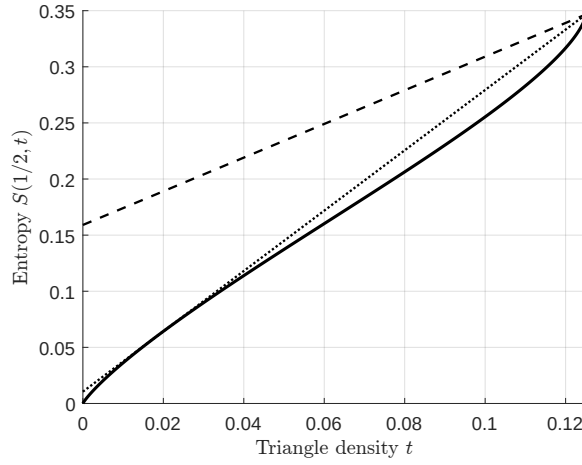
In this subsection we quantify the length of this shadow. Combining these results with Theorem 4.4 and with [14, Theorem 6.2], it appears that the only ERGM-visible points in the Razborov triangle are

- On the Erdős-Rényi curve itself,
- Near the interval $t = 0$, $0 < e < 1/2$, or
- In small neighborhoods of the cusps where two scallops meet.

We start on the interval $e = 1/2$, $t < e^3$, where we have a closed-form formula for the Boltzmann entropy $B(e, t)$, namely

$$B(e, t) = H\left(\frac{1}{2} + \delta\right),$$

where $t = e^3 - \delta^3$. Figure 4.5 shows the behavior of $B(e, t)$ as a function of t for t close to e^3 . Wherever $B(e, t)$ is concave up as a function of t , so is $B(e, t) - \frac{t}{3}$,

Figure 4.6 Comparing $B(e, t)$ to linear functions of t for $t \leq 1/8$.

whose second derivative with respect to t is the same as the second derivative of $B(e, t)$. By the second derivative test, such values of t cannot be local maxima of $B(e, t) - \frac{\beta}{3}t$, no matter what β is. In other words, they are ψ -invisible.

Now consider the form of $B(1/2, t)$ for all t between 0 and $1/8$, as shown in Figure 4.6. If $0 < \beta < \beta^* \approx 8.065017$, then the function $B(1/2, t) - \frac{\beta}{3}t$ is maximized at $t = 1/8$. For instance, the dashed line of slope 1.5 touches the graph of $B(1/2, t)$ at $t = 1/8$ and lies above the graph everywhere else.

When β reaches β^* , there are two values of t that maximize $B(1/2, t) - \frac{\beta}{3}t$, one at $t = t_0 \approx 0.01848362$ and the other at $t = 0.125$. That is, the dotted line of Figure 4.6 touches the graph twice. When $\beta > \beta^*$, the maximum is achieved at a single value of t , a value that is less than t_0 . As a result, the entire interval $(t_0, 1/8) \approx (0.018, 0.125)$ is ψ -invisible.

The behavior is similar for all $e < 1/2$. While we do not have a precise formula for $B(e, t)$ when $e \neq 1/2$, we have upper bounds on $B(e, t)$ from equation (3.4) and we know the behavior of $B(e, t)$ near $t = 0$, where Theorem 4.1 applies. To illustrate the method, we redo our calculation at $e = 1/2$ using only those facts.

Figure 4.7 shows the graph of $B(1/2, t)$ together with a straight line between the points $(0, 0)$ and $(1/8, H(1/2))$. This line also intersects the graph of $B(e, t)$ at $t \approx 0.047$. At all greater values of t , $B(e, t) < 8H(1/2)t$. If $t > 0.47$ and $\beta \leq 24H(1/2)$, then $B(1/2, 1/8) - \beta/24 > B(1/2, t) - \beta t/3$, while if $\beta \geq 24H(1/2)$, then $B(1/2, 0) - 0\beta/3 > B(1/2, t) - \beta t/3$. Such a value of t does not maximize $B(1/2, t) - \beta t/3$ for any value of β and so is ψ -invisible.

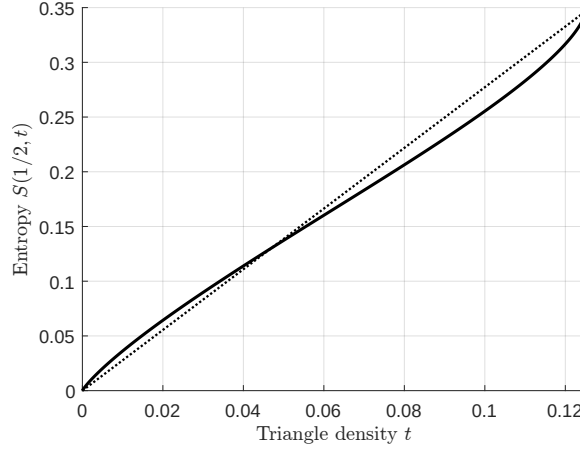


Figure 4.7 Finding invisible points using only elementary geometry

This argument still used our knowledge of the actual graph of $B(1/2, t)$. But even just using the upper bound

$$B \leq H(1/2) - \delta^2$$

from (3.4), we see that $B(e, t) < 8H(1/2)t = H(1/2)(1 - 8\delta^3)$ whenever $\delta < (8H(1/2))^{-1} = (4 \ln(2))^{-1} \approx 0.36067$. This establishes an upper bound of

$$\frac{1}{8} - \frac{1}{(4 \ln(2))^3} \approx 0.07808$$

for ψ -visible points with $e = 1/2$ and $t < e^3$.

While this last estimate is very far from being sharp, yielding an upper bound of 0.078 rather than 0.018, it gives us a robust method for computing upper bounds for ψ -visible values of t at all values of $e \leq 1/2$. The following is related to [14, Theorem 6.1]; its main added value is its focus on the geometry of the graph of the Boltzmann entropy.

Theorem 4.6 *For all $e \in (0, 1)$ and all sufficiently small values of $\delta > 0$, $t = e^3 - \delta^3$ is ψ -invisible when we fix $\varepsilon(g) = e$. Furthermore, (e, t) is ERGM-invisible.*

Proof From equation (3.4) and Theorem 3.3, we know that $B(e, 0) = \frac{1}{2}H(2e)$ and $B(e, t) \leq H(e) - \delta^2$. In the t - B plane, the straight line from $(0, \frac{1}{2}H(2e))$

to $(e^3, H(e))$ has equation

$$B = H(e) + K_2(t - e^3) = H(e) - K_2\delta^3,$$

where

$$K_2 = \frac{H(e) - \frac{1}{2}H(2e)}{e^3}.$$

Whenever $\delta < 1/K_2$, the upper bound $H(e) - \delta^2$ is lower than the linear function $H(e) - K_2\delta^3$. If $\beta \leq K_2/3$, then

$$B(e, t) - \beta t/3 < B(e, e^3) - \beta e^3/3,$$

while if $\beta \geq K_2/3$, then

$$B(e, t) - \beta t/3 < B(e, 0) - 0\beta/3.$$

Either way, the maximum of $B(e, t) - \frac{\beta}{3}t$ is not achieved at $t = e^3 - \delta^3$, so $t = e^3 - \delta^3$ is ψ -invisible.

When $e > 1/2$, the formulas for the minimum value t_{min} and of $H(e, t_{min})$ are more complicated. See Theorems 3.4 and 3.5. However, the line from $(t_{min}, B(e, t_{min}))$ to $(e^3, H(e))$ still has bounded slope m and the upper bound $H(e) - \delta^2$ still lies below the line when δ is small. Depending on whether β is greater than, less than, or equal to $3m$, $B(e, t) - \beta t/3$ is less than $B(e, t_{min}) - \beta t_{min}/3$, less than $H(e) - \beta e^3/3$, or both.

Since ψ -invisibility implies ERGM-invisibility, we are done. $\square$

Our proof provided actual estimates for how small δ needs to be for $t = e^3 - \delta^3$ to be ψ -invisible. Using the actual maximum value of $f(e, e')$ in equation (3.4) rather than the universal bound -1 yields a sharper estimate.

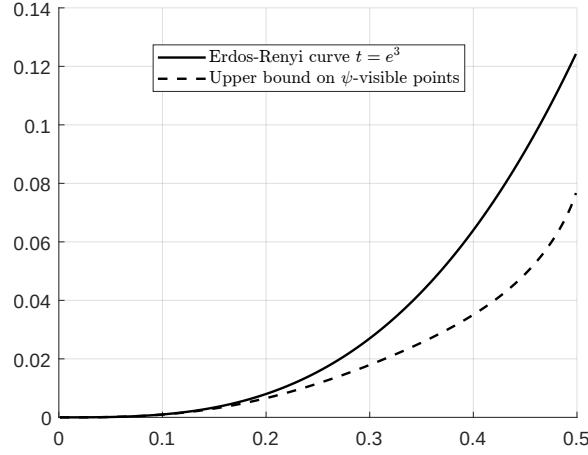
Corollary 4.7 *When $e < 1/2$, all values of $t = e^3 - \delta^3$ with*

$$\delta < \frac{e^3 H'(e)}{(1 - 2e)(H(e) - \frac{1}{2}H(2e))}$$

are ψ -invisible.

Figure 4.8 shows the result of this calculation. The upper (solid) curve is the Erdős-Rényi curve $t = e^3$. The lower (dashed) curve is our upper bound for ψ -visible points with $t < e^3$. Everything in between is ψ -invisible, and so is ERGM-invisible.

The bounds in Corollary 4.7, shown in Figure 4.8, are not sharp. We used a straight line from $(0, B(e, 0))$ to $(e^3, H(e))$ rather than the convex hull of the graph of $B(e, t)$. We used an upper bound for $B(e, t)$ rather than the actual function. We proved ψ -invisibility instead of the weaker ERGM-invisibility. In fact,

Figure 4.8 Bounding the ψ -visible points when $e < 1/2$

numerical estimates of $B(e, t)$ suggest that the picture is much more restrictive. When $e < 1/2$, it appears that only a small neighborhood of the bottom boundary is ψ -visible. When $e > 1/2$, it appears that only small neighborhoods of the cusps are ERGM-visible.

4.7.2 ERGM-invisibility above the Erdős-Rényi curve

Chatterjee and Diaconis [14, Theorem 4.1] already showed that the entire region above the Erdős-Rényi curve is ERGM-invisible. In this subsection, we propose a simple mechanism for this.

For each e , the entropy is maximized at $t = e^3$. However, increasing t above e^3 comes at an entropy cost that scales as $(t - e^3)$, not as $(t - e^3)^2$. In particular, whenever $0 > \beta > 3K$, where K is the constant of equation (3.4), $S(g) - \beta\tau(g)/3$ achieves its maximum value of $H(e) - \beta e^3/3$ at the constant graphon $g(x, y) = e$. The only possibility for getting a ψ -visible point (e, t) with $t > e^3$ is if β is large and negative.

However, every point with $t > e^3$ lies inside the convex hull of the Erdős-Rényi curve. Since β is negative, there are points (e', t') on the Erdős-Rényi curve, shown in red in Figure 4.9, for which $-\alpha e' - \beta t'/3 > -\alpha e - \beta t/3$. When β is large and negative, this difference is large enough to overcome any possible difference in entropy between the best graphon at (e, t) and the constant graphon $g(x, y) = e'$. In particular, either the empty graphon $g(x, y) = 0$ or the

complete graphon $g(x, y) = 1$ (or both) is guaranteed to do better than the best graphon at (e, t) .

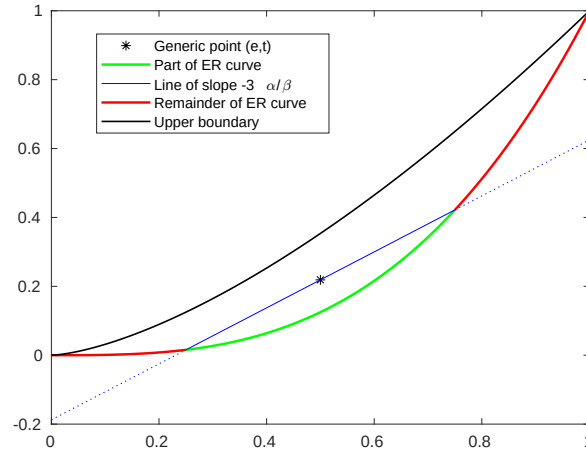


Figure 4.9 The reason why points above the Erdős-Rényi curve are ERGM-invisible. Points (e', t') in the “wings” of the Erdős-Rényi curve, marked in red, have $-\alpha e' - \beta t'/3 > -\alpha e - \beta t/3$. They dominate the point (e, t) when $|\beta|$ is large. One can shrink or eliminate one wing of the Erdős-Rényi curve by varying α and β , but only by enlarging the other.

It takes careful estimates to combine these two observations into a complete proof, showing that the first mechanism forces $|\beta|$ to be big enough for the second mechanism to exclude (e, t) . That is beyond the scope of this book. For a full (and very different) proof of invisibility above the Erdős-Rényi curve we refer the interested reader to the original work of Chatterjee and Diaconis [14].

The edge-2star model

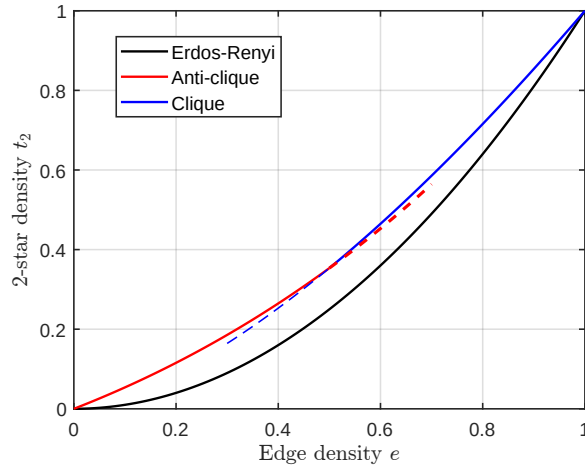
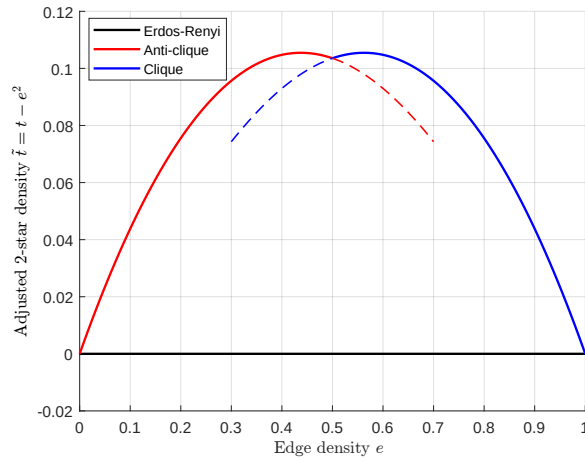
In this chapter we take a break from our study of the edge-triangle model to study the edge-2star model. (2stars are a special case of kstars, defined formally in the next section.) There are three motivations for this digression.

1. Pedagogical value. The edge-2star model is technically simpler than the edge-triangle model. It provides a good sandbox for developing a number of techniques that will eventually be brought to bear on the edge-triangle model.
2. Technical value. Near the Erdős-Rényi curve, the edge-triangle model is well-approximated by the edge-2star model. We will eventually prove properties of the edge-triangle model via analogous properties of the edge-2star model.
3. Scientific value. The edge-2star model has a number of properties that are of independent interest.

We should emphasize that many of the results of this chapter were first proven together with Kenyon and Ren in [26], by different methods.

We begin by determining the extent of the edge-2star phase space, proved in [1]. In terms of the 2star density t_2 , this is shown in Figure 5.1. It is convenient to define a *reduced 2star density* $\tilde{t}_2 := t_2 - e^2$. The possible values of $(e, \tilde{t}_2)$ are shown in Figure 5.2. The correctness of these figures is proven in Theorem 5.1.

We then study entropy-optimizing graphons in the interior of our phase space. In Section 5.2 we show that all optimal graphons are multipodal, with a uniform bound on the podality. We then show that the optimal graphons just above Erdős-Rényi are bipodal and vary analytically with $(e, \tilde{t}_2)$. The analysis is different for $e < 1/2$, for $e > 1/2$, and for e close to $1/2$, but the results indicate a single phase covering the entire width of the phase diagram. Near the top of the phase diagram, things are different. We prove that for large enough

Figure 5.1 Possible values of (e, t_2) .Figure 5.2 Possible values of $(e, \tilde{t}_2)$.

$\tilde{t}_2$, there is a phase transition at $e = 1/2$, with a unique optimal graphon to the left of that line and a very different optimal graphon to the right.

These results are displayed in Figure 5.3. Theorem 5.8 applies to the green region above Erdős-Rényi. Theorem 5.14 describes the red and blue regions near the cusp. The conjectures and open problems of Section 5.5 concern the dotted line of symmetry at $e = 1/2$. On that line, symmetric graphons are not

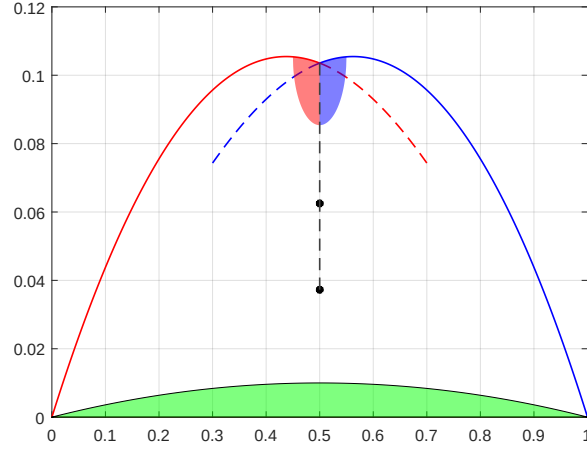


Figure 5.3 The regions where the various theorems apply. The bottom of the phase diagram is governed by Theorem 5.8 and the top by Theorem 5.14. The conjectured critical point lies on the dashed line of symmetry.

stable above the lower dot. Symmetric graphons do not even exist above the upper dot.

5.1 Definitions and the geometry of the phase space

A k star (often written k -star) is a graph with k edges and $k + 1$ vertices, in which one central vertex is connected to all the others. In particular, a 2star (sometimes called a “cherry”) is a graph with one central vertex connected to two other vertices. The density of 2stars in a graph G depends only on the degrees of the vertices of G , since the number of graph homomorphisms sending the center of a 2star to a vertex of degree m is exactly m^2 . In terms of graphons, let

$$d(x) = \int_0^1 g(x, y) dy$$

be the *degree function* of the graphon g . The density t_2 of 2stars is then

$$t_2 = \tau_2(g) = \int_0^1 d(x)^2 dx$$

and the reduced 2star density is

$$\tilde{t}_2 = t_2 - e^2 = \int_0^1 (d(x) - e)^2 dx.$$

That is, $\tilde{t}_2$ is the variance of the degree function. The fact that t_2 and $\tilde{t}_2$ depend only on the degree function, and not on the details of the graphon itself, will greatly simplify our analysis.

As with the edge-triangle model, the most basic question is to determine all the possible values of (e, t_2) or $(e, \tilde{t}_2)$.

Theorem 5.1 *For each $e \in (0, 1)$, the set of possible values of t_2 is a closed interval $[e^2, t_{\max}]$, where*

$$t_{\max} = \begin{cases} (1-e)^{3/2} + 2e - 1 & e \leq 1/2, \\ e^{3/2} & e \geq 1/2. \end{cases}$$

Equivalently, the set of possible values of $\tilde{t}_2$ is a closed interval $[0, \tilde{t}_{\max}]$, where

$$\tilde{t}_{\max} = \begin{cases} (1-e)^{3/2} - (1-e)^2 & e \leq 1/2, \\ e^{3/2} - e^2 & e \geq 1/2. \end{cases}$$

Proof As with the proof of Theorem 3.1, the existence of maximum and minimum values of t_2 follows from the compactness of $\mathcal{G}_e$. The fact that all values of t_2 between the maximum and the minimum are achieved follows from continuity, as we linearly interpolate between a graphon that achieves the minimum value to a graphon that achieves the maximum value.

The minimum value of $\tilde{t}_2$ is obviously zero and is achieved by any graphon with constant degree. Among such graphons, the entropy is maximized by the Erdős-Rényi graphon $g(x, y) = e$. As a result, the minimum value of t_2 is e^2 .

Now consider the graphons

$$g_c(x, y) = \begin{cases} 1 & x, y < \sqrt{e}, \\ 0 & \text{otherwise} \end{cases}$$

and

$$g_{ac}(x, y) = \begin{cases} 0 & x, y < \sqrt{1-e}, \\ 1 & \text{otherwise.} \end{cases}$$

The first graphon, called a *clique*, has $t_2 = e^{3/2}$ and $\tilde{t}_2 = e^{3/2} - e^2$. The second graphon, called an *anticlique*, has $t_2 = (1-e)^{3/2} + 2e - 1$ and $\tilde{t}_2 = (1-e)^{3/2} - (1-e)^2$. It is easy to check that the anticlique has a higher value of t_2 when $e < 1/2$, that the clique has the higher value when $e > 1/2$, and that they both have $t_2 = \sqrt{2}/4 \approx 0.3535$ and $\tilde{t}_2 = (\sqrt{2} - 1)/4 \approx 0.1035$ when $e = 1/2$.

In 1978, Ahlswede and Katona [1] proved that every t_2 -maximizing graph is equivalent to a clique or an anticlique. $\square$

There is an obvious symmetry relating cliques and anticliques. If we replace a graphon g with $1 - g$, then the edge density changes from e to $1 - e$ and the degree function changes from $d(x)$ to $1 - d(x)$. However $\tilde{t}_2$, which is the variance of the degree function, does not change at all. This explains the symmetry of Figure 5.2. Once we know that $\tilde{t}_2$ is maximized by cliques when $e \geq 1/2$, it automatically follows that $\tilde{t}_2$ is maximized by anticliques when $e \leq 1/2$.

Let $\mathcal{G}_{e, \tilde{t}_2}$ be the set of graphons with edge density e and reduced 2star density $\tilde{t}_2$. As usual, the Boltzmann entropy $B(e, \tilde{t}_2)$ is the maximum value of $S(g)$ on $\mathcal{G}_{e, \tilde{t}_2}$.

Theorem 5.2 *The function $B(e, \tilde{t}_2)$ is continuous. For fixed e , it is a strictly decreasing function of $\tilde{t}_2$, and in particular is positive except at the upper boundary of phase space.*

Proof Fix the edge density e . If $0 \leq \tilde{t}'_2 < \tilde{t}_2$ and a graphon g maximizes the entropy at $(e, \tilde{t}_2)$, then we can achieve a graphon at $(e, \tilde{t}'_2)$ with more entropy than that of g by taking a weighted average of g and an Erdős-Rényi graphon of the same edge density. This shows that $B(e, \tilde{t}_2)$ is a strictly decreasing function of $\tilde{t}_2$. To show that it is a continuous function of $\tilde{t}_2$, we must show that for any fixed number $\tilde{t}$,

$$\lim_{\tilde{t}_2 \rightarrow \tilde{t}^-} B(e, \tilde{t}_2) = B(e, \tilde{t}) = \lim_{\tilde{t}_2 \rightarrow \tilde{t}^+} B(e, \tilde{t}_2).$$

The first equality follows from the semicontinuity of the entropy function on the space of all graphons. If we take a sequence of entropy-maximizing graphons at $(e, \tilde{t}_2)$, as $\tilde{t}_2$ approaches $\tilde{t}$ from below, there is a (subsequential) limiting graphon with densities $(e, \tilde{t})$ whose entropy is at least the limit of the entropies of the graphons in the sequence, in other words at least $\lim_{\tilde{t}_2 \rightarrow \tilde{t}^-} B(e, \tilde{t}_2)$. However, $B(e, \tilde{t}_2)$ is a decreasing function of $\tilde{t}_2$, so $B(e, \tilde{t})$ cannot be any greater than $\lim_{\tilde{t}_2 \rightarrow \tilde{t}^-} B(e, \tilde{t}_2)$.

For the second equality, we must show that it is possible to take a graphon that maximizes entropy at $(e, \tilde{t})$ and increase the 2star density (by an arbitrarily small amount) at an arbitrarily small cost to the entropy. Let g be a graphon that maximizes the entropy at $(e, \tilde{t})$. Since that entropy is positive, there must be regions of positive measure where $g(x, y)$ is neither 0 nor 1. If the degree function $d(x)$ is not constant on those regions, we can increase the 2star density by increasing $g(x, y)$ where $d(x)$ is larger and decreasing $g(x, y)$ where it is smaller. If $d(x)$ is constant on the region, break it in half and increase $g(x, y)$ on one half and decrease $g(x, y)$ on the other half. In that case the increase in the 2star density will be second order in the change in g , but we can still achieve

an increase to the 2star density via an arbitrarily small change to g , and thus an arbitrarily small decrease in the entropy.

Now that we know that $B(e, \tilde{t}_2)$ is continuous in $\tilde{t}_2$, we consider changes in both the edge and 2star densities. If there is a path in $(e, \tilde{t}_2)$ space approaching a point $(e_0, \tilde{t}_0)$, compactness and the semicontinuity of the entropy functional in the cut metric implies that $B(e_0, \tilde{t}_0)$ is at least the limit of $B(e, \tilde{t}_2)$ along that path. To show that it is at most the limit of $B(e, \tilde{t}_2)$, we must show that it is possible to adjust the densities of an optimal graphon g at $(e_0, \tilde{t}_0)$ to achieve points along that path without having any discontinuous changes in the entropy $S(g)$. By adjusting the value of $g(x, y)$ (say, increasing it by an amount proportional to $1 - g(x, y)$, or decreasing it by an amount proportional to $g(x, y)$) we can change the edge density in either direction. Combined with our previous moves to change the 2star density without changing the edge density, we can vary both e and $\tilde{t}_2$. $\square$

Now that we know that $B(e, \tilde{t}_2)$ is well-defined and continuous, we consider its symmetry.

Theorem 5.3 *For any achievable values $(e, \tilde{t}_2)$, $(1 - e, \tilde{t}_2)$ is also achievable. Moreover, $B(1 - e, \tilde{t}_2) = B(e, \tilde{t}_2)$. Furthermore, the entropy-maximizing graphons at $(1 - e, \tilde{t}_2)$ are related to the entropy-maximizing graphons at $(e, \tilde{t}_2)$ by the $g \leftrightarrow 1 - g$ symmetry.*

Proof Since $H(1 - u) = H(u)$, the $g \leftrightarrow 1 - g$ symmetry preserves the entropy $S(g)$. If g is an entropy-maximizing graphon with densities $(e, \tilde{t}_2)$, then $1 - g$ has densities $(1 - e, \tilde{t}_2)$ and

$$S(1 - g) = S(g) = B(e, \tilde{t}_2).$$

This shows that $B(1 - e, \tilde{t}_2) \geq B(e, \tilde{t}_2)$. Reversing the roles of e and $1 - e$, we have $B(e, \tilde{t}_2) \geq B(1 - e, \tilde{t}_2)$, so $B(e, \tilde{t}_2) = B(1 - e, \tilde{t}_2)$, as claimed. Furthermore, $1 - g$ is an entropy-maximizing graphon with densities $(1 - e, \tilde{t}_2)$. $\square$

Now consider what happens when $e = 1/2$ and $\tilde{t}_2$ is fixed. By Theorem 5.3, the symmetry $g \leftrightarrow 1 - g$ must send the set of entropy maximizers to itself. This leaves two possibilities.

1. Symmetry is preserved. Every entropy-maximizer (and there may only be one) is mapped to a graphon that is equivalent to itself. For instance, a

graphon of the form

$$g_p = \begin{cases} p & x, y < 1/2, \\ 1-p & x, y > 1/2, \\ 1/2 & x < 1/2 < y, \\ 1/2 & y < 1/2 < x, \end{cases} \quad (5.1)$$

gets mapped to g_{1-p} . However, g_p and g_{1-p} are equivalent: just swap the intervals $(0, 1/2)$ and $(1/2, 1)$.

2. Symmetry is broken, as is uniqueness. There are at least two inequivalent entropy-maximizing graphons, related by the $g \leftrightarrow 1-g$ symmetry.

We will soon show that symmetry is preserved when $\tilde{t}_2$ is small, with a unique entropy maximizer of the form g_p . However, the situation is different when $\tilde{t}_2$ is close to its maximum value.

Theorem 5.4 *There exists a constant $\tilde{t}_M < (\sqrt{2} - 1)/4$ such that, whenever $e = 1/2$ and $\tilde{t}_2 > \tilde{t}_M$, symmetry is broken.*

Proof Suppose otherwise. Then there is a sequence of values of $\tilde{t}_2$, converging to $(\sqrt{2} - 1)/4$, for which there is an entropy maximizer that is invariant under the $g \leftrightarrow 1-g$ symmetry. By compactness, the sequence of entropy maximizers has a limit that is also invariant under $g \leftrightarrow 1-g$ and whose densities are $(1/2, (\sqrt{2} - 1)/4)$. However, the only graphons with those densities are cliques and anticliques, neither of which is invariant under $g \leftrightarrow 1-g$. Contradiction. $\square$

Note that the proof never used the fact that our symmetric graphons were entropy-maximizing. It merely used the (presumed) existence of symmetric graphons with $\tilde{t}_2$ arbitrarily close to $(\sqrt{2} - 1)/4$ to derive a contradiction. The contradiction, together with the fact that symmetry-preserving graphons must have $e = 1/2$, then shows

Corollary 5.5 *There is a neighborhood U of $(1/2, (\sqrt{2} - 1)/4)$ such that, whenever $(e, \tilde{t}_2) \in U$, $\mathcal{G}_{e, \tilde{t}_2}$ does not contain any $g \leftrightarrow 1-g$ symmetric graphons.*

5.2 Multipodality

Recalling the definition of multipodality in Section 2.7, we show that in the edge-2star model, multipodal graphons are ubiquitous:

Theorem 5.6 *Let g be an entropy-maximizing graphon for an achievable pair $(e, \tilde{t}_2)$ of densities in the edge/2star model. Then g is multipodal.*

Proof We already know that on the top boundary the only possible graphons are cliques and ant cliques, both of which are bipodal. On the bottom boundary, the entropy-maximizing graphon is Erdős-Rényi, which is 1-podal. We can therefore restrict our attention to the interior of our phase space.

We use the method of Lagrange multipliers. Unlike in the edge-triangle model, it is enough to consider small pointwise changes, for which the usual formalism of functional derivatives applies. These changes are “reversible” as long as $g(x, y)$ is neither 0 nor 1. We do not have to consider an analogue of the worth functional $W(C)$.

We compute the functional derivatives of $S(g)$, $\varepsilon(g)$ and $\tau_2(g)$ and look for constants α and β (depending on e and t_2) such that

$$\frac{\delta S(g)}{\delta g(x, y)} = \frac{\alpha}{2} \frac{\delta \varepsilon(g)}{\delta g(x, y)} + \frac{\beta}{2} \frac{\delta \tau(g)}{\delta g(x, y)}. \quad (5.2)$$

Wherever $g(x, y) = 0$, the equality is replaced by $\leq$, as the only possible change is irreversible, with $g(x, y)$ increasing. Wherever $g(x, y) = 1$, the equality is replaced by $\geq$.

We next compute those functional derivatives:

$$\begin{aligned} \frac{\delta S(g)}{\delta g(x, y)} &= H'(g(x, y)) = -\frac{1}{2} \ln \left(\frac{g(x, y)}{1 - g(x, y)} \right) \\ \frac{\delta \varepsilon(g)}{\delta g(x, y)} &= 1 \\ \frac{\delta \tau_2(g)}{\delta g(x, y)} &= d(x) + d(y). \end{aligned}$$

As in the edge-triangle model, we must avoid the possibility that singular entropy maximizers, taking the values 0 or 1 on a set of positive measure, appear in the interior of our phase space. This is proven in [25, Lemma 3.5], whose proof we sketch here:

- After applying a measure-preserving transformation of $[0, 1]$, an entropy maximizing graphon has to be doubly monotonic, with $g(x, y)$ non-decreasing in x for fixed y and non-decreasing in y for fixed x .
- A singular entropy maximizing graphon g must then equal 1 on a neighborhood of $(1, 1)$ or equal 0 on a neighborhood of $(0, 0)$. By increasing $g(x, y)$ near $(0, 0)$ and decreasing it near $(1, 1)$, we can increase the entropy by $O(\epsilon \ln(1/\epsilon))$ at the cost of decreasing $\tilde{t}_2$ by $O(\epsilon)$.
- To be an entropy maximizer, the graphon must have positive entropy and so must take on values other than 0 or 1 on a set of positive measure. We

vary the graphon in this region to return $\tilde{t}_2$ to its original value at the cost of decreasing the entropy by $O(\epsilon)$.

- The net result of this sequence of moves is to increase the entropy while leaving $\tilde{t}_2$ (and e) fixed, contradicting the optimality of g .

The functional derivatives of ε and τ_2 are linearly independent unless the degree function is constant. But the degree function is only constant when $\tilde{t}_2 = 0$. This establishes the existence of finite Lagrange multipliers α and β , with variational equations that apply almost everywhere, for all entropy maximizers in the interior of our phase space.

Note that the functional derivative of $S(g)$ diverges to $\pm\infty$ as $g(x, y)$ approaches 0 or 1, while the functional derivatives of $\varepsilon(g)$ and $\tau(g)$ are bounded. This means that, for fixed $(e, \tilde{t}_2, \alpha, \beta)$, $g(x, y)$ is uniformly bounded away from 0 and 1.

We use the equation (5.2) to solve for $g(x, y)$ in terms of the degree function.

$$\begin{aligned} -\frac{1}{2} \ln \left(\frac{g(x, y)}{1 - g(x, y)} \right) &= \frac{\alpha}{2} + \frac{\beta}{2} (d(x) + d(y)) \\ \ln \left(\frac{1}{g(x, y)} - 1 \right) &= \alpha + \beta (d(x) + d(y)) \\ \frac{1}{g(x, y)} &= 1 + \exp(\alpha + \beta (d(x) + d(y))) \\ g(x, y) &= \frac{1}{1 + \exp(\alpha + \beta (d(x) + d(y)))}. \end{aligned} \quad (5.3)$$

We integrate the last line of (5.3) over y to get a consistency condition for the degree function.

$$d(x) = \int_0^1 \frac{dy}{1 + \exp(\alpha + \beta (d(x) + d(y)))}. \quad (5.4)$$

Equivalently, every value of $d(x)$ is a root of the function

$$F(z) = z - \int_0^1 \frac{dy}{1 + \exp(\alpha + \beta (z + d(y)))}. \quad (5.5)$$

We claim that $F(z)$ is an analytic function of z . The distribution of $d(y)$ induces a distribution of $D = d(y)$, and our integral becomes

$$\int \frac{d\mu(D)}{1 + \exp(\alpha + \beta (z + D))}.$$

This is the convolution of an analytic function of z by a bounded measure. Since the Fourier transform of an analytic function decays exponentially at infinity and the Fourier transform of an integrable measure is bounded, the Fourier transform of the convolution decays exponentially at infinity, so the

convolution itself is an analytic function of z . Thus $F(z)$ is an analytic function of z .

Note that the function $F(z)$ is not constant, being negative for $z < 0$ and positive for $z > 1$. Being analytic, it can only have a finite number of roots between 0 and 1. That is, the degree function $d(x)$ can only take on a finite number of values. Let I_i be the subset of the unit interval where $d(x)$ takes on the i -th such value. Since the degree function determines $g(x, y)$ via equation (5.3), our graphon g must be multipodal.

This establishes multipodality on the interior of phase space. On the bottom boundary $\tilde{t}_2 = 0$ the entropy maximizing graphons are Erdős-Rényi, hence 1-podal. On the top boundary they are cliques or anticliques, hence bipodal. $\square$

Not only are all entropy maximizers multipodal, but we now show that there is a uniform bound on the podality. As a result, the space of all possible entropy maximizing graphons, for all possible values of $(e, \tilde{t}_2)$, is finite-dimensional. All of the subtleties of functional analysis disappear, leaving us with problems of finite dimensional calculus.

Theorem 5.7 *There is a number M such that all entropy maximizers at all possible points $(e, \tilde{t}_2)$ are multipodal with podality M or less.*

Proof If $f(z)$ is an analytic function on a compact interval with m roots (counted with multiplicity), then any C^m -small perturbation of $f(z)$ will also have at most m roots on the interval. Thus if g is a graphon with associated function $F(z)$ with m roots, and if $\hat{g}$ is an optimal graphon whose degree function is L^1 -close to that of g , then from the convolution, the associated function $\hat{F}(z)$ of $\hat{g}$ will be a C^m -small perturbation of $F(z)$ and so will have at most m roots, and $\hat{g}$ will be at most m -podal.

Now suppose there is no universal bound M on the podality of optimal graphons. Let $g_1, g_2, \dots$ be a sequence of optimal graphons (perhaps with different values of $(e, \tilde{t}_2)$) with the podality going to infinity, and let $F_i(z)$ be the associated functions for these graphons. Since the space of graphons is compact, there is a subsequence that converges to an entropy-maximizing graphon g_∞ . There are two possibilities, depending on whether the limiting value of β is finite or infinite.

If the limiting value of β is finite, then the associated function $F_\infty(z)$ of g_∞ is analytic, and so has only a finite number M_∞ of roots, counted with multiplicity. But then for large i , $F_i(z)$ has at most M_∞ roots, so g_i is at most M_∞ -podal, which is a contradiction.

If the limiting value of β is infinite, then we are on the boundary of our phase space. We can no longer speak of the function $F_\infty(z)$, but we can still speak of

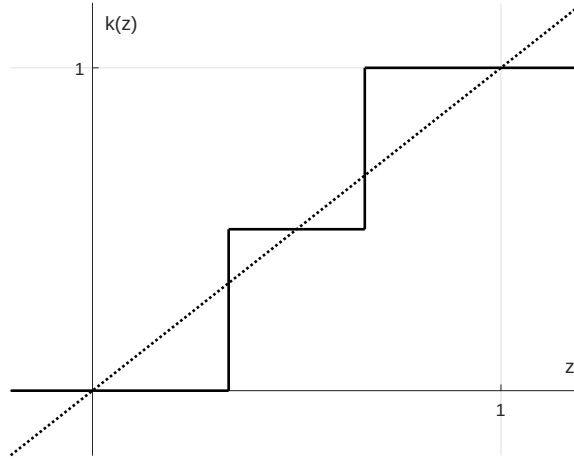


Figure 5.4 On the boundary of phase space, there are at most five solutions to $k(z) = z$, of which only three can be strictly between 0 and 1.

the limit of the graph of

$$k(z) := \int_0^1 \frac{dy}{1 + \exp(\alpha + \beta(z + d(y)))} \quad (5.6)$$

as $\beta \rightarrow \infty$, with α depending on β , and study where this graph intersects the graph of the function $f(z) = z$. See Figure 5.4. As $\beta \rightarrow \infty$, the logistic function $(1 + \exp(\alpha + \beta z))^{-1}$ approaches a step function. The convolution of this with a measure supported at 1 or 2 points (insofar as the optimal graphons on the boundary are either 1-podal or bipodal) is then a step function with 1 or 2 jumps. The graph then has 2 or 3 horizontal segments and 1 or 2 vertical segments, and so can intersect a straight line at most 5 times. Moreover, the intersections are transversal. After making an L^1 -small adjustment to the degree function, we get a function that equals z at most 5 times, so g_i is at most 5-podal, which is again a contradiction. $\square$

5.3 A bipodal phase just above Erdős-Rényi

We now turn to finding entropy-maximizing graphons for small values of $\tilde{t}_2$, using the Method of Competing Inequalities of Section 3.1. We do this first for $e < 1/2$. Thanks to the $g \leftrightarrow 1 - g$ symmetry, that also solves the problem

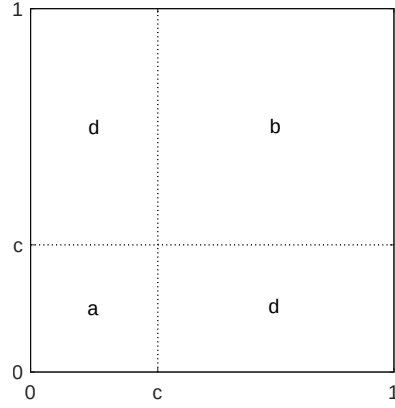


Figure 5.5 The parameters (a, b, c, d) that describe a general bipodal graphon.

for $e > 1/2$. We then study what happens when $e \approx 1/2$, filling the gap in the previous analysis. Taken together, we will prove

Theorem 5.8 *There is an open set in the $(e, \tilde{t}_2)$ plane, whose lower boundary is the line segment $\tilde{t}_2 = 0$, $0 \leq e \leq 1$, on which*

- *There is a unique entropy-maximizing graphon for each point $(e, \tilde{t}_2)$.*
- *The entropy-maximizing graphon is bipodal.*
- *The parameters describing the bipodal graphon vary analytically with e and $\tilde{t}_2$.*
- *The two values of the degree function are given, to leading order, by*

$$\frac{1}{2} \pm \sqrt{\left(e - \frac{1}{2}\right)^2 + \tilde{t}_2}.$$

- *The precise asymptotics of the bipodal parameters depend on the relative sizes of $\tilde{t}_2$ and $\left(e - \frac{1}{2}\right)^2$, as detailed in Theorems 5.9 and 5.11.*

In bipodal graphons, we use the letter c to denote the size of the first node and the letters a, b and d for values of the graphon. Specifically,

$$g(x, y) = \begin{cases} a & x, y < c, \\ b & x, y > c, \\ d & x < c < y \text{ or } y < c < x, \end{cases}$$

as illustrated in Figure 5.5. The graphons of Theorem 5.8 must be L^2 -close to

constant, so either c is small and b is close to e , or $1 - c$ is small and a is close to e , or a , b and d are all close to e .

5.3.1 Edge density below 1/2

Theorem 5.9 *Suppose that $0 < e < 1/2$ and that $\tilde{t}_2$ is sufficiently small (where the definition of “sufficiently small” can depend on e). Then there is a unique entropy-maximizing graphon. This graphon is bipodal. The parameters (a, b, c, d) vary analytically with e and $\tilde{t}_2$ and have the asymptotic form*

$$\begin{aligned} a &= \frac{(1-e)^3}{e^3 + (1-e)^3} + O(\tilde{t}_2) \\ b &= e + O(\tilde{t}_2) \\ c &= \frac{\tilde{t}_2}{(1-2e)^2} + O(\tilde{t}_2^2) \\ d &= 1 - e + O(\tilde{t}_2) \end{aligned} \tag{5.7}$$

as $\tilde{t}_2 \rightarrow 0$.

Proof We apply the Method of Competing Inequalities. As with the edge-triangle model, we define $g_1(x, y) = g(x, y) - e$ and try to

1. Minimize $\|g_1\|_2$ for a given $\tilde{t}_2$ and
2. Minimize the entropy cost for a given $\|g_1\|_2$.

Any graphon $g(x, y)$ can be expanded as

$$g(x, y) = e + (d(x) - e) + (d(y) - e) + g_2(x, y),$$

where g_2 is a function with zero marginals:

$$\int_0^1 g_2(x, y) dx = \int_0^1 g_2(x, y) dy = 0.$$

The four terms are all orthogonal in $L^2([0, 1]^2)$, so

$$\|g_1\|_2^2 = 2\|d - e\|_2^2 + \|g_2\|_2^2 = 2\tilde{t}_2 + \|g_2\|_2^2.$$

Minimizing $\|g_1\|_2$ is the same as setting $g_2 = 0$.

Maximizing the entropy for fixed $\|g_1\|_2$ works exactly the same as with the edge-triangle model. We define the function

$$f(e, e') = \begin{cases} \frac{H(e') - H(e) - (e' - e)H'(e)}{(e - e')^2} & e' \neq e, \\ \frac{1}{2}H''(e) & e' = e \end{cases}$$

and note that

$$\Delta S(g) := S(g) - H(e) = \iint f(e, g(x, y)) g_1(x, y)^2 dx dy.$$

This is the integral of the function $f(e, g(x, y))$ using the measure $g_1(x, y)^2 dx dy$. To maximize this, we want to concentrate the measure near the maximum value of $f(e, g(x, y))$. Viewed as a function of e' , $f(e, e')$ achieves a maximum value of $-H'(e)/(1-2e)$ at $e' = 1-e$. (See Appendix C for a proof of this and other properties of $f(e, e')$.) $\Delta S(g)$ is thus at most $-H'(e)\|g_1\|^2/(1-2e)$, which in turn is at most $-2H'(e)\tilde{t}_2/(1-2e)$.

In short, the recipe for maximizing the entropy for a given value of $\tilde{t}_2$ is to have $d(x)$ take on two values, one close to $1-e$ (on a set of small measure) and the other close to e , to have $g(x, y) = d(x) + d(y) - e$ everywhere, and to have $g(x, y)$ be close to e or $1-e$ everywhere. The last two conditions are contradictory, of course, since if $g(x, y) = d(x) + d(y) - e$, then there is a small square on which $g(x, y) \approx 2-3e$, which may not even be between 0 and 1! However, we can come close to achieving each condition with the graphon

$$g_a(x, y) = \begin{cases} e - \frac{(1-2e)(2c-c^2)}{(1-c)^2} & x, y > c \\ 1-e & \text{otherwise.} \end{cases}$$

The subscript a stands for “ansatz”. With this graphon, we compute

$$\begin{aligned} d(x) - e &= \begin{cases} 1-2e & x < c \\ -\frac{c}{1-c}(1-2e) & x > c, \end{cases} \\ \tilde{t}_2 &= \frac{c}{1-c}(1-2e)^2 \\ \|g_1\|_2^2 &= \frac{(1-2e)(2c-c^2)}{(1-c)^2} \\ &= \frac{1-2e}{1-c} \left(2c + \frac{c^2}{1-c} \right) \\ &= 2\tilde{t}_2 + \frac{\tilde{t}_2^2}{(1-2e)^2}. \end{aligned}$$

That is, $\|g_1\|_2$ is within a factor of $(1+O(\tilde{t}_2))$ of the minimum for the given $\tilde{t}_2$, and ΔS is within a factor of $(1+O(\tilde{t}_2))$ of the maximum for the given $\|g_1\|_2$, so ΔS is within a factor of $(1+O(\tilde{t}_2))$ of $-2H'(e)\tilde{t}_2/(1-2e)$.

Since the actual entropy maximizing graphon has at least as much entropy as g_a , it must come with a factor $(1+O(\tilde{t}_2))$ of minimizing $\|g_1\|$ and all but an $O(\tilde{t}_2)$ fraction of $\|g_1\|_2^2$ must come from values of $g_1(x, y)$ close to $1-2e$. This implies that the degree function $d(x)$ must take on a value close to $1-e$ on a set of measure close to $\tilde{t}_2/(1-2e)^2$.

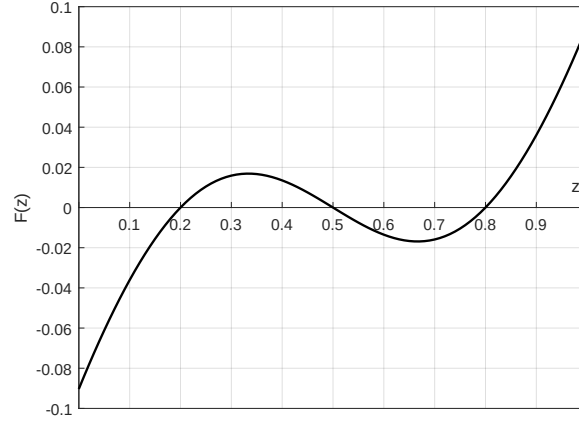


Figure 5.6 The function $F(z)$ of equation (5.5) when $e = 0.2$ and $\tilde{t}_2$ is small.

Now consider equation (5.4). Since $d(y)$ is close to e on all but a set of small measure, the right hand side is well-approximated by the logistic function

$$\frac{1}{1 + \exp(\alpha + \beta(z + e))},$$

where $z = d(x)$. A straight line can only intersect the graph of a logistic curve at three points. The fact that two of these points are near $z = e$ and $z = 1 - e$ determines the values of α and β and forces the third intersection point to be close to $z = 1/2$. The intersections are transverse, so making a small correction to the measure of $d(y)$ cannot change the number of roots of $F(z)$ or their approximate locations. See Figure 5.6.

The upshot is that any entropy maximizing graphon must be multipodal with podality at most 3, with the degree function taking on a value close to e , a value close to $1 - e$, and possibly a value close to $1/2$. Note that equation (5.4) does not say that the degree function **must** take on a value close to $1/2$, only that it **can**. To determine whether it does, we estimate the entropy cost (or gain) per unit width of having a small node with degree function close to $1/2$.

The reasoning behind this procedure is essentially the same as the proof of Theorem 2.12. In the edge-triangle model, the worth functional determines the change in entropy (relative to the change in edge and triangle densities) associated with changing the size of a node. In the absence of an analogous quantity for the edge-2star model, we will compute the first-order change to the entropy by hand.

So imagine that we have three nodes: A small node of width c_1 with $d(x)$ close to $1 - e$, a small node of width c_2 with $d(x)$ close to $1/2$, and a large node

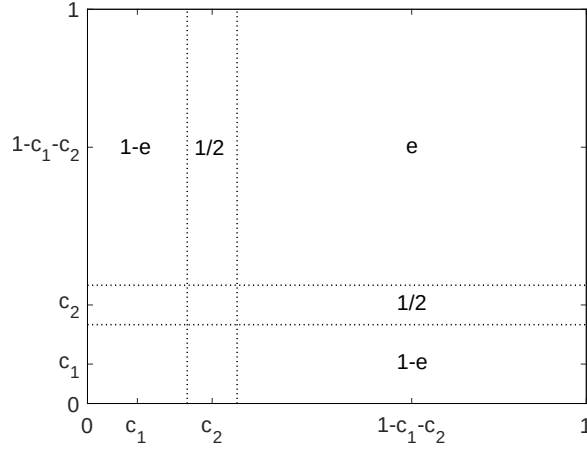


Figure 5.7 The approximate values for a tripodal graphon satisfying conditions (5.8). The unmarked regions are too small to materially affect the degree function.

of width $1 - c_1 - c_2$ with $d(x)$ close to e . Let the three intervals be denoted I_1 , I_2 and I_3 . The graphon must then take on the approximate values

$$\begin{aligned} 1 - e & \text{ on } I_1 \times I_3 \cup I_3 \times I_1, \\ 1/2 & \text{ on } I_2 \times I_3 \cup I_3 \times I_2, \\ e & \text{ on } I_3 \times I_3. \end{aligned} \tag{5.8}$$

We now consider, to first order in c_2 , the effect of having c_2 nonzero. Shifting some of the measure of $g_1(x, y)^2 dx dy$ from $I_1 \times I_3 \cup I_3 \times I_1$ to $I_2 \times I_3 \cup I_3 \times I_2$ changes the entropy for fixed $\|g_1\|_2$ by approximately

$$c_2(f(e, 1/2) - f(e, 1 - e))(1 - 2e)^2/2,$$

since the area of $I_2 \times I_3 \cup I_3 \times I_2$ is approximately $2c_2$ and $(\frac{1}{2} - e)^2 = (1 - 2e)^2/4$. That is, there is an entropy cost to first order in c_2 . To first order in c_2 , there is no compensating improvement to the value of $\|g_1\|$ for fixed $\tilde{t}_2$. To maximize entropy, we should therefore set $c_2 = 0$. That is, an entropy-maximizing graphon must actually be bipodal, not tripodal.

Before continuing, we take stock of what we have proven so far and plan our next steps. We know that an optimal graphon is bipodal with d close to $1 - e$ and with c close to $\tilde{t}_2/(1 - 2e)^2$. This means that we have reduced the space of possible entropy maximizers from an infinite-dimensional space to a 4-dimensional space with two constraints. We will express the most general

bipodal graphon with the given densities $(e, \tilde{t}_2)$ in terms of two free parameters (c, ν) .

The remainder of the proof is just a calculus problem. The calculations are grungy but not intrinsically difficult. We compute the partial derivatives $\partial_c S$ and $\partial_\nu S$. Setting these equal to zero gives the asymptotic expressions in equation (5.7). We then compute the Hessian

$$\begin{pmatrix} \partial_{cc}^2 S & \partial_{c\nu}^2 S \\ \partial_{c\nu}^2 S & \partial_{\nu\nu}^2 S \end{pmatrix}$$

and show that it is negative-definite, with negative trace and positive determinant, in the region where a solution might be found. This shows that there is a unique maximum to the entropy.

The same calculation also shows that small changes to e and $\tilde{t}_2$ can only give rise to small changes in the solution. Since the solutions to analytic equations are analytic everywhere except at singularities, this proves that (c, ν) , and therefore (a, b, c, d) , are analytic functions of e and $\tilde{t}_2$.

For a given value of c , consider the function

$$\nu(x) = \begin{cases} \sqrt{\frac{1-c}{c}} & x < c \\ -\sqrt{\frac{c}{1-c}} & x > c. \end{cases}$$

Any bipodal graphon with edge density e and node size c can be written in the form

$$g(x, y) = e + \mu (\nu(x) + \nu(y)) + \nu\nu(x)\nu(y),$$

where μ and ν are parameters. The degree function is then $d(x) = e + \mu\nu(x)$ and the reduced 2star density is

$$\tilde{t}_2 = \mu^2.$$

With e fixed and $\mu = \sqrt{\tilde{t}_2}$, our two free variables are c and ν . The parameters (a, b, d) are then

$$\begin{aligned} a &= e + 2\mu\sqrt{\frac{1-c}{c}} + \nu\frac{1-c}{c} \\ b &= e - 2\mu\sqrt{\frac{c}{1-c}} + \nu\frac{c}{1-c} \\ d &= d + \mu\left(\sqrt{\frac{c}{1-c}} - \sqrt{\frac{1-c}{c}}\right) - \nu. \end{aligned}$$

For d to be close to $1 - e$ and for a to remain between 0 and 1, we must have

$$c \approx \frac{\mu^2}{(1-2e)^2} = \frac{\tilde{t}_2}{(1-2e)^2}, \quad \mu \approx \sqrt{c}(1-2e), \quad \nu = O(c).$$

We compute the partial derivatives of a , b and d with respect to ν .

$$\begin{aligned} \partial_\nu a &= \frac{1-c}{c}, & \partial_\nu b &= \frac{c}{1-c}, & \partial_\nu d &= -1 \\ \partial_{\nu\nu}^2 a &= 0, & \partial_{\nu\nu}^2 b &= 0, & \partial_{\nu\nu}^2 d &= 0 \\ \partial_{\nu c}^2 a &= -1/c^2, & \partial_{\nu c}^2 b &= 1/(1-c)^2, & \partial_{\nu c}^2 d &= 0. \end{aligned}$$

Derivatives with respect to c are a bit more complicated:

$$\begin{aligned} \partial_c a &= -\mu c^{-3/2}(1-c)^{-1/2} - \nu c^{-2} &= O(c^{-1}) \\ \partial_c b &= -\mu c^{-1/2}(1-c)^{-3/2} + \nu(1-c)^{-2} &= O(1) \\ \partial_c d &= -\frac{\mu}{2} c^{-3/2}(1-c)^{-3/2} &= O(c^{-1}) \\ \partial_{cc}^2 a &= \frac{\mu}{2} (-3c^{-1/2}(1-c)^{-5/2} + c^{-3/2}(1-c)^{-3/2}) + \frac{2\nu}{c^3} &= O(c^{-2}) \\ \partial_{cc}^2 b &= \frac{\mu}{2} (3c^{-5/2}(1-c)^{-1/2} - c^{-3/2}(1-c)^{-3/2}) + \frac{2\nu}{(1-c)^3} &= O(c^{-1}) \\ \partial_{cc}^2 d &= \frac{3\mu}{4} (c^{-5/2}(1-c)^{-1/2} - c^{-1/2}(1-c)^{-5/2}) &= O(c^{-2}) \end{aligned}$$

With these in hand, we compute the derivatives of

$$S = c^2 H(a) + (1-c)^2 H(b) + 2c(1-c)H(d).$$

The first derivatives are

$$\begin{aligned} \partial_\nu S &= c(1-c)(H'(a) + H'(b) - 2H'(d)), \\ \partial_c S &= 2cH(a) - 2(1-c)H(b) + 2(1-2c)H(d) \\ &\quad + c^2 H'(a) \partial_c a + (1-c)^2 H'(b) \partial_c b + 2c(1-c)H'(d) \partial_c d \\ &= 2(H(d) - H(b)) + 2c(H(a) + H(b) - 2H(d)) + \nu(H'(b) - H'(a)) \\ &\quad - \frac{\mu}{\sqrt{c(1-c)}} (cH'(a) + (1-c)H'(b) + H'(d)) \end{aligned}$$

The second and third terms in the second-to-last line are $O(c)$, as are some of the contributions to the last line. The contributions to $\partial_c S$ that aren't $O(c)$ add up to

$$2(H(d) - H(b)) - \frac{\mu}{\sqrt{c(1-c)}} (H'(b) + H'(d)).$$

Note that this is zero when $b = e$ and $d = 1 - e$, as $H(1 - e) = H(e)$ and $H'(1 - e) = -H'(e)$. When c is nonzero, the additional terms will force b and d to change by $O(c)$, thus giving us the asymptotic forms of b and d shown in equation (5.7). Meanwhile, setting $\partial_\nu S = 0$, we get that

$$H'(a) = 2H'(d) - H'(b).$$

Since $b = e + O(\tilde{t})$ and $d = 1 - e + O(\tilde{t})$, $2H'(d) - H'(b) = -3H'(e) + O(\tilde{t})$. Thus

$$\ln\left(\frac{1-a}{a}\right) = -3\ln\left(\frac{1-e}{e}\right) + O(\tilde{t}).$$

Exponentiating both sides and solving for a then gives

$$a = \frac{(1-e)^3}{e^3 + (1-e)^3} + O(\tilde{t}).$$

All that remains is showing that the Hessian is negative-definite. The simplest term is

$$\partial_{vv}^2 S = (1-c)^2 H''(a) + c^2 H''(b) + 2c(1-c)H''(d).$$

This is a weighted average of the values of $H''(u)$ at $u = a, b$ and d . Since $H''(u)$ is bounded above by $H''(1/2) = -2$, this quantity is at most -2 . Next we consider the mixed term

$$\begin{aligned} \partial_{vc}^2 S &= (1-2c)(H'(a) + H'(b) - 2H'(d)) \\ &\quad + c(1-c)(H''(a)\partial_c a + H''(b)\partial_c b - 2H''(d)\partial_c d). \end{aligned}$$

Since $\partial_c(a)$ and $\partial_c(d)$ are $O(c^{-1})$ and $\partial_c(b) = O(1)$, the entire expression is $O(1)$. Finally, we consider the most singular part of $\partial_{cc}^2 S$, which scales as c^{-1} . These terms are:

$$\left[2H'(d) - \mu c^{-1/2}H''(d)\right]\partial_c d + \frac{\mu}{2}(H'(b) + H'(d))c^{-3/2}.$$

However, $H'(b) + H'(d) = H'(e) + H'(1-e) + O(c) = O(c)$, so the second term is actually $O(1)$. Since $\mu = \sqrt{c}(1-2e)$ (to leading order), the term in brackets is approximately

$$2 \int_{1/2}^{1-e} H''(u) - H''(d) du,$$

by the fundamental theorem of calculus. This integral is positive, since $H''(u)$ is a decreasing function on $[1/2, 1-e]$. Thus $\partial_{cc}^2 S$ scales as a positive multiple of

$$\partial_c d = -\mu c^{-3/2}/2 + O(1).$$

In particular, $\partial_{cc}^2 S$ is negative and scales as c^{-1} .

Since $\partial_{vv}^2 S$ is negative and bounded away from zero, $\partial_{cc}^2 S$ is negative and scales as c^{-1} , and $\partial_{vc}^2 = O(1)$, the Hessian matrix is negative definite. $\square$

Corollary 5.10 *The claims in Theorem 5.9 are true when $e > 1/2$ as well as when $e < 1/2$.*

Proof Suppose that $e > 1/2$. By the $g \leftrightarrow 1 - g$ symmetry, $B(e, \tilde{t}_2) = B(1 - e, \tilde{t}_2)$ and the entropy-maximizing graphon at $(e, \tilde{t}_2)$ is obtained by subtracting the (unique and bipodal by Theorem 5.9) entropy-maximizing graphon at $(1 - e, \tilde{t}_2)$ from 1. This is exactly of the form (5.7), only now with $e > 1/2$. $\square$

5.3.2 Behavior near $e = 1/2$

Thanks to Theorem 5.9 and Corollary 5.10, we understand the situation when $e \neq 1/2$ and $\tilde{t}_2$ is sufficiently small. However, our method of proof gave no indication of how small $\tilde{t}_2$ needs to be for the results to apply. Worse, the maximum possible value of $\tilde{t}_2$ in our construction goes to zero as $e \rightarrow 1/2$, since we have $\tilde{t}_2 \approx c^2(1 - 2e)^2 < (1 - 2e)^2$. [Note: this does not imply that the phase we have proven to exist is limited to a small region, only that the formulas (5.7) are only useful in a small region.]

The following theorem bridges the gap between Theorem 5.9 and Corollary 5.10, thereby finishing the proof of Theorem 5.8.

Theorem 5.11 *Let $\zeta = \sqrt{\tilde{t}_2 + \left(e - \frac{1}{2}\right)^2}$. For sufficiently small ζ , the entropy-maximizing graphon is unique and bipodal with parameters*

$$\begin{aligned} a &= 1 - e - 2\zeta + O(\zeta^2), \\ b &= 1 - e + 2\zeta + O(\zeta^2), \\ d &= 1 - e + O(\zeta^2), \\ c &= \frac{1}{2} \left(1 - \frac{e - \frac{1}{2}}{\zeta} \right) + O(\zeta). \end{aligned} \tag{5.9}$$

Furthermore, the exact values of the parameters a , b , c , and d are analytic functions of $(e, \tilde{t})$ everywhere except at the singular point $(\frac{1}{2}, 0)$.

Note that the degree function of the entropy-maximizing graphon is approximately $\frac{1}{2} - \zeta$ on the pole of size c and $\frac{1}{2} + \zeta$ on the pole of size $1 - c$. For fixed $e > 1/2$, this agrees, to leading order in $\tilde{t}_2$, with the results of Corollary 5.10. When $e < 1/2$, it agrees with the asymptotic results of Theorem 5.9 up to a symmetry that changes c to $1 - c$ and swaps the roles of a and b . When $e = 1/2$, we have the symmetric graphon that we previously called g_p , with $p = \frac{1}{2} \pm 2\zeta$.

Proof We combine the Method of Competing Inequalities with the same power series analysis that we used to prove Theorem 3.7 to show that any actual entropy maximizing graphon is close to an explicit bipodal graphon g_0 with

$$a = 1 - e - 2\zeta,$$

$$\begin{aligned} b &= 1 - e + 2\zeta, \\ d &= 1 - e, \\ c &= \frac{1}{2} \left(1 - \frac{e - \frac{1}{2}}{\zeta} \right). \end{aligned}$$

As noted in the proof of Theorem 5.9, any graphon g with degree function $d(x)$ can be uniquely written as

$$g(x, y) = d(x) + d(y) - e + g_2(x, y),$$

where g_2 has zero marginals. We will show that $g_2(x, y)$ is pointwise $O(\zeta^2)$ and that $d(x) - \frac{1}{2}$ only takes on two values, within $O(\zeta^2)$ of $\pm\zeta$. This implies that g is bipodal and follows the estimates (5.9). The analyticity of (a, b, c, d) then follows from the implicit function theorem.

We do this in three steps:

1. We estimate the entropy of an arbitrary graphon and of the ansatz graphon g_0 . We will show that, in an entropy-maximizing graphon, the variance of $(d(x) - \frac{1}{2})^2$ must be $O(\zeta^6)$.
2. We show that $F(z) = z - k(z)$ has exactly three roots, counted with multiplicity. This implies that our graphon is either bipodal with parameter values close to those of g_0 , or tripodal with a very small third node. We rule out the latter possibility.
3. We examine the variational equations on the space of bipodal graphons in a neighborhood of g_0 and show that there is a unique solution that depends analytically on $(e, \tilde{t})$.

Step 1: For any graphon, let

$$\nu_k = \int_0^1 \left(d(x) - \frac{1}{2} \right)^k dx$$

and

$$\mu_k = \iint_{[0,1]^2} \left(g(x, y) - \frac{1}{2} \right)^k dx dy$$

be the moments of $d(x) - 1/2$ and $g(x, y) - 1/2$. As in the proof of Theorem 3.7, the entropy $S(g)$ can be expanded as

$$S(g) = H(1/2) + \sum_{k=1}^{\infty} \frac{H^{(2k)}(1/2)}{(2k)!} \mu_{2k}.$$

Since all of the even derivatives $H^{(2k)}(1/2)$ are negative, we maximize the entropy by minimizing the moments μ_{2k} .

Note that the first two moments of $d(x) - 1/2$ are determined by e and $\tilde{t}$:

$$\nu_1 = e - \frac{1}{2} \text{ and } \nu_2 = \zeta^2 = \tilde{t} + \left(e - \frac{1}{2}\right)^2.$$

Higher even moments are bounded from below:

$$\nu_{2k} \geq \nu_2^k,$$

with equality if and only if $\left(d(x) - \frac{1}{2}\right)^2$ is constant. In particular, $\nu_4 - \nu_2^2$ is the variance of $\left(d(x) - \frac{1}{2}\right)^2$.

To obtain the moments μ_{2k} , we expand $g(x, y) - 1/2$ as

$$g(x, y) - \frac{1}{2} = \left(d(x) - \frac{1}{2}\right) + \left(d(y) - \frac{1}{2}\right) - \left(e - \frac{1}{2}\right) + g_2(x, y),$$

expand $(g(x, y) - \frac{1}{2})^{2k}$ in terms of these pieces, and integrate. The first two non-trivial moments of $g - 1/2$ are:

$$\begin{aligned} \mu_2 &= 2\nu_2 - \left(e - \frac{1}{2}\right)^2 + \|g_2\|_2^2, \\ \mu_4 &= 2\nu_4 + 6\nu_2^2 - 12\left(e - \frac{1}{2}\right)^2 \nu_2 + 5\left(e - \frac{1}{2}\right)^4 \\ &\quad + 24 \iint g_2(x, y) \times \\ &\quad \left[\left(d(x) - \frac{1}{2}\right)^2 \left(d(y) - \frac{1}{2}\right) - \left(e - \frac{1}{2}\right) \left(d(x) - \frac{1}{2}\right) \left(d(y) - \frac{1}{2}\right) \right] dx dy \\ &\quad + 12 \iint g_2(x, y)^2 \left[\left(d(x) - \frac{1}{2}\right)^2 - \left(d(x) - \frac{1}{2}\right) \left(d(y) - \frac{1}{2}\right) \right] dx dy \\ &\quad + 4 \iint g_2(x, y)^3 \left[2\left(d(x) - \frac{1}{2}\right) - \left(e - \frac{1}{2}\right) \right] dx dy \\ &\quad + \iint g_2(x, y)^4 dx dy + 6\left(e - \frac{1}{2}\right)^2 \|g_2\|_2^2. \end{aligned}$$

For the graphon g_0 , this simplifies to

$$\begin{aligned} \mu_2 &= 2\nu_2 - \left(e - \frac{1}{2}\right)^2 \\ \mu_4 &= 2\nu_2^2 - 12\nu_2 \left(e - \frac{1}{2}\right)^2 + 5\left(e - \frac{1}{2}\right)^4 \\ &= 2\zeta^4 + 6 \left[\zeta^2 - \left(e - \frac{1}{2}\right)^2 \right]^2 - \left(e - \frac{1}{2}\right)^4. \end{aligned}$$

Comparing these expressions for g and g_0 , we see that there is an $H''(1/2)\|g_2\|_2^2$ cost in having g_2 nonzero and a $H''''(1/2)(\nu_4 - \nu_2^2)/12$ cost in having $(d(x) - \frac{1}{2})^2$ not be constant. There are also costs in μ_4 from terms proportional to g_2^2 or g_2^4 .

The lowest-order benefits from having g_2 nonzero are terms proportional to

$$\iint g_2(x, y) \left(d(x) - \frac{1}{2}\right)^2 \left(d(y) - \frac{1}{2}\right) dx dy \quad \text{and} \\ \iint g_2(x, y) \left(e - \frac{1}{2}\right) \left(d(x) - \frac{1}{2}\right) \left(d(y) - \frac{1}{2}\right) dx dy$$

By Cauchy-Schwarz, the first term is $O(\zeta\|g_2\|_2\sqrt{\nu_4})$, while the second is $O(\zeta^3\|g_2\|_2)$. Since ν_4 cannot be much greater than ν_2^2 , the maximum possible benefit is $O(\zeta^3\|g_2\|_2)$.

Through μ_4 , those are the only possible benefits of having g_2 nonzero. Any benefits from the expansion of μ_6 and higher are higher order in ζ , $\|g_2\|_2$, or both, so the total benefit of having g_2 nonzero is $O(\zeta^3\|g_2\|_2)$. With a cost proportional to $\|g_2\|_2^2$ and benefits that are $O(\zeta^3\|g_2\|_2)$, $\|g_2\|_2$ must itself be $O(\zeta^3)$ and the net benefit of having g_2 nonzero is $O(\zeta^6)$.

In computing μ_6 and higher, there are potential benefits to having $(d(x) - 1/2)^2$ not be constant, such as a term proportional to $(e - \frac{1}{2})\nu_2\nu_3$ in the expansion of μ_6 . However, these are all $O(\zeta^6)$ or smaller. The upshot is that the sum of *all* of the benefits of having g be different from g_0 is $O(\zeta^6)$.

By assumption, the graphon g maximizes the entropy and in particular has more entropy than g_0 . Since the entropy benefits from having g different from g_0 are $O(\zeta^6)$, the costs (which by assumption are smaller than the benefits) must also be $O(\zeta^6)$. In particular, the variance of $(d(x) - \frac{1}{2})^2$ must be $O(\zeta^6)$.

Step 2: Since $\int (d(x) - 1/2)^2 dx = \zeta^2$ and the variance of $(d(x) - 1/2)^2$ is $O(\zeta^6)$, $d(x)$ must take on values within $O(\zeta^2)$ of $\frac{1}{2} \pm \zeta$. In particular, $F(z)$ must have two roots close to $z = 1/2$. Put another way, the line $y = z$ is nearly tangent to the graph $y = k(z)$ at $z = 1/2$. However, the third derivative of the logistic function is positive near the point of inflection. After convolving by a degree function that is localized near $d = 1/2$, the third derivative is still positive, meaning that the graph of $k(z)$ (and therefore the graph of $F(z)$) also has only one point of inflection. By Rolle's Theorem, this implies that $F(z)$ has at most three roots, counted with multiplicity. Thus our optimal graphon is at most tripodal.

We recall some facts about cubic polynomials. Suppose that

$$f(x) = x^3 + a_2x^2 + a_1x + a_0$$

has three real roots. The sum of the roots is $-a_2$ and the average of the roots

is $-a_2/3$, which is also the unique point where $f''(x) = 0$. Now consider the following convolution of f made with the function $d(y)$:

$$\tilde{f}(x) = \int_0^1 f(x + d(y)) dy.$$

This is still a cubic polynomial. If $\tilde{f}$ has three roots, then the sum of the roots is $-3e - a_2$, where $e = \int_0^1 d(y) dy$, and the average of the roots is $-e - a_2/3$. Furthermore, a simple algebraic calculation shows that

$$\tilde{f}(-e - \frac{a_2}{3}) = f(-\frac{a_2}{3}) + \int_0^1 (d(y) - e)^3 dy.$$

Now consider the function

$$\frac{1}{1 + \exp(\alpha + \beta z)}$$

Near the point of inflection $z = -\alpha/\beta$, this function is approximately cubic, with corrections of order $(z + \alpha/\beta)^5$. The value of this function at the point of inflection is exactly $1/2$. Convolving this function by the degree function $d(y)$ we obtain the function $k(z)$. The solutions to $k(z) = z$ have average value $\bar{d} = -e - \frac{\alpha}{\beta}$. The value of $k(\bar{d})$ is $\frac{1}{2} + \int_0^1 (d(x) - e)^3 dx$, plus $O((\bar{d} - e)^5)$ corrections due to $k(z)$ not being exactly cubic.

We have already shown that two of the three roots of $k(z) - z$ are $\frac{1}{2} \pm \zeta + O(\zeta^2)$. Since $\int_0^1 d(x)^3 dx$ is then $O(\zeta^3)$, this implies that the average of the three roots is $\frac{1}{2} + O(\zeta^3)$, which implies that the third root must be $\frac{1}{2} + O(\zeta^2)$. Let c_3 be the size of the corresponding node.

We now compute the cost

$$v_4 - v_2^2 = \int_0^1 ((d(x) - \frac{1}{2})^2 - \zeta^2)^2 dx \approx c_3 \zeta^4.$$

associated with having c_3 nonzero. The derivative of the entropy with respect to c_3 has a positive term of order ζ^4 . Since $d(x) - 1/2$ is pointwise $O(\zeta)$, all of the contributions to μ_6 and higher have order ζ^6 and higher, and cannot overcome this quartic cost. Nor can the cross-terms with g_2 , which we have shown to be $O(\zeta^6)$. Since the derivative of the entropy with respect to c_3 is positive, the entropy is maximized when $c_3 = 0$. That is, the optimal graphon is bipodal, not tripodal, with the degree function on the two nodes being $\frac{1}{2} \pm \zeta + O(\zeta^2)$.

Step 3: Bipodal graphons are described by four parameters (a, b, c, d) . The edge density, 2star density, and entropy are all analytic functions of (a, b, c, d) , so maximizing the entropy for each $(e, \tilde{t}_2)$ gives a 2-dimensional analytic variety of solutions in $(a, b, c, d, e, \tilde{t}_2)$ -space. To see that this analytic variety makes

(a, b, c, d) into analytic functions of e and t , we need only check that the tangent space does not degenerate. That is, we must check that on this family of solutions, the partial derivatives of (a, b, c, d) with respect to e and $\tilde{t}_2$ are finite and linearly independent.

We show this first for the approximate solutions g_0 , except at the singularity $\zeta = 0$ (where the derivative of ζ with respect to $\tilde{t}$ diverges). Note that $\partial d / \partial e = -1$, while $\partial d / \partial \tilde{t} = 0$. However, the partial derivatives of a, b and c with respect to $\tilde{t}$ are all nonzero, showing that $\partial_e(a, b, c, d)$ and $\partial_{\tilde{t}}(a, b, c, d)$ are linearly independent vectors. In particular, the minor

$$\frac{\partial a}{\partial e} \frac{\partial d}{\partial \tilde{t}_2} - \frac{\partial d}{\partial e} \frac{\partial a}{\partial \tilde{t}_2} = \frac{1}{\zeta}.$$

The difference between the true optimal graphon g and the approximant g_0 is small, in particular being $O(\zeta^2)$ for a and d , with the derivatives of these terms with respect to ζ being $O(\zeta)$. Since

$$\frac{\partial \zeta}{\partial e} = \frac{e - \frac{1}{2}}{\zeta} = O(1) \text{ and } \frac{\partial \zeta}{\partial \tilde{t}_2} = \frac{1}{2\zeta},$$

these extra terms can only change $\partial_e(a)$ and $\partial_e(d)$ by $O(\zeta)$ and $\partial_{\tilde{t}}(a)$ and $\partial_{\tilde{t}}d$ by $O(1)$, resulting in an $O(1)$ change in $\partial_e a \partial_{\tilde{t}_2} d - \partial_e d \partial_{\tilde{t}_2} a$, which remains nonzero for small ζ . □

Corollary 5.12 *For $e = 1/2$ and sufficiently small values of $\tilde{t}_2$, there is no symmetry breaking. Instead, the entropy-maximizing graphon is unique and symmetric, of the form g_p of equation (5.1), with $p = \frac{1}{2} \pm \sqrt{4\tilde{t}_2}$.*

Proof By Theorem 5.11, the entropy-maximizing graphon is bipodal and unique. Thus, if g is an entropy-maximizing graphon, then $1 - g$, which is also entropy-maximizing with the same value of $\tilde{t}_2$, must be equivalent to g . For a bipodal graphon g to be equivalent to $1 - g$, we must have $c = d = 1/2$ and $a + b = 1$. That is, we must have g_p . Since $\tilde{t}_2(g_p) = (p - \frac{1}{2})^2/4$, p must equal $\frac{1}{2} \pm \sqrt{4\tilde{t}_2}$. □

5.4 Symmetry-breaking and non-uniqueness

Consider the line segment $e = 1/2$. Near the bottom of the segment, Corollary 5.12 tells us that symmetry is preserved with the optimizing graphon being of the form g_p . Near the top of the segment, Theorem 5.4 tells us that symmetry is broken. This begs two questions:

- What is the nature of the symmetry breaking for large $\tilde{t}_2$?
- What happens for intermediate values of $\tilde{t}_2$?

In this section we give a complete answer to the first question. The second question remains open at this time.

Let ϵ be a parameter, strictly less than half the cut metric distance between the clique with $(e, \tilde{t}_2) = (1/2, (\sqrt{2} - 1)/4)$ and the anticlique with the same densities. We call a graphon “cliquelike” if it is within ϵ of a clique in the cut metric and “anticliquelike” if it is within ϵ of an anticlique.

Lemma 5.13 *There is a neighborhood U of $(1/2, (\sqrt{2} - 1)/4)$ such that every graphon with $(e, \tilde{t}_2) \in U$ is either cliquelike or anticliquelike.*

Proof Suppose otherwise. Then there is a sequence of graphons g_i , with edge and reduced 2star densities converging to $(1/2, (\sqrt{2} - 1)/4)$, that are neither cliquelike nor anticliquelike. By compactness, there exists a subsequential limit with edge and reduced 2star densities equal to $(1/2, (\sqrt{2} - 1)/4)$. This limiting graphon is neither a clique nor an anticlique, contradicting the theorem of [1] that all $\tilde{t}_2$ maximizers are cliques or anticliques. $\square$

For each $(e, \tilde{t}_2) \in U$, let $B_c(e, \tilde{t}_2)$ be the maximum entropy among cliquelike graphons and let $B_a(e, \tilde{t}_2)$ be the maximum among anticliquelike graphons. (These quantities measure the rate of growth in the number of cliquelike and anticliquelike graphs with densities $(e, \tilde{t}_2)$, but that interpretation is not needed in this proof.) If there are no cliquelike graphons with densities $(e, \tilde{t}_2)$, then we set $B_c(e, \tilde{t}_2) = 0$, and likewise for anticliquelike graphons and B_a .

Theorem 5.14 *There is an open set U in $(e, \tilde{t})$ -space containing the point $(\frac{1}{2}, \frac{\sqrt{2}-1}{4})$ such that*

- When $e > \frac{1}{2}$, the entropy-optimizing graphon is cliquelike, bipodal and unique,
- When $e < \frac{1}{2}$, the entropy-optimizing graphon is anticliquelike, bipodal and unique.
- When $e = \frac{1}{2}$, there are two entropy-optimizing graphons, one cliquelike and the other anticliquelike, and both bipodal.

Proof We begin by showing that cliquelike graphons dominate when $e > 1/2$ and anticliquelike graphons dominate when $e < 1/2$. Thanks to the $g \leftrightarrow 1 - g$ symmetry, $B_c(e, \tilde{t}_2) = B_a(1 - e, \tilde{t}_2)$, and in particular $B_c(1/2, \tilde{t}_2) = B_a(1/2, \tilde{t}_2)$.

Lemma 5.15 *For fixed $\tilde{t}_2$, and in a sufficiently small neighborhood of $(1/2, (\sqrt{2}-1)/4)$, $B_c(e, \tilde{t}_2)$ is an increasing function of e .*

Proof of lemma Let g be the graphon that maximizes the entropy among cliquelike graphons with density $(e, \tilde{t}_2)$. Since $(e, \tilde{t}_2)$ is close to the upper boundary, $g(x, y)$ must take on values both very close to 1 and very close to 0. By equation (5.3), this implies that there are values of (x, y) where $\exp(\alpha + \beta(d(x) + d(y)))$ is close to zero and other values where $\exp(\alpha + \beta(d(x) + d(y)))$ is large and positive. Since $d(x) + d(y)$ is always between 0 and 2, this is only possible if β is large (and negative, since larger values of $d(x) + d(y)$ are associated with larger values of $g(x, y)$), approaching $-\infty$ as we approach the upper boundary.

Now consider a change to the graphon in which we expand the size c of a podes with $d(x)$ close to $\sqrt{2}/2$ at the expense of a podes with $d(x)$ close to 0. Since the derivative of the entropy is a linear combination of the derivatives of the edge and 2star densities with coefficients $\alpha/2$ and $\beta/2$, we must have

$$\frac{dS}{dc} = \frac{\alpha}{2} \frac{de}{dc} + \frac{\beta}{2} \frac{dt_2}{dc}.$$

However,

$$\frac{de}{dc} \approx 2\sqrt{e}, \quad \frac{dt_2}{dc} \approx 3e, \quad \text{and} \quad \frac{dS}{dc} = o(1).$$

The last claim is true because we are not changing the values of $g(x, y)$, merely changing the area of the region where each value is achieved. We are expanding the area where $g(x, y) \approx 1$ by $O(c)$ at the expense of the area where $g(x, y) \approx 0$. Since $H(0) = H(1) = 0$, so dS/dc is small.

Since β is large but $\sqrt{e}\alpha + \frac{3}{2}e\beta$ is not, we must have

$$\frac{\alpha}{\beta} \approx \frac{-3\sqrt{e}}{2}. \quad (5.10)$$

Since β is negative, α must be positive. But $\alpha = 2\partial_e B_c(e, \tilde{t}_2)$, so $B_c(e, \tilde{t}_2)$ is an increasing function of e . $\square$

Since $B_c(e, \tilde{t}_2)$ is an increasing function of e , $B_a(e, \tilde{t}_2) = B_c(1 - e, \tilde{t}_2)$ must be a decreasing function of e , so $B_c(e, \tilde{t}_2) - B_a(e, \tilde{t}_2)$ must be an increasing function of e . Since this difference is zero when $e = 1/2$, it must be negative when $e < 1/2$ and positive when $e > 1/2$. That is, cliquelike graphons have more entropy when $e > 1/2$, anticliquelike graphons have more entropy when $e < 1/2$, and it's a tie when $e = 1/2$.

What remains for the proof of Theorem 5.14 is showing that the entropy-optimizing graphons are unique and bipodal. Thanks to the $g \leftrightarrow 1-g$ symmetry,

it is sufficient to show this for cliquelike graphons with $e \geq 1/2$. Let g be such an entropy-maximizing cliquelike graphon. We prove each of these properties of g in turn.

1. The degree function $d(x)$ only takes values close to 0, $\sqrt{e}/2$, and $\sqrt{e}$.
2. The degree function $d(x)$ only takes values close to 0 or $\sqrt{e}$.
3. The degree function $d(x)$ only takes two values, one close to 0 and one close to $\sqrt{e}$. That is, g is bipodal.
4. The parameters that define this bipodal graphon are uniquely determined.

Plugging equation (5.10) into equations (5.3), (5.4) and (5.6) gives

$$\begin{aligned} g(x, y) &= \frac{1}{1 + \exp\left(-\beta(d(x) + d(y) - \frac{3}{2}\sqrt{e} + o(1))\right)}, \\ d(x) &= \int_0^1 \frac{dy}{1 + \exp\left(-\beta(d(x) + d(y) - \frac{3}{2}\sqrt{e} + o(1))\right)}, \\ k(z) &= \int_0^1 \frac{dy}{1 + \exp\left(-\beta(z + d(y) - \frac{3}{2}\sqrt{e} + o(1))\right)}. \end{aligned}$$

Since β is large and negative, $g(x, y)$ is close to 1 whenever $d(x) + d(y)$ is bigger than a threshold value that is $\frac{3}{2}\sqrt{e} + o(1)$, is close to 0 whenever $d(x) + d(y)$ is smaller than that threshold, and only takes values substantially different from 0 or 1 when $d(x) + d(y)$ is very close to that threshold. Note that $\sqrt{e} \approx \sqrt{2}/2 > 2/3$, so the threshold is greater than 1. Since the function $d(y)$ is close to 0 on a set of measure approximately $1 - \sqrt{e}$ and close to $\sqrt{e}$ on a set of measure approximately $\sqrt{e}$, and only takes on other values on sets of small measure, the function $k(z)$ is approximately a step function, as shown in Figure 5.8.

Of course the function $k(z)$ isn't exactly a step function. However, changing the graph of $k(z)$ slightly by having β finite and making L^1 -small changes to $d(y)$ can't create intersection points far from where they already are. The only possible values of z are close to 0, close to $\sqrt{e}/2$ or close to $\sqrt{e}$, as claimed. This completes the first step.

Let I_1 be the union of all the podes where $d(x)$ is close to $\sqrt{e}$, let I_2 be the union of all the podes where $d(x)$ is close to $\frac{1}{2}\sqrt{e}$, and let I_3 be the union of all the podes where $d(x)$ is close to 0. By the definition of clique-like, the measure of I_1 must be close to $\sqrt{e}$, the measure of I_3 must be close to $1 - \sqrt{e}$ and the measure of I_2 must be close to 0.

Note that $d(x) + d(y)$ is above the threshold of $\frac{3}{2}\sqrt{e}$ on $I_1 \times I_1$, is close to the threshold on $I_1 \times I_2 \cup I_2 \times I_1$, and is below the threshold everywhere else.

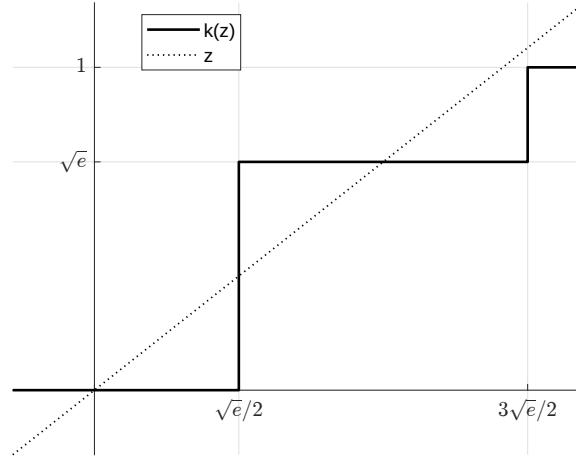


Figure 5.8 When $e \approx 1/2$ and $\tilde{t}_2$ is nearly maximal, there are only two solutions to $k(z) = z$.

This implies that $g(x, y)$ is pointwise close to 1 on $I_1 \times I_1$, takes on the average value $1/2$ on $I_1 \times I_2$ (in order for the degree function to be $\frac{1}{2}\sqrt{e}$) and is close to zero everywhere else.

Now consider what happens as we vary the size of I_2 while keeping e fixed. The entropy associated with the region $I_1 \times I_2 \cup I_2 \times I_1$ is linear in the size of I_2 , as is the extent to which $\tilde{t}$ (which is the variance of the degree function) is reduced from the maximum. That is, β must be $O(1)$. However, β must diverge as t approaches the maximum value, as otherwise the graphon would not approach 0 on $I_1 \times I_1$ and 1 on $I_3 \times I_3$. This contradiction implies that the size of I_2 is in fact zero, completing the second step.

Next we consider the solutions of $k(z) = z$ near $z = 0$ and $z = \sqrt{e}$. Having multiple nodes with $d(x)$ close to 0, or multiple nodes with $d(x)$ close to $\sqrt{e}$, could smear the vertical part of the step function somewhat, but the portions of the graph near 0 and $\sqrt{e}$ are nearly flat. (If there were any nodes with $d(x)$ close to $\sqrt{e}/2$, that would introduce small steps near $z = \sqrt{e}$, insofar as the threshold is $\frac{3}{2}\sqrt{e}$, but we just ruled out the existence of such nodes.) Since $k'(z)$ is never greater than 1 near $z = 0$ or $z = \frac{1}{2}\sqrt{e}$, there can only be one point near 0 and only one point near $\sqrt{e}$ where $k(z) = z$. That is, the graphon must be bipodal, completing the third step.

Consider the parameters (a, b, c, d) of this bipodal graphon. Since we are looking for clique-like graphons, we want $a \approx 1$, $b \approx 0$, $c \approx \sqrt{e}$, $d \approx 0$. We compute the gradient of the edge density, 2star density and entropy with respect

to (a, b, c, d) and set

$$2\nabla S = \alpha \nabla e + \beta \nabla t_2.$$

The four components of this vector equation, plus the constraints on e and t_2 , give six equations in six unknowns. The system of equations is non-degenerate and yields a single family of solutions with $a \approx 1$, $b \approx 0$, $c \approx \sqrt{e}$, and $d \approx 0$, namely

$$\begin{aligned} a &= 1 - \delta + O(\delta^2) \\ b &= O(\delta^3) \\ c &= \sqrt{e} + \left(\frac{3}{2}\sqrt{e} - 1\right)\delta + O(\delta^2) \\ d &= \delta + O(\delta^2), \end{aligned}$$

where δ is a small parameter. □

5.5 Conjectures and open problems

So far, the results of this chapter have all been rigorous mathematics. We have proven that entropy maximizers are all multipodal, that there is a uniform bound on the podality, that there is an unbroken bipodal phase (with unique maximizing graphons that depend analytically on e and $\tilde{t}_2$) when $\tilde{t}_2$ is small, and that there is a symmetry-breaking phase transition at $e = 1/2$ when $\tilde{t}_2$ is large.

Our understanding is also enhanced by numerical experiments, searching the sample space of multipodal graphons to find the best one for each point $(e, \tilde{t}_2)$. Based on this numerical evidence, we present three conjectures for future research:

Conjecture 5.16 *For every achievable pair $(e, \tilde{t}_2)$, the entropy-maximizing graphons are all either bipodal (when $\tilde{t}_2 > 0$) or Erdős-Rényi (when $\tilde{t}_2 = 0$).*

Conjecture 5.17 *There is a value $\tilde{t}_2^* \in (0, (\sqrt{2} - 1)/4)$ such that the entropy-maximizing graphon is unique, with parameters that depend analytically on $(e, \tilde{t}_2)$, everywhere except when $e = 1/2$ and $\tilde{t}_2 > \tilde{t}_2^*$. When $e = 1/2$ and $\tilde{t}_2 > \tilde{t}_2^*$, there are two distinct entropy maximizing graphons, one the limit of the entropy maximizer from the left and the other the limit from the right.*

This situation may seem peculiar, as we have a phase *transition* crossing the line segment $e = 1/2$, $\tilde{t}_2 > \tilde{t}_2^*$, but we only have one *phase*, insofar as the two sides of the phase transition are connected by a path that goes around

the “critical point” $(1/2, \tilde{t}_2^*)$. This is exactly what happens with fluids in the real world. For instance, water has a critical point at a temperature of 374° Celsius and a pressure of 217.75 atmospheres. We usually think of liquid water and steam as belonging to separate phases, as we are familiar with the phase transition that occurs when we heat water above 100° Celsius at one atmosphere of pressure. However, we can also convert water to steam continuously, without any sort of phase transition, by going around the critical point, for instance by raising the pressure above 217.75 atmospheres, then increasing the temperature above 374° , then lowering the pressure, and finally lowering the temperature. In truth, there is only one “fluid” phase that encompasses both liquid and gas; those two terms are primarily useful when liquid and gas are in contact.

Conjecture 5.18 *The critical value $\tilde{t}_2^*$ is the same as the “bifurcation point” at which the graphon g_p becomes unstable to small perturbations in the node size c . That is, for $e = 1/2$ and $\tilde{t}_2 \leq \tilde{t}_2^*$ the entropy maximizing graphon is unique and is g_p , but there is a bifurcation at $\tilde{t}_2 = \tilde{t}_2^*$, after which the parameter c veers away from $1/2$.*

The bifurcation point is known [56]. It satisfies an explicit transcendental equation and is approximately 0.0372763703. What is not known is whether this bifurcation point is actually a critical point, as described in Conjecture 5.17.

Edge- k star models

A k star (often written k -star) is a graph with k edges and $k + 1$ vertices, in which one central vertex is connected to all the others. We denote the k star density as $t_k = \tau_k(g)$ and the reduced k star density $t_k - e^k$ as $\tilde{t}_k = \tilde{\tau}_k(g)$. In terms of the degree function $d(x)$, we have

$$t_k = \int_0^1 d(x)^k dx.$$

In Chapter 5, we studied the edge-2star model and proved a number of important properties about it:

- The bottom of the phase space is marked by Erdős-Rényi graphons, while the top is marked by cliques and anticliques.
- The Boltzmann entropy function $B(e, \tilde{t}_2)$ is continuous. For fixed e , it is a strictly decreasing function of $\tilde{t}_2$.
- All entropy-maximizing graphons are multipodal. In fact, there is a universal bound on the podality of entropy maximizers. (We conjecture that this bound is in fact 2.)
- Just above the Erdős-Rényi line, the optimal graphon is unique and bipodal, with parameters that vary analytically with $(e, \tilde{t}_2)$.
- There is a phase transition from anticliquelike to cliquelike graphons near the top of the phase space.

In this chapter, we prove similar results for the edge- k star model where $k \geq 3$. In some cases, the arguments are largely a repetition of what we already did for 2stars, in which case we will skip the details. We will still provide details when the changes are significant, or when the result has not previously appeared in the literature.

We will also consider models where we constrain the density of edges and of

a positive linear combinations of k star densities. This may seem peculiar, but every model involving the density of edges and one other subgraph H can be approximated, just above the Erdős-Rényi curve, by a model involving such a linear combination of k stars. Thanks to the results in this chapter about linear combinations of k stars, we will be able to prove universal results, applying to an arbitrary graph H , in Chapter 7.

Some of the results in this chapter have been previously published, and we will acknowledge this; our proofs are meant to meld coherently with the proofs in the book.

6.1 Phase space

As always, the most basic question is to determine all the possible values of $(e, \tilde{t}_k)$.

Theorem 6.1 *Suppose that $k \geq 2$. For each $e \in (0, 1)$, the set of possible values of $\tilde{t}_k$ is a closed interval $[0, \tilde{t}_{\max}]$, where*

$$\tilde{t}_{\max} = \max \left(e^{(k+1)/2} - e^k, (1 - \sqrt{1-e})^k - (1 - \sqrt{1-e})^{k+1} + 1 - \sqrt{e} - e^k \right).$$

The minimum is achieved by graphons (such as the Erdős-Rényi graphon) that have $d(x) = e$ for all x . The maximum is achieved by a clique or an anticlique (or, for one value of e , by both), and not by any other graphons.

Proof As with the proof of Theorem 3.1, the existence of maximum and minimum values of $\tilde{t}_k$ follows from the compactness of $\mathcal{G}_e$. The fact that all values of $\tilde{t}_k$ between the maximum and the minimum are achieved follows from continuity, as we linearly interpolate between a graphon that achieves the minimum value to a graphon that achieves the maximum value.

The minimum value of e^k is achieved by an Erdős-Rényi graphon. To see that no smaller values are possible, note that the difference between two k th powers can be written as

$$a^k - b^k = kb^{k-1}(a-b) + (a-b)^2 \left(a^{k-2} + 2a^{k-3}b + \cdots + (k-1)b^{k-2} \right),$$

so

$$\begin{aligned} \tilde{t}_k &= \int_0^1 d(x)^k - e^k dx \\ &= \int_0^1 ke^{k-1}(d(x) - e) \\ &\quad + (d(x) - e)^2 \left(d(x)^{k-2} + 2ed(x)^{k-3} + \cdots + (k-1)e^{k-2} \right) dx. \end{aligned}$$

The integral of the first term is zero and the integral of the second is non-negative, equalling zero if and only if $d(x) = e$ almost everywhere. (The same result also follows easily from Holder's inequality, but we prefer an explicit demonstration.)

As for the maximum, the proof for $k = 2$ first appeared in [1], while that for $2 < k \leq 30$ first appeared in [25]. Cliques have $t_k = e^{(k+1)/2} - e^k$ and do better than anticliques when e is large. Anticliques have

$$\tilde{t}_k = (1 - \sqrt{1 - e})^k - (1 - \sqrt{1 - e})^{k+1} + 1 - \sqrt{e} - e^k$$

and do better when e is small. For each k , there is a crossover point where the two expressions are equal and there are two inequivalent t_k -maximizing graphons. This crossover point is $e = 1/2$ when $k = 2$, is $e = 3/4$ when $k = 3$, and approaches 1 as k grows.

The analysis in [25] shows that a $\tilde{t}_k$ -maximizing graphon is either a clique, an anticlique, or a tripodal graphon of a particular form. Eliminating the third possibility is equivalent to showing that a certain algebraic function (which depends on k) has no roots on the unit interval. This required a separate calculation for each value of k ; the authors of [25], two of whom are writing this book, only had enough patience to check it for k up to 30. The result was subsequently extended to all values of k in [59].

□

6.2 Entropy and multipodality

The results in this section for the edge- k star model are essentially the same as those we already obtained for the edge-2star model.

Theorem 6.2 *The Boltzmann entropy $B(e, \tilde{t}_k)$ is continuous. For fixed e , it is a strictly decreasing function of $\tilde{t}_k$, and in particular is positive except at the upper boundary of phase space.*

Proof The proof is word-for-word identical to the proof of Theorem 5.2. The listed moves that increase or decrease the density of 2stars also increase or decrease the density of k stars. □

Theorem 6.3 *Let g be a graphon that maximizes the entropy $S(g)$ among graphons with the same edge and (reduced) k star densities $(e, \tilde{t}_k)$ as g . Then g is multipodal.*

Proof At the bottom of the phase space, the optimal graphons are Erdős-Rényi. At the top, the only possible graphons are cliques or anticliques. What remains is the interior of our phase space.

As with 2stars, we use the method of Lagrange multipliers, only now the functional derivative of the k star density is

$$\frac{\delta \tau_k(g)}{\delta g(x, y)} = \frac{k}{2} \left(d(x)^{k-1} + d(y)^{k-1} \right).$$

The functional derivatives of ε and τ_k are linearly independent except where the degree function is constant, that is at $\tilde{t}_k = 0$. We eliminate the possibility of singular entropy maximizers in the interior of phase space exactly as for 2stars. Picking Lagrange multipliers α and β such that $dS = \frac{\alpha}{2} de + \frac{\beta}{2} dt_k$, we have

$$\begin{aligned} \ln \left(\frac{1}{g(x, y)} - 1 \right) &= \alpha + \frac{\beta k}{2} \left(d(x)^{k-1} + d(y)^{k-1} \right) \\ \frac{1}{g(x, y)} &= 1 + \exp \left(\alpha + \frac{\beta k}{2} \left(d(x)^{k-1} + d(y)^{k-1} \right) \right) \\ g(x, y) &= \frac{1}{1 + \exp \left(\alpha + \frac{\beta k}{2} \left(d(x)^{k-1} + d(y)^{k-1} \right) \right)} \\ d(x) &= \int_0^1 \frac{dy}{1 + \exp \left(\alpha + \frac{\beta k}{2} \left(d(x)^{k-1} + d(y)^{k-1} \right) \right)}. \end{aligned} \quad (6.1)$$

Equivalently, defining

$$k(z) = \int_0^1 \frac{dy}{1 + \exp \left(\alpha + \frac{\beta k}{2} \left(z^{k-1} + d(y)^{k-1} \right) \right)},$$

every value of $d(x)$ is a root of the function $F(z) := z - k(z)$.

Showing that $F(z)$ is an analytic function of z is slightly trickier than before, since $k(z)$ is not itself the convolution of a fixed function of z . Instead, let $W = \frac{\beta k}{2} z^{k-1}$ and let $Y = \frac{\beta k}{2} d(y)^{k-1}$. The distribution of $d(y)$ induces a distribution of Y , and $k(z)$ becomes

$$\int \frac{d\mu(Y)}{1 + \exp(\alpha + W + Y)}.$$

This is the convolution of an analytic function of W by a bounded measure, and so is an analytic function of W . Since W is itself an analytic function of z , $k(z)$ is an analytic function of z , and so is $F(z)$. As before, $F(z)$ can only have finitely many roots on $[0, 1]$, counted with multiplicity, so g must be multipodal. $\square$

Theorem 6.4 *For each k there is a number M_k such that all entropy maximizers at all possible points (e, t_k) are multipodal with podality M_k or less.*

Proof This is essentially the same proof as for $k = 2$. If there were a sequence of entropy maximizers with increasing podality, then the limit would have to be a graphon for which $F(z)$ had an infinite number of roots, counted with multiplicity. This cannot happen on the boundary, since entropy-maximizing boundary graphons are at most tripodal. It cannot happen on the interior with β finite, since by the proof of Theorem 6.3 $F(z)$ is analytic and not identically zero. All that remains is excluding the possibility that the limiting graphon has β infinite despite being in the interior.

If we are in the interior, then the entropy must be positive by Theorem 6.2. This means that $g(x, y)$ must be strictly between 0 and 1 on at least one rectangle. But then, by varying $g(x, y)$ in this rectangle, we can increase t_k infinitesimally with the limit of $\Delta S / \Delta t_k$ finite as $\Delta t_k \rightarrow 0$. That is, we must have β finite. $\square$

6.3 Uniqueness and bipodality just above Erdős-Rényi

Our goal in this section is to prove that there is a bipodal phase in the edge- k star model just above the Erdős-Rényi line. Before we precisely state and prove the theorems, let's do a heuristic investigation of what the entropy-maximizing graphon(s) should look like when $\tilde{t}_k$ is small. Such a graphon would have to be

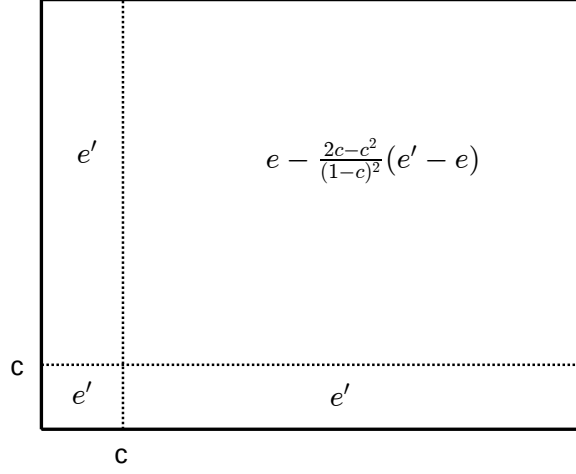
- Multipodal, by Theorem 6.3.
- Close, in an L^2 sense, to the Erdős-Rényi graphon with edge density e .
- Having large podes with $d(x) \approx e$ and small podes with $d(x)$ substantially different from e .
- Having $g(x, y) \approx d(x)$ wherever y is in one of the large podes.

To explore these possibilities, pick a number $e' \in (0, 1)$ and consider the graphon

$$g(x, y) = \begin{cases} e - \frac{2c-c^2}{(1-c)^2}(e' - e) & x, y > c, \\ e' & \text{otherwise,} \end{cases}$$

where c is a small parameter. See Figure 6.1. The degree function is then

$$d(x) = \begin{cases} e' & x < c, \\ e - c \frac{e' - e}{1 - c} & x > c. \end{cases}$$

Figure 6.1 A bipodal graphon with degrees e' and $e - \frac{c}{1-c}(e' - e)$

To leading order in c , the change in the entropy and k star density, relative to the Erdős-Rényi graphon, are

$$\begin{aligned}\tilde{t}_k &= c \left[(e')^k - e^k - k e^{k-1} (e' - e) \right] + O(c^2), \\ S(g) - H(e) &= 2c \left[H(e') - H(e) - (e' - e)H'(e) \right] + O(c^2).\end{aligned}$$

To get the greatest entropy for a given value of $\tilde{t}_k$, we want to maximize the ratio

$$f_k(e, e') := 2 \frac{H(e') - H(e) - (e' - e)H'(e)}{(e')^k - e^k - k e^{k-1} (e' - e)}. \quad (6.2)$$

The function $f_k(e, e')$ has a removable singularity at $e' = e$, which we fill by defining

$$f_k(e, e) = \frac{2H''(e)}{k(k-1)e^{k-2}}.$$

The function $f_k(e, e')$ was investigated in [26], where it was proved that

- For fixed e , $f_k(e, e')$ has a unique maximum at a point we denote $e' = \gamma_k(e)$.
- The function γ_k is decreasing has a unique fixed point at $e' = (k-1)/k$.
- The function γ_k is an involution: $\gamma_k(\gamma_k(e)) = e$.

For a review of these results, see Appendix C. As a result, the best bipodal graphon when $e \neq (k-1)/k$ and $\tilde{t}_k$ is small should have a small node with $d(x)$ close to $\gamma_k(e)$ and a large node with $d(x)$ close to e . This means that

the parameters d and b should be approximately $\gamma_k(e)$ and $e - 2c(\gamma_k(e) - e)$, respectively. By equation (6.1), a must satisfy

$$H'(a) = 2H'(d) - H'(b) \approx 2H'(\gamma_k(e)) - H'(e).$$

But what about multipodal graphons with more podes? If a graphon has multiple small podes, with widths $c_1, c_2, \dots$ and degree functions equalling $e_1, e_2, \dots$, then $\tilde{t}_k$ and the change in entropy is, to first order, just the sum of the contributions from each pode.

$$\begin{aligned} \tilde{t}_k &\approx \sum c_i [(e_i)^k - e^k - k e^{k-1} (e_i - e)], \\ S(g) - H(e) &\approx \sum 2c_i [H(e_i) - H(e) - (e_i - e)H'(e)]. \end{aligned}$$

The way to maximize $S(g)$ is to have all values of c_i equal zero, except for the pode whose value of e_i maximizes $f_k(e, e_i)$. That is, a single small pode with the right value of e_i is better than multiple small podes.

With these calculations in mind, we are ready to state a precise theorem and prove it.

Theorem 6.5 *Suppose that $0 < e < 1$ and $e \neq (k-1)/k$. Then, for sufficiently small $\tilde{t}_k$, there is a unique entropy maximizing graphon with densities $(e, \tilde{t}_k)$. This graphon is bipodal, with parameters that are analytic functions of $(e, \tilde{t}_k)$. As $\tilde{t}_k \rightarrow 0$, these parameters take the asymptotic form*

$$\begin{aligned} a &= a_0 + O(\tilde{t}_k) \\ b &= e + O(\tilde{t}_k) \\ c &= \frac{\tilde{t}_k}{\gamma_k(e)^k - e^k - k e^{k-1} (\gamma_k(e) - e)} + O(\tilde{t}_k^2) \\ d &= \gamma_k(e) + O(\tilde{t}_k), \end{aligned} \tag{6.3}$$

where $H'(a_0) = 2H'(\gamma_k(e)) - H'(e)$.

Proof The function $(1 + \exp(\alpha + \frac{\beta k}{2}(z^{k-1} + e^{k-1})))^{-1}$ has at most one point of inflection between 0 and 1. This condition is robust, so for any degree function L^1 -close to being constant, the function $k(z)$ still has at most one point of inflection, so $F(z)$ has at most three roots. That is, our graphon is at most tripodal. One of the values of the degree function is close to e , another must be close to $\gamma_k(e)$, and this determines α and β and the remaining root of $F(z)$. In particular, the third root is not infinitesimally close to e , so we are in the situation of having one large pode and two (or fewer) small podes.

As noted above, we can get more entropy per change in $\tilde{t}_k$ if the third pode has size zero, in other words if our graphon is bipodal. As $\tilde{t}_k \rightarrow 0$, d must then approach $\gamma_k(e)$, a must always satisfy $H'(a) = 2H'(d) - H'(b)$,

and the value of c is determined by the value of $\tilde{t}_k$. As with the edge-2star model, the solutions to the variational equations define a 2-dimensional variety, implying that (a, b, c, d) are analytic functions of $(e, \tilde{t}_k)$ except where e and $\tilde{t}_k$ cannot be independently varied to first order. The calculation showing that this degeneration never happens is almost identical to the one we already did for 2stars.

All that remains is estimating the difference between (a, b, c, d) and the leading-order terms in equation (6.3). The leading corrections to (b, d) are due to having a small pole of size $c = O(\tilde{t}_k)$, and are thus a factor of $O(\tilde{t}_k)$ smaller than the leading terms. This induces an $O(\tilde{t}_k)$ change in a . Finally, a short calculation gives

$$\tilde{t}_k = c\gamma_k(e)^k - e^k - ke^{k-1}(\gamma(e) - e) + O(c^2),$$

so

$$\begin{aligned} c &= \frac{\tilde{t}_k}{\gamma_k(e)^k - e^k - ke^{k-1}(\gamma(e) - e)} + O(c^2) \\ &= \frac{\tilde{t}_k}{\gamma_k(e)^k - e^k - ke^{k-1}(\gamma(e) - e)} + O(\tilde{t}_k^2). \end{aligned}$$

□

We also consider constraining positive linear combinations of k star densities. While this may seem artificial, it provides an important tool for understanding general edge- H models. We assign a finite number of non-negative weights $w_2, w_3, \dots, w_K$ and constrain the edge density e and $\tilde{t} = \sum_{i=2}^K w_i \tilde{t}_i$. The results are similar to Theorem 6.5, only with a finite set of bad values of e instead of just one.

Theorem 6.6 *Fix a finite set of positive weights w_i and let $\tilde{t} = \sum_{i=2}^K w_i \tilde{t}_i$. For all but finitely many values of e , and for all sufficiently small $\tilde{t}$, there is a unique entropy maximizing graphon with densities $(e, \tilde{t})$. This graphon is bipodal, with parameters that are analytic functions of $(e, \tilde{t})$.*

Proof As usual, we use the method of Lagrange multipliers. The functional derivative of $\tau(g)$ is

$$\frac{1}{2} \sum_k k w_k (d(x)^{i-1} + d(y)^{i-1}).$$

The Euler-Lagrange equations then become

$$\ln \left(\frac{1}{g(x, y)} - 1 \right) = \alpha + \beta \sum_k \frac{k w_k}{2} (d(x)^{k-1} + d(y)^{k-1}),$$

$$\begin{aligned}\frac{1}{g(x, y)} &= 1 + \exp\left(\alpha + \beta \sum_k \frac{k w_k}{2} (d(x)^{k-1} + d(y)^{k-1})\right), \\ g(x, y) &= \frac{1}{1 + \exp\left(\alpha + \beta \sum_k \frac{k w_k}{2} (d(x)^{k-1} + d(y)^{k-1})\right)}, \\ d(x) &= \int_0^1 \frac{dy}{1 + \exp\left(\alpha + \beta \sum_k \frac{k w_k}{2} (d(x)^{k-1} + d(y)^{k-1})\right)}.\end{aligned}$$

Equivalently, defining

$$k(z) = \int_0^1 \frac{dy}{1 + \exp\left(\alpha + \beta \sum_k \frac{k w_k}{2} (z^{k-1} + d(y)^{k-1})\right)},$$

every value of $d(x)$ is a root of the function $F(z) := z - k(z)$.

In the limit of $\tilde{t} \rightarrow 0$, the function $F(z)$ approaches

$$z - \left(1 + \exp\left(\alpha + \beta \sum_k \frac{k w_k}{2} (z^{k-1} + e^{k-1})\right)\right)^{-1}.$$

This is analytic and only has a finite number of roots, counted with multiplicity. Thus for small enough values of $\tilde{t}$, the entropy-maximizing graphon must be multipodal. Furthermore, podes with degrees significantly different from e must be small.

As with k stars, each small pde contributes both to $\tilde{t}$ and to $S(g) - H(e)$ to first order in its width. The efficiency, or change in S per change in $\tilde{t}$, for a pde with degree equal to e' , is given by

$$f(e, e') := 2 \frac{H(e') - H(e) - (e' - e)H'(e)}{\sum_k w_k [(e')^k - e^k - k e^{k-1}(e' - e)]}$$

for $e' \neq e$, where we fill in the singularity by defining

$$f(e, e) = \frac{2H''(e)}{\sum_k k(k-1)w_k e^{k-2}}.$$

As long as $f(e, e')$ has a unique maximum, there can only be one pde with degree function different from e , namely one whose degree function maximizes f .

Setting $\partial_{e'} f(e, e') = 0$ gives a set of analytic equations that depend on e . For all but finitely many values of e , the minimum of $f(e, e')$ will be achieved at a unique point that is not equal to e , and whose value depends analytically on e . Furthermore, for all but finitely many values of e , the function $k(z)$ does not have a point of inflection at e , so there is only one pde with degree function close to e . Thus our optimizing graphon is bipodal, with the degree

function taking values close to e and e' , and with those values (and indeed all the parameters of our bipodal graphon) varying analytically with e and $\tilde{t}$. $\square$

Theorems 6.5 and 6.6 leave open the question of whether there is a single bipodal phase just above the Erdős-Rényi line, as was the case for 2stars, or whether instead there are multiple phases, separated at the value(s) of e where Theorems 6.5 and 6.6 do not apply. In the absence of a rigorous proof, we leave our expectations in the form of conjectures and questions.

Conjecture 6.7 *For all $(e, \tilde{t}_k)$ with $\left(e - \frac{k-1}{k}\right)^2 + \tilde{t}_k$ sufficiently small, the entropy maximizing graphon is unique and bipodal, with parameters that vary analytically with e and $\tilde{t}_k$. Where this neighborhood overlaps with the region where Theorem 6.5 applies, the graphons indicated by the two theorems agree. In particular, there is a single bipodal phase in the edge- k star model whose lower boundary is the entire Erdős-Rényi line.*

Conjecture 6.8 *For any finite set of positive weights $\{w_i\}$, and for $t := \sum w_i t_i$, there is a single bipodal phase, whose lower boundary is the entire Erdős-Rényi line $\tilde{t} = 0$, in the model where we constrain e and t .*

As we have noted, the edge- H model, where H is an arbitrary fixed graph, behaves like a mixed k star model just above Erdős-Rényi. If Conjecture 6.8 is true, then it is reasonable to ask:

Question 6.9 *Is there a graph H for which the edge- H model has more than one phase above the Erdős-Rényi curve?*

There is a big difference between studying supersaturated graphs and under-saturated graphs. There are many graphs, including the basic triangle, that have multiple phases **below** Erdős-Rényi. It remains to be seen whether the phase portrait above Erdős-Rényi can be similarly interesting.

6.4 Phase transition from anticliques to cliques

Pick an integer $k \geq 2$. The top of the edge- k star phase space is defined by the maximum of two functions of e . Let $\tilde{t}_{a,\max}(e)$ and $\tilde{t}_{c,\max}(e)$ be the maximum values of $\tilde{t}_k$ achievable by anticliques or cliques, respectively. Let p_k be the point where these two curves meet. How does the edge- k star model behave in a neighborhood of p_k ?

We have already solved this problem for $k = 2$. See Figure 5.3. In that case, $p_2 = (1/2, (\sqrt{2} - 1)/4)$ and a neighborhood U of p_2 divides neatly into two

pieces. Where $e < 1/2$, graphons that maximize entropy are anticliquelike, while where $e > 1/2$ they are cliquelike. Along the line $e = 1/2$, with $\tilde{t}$ close to $(\sqrt{2} - 1)/4$, there are multiple optimal graphons.

When $k > 2$, the situation isn't quite so simple, in that there isn't a symmetry sending cliquelike graphons near p_k to anticliquelike graphons, and vice-versa. However, there is still a phase transition.

Theorem 6.10 *For each integer $k \geq 2$, there is a neighborhood U of p_k and a curve extending down from p_k , such that the entropy maximizing graphon is anticliquelike to the left of the curve and cliquelike to the right, and such that there is more than one entropy maximizing graphon along the curve.*

Proof As in the proof of Theorem 5.14, there is a neighborhood U of p_k on which all graphons can be classified as cliquelike or anticliquelike. For each $(e, \tilde{t}_k) \in U$, let $B_a(e, \tilde{t}_k)$ and $B_c(e, \tilde{t}_k)$ be the maximum entropy achievable by an anticliquelike (resp. cliquelike) graphon, or 0 if such a graphon does not exist. We will show that U decomposes into two pieces divided by a curve, with $B_a > B_c$ to the left of the curve and $B_a < B_c$ to the right.

Let e_0 be the value of e at p_k . When $e < e_0$ and $\tilde{t}_k = \tilde{t}_{c,\max}(e)$ (the dashed blue curve in Figure 6.2), there exist positive entropy anticliquelike graphons but no positive entropy cliquelike graphons, so $B_a - B_c > 0$. Similarly, $B_a - B_c < 0$ when $e > e_0$ and $\tilde{t}_k = \tilde{t}_{a,\max}(e)$ (the dashed red curve). Since $B_a - B_c$ is a continuous function of $(e, \tilde{t}_k)$, any path within U from the dashed blue line to the dashed red line must pass through a point where $B_a - B_c = 0$. Thus there must be a curve, extending down from the point p_k , on which $B_a - B_c = 0$. This is shown as a dotted black line in Figure 6.2.

In fact, the set of points where $B_a = B_c$ must be an analytic variety. Both the optimal cliquelike graphons and the optimal anticliquelike graphons are multipodal with podality $\leq M_k$. The Euler-Lagrange equations are analytic in the finitely many parameters needed to describe these graphons, and so have analytic solutions. We next show that this analytic variety is smooth by showing that the derivative of $B_a - B_c$ never vanishes.

In fact, we will find a fixed direction such that, when we move uniformly in that direction, $B_a - B_c$ increases at a rate that is bounded below.¹

Let m_a and m_c be the derivatives of $\tilde{t}_{a,\max}$ and $\tilde{t}_{c,\max}$ with respect to e at the point where these functions are equal. In other words, m_a and m_c are the

¹ This is *almost* the same as saying that the directional derivative is always positive. However, there may be points where the directional derivative jumps from one positive value to another. At such points, the derivative does not actually exist. This is why we speak instead of growth that is at least linear.

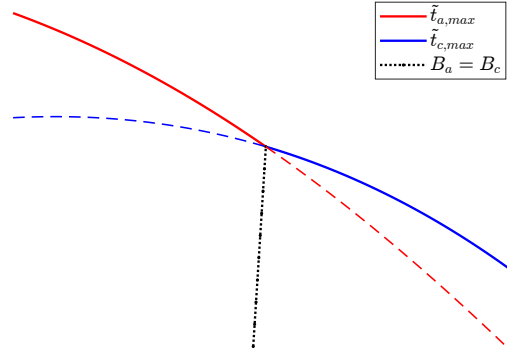


Figure 6.2 Any local path from the dashed blue curve on the left, where $B_a > 0$ and $B_c = 0$, to the dashed red curve on the right, where $B_a = 0$ and $B_c > 0$, must cross the dotted black curve where $B_a = B_c$.

slopes of the two pieces of the upper boundary of the phase space at p_k . Note that $m_a < m_c$.

Any cliquelike stationary point of the entropy must have some podes with degree function close to $\sqrt{e}$ and some other podes with degree function close to 0. By increasing the size of the first kind of podes at the expense of the second kind, we change $\tilde{t}_k$ by approximately m_c times the change in e , but the change in the entropy is very small. We conclude that, if the Lagrange multipliers for this graphon are α_c and β_c , then

$$\alpha_c/\beta_c = m_c + o(1).$$

(The details of this argument are exactly the same as in the proof of Theorem 5.14.) Similarly, in an anticliquelike graphon we can vary the size of the podes with degree close to $1 - \sqrt{1 - e}$ at the expense of those with degree close to 1 with a small change to the entropy, showing that its Lagrange multipliers (α_a, β_a) must satisfy

$$\alpha_a/\beta_a = m_a + o(1).$$

This implies that the level curves for B_a have slope $m_a + o(1)$, while those for B_c have slope $m_b + o(1)$. If we pick a number m such that $m_a < m < m_c$ and move rightwards along a line of slope m , B_c increases linearly while B_a decreases linearly, so $B_a - B_c$ is decreasing. This is illustrated in Figure 6.3,

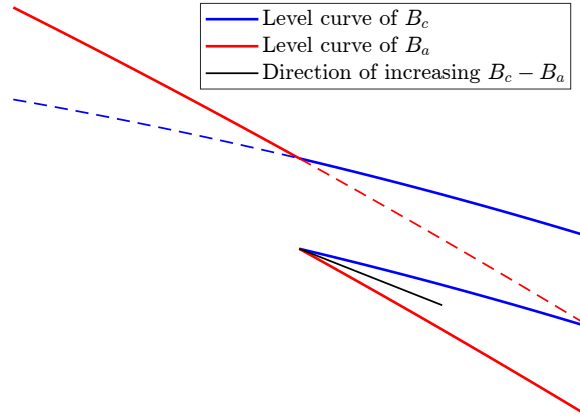


Figure 6.3 Starting at any point in U , the level curve for B_c is roughly parallel to the graph of $\tilde{t}_{c,\max}$, the level curve for B_a is roughly parallel to the graph of $\tilde{t}_{a,\max}$, and moving right along a line of intermediate slope m will increase $B_c - B_a$ to first order.

where the red curves are level curves of B_a , the blue curves are level curves of B_c , and $B_c - B_a$ increases linearly along the black curve. $\square$

Supersaturated graphs in the edge-triangle model

Having sharpened our tools on the edge-2star and edge- k star models, we return to our main focus, the edge-triangle model. In this chapter we prove the existence of a phase, labeled $F(1,1)$, covering the region where $t - e^3$ is small and positive. More precisely, we show that in this region the entropy-maximizing graphon is unique and bipodal, with the parameters (a, b, c, d) being analytic functions of (e, t) . It is probable (although as yet unproven) that this is a portion of the same phase that we established near the upper boundary in Theorem 4.3.

This behavior near the Erdős-Rényi curve is very similar to what we already saw in the edge-2star model. Indeed, we will use our results about the edge-2star model, together with an approximation that relates the triangle count to the 2star count, to prove that, for all $e \neq 1/2$, the entropy maximizing graphon is approximately bipodal. Showing that it is *exactly* bipodal will require a completely different argument.

The similarity raises the question of whether we can make universal statements about models in which we constrain the edge density and the density of one other graph. Indeed we can. In Section 7.4 we will show that, just above Erdős-Rényi, every such model behaves like a model where we constrain a positive linear combination of k stars, which we already solved in Chapter 6.

The edge-triangle model with $t < e^3$ (undersaturated graphs) is fundamentally different, and more subtle, than the model with $t > e^3$ (supersaturated graphs). We will take up that problem, and the infinitely many different phases that arise, in Chapter 8.

7.1 Methods

The proofs in this chapter are not easy. They involve delicate estimates developed over many years of work, up to and including the time when this book was

written. There isn't any one simple idea that achieves all of our goals. However, there are a number of themes, or tricks of the trade, that recur in many of the proofs.

1. Asymptotic analysis. We wish to show that the optimal graphon takes a certain form when $\tilde{t} := t - e^3$ is sufficiently small. We show that various quantities, such as $\|g - e\|_2$, go to zero at a certain rate as $t \rightarrow e^3$. We then estimate the entropy of different structures, only keeping terms up to a certain order in $\tilde{t}$ or $\|g - e\|_2$ or something similar. If one structure does better than the rest at a certain order, then that structure is best whenever $\tilde{t}$ is sufficiently small.
2. Comparison to results for k stars, and especially for 2stars. This is specific to the region just above Erdős-Rényi. There, the density of triangles is closely related to the density of 2stars. In fact, the density of *any* graph H is closely related to a positive linear combination of k star densities. This allows us to treat the problem of maximizing the entropy for fixed (e, t) as a perturbation of a previously-solved problem involving k stars.
3. Perturbing around bipodal graphons. Unlike with k star models, there is no simple *a priori* reason why the entropy maximizing graphon in the edge-triangle model must be multipodal, much less bipodal.

As a result, most of our proofs require two separate stages. First we must show that the entropy maximizing graphon can be well-approximated by a bipodal graphon. That is, that we can write

$$g(x, y) = g_b(x, y) + g_2(x, y),$$

where g_b is a bipodal graphon whose parameters we can estimate, and $g_2(x, y)$ is a pointwise-small perturbation that averages to zero on each quadrant. We then estimate the ratio

$$\frac{S(g) - S(g_b)}{\tau(g) - \tau(g_b)}.$$

If g_2 is nonzero, and if this ratio is larger in magnitude than what could be obtained by varying the bipodal parameters of g_b , then we could increase the entropy of g by first replacing g by g_b and then adjusting the bipodal parameters to restore the original values of t . Since this contradicts the assumption that g is an entropy maximizing graphon, we conclude that $g_2 = 0$ and that g is already bipodal.

4. Power series around $g = \frac{1}{2}$. The Taylor series for $H(u)$ around $u = 1/2$ converges absolutely on the entire interval $[0, 1]$. The odd terms are zero,

by symmetry, so

$$H(g(x, y)) = H(1/2) + \sum_{k=1}^{\infty} \frac{H^{(2k)}(1/2)}{2k!} \left(g(x, y) - \frac{1}{2}\right)^{2k}.$$

Integrating over $[0, 1]^2$, we have

$$S(g) = H(1/2) + \sum_{k=1}^{\infty} \frac{H^{(2k)}(1/2)}{2k!} \mu_{2k},$$

where

$$\mu_{2k} = \iint \left(g(x, y) - \frac{1}{2}\right)^{2k} dx dy.$$

All of the even derivatives $H^{(2k)}(1/2)$ are negative, so we seek to minimize as many moments μ_{2k} as possible. When $e \approx \frac{1}{2}$, the moments decay rapidly with k , and it is possible to obtain valuable estimates just by looking at μ_2 and μ_4 .

5. Diagonalization. A graphon can be viewed as an integral operator on $L^2([0, 1])$. It is symmetric and Hilbert-Schmidt, and therefore diagonalizable, so we can write

$$g(x, y) = \sum_{j=1}^{\infty} \lambda_j v_j(x) v_j(y),$$

where the functions v_j are orthonormal in $L^2([0, 1])$. We are interested in comparing $t = \text{tr}(g^3) = \sum_j \lambda_j^3$ to $\|g\|_2^2 = \sum \lambda_j^2$. For a fixed value of $t = \sum_j \lambda_j^3$, $\|g\|_2^2$ achieves a the minimum value of $t^{2/3}$ when $\lambda_1 > 0$ and all the other eigenvalues λ_j are zero. If $\|g\|_2$ is close to the minimum value, then $\sum_{j=2}^{\infty} \lambda_j^2$ must be small and we can treat g as a perturbation of a rank one operator.

Before stating and proving our theorems about the F(1,1) phase, let's examine how the triangle density t of a supersaturated graphon and the 2star density t_2 are related. If g is any graphon, then

$$\tau_2(g) = \int_0^1 d(x)^2 dx = e^2 + \int_0^1 (d(x) - e)^2 dx,$$

where

$$d(x) = \int_0^1 g(x, y) dy$$

is the degree function of g . In particular, the reduced 2star density $\tilde{t}_2 = \tau(g) - e^2$ is the variance of $d(x)$.

Writing $g(x, y) = e + g_1(x, y)$, we expand

$$\begin{aligned}
 \tau(g) &= \iiint g(x, y)g(y, z)g(z, x) dx dy dz \\
 &= \iiint (e + g_1(x, y))(e + g_1(y, z))(e + g_1(x, z)) dx dy dz \\
 &= e^3 + \iiint 3eg_1(x, y)g_1(x, z) + g_1(x, y)g_1(y, z)g_1(z, x) dx dy dz \\
 &= e^3 + 3e\tilde{t}_2 + \text{tr}(g_1^3).
 \end{aligned}$$

A priori, the $\text{tr}(g_1^3)$ term is of order $\|g_1\|_2^3$. In practice, when maximizing the entropy for $\tilde{t}_2 > 0$, it is usually $O(\|g_1\|_2^4)$. Either way, it is a small perturbation of the $3e\tilde{t}_2$ term, which can go as $\|g_1\|_2^2$. The upshot is that maximizing the entropy for fixed (e, t) is almost the same as maximizing the entropy for fixed $(e, \tilde{t}_2)$, which we know is achieved with bipodal graphons.

So far this is just a heuristic argument, but we can sharpen it by examining the functional derivative

$$\begin{aligned}
 \frac{\delta\tau(g)}{\delta g(x, y)} &= 3 \int_0^1 g(x, z)g(y, z) dz \\
 &= 3 \int_0^1 (e + g_1(x, z))(e + g_1(y, z)) dz \\
 &= 3 \left(e^2 + e(d(x) - e) + e(d(y) - e) + \int_0^1 g_1(x, z)g_1(y, z) dz \right) \\
 &= 3e(d(x) + d(y) - e) + 3 \int_0^1 g_1(x, z)g_1(y, z) dz.
 \end{aligned}$$

As usual, we set the functional derivative of $S(g)$ equal to a linear combination of the functional derivatives of $\varepsilon(g)$ and $\tau(g)$:

$$\begin{aligned}
 \frac{\delta S(g)}{\delta g(x, y)} &= \alpha \frac{\delta \varepsilon(g)}{\delta g(x, y)} + \frac{\beta}{3} \frac{\delta \tau(g)}{\delta g(x, y)} \\
 \frac{1}{2} \ln \left(\frac{1 - g(x, y)}{g(x, y)} \right) &= \alpha + \beta e(d(x) + d(y) - e) + \beta G_1(x, y),
 \end{aligned}$$

where

$$G_1(x, y) = \int_0^1 g_1(x, z)g_1(y, z) dz,$$

in analogy to the way that $G(x, y)$ is computed from $g(x, y)$. Solving for $g(x, y)$ in terms of the right hand side of the second line and then integrating, we obtain the following analogues of equations (5.3) and (5.4):

$$g(x, y) = \frac{1}{1 + \exp \left(\alpha_2 + \beta_2(d(x) + d(y) + \frac{1}{e}G_1(x, y)) \right)}, \quad (7.1)$$

$$d(x) = \int_0^1 \frac{dy}{1 + \exp\left(\alpha_2 + \beta_2(d(x) + d(y) + \frac{1}{e}G_1(x, y))\right)}.$$

Here

$$\alpha_2 = 2(\alpha - e^2\beta) \quad \text{and} \quad \beta_2 = 2e\beta.$$

The key to proving similarities between the edge-triangle model with Lagrange multipliers (α, β) and the edge-2star model with Lagrange multipliers (α_2, β_2) will be establishing effective bounds on $G_1(x, y)$.

7.2 Supersaturated graphs with $e \neq 1/2$

As with the edge-2star model, we first study what happens when $e \neq 1/2$, which is relatively straightforward, and then take up the region near $e = 1/2$, which is more difficult.

Theorem 7.1 *Suppose that $e \neq 1/2$ and that $\tilde{t} := t - e^3$ is non-negative and sufficiently small (where the definition of “sufficiently small” can depend on e). Then there is a unique entropy-maximizing graphon. This graphon is bipodal. The parameters (a, b, c, d) vary analytically with e and $\tilde{t}$ and have the asymptotic form*

$$\begin{aligned} a &= a_0 + O(\tilde{t}) \\ b &= e + O(\tilde{t}) \\ c &= \frac{\tilde{t}}{3e(1-2e)^2} + O(\tilde{t}^2) \\ d &= 1 - e + O(\tilde{t}) \end{aligned} \tag{7.2}$$

as $\tilde{t} \rightarrow 0^+$, where a_0 is the solution to

$$H'(a_0) = \frac{e-2}{e}H'(e). \tag{7.3}$$

Proof We begin with equation (7.1). For entropy-maximizing graphons, the Lagrange multipliers α and β are related to the partial derivatives of the Boltzmann entropy $B(e, t)$. From the entropy estimates in the proof of Theorem 3.6, we see that neither α nor β diverge as $\tilde{t} \rightarrow 0^+$. This means that we can rewrite equation (7.1) as

$$g(x, y) = \frac{1}{1 + \exp\left(\alpha_2 + \beta_2(d(x) + d(y))\right)} + O(G_1(x, y)).$$

Integrating, we get

$$d(x) = \int_0^1 \frac{dy}{1 + \exp(\alpha_2 + \beta_2(d(x) + d(y)))} + O\left(\int_0^1 G_1(x, y) dy\right). \quad (7.4)$$

Now we estimate the size of $G_1(x, y)$. Since $\Delta S(g) := S(g) - H(e)$ scales as $\|g_1\|_2^2$ and also as $\tilde{t}$, we must have

$$\int_0^1 \int_0^1 g_1(x, z)^2 dz dx = O(\tilde{t}).$$

Let I_0 be the set of x 's for which $\int_0^1 g_1(x, z)^2 dz$ is not bounded by a fixed (but arbitrary) multiple of $\tilde{t}^{2/3}$. The measure of I_0 is then $O(\tilde{t}^{1/3})$. By Cauchy-Schwarz,

$$G_1(x, y) \leq \sqrt{\int_0^1 g_1(x, z)^2 dz \int_0^1 g_1(y, z)^2 dz}.$$

That is $O(\tilde{t}^{1/3})$ except on $I_0 \times I_0$. Thus, for any x whatsoever,

$$\begin{aligned} \left| \int_0^1 G_1(x, y) dy \right| &\leq \left| \int_{I_0} G_1(x, y) dy \right| + \left| \int_{I_0^c} G_1(x, y) dy \right| \\ &= O(\tilde{t}^{1/3}) + O(\tilde{t}^{1/3}) = O(\tilde{t}^{1/3}). \end{aligned}$$

The bound on the first integral comes from the size of I_0 , while the bound on the second integral comes from the size of $G_1(x, y)$ for $y \notin I_0$.

Plugging this result into equation (7.4), we get

$$d(x) = \int_0^1 \frac{dy}{1 + \exp(\alpha_2 + \beta_2(d(x) + d(y)))} + O(\tilde{t}^{1/3}).$$

Defining

$$k(z) = \int_0^1 \frac{dy}{1 + \exp(\alpha_2 + \beta_2(z + d(y)))},$$

exactly as in the edge-2star model, we conclude that all values of $d(x)$ are close to solutions to $k(z) = z$.

We now repeat the reasoning of the proof of Theorem 5.9, which you can refer back to for details, showing that:

- For small values of $\tilde{t}$, and hence graphons with $d(y)$ concentrated near e , the graph of $k(z)$ has only one point of inflection.
- There can only be three solutions to $k(z) = z$.
- Two of these must be near $z = e$ and $z = 1 - e$.

- This then fixes the values of α_2 and β_2 and the third root near $z = 1/2$.

Since the intersections are transverse, adding an $O(\tilde{t}^{1/3})$ term to $k(z)$ can only change the values of $d(x)$ by $O(\tilde{t}^{1/3})$. That is, all values of $d(x)$ must be within $O(\tilde{t}^{1/3})$ of e , $1 - e$ or $1/2$. This does **not** by itself prove that our solution is multipodal, since different values of x could have slightly different values of $\int G_1(x, y) dy$, resulting in slightly different values of $d(x)$. However, it does mean that the graphon must be approximately multipodal, with a small interval of size c_1 where the degree function is close to $1 - e$, potentially a region of size c_2 where the degree function is close to $1/2$, and a large region of size $1 - c_1 - c_2$ where the degree function is close to e .

The argument for why $c_2 = 0$ is exactly as in the proof of Theorem 5.9. To leading order in c_2 , having such a pole costs too much entropy for the number of 2stars (and therefore the number of triangles) it produces. We conclude that g is approximately bipodal, with a pole I_1 where $d(x) \approx 1 - e$ and a pole $I_2 = I_1^c$ where $d(x) \approx e$. We then average g over each quadrant to get a graphon g_b that is exactly bipodal and write

$$g(x, y) = g_b(x, y) + g_2(x, y),$$

where

$$\iint_{I_1 \times I_1} g_2(x, y) dx dy = \iint_{I_1 \times I_2} g_2(x, y) dx dy = \iint_{I_2 \times I_2} g_2(x, y) dx dy = 0.$$

Without loss of generality we can take $I_1 = [0, c]$ and $I_2 = [c, 1]$.

We now show that $g_2(x, y)$ is pointwise small, beginning on $I_2 \times I_2$.

We have already established that the function $d(x)$ only varies by $O(\tilde{t}^{1/3})$ on I_2 . What's more, on $(I_2 \times I_2) \cap (I_0 \times I_0)^c$, $G(x, y)$ is $O(\tilde{t}^{1/3})$. Since the right hand side of equation (7.1) is within $O(\tilde{t}^{1/3})$ at all points of $(I_2 \times I_2) \cap (I_0 \cap I_0)^c$, the value of $g(x, y)$ can only vary by $O(\tilde{t}^{1/3})$ in this region.

Moreover, since the average value of $g_1(x, y)$ in this region is $O(\tilde{t}^{1/3})$, $g_1(x, y)$ must be pointwise $O(\tilde{t}^{1/3})$ on $(I_2 \times I_2) \cap (I_0 \cap I_0)^c$.

That, in turn, implies that $G(x, y)$ is $O(\tilde{t}^{1/3})$ whenever $x \in I_2$ (and likewise whenever $y \in I_2$), since $g_1(x, z)$ is $O(\tilde{t}^{1/3})$ for all $z \in I_2 \cap I_0^c$. That is, our set $I_0 \cap I_0$ on which $G(x, y)$ is bigger than $O(\tilde{t}^{1/3})$ is actually just $I_1 \times I_1$.

On $I_1 \times I_2$ (and likewise on $I_2 \times I_1$), $d(x)$ is constant to within $O(\tilde{t}^{1/3})$, $d(y)$ is constant to within $O(\tilde{t}^{1/3})$, and $G(x, y)$ is everywhere $O(\tilde{t}^{1/3})$, so all points have the same right hand side to equation (7.1) to within $O(\tilde{t}^{1/3})$, so all points have the same value of $g(x, y)$ to within $O(\tilde{t}^{1/3})$.

Finally, we turn to $I_1 \times I_1$. Here $G(x, y)$ is **not** small, since $g_1(x, z)$ and $g_2(y, z)$ are approximately $(1 - 2e)$ whenever $z \in I_2$. However, $G_1(x, y)$ is

nearly constant on $I_1 \times I_1$:

$$\begin{aligned} G_1(x, y) &= \left(\int_{I_2} g_1(x, z) g_1(y, z) dz + \int_{I_1} g_1(x, z) g_1(y, z) dz \right) \\ &= \left(\int_{I_2} (1 - 2e + O(\tilde{t}^{1/3})) (1 - 2e + O(\tilde{t}^{1/3})) dz \right) + O(\tilde{t}) \\ &= (1 - 2e)^2 + O(\tilde{t}^{1/3}). \end{aligned}$$

Since $d(x)$, $d(y)$ and $G_1(x, y)$ only vary by $O(\tilde{t}^{1/3})$ from point to point, $g(x, y)$ can only vary by $O(\tilde{t}^{1/3})$.

Note that this calculation also determines the value of a_0 . Setting $\delta S(g)/\delta(x, y)$ equal to a linear combination of $\delta \varepsilon(g)/\delta g(x, y)$ and $\delta \tau(g)/\delta g(x, y)$ at three points (one in $I_2 \times I_2$, one in $I_1 \times I_2$ and one in $I_1 \times I_1$ and taking a limit as $\tilde{t} \rightarrow 0^+$ gives the equations:

$$\begin{aligned} H'(e) &= \frac{\alpha_2}{2} + \frac{\beta_2}{2} * 2e \\ H'(1 - e) &= \frac{\alpha_2}{2} + \frac{\beta_2}{2} * 1 \\ H'(a_0) &= \frac{\alpha_2}{2} + \frac{\beta_2}{2} * \left(2(1 - e) + \frac{(1 - 2e)^2}{e} \right) \end{aligned}$$

The first two equations, together with the fact that $H'(1 - e) = -H'(e)$, give us α_2 and β_2 in terms of $H'(e)$:

$$\alpha_2 = -\frac{2(2e + 1)}{2e - 1} H'(e); \quad \beta_2 = \frac{4}{2e - 1} H'(e).$$

Plugging these values into the third equation yields equation (7.3).

We now show that $g_2(x, y)$ is in fact identically zero by comparing the entropy and triangle densities for g_b and g . We will show that having $g_2(x, y)$ nonzero can increase the triangle count (relative to that of g_b), but at the cost of decreasing the entropy. The change in entropy per change in $\tilde{t}$ achieved by having $g_2(x, y)$ nonzero will turn out to be more negative than the ratio achieved by keeping the graphon bipodal and making small changes to the parameters (a, b, c, d) for g_b . This will imply that any graphon with $g_2(x, y)$ nonzero is **not** entropy maximizing, and hence that the actual entropy maximizer is bipodal.

Lemma 7.2 *For any pointwise small function $g_2(x, y)$,*

$$\frac{S(g) - S(g_b)}{\tau(g) - \tau(g_b)} \leq \frac{H''(e)}{3e} + o(1).$$

Proof of lemma Since we are only computing the ratio to within $o(1)$ errors, we only need to work to leading (that is, quadratic) order in $\|g_2\|_2$. Note that

$$\tau(g) - \tau(g_b) = \text{tr}((g_b + g_2)^3) - \text{tr}(g_b^3)$$

$$= 3 \operatorname{tr}(g_b^2 g_2) + 3 \operatorname{tr}(g_b g_2^2) + \operatorname{tr}(g_2^3).$$

The first term on the last line is zero, since g_b^2 is constant on each quadrant and g_2 integrates to zero on each quadrant. The last term is bounded by $\|g_2\|_2^3$. The term that matters is the middle one.

On each quadrant, we write $g_2(x, y)$ as a sum of three terms:

- A function of x that averages to zero on the quadrant,
- A function of y that averages to zero on the quadrant, and
- A function $g_3(x, y)$ that has zero marginals on the quadrant.

Specifically, we write

$$g_2(x, y) = \begin{cases} \frac{\eta_1(x)}{c} + \frac{\eta_1(y)}{c} + g_3(x, y) & \text{on } I_1 \times I_1, \\ \frac{\eta_2(x)}{1-c} + \frac{\eta_3(y)}{c} + g_3(x, y) & \text{on } I_1 \times I_2, \\ \frac{\eta_2(y)}{1-c} + \frac{\eta_3(x)}{c} + g_3(x, y) & \text{on } I_2 \times I_1, \\ \frac{\eta_4(x)}{1-c} + \frac{\eta_4(y)}{1-c} + g_3(x, y) & \text{on } I_2 \times I_2, \end{cases}$$

where

$$\int_0^c \eta_1(x) dx = \int_0^c \eta_2(x) dx = \int_c^1 \eta_3(x) dx = \int_c^1 \eta_4(x) dx = 0.$$

This essentially defines the functions $\eta_i(x)$. The normalizing factors of c^{-1} and $(1-c)^{-1}$ are chosen to simplify the formulas for $\tau(g) - \tau(g_b)$ and $S(g) - S(g_b)$.

There are no contributions from g_3 in expanding $3 \operatorname{tr}(g_b g_2^2)$, since if a term involves $g_3(x, y)$, then the remaining factors are either independent of x or independent of y . Either way, we get zero after integrating over x and y . The only contributions to $3 \operatorname{tr}(g_b g_2^2)$ are those involving two factors of $\eta_i(x)$. Specifically,

$$\begin{aligned} 3 \operatorname{tr}(g_b g_2^2) &= \int_0^c 3a\eta_1(x)^2 + 6d\eta_1(x)\eta_2(x) + 3b\eta_2(x)^2 dx \\ &\quad + \int_c^1 3a\eta_3(x)^2 + 6d\eta_3(x)\eta_4(x) + 3b\eta_4(x)^2 dx. \end{aligned} \quad (7.5)$$

Meanwhile, since $g_2(x, y)$ is pointwise small, we can compute $S(g) - S(g_b)$ using a 2nd-order Taylor series for $H(g)$ in each quadrant.

$$\begin{aligned} S(g) - S(g_b) &= \int_0^c \frac{H''(a)}{c} \eta_1(x)^2 + \frac{H''(d)}{1-c} \eta_2(x)^2 dx \\ &\quad + \int_c^1 \frac{H''(d)}{c} \eta_3(x)^2 + \frac{H''(b)}{1-c} \eta_4(x)^2 dx \\ &\quad + o(\|g_2\|^2). \end{aligned} \quad (7.6)$$

We have ignored any costs associated with $g_3(x, y)$ in (7.6) since there is no benefit to having g_3 nonzero. For the rest of the calculation, we set $g_3 = 0$.

Applying Cauchy-Schwartz to the cross-terms between η_1 and η_2 in (7.5), and likewise between η_3 and η_4 , shows that η_1 and η_3 should be scalar multiples of η_2 and η_4 , respectively. However, the costs associated with η_1 and η_3 are divergent, indicating that these multiples should be $o(1)$. (In fact, they work out to be $O(c) = O(\tilde{t})$.) This then makes

$$3 \operatorname{tr}(g_b g_2^2) = (3e + o(1)) \int_0^c \eta_2(x)^2 dx + (3e + o(1)) \int_c^1 \eta_4(x)^2 dx,$$

where we have used the fact that $b = e + o(1)$. Meanwhile,

$$S(g) - S(g_b) = (H''(e) + o(1)) \int_0^c \eta_2(x)^2 dx + (H''(e) + o(1)) \int_c^1 \eta_4(x)^2 dx,$$

where we have used the fact that $d = 1 - e + o(1)$, hence that $H''(d) = H''(e) + o(1)$. That is, after optimizing the form of $g_2(x, y)$ by eliminating g_3 and setting η_1 and η_3 equal to $o(1)$ multiples of η_2 and η_4 , we obtain a ratio of

$$\frac{S(g) - S(g_b)}{\tau(g) - \tau(g_b)} = \frac{H''(e)}{3e} + o(1),$$

as claimed. $\square$

Now we turn to varying the parameters of the bipodal graphon g_b . We will fix a and d , let c vary, and adjust b to keep the overall edge density equal to e . That is, we set

$$b = e - \frac{c^2(a - e) + 2c(1 - c)(d - e)}{(1 - c)^2},$$

which then gives us

$$\frac{db}{dc} = -2(d - e) + O(c) = -2(1 - 2e) + o(1).$$

Lemma 7.3 *In this family of graphons,*

$$\frac{dS/dc}{d\tilde{t}/dc} = \frac{1}{3e} \left(\frac{H'(e)}{e - \frac{1}{2}} \right) + o(1).$$

Proof of lemma We compute the numerator and denominator separately, using the facts that c is small and that $d = 1 - e + o(1)$. Since

$$S = c^2 H(a) + 2c(1 - c)H(d) + (1 - c)^2 H(b),$$

we have

$$\begin{aligned}\frac{dS}{dc} &= 2cH(a) + 2(1-2c)H(d) - 2(1-c)H(b) + (1-c)^2H'(b)\frac{db}{dc} \\ &= 2(H(d) - H(b)) - 2H'(b)(d-e) + O(c) \\ &= -2H'(e)(1-2e) + o(1).\end{aligned}$$

Meanwhile,

$$\tilde{t} = c^3a^3 + 3c^2(1-c)ad^2 + 3c(1-c)^2bd^2 + (1-c)^3b^3 - e^3,$$

so

$$\begin{aligned}\frac{d\tilde{t}}{dc} &= 3c^2a^3 + (6c - 9c^2)ad^2 + 3(1-4c+3c^2)bd^2 - 3(1-c)^2b^3 \\ &\quad + [3c(1-c)^2d^2 + 3(1-c)^3b^2] \frac{db}{dc} \\ &= 3bd^2 - 3b^3 + 3b^2\frac{db}{dc} + O(c) \\ &= 3bd^2 - 3b^3 - 6b^2(1-2e) + o(1) \\ &= 3e(1-2e)^2 + o(1).\end{aligned}$$

The ratio is then

$$\frac{dS/dc}{d\tilde{t}/dc} = \frac{-2H'(e)(1-2e) + o(1)}{3e(1-2e)^2 + o(1)} = \frac{1}{3e} \left(\frac{H'(e)}{e - \frac{1}{2}} \right) + o(1).$$

□

Returning to the proof of Theorem 7.1, we must compare

$$\frac{H''(e)}{3e} \quad \text{and} \quad \frac{1}{3e} \left(\frac{H'(e)}{e - \frac{1}{2}} \right)$$

When $e > 1/2$, the difference between them is

$$\frac{1}{3e(e - \frac{1}{2})} \int_{1/2}^e (H''(e) - H''(u)) du.$$

The integrand is negative, since $H''(u)$ is a decreasing function on $[\frac{1}{2}, e]$, so

$$\frac{H''(e)}{3e} < \frac{1}{3e} \left(\frac{H'(e)}{e - \frac{1}{2}} \right)$$

when $e > 1/2$. Since $H''(e)$ and $\frac{H'(e)}{e - \frac{1}{2}}$ are both even functions of $e - \frac{1}{2}$, the inequality also holds when $e < 1/2$.

This comparison shows that, starting at g_b and attempting to increase $\tilde{t}$, increasing c yields more entropy per increase in $\tilde{t}$ than adding $g_2(x, y)$ to g_b .

We conclude that g_2 must in fact be zero, and that our optimal graphon is in fact bipodal, with (a, b, c, d) having the asymptotic form given in equation (7.2). Uniqueness of this graphon, and its analytic dependence on $(e, \tilde{t})$, then follows by exactly the same arguments as in Theorem 5.9.

The use of Lagrange multipliers in this proof is justified by the same argument as for the theorems in Chapter 4. By Lemma 2.13, the entropy maximizers are nonsingular for all t outside a set of measure zero. The above calculations then establish that these nonsingular optimal graphons are bipodal, with an entropy that is differentiable with respect to t . Theorem 2.14 then says that the set of measure zero is in fact empty, and that our arguments apply for *all* sufficiently small $\tilde{t}$.

□

7.3 Supersaturated graphs with $e \approx 1/2$

Theorem 7.1 only covers what happens when $e < 1/2$ or $e > 1/2$. As with the edge-2star model, we need a separate calculation near $e = 1/2$ to bridge the two regions and show that there is a single bipodal phase whose lower boundary is the entire Erdős-Rényi curve.

Let $\zeta^2 = et^{1/3} - e^2 + \left(e - \frac{1}{2}\right)^2$. This is nearly the same parameter as in the edge-2star model, only adapted to the situation where $e(t^{1/3} - e)$, rather than $t_2 - e^2$, measures how far we are above the Erdős-Rényi curve.

Theorem 7.4 *For all (e, t) with $t > e^3$ positive and ζ sufficiently small, the entropy maximizing graphon is unique and bipodal, with parameters that are analytic functions of (e, t) .*

Proof The proof follows a similar strategy as the proof of Theorem 7.1, with the use of Lagrange multipliers justified by Lemma 2.13 and Theorem 2.14. First we show that the entropy maximizing graphon must be approximately bipodal of a certain form. We then write

$$g(x, y) = g_b(x, y) + g_2(x, y),$$

where g_b is a bipodal graphon (whose parameters we know to within a small error) and $g_2(x, y)$ averages to zero on each quadrant, and estimate

$$\frac{S(g) - S(g_b)}{\tau(g) - \tau(g_b)}.$$

We show that this is less than what can be achieved by varying the bipodal parameters associated with g_b .

We express the steps as a series of lemmas:

Lemma 7.5 *The entropy maximizing graphon is nearly rank-1. Specifically, the estimate*

$$g(x, y) = \frac{d(x)d(y)}{e} + O(\zeta^3),$$

where $d(x)$ is the degree function, applies pointwise.

Lemma 7.6 *The degree function of the entropy maximizing graphon can be approximated, pointwise to within $O(\zeta^2)$, by a step function taking two values: $\frac{1}{2} \pm \zeta$.*

Given Lemma 7.6, let I_1 be the subset of the unit interval on which $d(x)$ is close to $\frac{1}{2} - \zeta$. Without loss of generality we can take this to be the interval $[0, c]$ for some positive number c . Let $I_2 = [c, 1]$. If g is not already bipodal, we average $g(x, y)$ over $I_1 \times I_1$, $I_1 \times I_2$, $I_2 \times I_1$ and $I_2 \times I_2$ to obtain a bipodal graphon g_b with $\tau(g_b)$ slightly less than $\tau(g)$ and $S(g_b)$ slightly greater than $S(g)$.

Lemma 7.7 *If the entropy maximizing graphon g is not already bipodal, then*

$$\frac{S(g) - S(g_b)}{\tau(g) - \tau(g_b)} \leq \frac{-2}{3e} \left(1 + 4\zeta^2 + O(\zeta^3) \right).$$

Lemma 7.8 *Working within the space of bipodal graphons, we can vary the parameters of g_b to increase t with*

$$\frac{dS}{dt} = \frac{-2}{3e} \left(1 + \frac{4}{3}\zeta^2 \right) + O(\zeta^3)$$

Since $\frac{4}{3} < 4$ and since the factor $-2/3e$ is negative, the derivative in Lemma 7.8 is strictly greater than the difference quotient in Lemma 7.7. That is, if g is not bipodal then we can increase the entropy of g , without changing t , by first averaging over each node to get g_b and then varying the bipodal parameters to recover the original value of t . This contradicts the assertion that g is entropy maximizing. We conclude that g is bipodal, with parameters close to those of a rank-1 graphon whose degree function only takes on the values $\frac{1}{2} \pm \zeta$. That is,

$$\begin{aligned} a &= \frac{(\frac{1}{2} - \zeta)^2}{e} + O(\zeta^2) \\ &= 1 - e - 2\zeta + O(\zeta^2), \\ b &= \frac{(\frac{1}{2} + \zeta)^2}{e} + O(\zeta^2) \\ &= 1 - e + 2\zeta + O(\zeta^2), \end{aligned}$$

$$\begin{aligned}
d &= \frac{\frac{1}{4} - \zeta^2}{e} + O(\zeta^2) \\
&= 1 - e + O(\zeta^2) \\
c &= \frac{1}{2} \left(1 - \frac{e - \frac{1}{2}}{\zeta} \right) + O(\zeta^2).
\end{aligned}$$

The precise values can be obtained by solving a calculus problem in the 4-dimensional space of bipodal graphons. As usual, the variational equations are nondegenerate, so the resulting parameters are analytic functions of (e, t) . $\square$

What remains is proving Lemmas 7.5–7.8. Let

$$\mu_{2k} = \iint (g(x, y) - \frac{1}{2})^{2k} dx dy, \quad \nu_k = \int_0^1 (d(x) - \frac{1}{2})^k dx$$

be the moments, centered at $1/2$, of the graphon $g(x, y)$ and the degree function $d(x) = \int_0^1 g(x, y) dy$. We expand the entropy using a power series for $H(u)$ around $u = 1/2$, with all of the odd terms being identically zero.

$$S(g) = H(1/2) + \sum_{k=1}^{\infty} \frac{H^{(2k)}(1/2)}{(2k)!} \mu_{2k}.$$

Since all of the even derivatives of $H(u)$ are negative at $u = 1/2$, maximizing the entropy is associated with minimizing the moments μ_{2k} . We will soon see that $\mu_2 = O(\zeta^2)$. Having values of $(g(x, y) - \frac{1}{2})^2$ that are anomalously large only serves to increase the higher moments, so

$$g(x, y) = \frac{1}{2} + O(\zeta)$$

throughout. This makes $\mu_{2k} = O(\zeta^{2k})$. To understand the entropy of our graphon to within $O(\zeta^6)$, we only need to compute μ_2 and μ_4 .

Proof of Lemma 7.5 We begin with a basic estimate from Chapter 3.

$$S(g) \leq S(e) + \frac{1}{2} H''(1/2) \|g - e\|_2^2 \leq H(1/2) + \frac{1}{2} H''(1/2) \left[(e - \frac{1}{2})^2 + \|g - e\|_2^2 \right].$$

We diagonalize our graphon $g(x, y)$

$$g(x, y) = \sum_{j=1}^{\infty} \lambda_j v_j(x) v_j(y),$$

where the functions $v_j(x)$ are orthonormal in $L^2([0, 1])$ and $|\lambda_1| \geq |\lambda_2| \geq \dots$. Let $g_1(x, y) = g(x, y) - \lambda_1 v_1(x) v_1(y)$ be the difference between $g(x, y)$ and a

rank one graphon. (Note that this is different from the $g_1(x, y)$ that appeared in the proof of Theorem 7.1.) Then

$$t = \text{tr}(g^3) = \sum_j \lambda_j^3,$$

while

$$\|g\|_2^2 = \sum_{j=1}^{\infty} \lambda_j^2$$

and

$$\mu_2 = \|g\|_2^2 - e^2 + (e - \frac{1}{2})^2.$$

The quantity $\|g\|_2^2$ is minimized when g has rank one. In that case,

$$\begin{aligned} \mu_2 &= t^{2/3} - e^2 + (e - \frac{1}{2})^2 \\ &= \left(\frac{e^2 + \zeta^2 - (e - \frac{1}{2})^2}{e} \right)^2 - e^2 + (e - \frac{1}{2})^2 \\ &= 2\zeta^2 - (e - \frac{1}{2})^2 + \frac{(\zeta^2 - (e - \frac{1}{2})^2)^2}{e^2}. \end{aligned}$$

As long as $v_1(x)$ stays within $O(\zeta)$ of 1, μ_4 and all higher moments are $O(\zeta^4)$ and our graphon comes within $O(\zeta^4)$ of achieving the maximum possible entropy. If another graphon $g(x, y)$ has higher entropy, then that other graphon must have

$$\mu_2 = t^{2/3} - e^2 + (e - \frac{1}{2})^2 + O(\zeta^4),$$

and so must have $\|g_1\|_2 = O(\zeta^2)$.

Now that we know that $\|g_1\|_2$ is small, we can compute the effect on the moments of having a nonzero g_1 . As we have seen, having g_1 nonzero results in a $\|g_1\|_2^2$ contribution to μ_2 , which we view as a cost. There is a slight benefit in reducing the value of λ_1 and thereby reducing μ_2 , but $\lambda_1^3 \geq t - \|g_1\|_2^3$, so this benefit is only $O(\|g_1\|_2^3)$. That is, the net effect on μ_2 is an increase of $\|g_1\|_2^2 + O(\|g_1\|_2^3)$.

To find any benefit for having g_1 nonzero, we have to look at

$$\begin{aligned} \mu_4 &= \iint (g - \frac{1}{2})^4 dx dy \\ &= \iint \left(\lambda_1 v_1(x) v_1(y) - \frac{1}{2} + g_1(x, y) \right)^4 dx dy. \end{aligned}$$

Expanding this in powers of g_1 , we can ignore quadratic and quartic terms

(which are positive and so count as costs, not benefits) and cubic terms (which are dominated by the quadratic terms). The key contribution is the linear term

$$\iint 4 \left(\lambda_1 v_1(x) v_1(y) - \frac{1}{2} \right)^3 g_1(x, y) dx dy.$$

Given an $O(\zeta^3 \|g_1(x, y)\|)$ benefit from μ_4 (and similar but smaller benefits from μ_6, μ_8 , and so on) and an $O(\|g_1\|^2)$ cost, the total is maximized by taking $g_1(x, y)$ to be $-\frac{1}{2}$ times the sum of the coefficients of the linear terms (projected onto the subspace orthogonal to $v_1(x)v_1(y)$) divided by the coefficient of the quadratic term. Since the linear coefficients are pointwise $O(\zeta^3)$ and the quadratic coefficient is $-H''(1/2)/2 = -1$, $g_1(x, y)$ is pointwise $O(\zeta^3)$ and contributes $O(\zeta^6)$ to the entropy.

Finally, since $g_1(x, y)$ is $O(\zeta^3)$, the contribution of $g_1(x, y)$ to the degree function is likewise $O(\zeta^3)$, so the degree function is

$$d(x) = \lambda_1 v_1(x) \int_0^1 v_1(y) dy + O(\zeta^3)$$

and

$$e = \int_0^1 d(x) dx = \lambda_1 \left(\int_0^1 v_1(x) dx \right)^2 + O(\zeta^3).$$

Rewriting our graphon in terms of $d(x)$ and e then yields

$$g(x, y) = \frac{d(x)d(y)}{e} + O(\zeta^3). \quad (7.7)$$

□

Proof of Lemma 7.6 Using equation 7.3, we compute μ_2 and μ_4 to within $O(\zeta^6)$ in terms of the moments v_k . We ignore contributions from g_1 , which only contribute at order ζ^6 . Note that

$$v_1 = e - \frac{1}{2},$$

while

$$\begin{aligned} v_2 &= \int_0^1 \left(d(x) - \frac{1}{2} \right)^2 dx \\ &= \int_0^1 d(x)^2 - e^2 + \left(e - \frac{1}{2} \right)^2 dx \\ &= e^{1/3} - e^2 + \left(e - \frac{1}{2} \right)^2 + O(\|g_1\|^3) \\ &= \zeta^2 + O(\zeta^9). \end{aligned}$$

Since we only care about terms through order ζ^5 , we can treat v_2 as equaling ζ^2 .

For computing μ_{2k} , it is convenient to write $d(x)d(y)/e$ as

$$\frac{1}{2e} \left[\left(d(x) - \frac{1}{2} \right) + \left(d(y) - \frac{1}{2} \right) + 2 \left(d(x) - \frac{1}{2} \right) \left(d(y) - \frac{1}{2} \right) - \left(e - \frac{1}{2} \right) \right].$$

At times, we will also write

$$\frac{1}{2e} = \frac{1}{1 + 2(e - \frac{1}{2})} = 1 - 2(e - \frac{1}{2}) + 4(e - \frac{1}{2})^2 + \dots$$

Squaring and integrating, we have

$$\begin{aligned} \mu_2 &= \frac{1}{4e^2} \left[2v_2 - (e - \frac{1}{2})^2 + 4v_2^2 + 8(e - \frac{1}{2})v_2 - 4(e - \frac{1}{2})^3 \right] \\ &= \frac{1}{4e^2} \left[2\zeta^2 - (e - \frac{1}{2})^2 + 4\zeta^4 + 8(e - \frac{1}{2})\zeta^2 - 4(e - \frac{1}{2})^3 \right]. \end{aligned}$$

Note that this expression depends only on e and t (and therefore ζ), and not on the actual degree function $d(x)$.

Taking the fourth power of the expansion of $(d(x)d(y) - \frac{1}{2})/e$ and integrating yields terms that scale as ζ^4 , ζ^5 , ζ^6 , ζ^7 and ζ^8 . The ζ^4 terms are:

$$\frac{1}{16e^4} \left(2v_4 + 6v_2^2 - 12v_2(e - \frac{1}{2})^2 + 5(e - \frac{1}{2})^4 \right).$$

The only term that isn't fixed is

$$2v_4 = 2\zeta^4 + 2 \int_0^1 \left((d(x) - \frac{1}{2})^2 - \zeta^2 \right)^2 dx.$$

This is minimized by having $(d(x) - \frac{1}{2})^2$ be everywhere equal to ζ^2 . More precisely, $d(x)$ should equal $\frac{1}{2} - \zeta$ on an interval of length

$$c = \frac{1}{2} \left(1 - \frac{e - \frac{1}{2}}{\zeta} \right)$$

and $\frac{1}{2} + \zeta$ on an interval of length $1 - c$. This yields $\int_0^1 d(x) dx = e$ as well as $v_2 = \int_0^1 (d(x) - \frac{1}{2})^2 = \zeta^2$.

We now consider the effect of the order ζ^5 terms, which are

$$\frac{1}{2e^4} \left(2(e - \frac{1}{2})v_4 + 6 \left(v_3 - (e - \frac{1}{2})v_2 \right) \left(v_2 - (e - \frac{1}{2})^2 \right) - (e - \frac{1}{2})^5 \right).$$

Minimizing v_4 fixes the value of v_3 , since

$$\int_0^1 (d(x) - \frac{1}{2})^3 dx = \zeta^2(e - \frac{1}{2}) + \int_0^1 (d(x) - \frac{1}{2}) \left((d(x) - \frac{1}{2})^2 - \zeta^2 \right) dx.$$

By Cauchy-Schwarz, the integral on the right hand side is bounded in magnitude by $\sqrt{v_2(v_4 - v_2^2)}$. To change v_3 with a minimal increase in v_4 , we need $(d(x) - \frac{1}{2})^2 - \zeta^2$ to be proportional to $d(x) - \frac{1}{2}$. This yields a quadratic equation, so $d(x)$ still only takes two values, but these values are no longer exactly $\frac{1}{2} \pm \zeta$.

Since the increase to μ_4 is $2(v_4 - v_2^2)$ and the decrease is bounded by a multiple of $v_2^{3/2} \sqrt{v_4 - v_2^2}$, the maximum is achieved when $\sqrt{v_4 - v_2^2}$ is of order $v_2^{3/2} = \zeta^3$. That is, we have

$$\begin{aligned} (d(x) - \frac{1}{2})^2 - \zeta^2 &= O(\zeta^3) \\ (d(x) - \frac{1}{2})^2 &= \zeta^2(1 + O(\zeta)) \\ d(x) &= \frac{1}{2} \pm \zeta + O(\zeta^2). \end{aligned}$$

Note that changing v_3 by $O(\zeta^3)$, and thereby changing v_4 by $O(\zeta^3)$, only changes the entropy by $O(\zeta^6)$.

Contributions from the higher-order terms in the expansion of μ_4 , as well as from the higher moments, likewise shift the degree function, but by less. Without examining the details of those terms we cannot immediately conclude that $d(x)$ only takes on two values, but these higher-order terms shift $d(x)$ by less than this leading $v_2 v_3$ term, so $d(x) = \frac{1}{2}$ remains within $O(\zeta^2)$ of $\frac{1}{2} \pm \zeta$ everywhere.

So far in this proof, we have neglected the $g_1(x, y)$ term in the expansion of $g(x, y)$. However, we have already proven that $g_1(x, y)$ is pointwise $O(\zeta^3)$. Since the claims in the lemma are only up to $O(\zeta^2)$ errors, adding $g_1(x, y)$ does not affect the results in any way. $\square$

The proofs of Lemmas 7.7 and 7.8 concern perturbations of the bipodal graphon g_b obtained by averaging our entropy maximizing graphon g over each quadrant. Unfortunately, we do not know the exact parameters of g_b . However, we have a good approximation, namely the rank one bipodal graphon g_0 with

$$\begin{aligned} a &= \frac{(\frac{1}{2} - \zeta)^2}{e}, \\ b &= \frac{(\frac{1}{2} + \zeta)^2}{e}, \\ d &= \frac{\frac{1}{4} - \zeta^2}{e}, \end{aligned}$$

$$c = \frac{1}{2} \left(1 - \frac{e - \frac{1}{2}}{\zeta} \right), \quad (7.8)$$

for which $d(x)$ takes on the two values $\frac{1}{2} \pm \zeta$. As we have seen, adjusting the bipodal parameters to allow $v_4 - v_2^2$ to be nonzero (thereby adjusting the values of v_3 and higher moments) only changes the entropy by $O(\zeta^6)$, as does adding a term $g_1(x, y)$ to increase the rank of the graphon. Since the entropy is given by an absolutely convergent power series, and since we are well within the radius of convergence, the derivative of the entropy with respect to ζ can be computed to within an $O(\zeta^5)$ error using our approximate graphon. Changes to the entropy and to t when we add $g_2(x, y)$ to g_b are similar. As a result, dS/dt for g_0 and $\frac{S(g_0+g_2)-S(g_0)}{\tau(g_0+g_2)-\tau(g_0)}$ come within $O(\zeta^4)$ of dS/dt for g_b and $\frac{S(g_b+g_2)-S(g_b)}{\tau(g_b+g_2)-\tau(g_b)}$.¹ Since Lemmas 7.7 and 7.8 only concern these quantities up to $O(\zeta^3)$ errors, that is accurate enough for us!

Proof of Lemma 7.7 As in the proof of Theorem 7.1, we write

$$g_2(x, y) = \begin{cases} \frac{\eta_1(x)}{c} + \frac{\eta_1(y)}{c} + g_3(x, y) & \text{on } I_1 \times I_1, \\ \frac{\eta_2(x)}{1-c} + \frac{\eta_3(y)}{c} + g_3(x, y) & \text{on } I_1 \times I_2, \\ \frac{\eta_2(y)}{1-c} + \frac{\eta_3(x)}{c} + g_3(x, y) & \text{on } I_2 \times I_1, \\ \frac{\eta_4(x)}{1-c} + \frac{\eta_4(y)}{1-c} + g_3(x, y) & \text{on } I_2 \times I_2, \end{cases}$$

where

$$\int_0^c \eta_1(x) dx = \int_0^c \eta_2(x) dx = \int_c^1 \eta_3(x) dx = \int_c^1 \eta_4(x) dx = 0,$$

and where $g_3(x, y)$ has zero marginals on each quadrant. As in our earlier calculation,

$$\begin{aligned} \tau(g) - \tau(g_b) &= 3 \operatorname{tr}(g_b g_2^2) + \operatorname{tr}(g_2^3) \\ &= \int_0^c 3a\eta_1(x)^2 + 6d\eta_1(x)\eta_2(x) + 3b\eta_2(x)^2 dx \\ &\quad + \int_c^1 3a\eta_3(x)^2 + 6d\eta_3(x)\eta_4(x) + 3b\eta_4(x)^2 dx + O(\|g_2\|_2^3). \end{aligned} \quad (7.9)$$

The quantity g_3 does not contribute at all to $\operatorname{tr}(g_b g_2^2)$. Also as in our earlier calculation,

$$S(g) - S(g_b) = \int_0^c \frac{H''(a)}{c} \eta_1(x)^2 + \frac{H''(d)}{1-c} \eta_2(x)^2 dx$$

¹ It is only $O(\zeta^4)$, and not $O(\zeta^5)$, because $dS/d\zeta$ and $dt/d\zeta$ are each $O(\zeta)$, so errors of size $O(\zeta^5)$ in each term result in errors of size $O(\zeta^4)$ in the ratio.

$$+ \int_c^1 \frac{H''(d)}{c} \eta_3(x)^2 + \frac{H''(b)}{1-c} \eta_4(x)^2 dx + o(\|g_2\|^2). \quad (7.10)$$

Both the increase in t (7.9) and the entropy cost (7.10) are the sum of a term involving η_1 and η_2 , and a term involving η_3 and η_4 . That is, we have two separate problems, one involving η_1 and η_2 and the other involving η_3 and η_4 .

We tackle the one involving η_1 and η_2 first. By Cauchy-Schwarz, the change in t is maximized for given $\|\eta_1\|_2$ and $\|\eta_2\|_2$ by having $\eta_2(x)$ be proportional to $\eta_1(x)$. If we define

$$N_1^2 = \int_0^c \eta_1(x)^2 dx, \quad N_2^2 = \int_0^c \eta_2(x)^2 dx,$$

then the change in t is given, to leading order in N , by the matrix product

$$\begin{pmatrix} N_1 & N_2 \end{pmatrix} \begin{pmatrix} 3a & 3d \\ 3d & 3b \end{pmatrix} \begin{pmatrix} N_1 \\ N_2 \end{pmatrix},$$

while the change in entropy is

$$\begin{pmatrix} N_1 & N_2 \end{pmatrix} \begin{pmatrix} \frac{H''(a)}{c} & 0 \\ 0 & \frac{H''(d)}{1-c} \end{pmatrix} \begin{pmatrix} N_1 \\ N_2 \end{pmatrix}.$$

Maximizing the ratio of these quantities is a generalized eigenvalue problem. We just have to find the root(s) of the determinant of

$$\begin{pmatrix} \frac{H''(a)}{c} & 0 \\ 0 & \frac{H''(d)}{1-c} \end{pmatrix} - \lambda \begin{pmatrix} 3a & 3d \\ 3d & 3b \end{pmatrix}.$$

We use the values of (a, b, c, d) given by the rank one bipodal matrix g_0 . With these values, $ab - d^2 = 0$, so the determinant is a linear function of λ , namely

$$-3\lambda \left(a \frac{H''(d)}{1-c} + b \frac{H''(a)}{c} \right) + \frac{H''(a)H''(d)}{c(1-c)}.$$

The lone root gives an upper bound for the change in entropy per change in t . There is no lower bound, since if $\begin{pmatrix} N_1 \\ N_2 \end{pmatrix}$ lies in the kernel of $\begin{pmatrix} 3a & 3d \\ 3d & 3b \end{pmatrix}$, then we decrease the entropy without any change to t .

Setting the determinant equal to zero gives

$$\lambda = \frac{H''(a)H''(d)}{3(acH''(d) + b(1-c)H''(a))}. \quad (7.11)$$

We approximate $H''(a)$ with a power series around $1/2$:

$$H''(a) = H''(1/2) + \frac{1}{2}H''''(1/2)(a - \frac{1}{2})^2 + O(a - \frac{1}{2})^4$$

$$= -2 - 8\left(a - \frac{1}{2}\right)^2 + O\left(a - \frac{1}{2}\right)^2,$$

with a similar approximation for $H''(d)$. Plugging in the values (7.8), the numerator of (7.11) becomes

$$4 + 32\left(\left(e - \frac{1}{2}\right)^2 + 2\zeta\left(e - \frac{1}{2}\right) + 2\zeta^2\right) + O(\zeta^4),$$

while the denominator becomes

$$\begin{aligned} & -6\left(e + 6\zeta^2 + 4\left(e - \frac{1}{2}\right)^2 + 8\left(e - \frac{1}{2}\right)\zeta\right) + O(\zeta^4) \\ &= -6e\left(1 + 12\zeta^2 + 8\left(e - \frac{1}{2}\right)^2 + 16\left(e - \frac{1}{2}\right)\zeta\right) + O(\zeta^3). \end{aligned}$$

The ratio (7.11) is then

$$\frac{-2}{3e}(1 + 4\zeta^2) + O(\zeta^3). \quad (7.12)$$

The calculation involving η_3 and η_4 is nearly identical, only with c replaced by $1 - c$ and with the roles of a and b reversed. This is equivalent to replacing ζ with $-\zeta$. Since the expression (7.12) is an even function of ζ (up to $O(\zeta^3)$ errors), the effect of having nonzero η_3 and η_4 is the same as that of having nonzero η_1 and η_2 . That is,

$$\frac{S(g_0 + g_2) - S(g_0)}{\tau(g_0 + g_2) - \tau(g_0)} \leq \frac{-2}{3e}(1 + 4\zeta^2) + O(\zeta^3).$$

Our calculations so far have involved the known graphon g_0 rather than the unknown graphon g_b . However, we already know that $g_0 - g_b$ is small, with their entropies differing by only $O(\zeta^6)$, and with g_b being within $O(\zeta^3)$ of having rank 1. This implies that the differences $S(g_b + g_2) - S(g_b)$ and $S(g_0 + g_2) - S(g_0)$ agree to within $O(\zeta^4)$, as do the differences $\tau(g_b + g_2) - \tau(g_b)$ and $\tau(g_0 + g_2) - \tau(g_0)$. We conclude that

$$\frac{S(g_b + g_2) - S(g_b)}{\tau(g_b + g_2) - \tau(g_b)} \leq \frac{-2}{3e}(1 + 4\zeta^2) + O(\zeta^3).$$

□

Proof of Lemma 7.8 As in the proof of Lemma 7.7, we work with the graphon g_0 rather than the unknown g_b . This changes the entropy by $O(\zeta^6)$ and the derivative of that entropy with respect to ζ by $O(\zeta^5)$, ultimately changing the ratio $\frac{dS/d\zeta}{d\tau/d\zeta}$ by $O(\zeta^4)$. Since we are only interested in the answer through order ζ^2 , this does not matter.

We compute μ_2 and μ_4 for the graphon g_0 . Since g_0 has rank 1, we already have that

$$\mu_2 = \frac{1}{4e^2} \left[2\zeta^2 - (e - \frac{1}{2})^2 + 4\zeta^4 + 8(e - \frac{1}{2})\zeta^2 - 4(e - \frac{1}{2})^3 \right],$$

so

$$\frac{d\mu_2}{d\zeta} = 2\zeta \left(2 + 8 \left(\zeta^2 - (e - \frac{1}{2})^2 \right) \right).$$

Meanwhile,

$$\begin{aligned} \mu_4 &= c^2(a - \frac{1}{2})^4 + (1 - c)^2(b - \frac{1}{2})^4 + 2c(1 - c)(d - \frac{1}{2})^4 \\ &= c^2((e - \frac{1}{2}) + 2\zeta)^4 + (1 - c)^2((e - \frac{1}{2}) - 2\zeta)^4 \\ &\quad + 2c(1 - c)((e - \frac{1}{2})^4) + O(\zeta^5) \\ &= (e - \frac{1}{2})^4 - 8(e - \frac{1}{2})^3\zeta(1 - 2c) + 24(1 - 2c(1 - c))(e - \frac{1}{2})^2\zeta^2 \\ &\quad - 32(e - \frac{1}{2})\zeta^3(1 - 2c) + 16(1 - 2c(1 - c))\zeta^4 + O(\zeta^5) \\ &= 5(e - \frac{1}{2})^2 - 12(e - \frac{1}{2})^2\zeta^2 + 8\zeta^4 + O(\zeta^5), \end{aligned}$$

where we have used the identities

$$\zeta(1 - 2c) = e - \frac{1}{2} \quad \text{and} \quad 4\zeta^2c(1 - c) = \zeta^2 - (e - \frac{1}{2})^2.$$

We then have

$$\frac{d\mu_4}{d\zeta} = 2\zeta \left(16\zeta^2 - 12(e - \frac{1}{2})^2 \right) + O(\zeta^4).$$

Since

$$\begin{aligned} S &= H(1/2) + \frac{1}{2}H''(1/2)\mu_2 + \frac{1}{24}H''''(1/2)\mu_4 + O(\zeta^6) \\ &= H(1/2) - \mu_2 - \frac{2}{3}\mu_4 + O(\zeta^6), \end{aligned}$$

we have

$$\begin{aligned} \frac{dS}{d\zeta} &= -\frac{d\mu_2}{d\zeta} - \frac{2}{3}\frac{d\mu_4}{d\zeta} + O(\zeta^5) \\ &= -4\zeta \left(1 + \frac{28}{3}\zeta^2 - 8(e - \frac{1}{2})^2 \right) + O(\zeta^4). \end{aligned}$$

Now we turn to the triangle density. Since

$$\zeta^2 = et^{1/3} - e^2 + (e - \frac{1}{2})^2,$$

we have

$$t = \frac{\left(e^2 + \zeta^2 - \left(e - \frac{1}{2}\right)^2\right)^3}{e^3},$$

and so

$$\begin{aligned} \frac{dt}{d\zeta} &= 6\zeta \frac{\left(e^2 + \zeta^2 - \left(e - \frac{1}{2}\right)^2\right)^2}{e^3} \\ &= 6\zeta \frac{e^4 + 2e^2(\zeta^2 - (e - \frac{1}{2})^2) + O(\zeta^4)}{e^3} \\ &= 6e\zeta \left(1 + \frac{2}{e^2}(\zeta^2 - (e - \frac{1}{2})^2)\right) + O(\zeta^5) \\ &= 6e\zeta(1 + 8\zeta^2 - 8(e - \frac{1}{2})^2) + O(\zeta^4). \end{aligned}$$

We then have

$$\begin{aligned} \frac{dS}{dt} &= \frac{dS/d\zeta}{dt/d\zeta} \\ &= \frac{-4\zeta \left(1 + \frac{28}{3}\zeta^2 - 8(e - \frac{1}{2})^2\right) + O(\zeta^4)}{6e\zeta(1 + 8\zeta^2 - 8(e - \frac{1}{2})^2) + O(\zeta^4)} \\ &= \frac{-2}{3e} \left(1 + \frac{4}{3}\zeta^2\right) + O(\zeta^3). \end{aligned}$$

□

While we have only proven that the entropy maximizing graphon is unique, bipodal, and smoothly varying on a small open set just above the Erdős-Rényi curve, numerical evidence suggests more.

Conjecture 7.9 *In the edge-triangle model there is a single phase covering the entire region above the Erdős-Rényi curve $t = e^3$. At each point in this region, the entropy maximizing graphon is unique and bipodal, with parameters that vary analytically with (e, t) .*

7.4 A universal F(1,1)-like phase

Just above Erdős-Rényi, the triangle count for an entropy maximizing graphon is closely related to its 2star count:

$$t - e^3 = 3e\tilde{t}_2 + O(\tilde{t}_2^{3/2}).$$

Similarly, the density t_T of tetrahedra is closely related to its 3star count:

$$t_T - e^6 = 4e^3 \tilde{t}_3 + O(\tilde{t}_3^{3/2}).$$

We will soon see that the density t_H of *any* finite graph H is closely related to a certain positive linear combination $\sum_k w_k \tilde{t}_k$ of k star densities. Maximizing the entropy for fixed t_H is nearly the same as maximizing the entropy for fixed $\sum_k w_k \tilde{t}_k$. But that is a problem we already solved in Theorem 6.6. We will now show that, for *any* graph H , the entropy maximizing graphon just above Erdős-Rényi, and for all but finitely many values of e , is unique and bipodal of a certain form.

Let H be any fixed finite graph with edge set $\mathcal{E}$ and vertex set $\mathcal{V} = \{1, 2, \dots, V\}$. We will write (i, j) to denote an edge connecting vertex i to vertex j . Let E be the total number of edges in H . For each integer $k > 1$, let m_k be the number of vertices of degree k .

For any graphon g , the associated density of H 's is

$$\tau_H(g) = \int_{[0,1]^V} \prod_{(i,j) \in \mathcal{E}} g(x_i, x_j) d^V x.$$

If g is an Erdős-Rényi graphon, then this is just e^E . We also define the reduced H -density

$$\tilde{\tau}(g) = \tau(g) - e^E.$$

Lemma 7.10 *Let $g(x, y) = e + g_1(x, y)$ be a graphon with overall edge density e . For each k , let $w_k = e^{E-k} m_k$ and let $\tilde{\tau}_w(g) = \sum_k w_k \tilde{\tau}_k(g)$, where $\tilde{\tau}_k(g) = \tau_k(g) - e^k$ is the reduced density of k stars. Then*

$$\tilde{\tau}(g) = \tilde{\tau}_w(g) + O(\|g_1\|_2^3).$$

Proof We plug $g = e + g_1$ into the formula for $\tau_H(g)$ and expand the product of factors $e + g_1(x_i, x_j)$. To each term in the expansion we associate a subset $\mathcal{E}' \subset \mathcal{E}$ of edges corresponding to the factors of g_1 . This gives

$$\tau(H) = \sum_{\mathcal{E}' \subset \mathcal{E}} e^{E-|\mathcal{E}'|} \int_{[0,1]^V} \prod_{(i,j) \in \mathcal{E}'} g_1(x_i, x_j) d^V x.$$

We then sort the subsets $\mathcal{E}'$ into classes:

1. When $\mathcal{E}'$ is empty, we get e^E .
2. When $\mathcal{E}'$ is a single edge, we get zero, since $\iint g_1(x_i, x_j) dx_i dx_j = 0$.
3. When $\mathcal{E}'$ consists of two or more edges, all touching vertex i with degree k_i , then we get a contribution that is potentially of order $\|g_1\|_2^2$. The sum of all of these terms is precisely e^{E-k_i} times $\tilde{\tau}_{k_i}(g)$, since this is exactly

the expansion we would get if we applied $(e + g_1)$ to all of the edges touching i and e to all of the edges not touching i (and then subtracted e^E , corresponding to $\mathcal{E}' = \emptyset$). Summing over i , we get $\sum_k w_k \tilde{\tau}_k(g)$.

4. We claim that all other subsets $\mathcal{E}'$ contribute $O(\|g\|_2^3)$. There are three cases to consider.

- If $\mathcal{E}'$ is connected and not all of the edges share a common vertex, then it must either contain a triangle (e.g., the edges $(1, 2)$, $(2, 3)$ and $(3, 1)$) or a chain of length 3 (e.g., the edges $(1, 2)$, $(2, 3)$, and $(3, 4)$). If it contains a triangle (say, with vertices 1, 2 and 3), then

$$\begin{aligned} \left| \int_{[0,1]^V} \prod_{(i,j) \in \mathcal{E}'} g_1(x_i, x_j) d^V x \right| &\leq \int_{[0,1]^V} \prod_{(i,j) \in \mathcal{E}'} |g_1(x_i, x_j)| d^V x \\ &\leq \int_{[0,1]^V} |g_1(x_1, x_2) g_1(x_2, x_3) g_1(x_3, x_1)| d^V x \\ &\leq \|g_1\|_2^3 \\ &= \|g_1\|_2^3. \end{aligned}$$

In the second line, we replaced all but three of the factors $|g_1(x_i, x_j)|$ with 1, since $|g_1(x_i, x_j)| \leq 1$.

- Likewise, if $\mathcal{E}'$ contains a chain from 1 to 2 to 3 to 4, then

$$\begin{aligned} \left| \int_{[0,1]^V} \prod_{(i,j) \in \mathcal{E}'} g_1(x_i, x_j) d^V x \right| &\leq \int_{[0,1]^V} \prod_{(i,j) \in \mathcal{E}'} |g_1(x_i, x_j)| d^V x \\ &\leq \int_{[0,1]^V} |g_1(x_1, x_2) g_1(x_2, x_3) g_1(x_3, x_4)| d^V x \\ &\leq \|g_1\|_2^3 \\ &= \|g_1\|_2^3. \end{aligned}$$

- If $\mathcal{E}'$ is disconnected, then the integral over $d^V x$ factors into a product of integrals, one for each connected component of $\mathcal{E}'$. Each integral is either zero (if the component consists of a single edge) or is $O(\|g_1\|_2^2)$ (if it contains two edges sharing a vertex). Thus the contribution from $\mathcal{E}'$ is either zero or $O(\|g_1\|_2^4)$.

□

Theorem 7.11 *Let H be a fixed finite graph. For all but finitely many values of e , and for all sufficiently small positive values of $\tilde{\tau}_H$, there is a unique entropy maximizing graphon with densities $(e, \tilde{\tau}_H)$. This graphon is bipodal, with parameters that are analytic functions of $(e, \tilde{\tau})$.*

Proof For each integer $k \geq 2$, let m_k be the number of vertices of H of degree k and let $w_k = e^{E-k} m_k$. Let $\tilde{t}_w = \sum_k w_k \tilde{t}_k$. As in the proof of Theorem 6.6, we consider the function

$$f_w(e, e') := 2 \frac{H(e') - H(e) - (e' - e)H'(e)}{\sum_k w_k [(e')^k - e^k - k e^{k-1}(e' - e)]}$$

for $e' \neq e$ and

$$f_w(e, e) = \frac{2H''(e)}{\sum_k k(k-1)w_k e^{k-2}}.$$

For all but finitely many values of e , there is a unique value of e' , which we denote $\tilde{e}$, that maximizes $f_w(e, e')$. Furthermore, for all but finitely many values of e , $\tilde{e} \neq e$. The ratio

$$\frac{S(g) - H(e)}{\tilde{\tau}_H(g)}$$

is bounded by a number that is within $O(\tilde{t}_H^{1/2})$ of the maximum value of

$$\frac{S(g) - H(e)}{\tilde{\tau}_w(g)}.$$

As $\tilde{t}_w$ (and therefore $\tilde{t}_H$) approaches zero from above, this maximum ratio approaches $f_w(e, \tilde{e})$, but only when $g(x, y)$ is close, in an L^2 -sense, to a bipodal graphon with $a \approx 2\tilde{e} - e$, $b \approx e$, and $d \approx \tilde{e}$.

The next stage in the proof is showing that g is pointwise close to a bipodal graphon. This follows much the same path as the proof of Theorem 7.1. We compare the functional derivative $\frac{\delta \tau_H(g)}{\delta g(x_i, x_j)}$ to the functional derivative $\frac{\delta \tau_w(g)}{\delta g(x_i, x_j)}$, only now using the same diagrammatic expansion as in the proof of Lemma 7.10. For these purposes, $\mathcal{E}'$ must contain the edge (i, j) .

- Terms where $\mathcal{E}'$ is a set of edges emerging from vertex i , or a set of edges emerging from vertex j , contribute exactly the same to the two functional derivatives. (These are the only contributions to $\frac{\delta \tau_w(g)}{\delta g(x_i, x_j)}$.)
- The contribution when $\mathcal{E}'$ consists of edges emerging from vertex i and edges emerging from vertex j and no other edges is a power of $(d(x_i) - e)$ times a power of $(d(x_j) - e)$ times a power of $G(x_i, x_j)$, where we get one factor of $(d(x_i) - e)$ for each edge from vertex i to a vertex that is not connected (via an edge in $\mathcal{E}'$) to vertex j , one factor of $G(x_i, x_j)$ for each edge from vertex i to a vertex that is connected to vertex j , and one factor of $(d(x_j) - e)$ for each edge from vertex j to a vertex that is not connected to vertex i , and a factor of e for each edge of H that is not in $\mathcal{E}'$. See Figure 7.1. This product is bounded by $G(x_i, x_j)$ if there are any triangles in $\mathcal{E}'$ containing (i, j) , and is bounded by $(d(x_i) - e)(d(x_j) - e)$ if there are not.

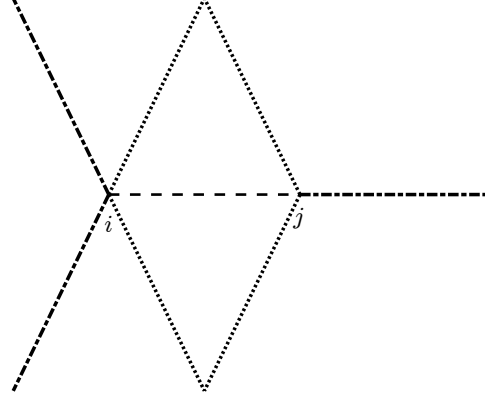


Figure 7.1 A graph $\mathcal{E}'$ contributing to $\frac{\delta \tau_H(g)}{\delta g(x_i, x_j)}$. Each diagonal dot-dashed edge contributes a factor of $d(x_i) - e$, the horizontal dot-dashed edge contributes a factor of $d(x_j) - e$, each pair of dotted edges contributes a factor of $G(x_i, x_j)$, and the edges not in $\mathcal{E}'$ give a factor of e^{E-8} .

- Whenever $\mathcal{E}'$ contains edges that are not connected to vertices i or j , the corresponding contribution to the functional derivative of $\tau(g)$ is $O(\|g_1\|_2)^2$.

The upshot is that $\frac{\delta \tau_H(g)}{\delta g(x, y)}$ is pointwise close to $\frac{\delta \tau_w(g)}{\delta g(x, y)}$ everywhere except on $I_1 \times I_1$ (where $(d(x) - e)(d(y) - e)$ is non-negligible) and on a set $I_0 \times I_0$, where I_0 has measure at most $O(\tilde{t}_H^{1/3})$. This implies that

$$\int_0^1 G(x, y) dy = O(\tilde{t}_H^{1/3})$$

everywhere. If we normalize our Lagrange multipliers as in the k star models, specifically as

$$dS = \frac{\alpha}{2} d\varepsilon + \frac{\beta}{2} d\tau_H,$$

then the degree function is everywhere within $O(\tilde{t}_H^{1/3})$ of the solutions to

$$d(x) = \int_0^1 \frac{dy}{1 + \exp\left(\alpha + \beta \frac{\delta \tilde{\tau}_w}{\delta g(x, y)}\right)}, \quad (7.13)$$

two of which are numbers close to e and $\tilde{e}$. (For all but finitely many values of e , there are only two such roots.)

We then eliminate the solutions to (7.13) that are not close to e or $\tilde{e}$. To leading

order in the sizes of those anomalous podes, these cost too much entropy for their production of H 's and we can improve our ratio of $S(g) - H(e)$ to $\tilde{t}_H$ by setting these podes sizes to zero. That is, the degree function is pointwise close to a step function with values e and $\tilde{e}$. This defines our podes with no ambiguity, since *every* point x either has $d(x) \approx \tilde{e}$ (in which case $x \in I_1$) or $d(x) \approx e$ (in which case $x \in I_2$). We then average g over each quadrant and write

$$g(x, y) = g_b(x, y) + g_2(x, y),$$

where g_b is exactly bipodal and g_2 averages to zero on each quadrant.

The bootstrap showing that $I_0 = I_1$, that $G(x, y)$ is small on $I_2 \times I_2$ and $I_1 \times I_2$, and that $G(x, y)$ is nearly constant on $I_1 \times I_1$, is word-for-word identical to the proof of Theorem 7.1. Since $d(x) - e$, $d(y) - e$ and $G(x, y)$ are nearly constant on each quadrant, $g(x, y)$ is then nearly constant on each quadrant, so $g_2(x, y)$ is pointwise small.

The final step is computing $\frac{S(g) - S(g_b)}{\tau_H(g) - \tau_H(g_b)}$ to leading order in $\|g_2\|_2$ and comparing that ratio to $f_w(e, \tilde{e})$. The value of $\tau_H(g) - \tau_H(g_b)$ comes from our usual diagrammatic expansion, only now with a factor of g_2 for every edge in $\mathcal{E}'$ and a factor of g_b for every edge not in $\mathcal{E}'$. Since g_2 is pointwise small, the leading contributions come when $\mathcal{E}'$ consists of exactly two edges sharing a vertex i . Suppose that these edges connect vertex i to vertices j_1 and j_2 .

What follows is a complication that must be addressed but that turns out not to matter at all. (Impatient readers may skip this and the following three paragraphs.) In our expansion, we assign a factor of g_b to all but the two edges connecting vertex i to vertices j_1 and j_2 . If instead we assigned a factor of g_b to the edges touching i and e to all of the edges not touching i , we would get e^{E-k_i} times the change (to second order in g_2) in the k_i star density, where k_i is the degree of vertex i . Adding g_2 changes the degree function from

$$\begin{cases} ca + (1-c)d & x \in I_1, \\ cd + (1-c)b & x \in I_2, \end{cases} \quad \text{to} \quad \begin{cases} ca + (1-c)d + \eta_1(x) + \eta_2(x) & x \in I_1, \\ cd + (1-c)b + \eta_3(x) + \eta_4(x) & x \in I_2, \end{cases}$$

thereby changing the k_i star density by

$$\frac{k_i(k_i - 1)}{2} \left[(ca + (1-c)d)^{k_i-2} \int_0^c (\eta_1(x) + \eta_2(x))^2 dx + (cd + (1-c)b)^{k_i-2} \int_c^1 (\eta_3(x) + \eta_4(x))^2 dx \right]$$

Multiplying by e^{E-k_i} and summing over i , we would get exactly the change, to second order in $\|g_2\|_2^2$, to t_w .

So does replacing $g_b(x, y)$ with e actually matter? The difference is pointwise

small except when both x and y are in I_1 . If we add or subtract $g_b - e$ from the factor for an edge between vertices j_3 and j_4 , then the extra term only matters with x_{j_3} and x_{j_4} are in $I_1 \times I_2$. This is a set of measure $O(\tilde{t}_H^2)$ and so only contributes $O(\tilde{t}_H^2 \|g_2\|_2^2)$. However, if j_3 or j_4 is the same as j_1 or j_2 , and if we are looking at the contribution of η_1 or η_3 , then we have already restricted x_{j_3} or x_{j_4} to lie in I_1 . We then get something of order $O(\tilde{t}_H \|g_2\|_2^2)$. This is still negligible.

However, if the edge (j_3, j_4) is exactly the edge (j_1, j_2) , then we get something of order $\|g_2\|_2^2$, which is *not* negligible. This is what we saw in the proof of Theorem 7.1, where the change in the triangle density was three (one for each vertex) times

$$\begin{aligned} & \int_0^c a\eta_1(x)^2 + 2d\eta_1(x)\eta_2(x) + b\eta_2(x)^2 dx \\ & + \int_c^1 a\eta_3(x)^2 + 2d\eta_3(x)\eta_4(x) + b\eta_4(x)^2 dx \end{aligned} \quad (7.14)$$

rather than three times

$$e \left[\int_0^c (\eta_1(x) + \eta_2(x))^2 dx + \int_c^1 (\eta_3(x) + \eta_4(x))^2 dx \right] \quad (7.15)$$

The result is that the contribution from a pair of edges (i, j_1) and (i, j_2) depends on whether vertices j_1 and j_2 are connected by an edge in $\mathcal{E}$ or not. If they are not, then we get

$$\begin{aligned} & \frac{e^{E-k_i}}{2} \left[(ca + (1-c)d)^{k_i-2} \int_0^c (\eta_1(x) + \eta_2(x))^2 dx \right. \\ & \quad \left. + (cd + (1-c)b)^{k_i-2} \int_c^1 (\eta_3(x) + \eta_4(x))^2 dx \right] \end{aligned}$$

If they are, then we get

$$\begin{aligned} & \frac{e^{E-1-k_i}}{2} \left[(ca + (1-c)d)^{k_i-2} \int_0^c a\eta_1(x)^2 + 2d\eta_1(x)\eta_2(x) + b\eta_2(x)^2 dx \right. \\ & \quad \left. + (cd + (1-c)b)^{k_i-2} \int_c^1 a\eta_3(x)^2 + 2d\eta_3(x)\eta_4(x) + b\eta_4(x)^2 dx \right] \end{aligned}$$

As noted earlier, this complication doesn't actually matter. The entropy cost for having g_2 nonzero is

$$\begin{aligned} S(g) - S(g_b) &= \int_0^c \frac{H''(a)}{c} \eta_1(x)^2 + \frac{H''(d)}{1-c} \eta_2(x)^2 dx \\ & \quad + \int_c^1 \frac{H''(d)}{c} \eta_3(x)^2 + \frac{H''(b)}{1-c} \eta_4(x)^2 dx \end{aligned}$$

$$+o(\|g_2\|^2).$$

Since c is infinitesimal, the cost of having η_1 or η_3 nonzero is enormous, so $\eta_1(x)$ and $\eta_3(x)$ are $o(1)$ multiples of $\eta_2(x)$ and $\eta_4(x)$, respectively. Furthermore, b and d are approximately e and $\tilde{e}$, respectively. The expressions (7.14) and (7.15) simplify to the same thing, and we have

$$\begin{aligned}\tau_H(g) - \tau_H(g_b) &= \sum_k w_k \frac{k(k-1)}{2} \tilde{e}^{k-2} \left(\int_0^c \eta_2(x)^2 dx \right) \\ &\quad + \sum_k w_k \frac{k(k-1)}{2} e^{k-2} \left(\int_c^1 \eta_4(x)^2 dx \right) + o(\|g_2\|_2^2) \\ S(g) - S(g_b) &= H''(\tilde{e}) \left(\int_0^c \eta_2(x)^2 dx \right) \\ &\quad + H''(e) \left(\int_c^1 \eta_4(x)^2 dx \right) + o(\|g_2\|_2^2)\end{aligned}$$

In short, the change in entropy per increase in $\tilde{t}_H$ is approximately $f_w(\tilde{e}, \tilde{e})$ when we have η_2 nonzero, it is approximately $f_w(e, e)$ when we have η_4 nonzero, and it is approximately $f_w(e, \tilde{e})$ when we vary c . By assumption $f_w(e, e')$ is maximized (for fixed e) at $e' = \tilde{e} \neq e$, so varying c is more efficient than having η_4 nonzero. All that is left is showing that $f_w(e, \tilde{e}) > f_w(\tilde{e}, \tilde{e})$.

First of all, we know that $f_w(e, \tilde{e})$ can never be less than $f_w(\tilde{e}, \tilde{e})$. If it were, then the exact same calculation for the model where we fix e and t_w would yield an entropy maximizing graphon that is not bipodal, contradicting Theorem 6.6. The difference $f_w(e, \tilde{e}) - f_w(\tilde{e}, \tilde{e})$ is thus an analytic function of e that is non-negative. This function is easily checked to be strictly positive when e is sufficiently close to 0 or 1, so it can only be zero at a finite set of values of e . Insofar as the claims of the theorem are only “for all but finitely many values of e ”, this exceptional set does not matter. $\square$

The final indirect argument that $f_w(e, \tilde{e})$ is greater than $f_w(\tilde{e}, \tilde{e})$ for all but finitely many values of e is perhaps unsatisfying. We conjecture that, with further work, it will turn out to be unnecessary.

Conjecture 7.12 *If for fixed e the function $f_w(e, \hat{e})$ is maximized at $\hat{e} = \tilde{e}$, and if $\tilde{e} \neq e$, then $f_w(\tilde{e}, \hat{e})$ is maximized at $\hat{e} = e$.*

In support of Conjecture 7.12, we present a proposition suggesting a symmetry between e and $\tilde{e}$.

Proposition 7.13

$$\frac{\partial f_w(e, e')}{\partial e'} = 0 \quad \text{if and only if} \quad \frac{\partial f_w(e', e)}{\partial e} = 0.$$

Proof Let $N(e, e')$ and $D(e, e')$ be the numerator and denominator of the fraction that defines $f_w(e, e')$. That is, let

$$\begin{aligned} N(e, e') &= H(e') - H(e) - (e' - e)H'(e), \\ D(e, e') &= \sum w_k [(e')^k - e^k - k(e' - e)e^{k-1}]. \end{aligned}$$

Note that

$$\begin{aligned} (e' - e) \frac{\partial N(e, e')}{\partial e'} - N(e, e') &= N(e', e), \\ (e' - e) \frac{\partial D(e, e')}{\partial e'} - D(e, e') &= D(e', e). \end{aligned}$$

Setting

$$0 = \frac{\partial f_w(e, e')}{\partial e'} = \frac{D(e, e') \frac{\partial N(e, e')}{\partial e'} - N(e, e') \frac{\partial D(e, e')}{\partial e'}}{D(e, e')^2},$$

is equivalent to setting

$$\frac{\frac{\partial N(e, e')}{\partial e'}}{\frac{\partial D(e, e')}{\partial e'}} = \frac{N(e, e')}{D(e, e')}. \quad (7.16)$$

But then

$$\frac{N(e, e')}{D(e, e')} = \frac{(e' - e) \frac{\partial N(e, e')}{\partial e'} - N(e, e')}{(e' - e) \frac{\partial D(e, e')}{\partial e'} - D(e, e')} = \frac{N(e', e)}{D(e', e)}.$$

Meanwhile, the function $\frac{\partial N(e, e')}{\partial e'} = H'(e') - H'(e)$ is odd under interchange of e and e' , as is $\frac{\partial D(e, e')}{\partial e'}$, so the ratio is invariant. Thus the equation (7.16) is invariant under interchange of e and e' . $\square$

Since $f_w(e, e')$ for fixed e is maximized at $e' = \tilde{e}$, we have shown that $f_w(e, \tilde{e}) = f_w(\tilde{e}, e)$ and that $f_w(\tilde{e}, \hat{e})$ has a critical point at $\hat{e} = e$. What remains is showing that this critical point is actually a global maximum. While this is known to be the case for k star models, it remains unproven for positive linear combinations of k stars, and therefore for edge- H models.

We close this chapter with three more conjectures on the edge- H model, suggesting directions for future work. It seems likely that the first two can be tackled with the methods presented in this book, while the third is considerably more speculative.

Conjecture 7.14 *There is only one value of e for which $\tilde{e} = e$. For all other values of e , $\tilde{e}$ is an analytic function of e . At each such value, and for t_H greater than but sufficiently close to Erdős-Rényi, the entropy maximizing graphon is bipodal with degree functions close to e and $\tilde{e}$.*

Conjecture 7.15 *For any fixed graph H , there is a single bipodal phase just above Erdős-Rényi, covering all values of e . In addition to the results of Conjecture 7.14, there is a construction similar to Theorem 7.4 to fill the gap at the point where $\tilde{e} = e$.*

Conjecture 7.16 *For any fixed graph H , there is a single bipodal phase covering the entire region above Erdős-Rényi. There may be phase transitions, as in the cliquelike-to-anticliquelike transition in the edge-2star model, but there is still an analytic path from any entropy maximizing graphon at any point (e, t_H) to any other entropy maximizing graphon at any other point.*

Undersaturated graphs in the edge-triangle model

In the edge-triangle model, and more generally in models where we constrain the edge density and one other density t_H , there is a sharp difference between the supersaturated regime ($t > e^3$, or more generally $t_H > e^E$, where H contains E edges) and the undersaturated regime ($t < e^3$, or $t_H < e^E$). We give here a sketch of distinguishing features, which are proven in the following sections. Some of the results in this chapter have been previously published, and we will acknowledge this; our proofs are meant to meld coherently with the proofs in the book.

- When t is slightly greater than e^3 , the edge-triangle model is a perturbation of the edge-2star model. Similarly, the edge- H model just above Erdős-Rényi is a perturbation of a model where we constrain a positive linear combination of k stars. However, there is no undersaturated regime for k stars. For any graphon g , $\tau_k(g) \geq e^k$.
- The region just above Erdős-Rényi consists of a single phase, which we denote $F(1,1)$. Indeed, it is believed that the entire region above Erdős-Rényi (and not just an infinitesimal neighborhood of the Erdős-Rényi curve) lies in a single phase. However, we will see that the region just below Erdős-Rényi contains at least three distinct phases, and may well contain infinitely many phases.
- In the edge-2star model, there is a symmetry that sends $(e, \tilde{t}_2)$ to $(1 - e, \tilde{t}_2)$. This symmetry has echoes in the edge-triangle model, with the behavior just above Erdős-Rényi for $e < 1/2$ being qualitatively similar to that for $e > 1/2$. But just below Erdős-Rényi, entropy maximizing graphons with $e > 1/2$ are very different from those with $e < 1/2$.
- The Boltzmann entropy function $B(e, t)$ is fairly well-behaved for $t > e^3$, with $\partial B / \partial t$ approaching a finite negative number as t approaches e^3 from above. However, $\partial B / \partial t$ diverges to $+\infty$ as t approaches e^3 from below.

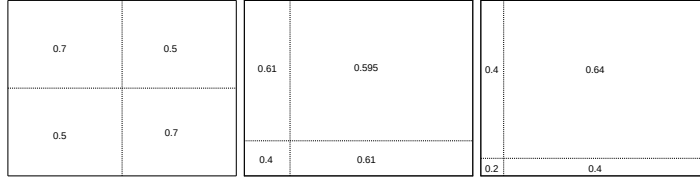


Figure 8.1 Three graphons with $e = 0.6$. The left graphon is symmetric bipodal, as found in the A(2,0) phase. The middle graphon is typical of the B(1,1) phase. The right graphon is typical of the supersaturated F(1,1) phase. Note that the L^2 difference between these graphons and a constant graphon is evenly distributed in the first graphon, is concentrated in the small square in the second, and is mostly in the rectangles in the third.

- In edge- H models, $\partial B / \partial t_H$ approaches a finite negative number as $t_H \rightarrow e^E$ from above, but $B(e, \tilde{t}_H)$ doesn't even have to be continuous as we approach Erdős-Rényi from below.

We explored the supersaturated regime in Chapter 7. Here we explore the undersaturated regime, mostly for the edge-triangle model but with a few additional results concerning edge- H models. Most of our effort will be towards proving the existence of two open sets in the edge-triangle model on which the optimal graphon is unique, bipodal, and depends analytically on (e, t) . We also show that there cannot be any analytic path connecting the two regions. That is, they represent genuinely different phases, which we denote B(1,1) and A(2,0). (Recall the nomenclature of phases in Figure 1.3.)

We compare the structure of the A(2,0), B(1,1) and F(1,1) phases in Figure 8.1 where these graphons are displayed for edge density $e = 0.6$.

In the A(2,0) phase, the entropy maximizing graphon is symmetric bipodal, meaning that $a = b = e - \delta$, $d = e + \delta$, and $c = 1/2$. We previously proved in Theorem (3.7) that the optimal graphon was of this form on the line segment $e = 1/2$, $0 < t < 1/8$, using the Method of Saturated Inequalities. Here we extend the analysis to an open set containing this segment, using the Method of Competing Inequalities.

The B(1,1) phase lies just below the Erdős-Rényi curve for $e > 1/2$. Let $\delta = (e^3 - t)^{1/3}$. We will show that the entropy maximizing graphon in the B(1,1) phase is bipodal with

$$\begin{aligned} a &= 1 - e + O(\delta), \\ b &= e + O(\delta^2), \\ d &= e + \delta + O(\delta^2), \\ c &= O(\delta) \end{aligned} \tag{8.1}$$

Similar results apply for $e > 1/2$ in the edge- H model, where H is an m -cycle with m odd. The only difference is that $\delta = (e^m - t_H)^{1/m}$ instead of $(e^3 - t)^{1/3}$.

The situation is different when $e < 1/2$. There we are able to prove bounds on the entropy as a function of t , but determining the form of the optimal graphon is more difficult. When $\frac{3-\sqrt{3}}{6} < e < \frac{1}{2}$ and t is slightly less than e^3 , the optimal graphon is believed to be symmetric bipodal, although this has not been proven. When $e < \frac{3-\sqrt{3}}{6}$, we will show that the optimal graphon is not bipodal. Understanding the nature of the optimal graphon in this region is an important open problem.

Finally, we will consider what happens just below Erdős-Rényi for a general edge- H model. (If H is bipartite undersaturated graphons do not exist, so we can restrict our attention to graphs H that contain odd cycles.) The analysis depends on the size of the smallest odd cycle, and also on the size of the smallest even cycle, if such even cycles exist. The behavior of $B(e, t_H)$ as t_H approaches the Erdős-Rényi curve from below is always singular. In some cases it is even discontinuous.

8.1 Methods

There are a few techniques that are specific to the undersaturated regime that we have not encountered before.

- Power series around $g = e$. If e is not close to $1/2$, then it usually is not practical to expand $H(g)$ around $g = 1/2$. Instead, we expand around $g = e$, obtaining

$$S(g) = H(e) + \sum_{k=2}^{\infty} \frac{H^{(k)}(e)}{k!} \mu_k,$$

where now

$$\mu_k = \iint (g(x, y) - e)^k dx dy$$

is a moment centered at $g = e$. If $e > 1/2$, then all of the derivatives $H^{(k)}(e)$ are negative and we want to minimize all of the moments μ_k . If $e < 1/2$, then the odd derivatives are positive, while the even derivatives are negative. In this case, we want to minimize the even moments while maximizing the odd moments. To illustrate this method, we prove a theorem about rank one perturbations of Erdős-Rényi graphons.

Theorem 8.1 Suppose that $e < 1/2$ and that $t < e^3$. Among all graphons of the form

$$g(x, y) = e + \lambda v(x)v(y),$$

where now

$$\mu_k = \int_0^1 v(x) dx = 0, \quad \int_0^1 v(x)^2 dx = 1,$$

the unique entropy maximizer (up to the usual equivalence) has

$$v(x) = \begin{cases} 1 & x < 1/2, \\ -1 & x > 1/2. \end{cases} \quad (8.2)$$

Proof Such a rank one perturbation of an Erdős-Rényi graphon has $t = e^3 + \lambda^3$, so $\lambda = (t - e^3)^{1/3}$ is negative. The moments are $\mu_k = \lambda^k \left(\int_0^1 v(x)^k dx \right)^2$. When k is odd, this is maximized when $\int_0^1 v(x)^k dx = 0$. When k is even, this is bounded from below by $\mu_2^{k/2}$, with equality if and only if $(g(x, y) - e)^2$ is a constant. That is, the function (8.2) results in maximal (namely zero) odd moments and minimal even moments, and so maximizes the entropy. Any other entropy maximizer must minimize the even moments, and so must equal 1 on a set of measure 1/2 and -1 on a set of measure 1/2, and so must be equivalent to (8.2). Thus, up to equivalence, the unique entropy maximizing graphon is symmetric bipodal with $\lambda = -(e^3 - t)^{1/3}$ and with $v(x)$ of the form (8.2). $\square$

- **Diagonalization.** Instead of diagonalizing the graphon g , we diagonalize $g - e$, or sometimes $g - \frac{1}{2}$. The triangle density is bounded below by $e^3 + \text{tr}((g - e)^3)$, with equality if the degree function is constant. Meanwhile, the change in entropy relative to Erdős-Rényi typically goes as $\|g - e\|_2^2$. The best graphon should therefore have $g - e$ close to being rank one with a negative eigenvalue, thereby achieving $\text{tr}(g - e)^3 \approx -\|g - e\|_2^3$.
- **Competing inequalities.** This is not a new technique, having first appeared in Chapter 3, but it comes into its own here. Finding the best graphon frequently means trying to optimize two quantities that cannot be simultaneously maximized. For instance, we might want $g - e$ to have rank one while giving $g - e$ values that cannot be achieved by a rank one operator. We derive an upper bound for the entropy by pretending that both goals can be achieved. Then we exhibit a graphon that comes close to achieving this upper bound, thereby providing a lower bound for the maximal entropy. Since the actual optimal graphon has entropy very close to the result of optimizing both quantities, we obtain estimates of how close it must come to satisfying each one.

8.2 Just below Erdős-Rényi with $e > 1/2$; the B(1,1) phase

In this section we prove the existence of the B(1,1) phase. That is, suppose that $\delta = (e^3 - t)^{1/3} > 0$.

Theorem 8.2 *For all $e > 1/2$, and for all δ positive and sufficiently small, the entropy maximizing graphon is unique and bipodal, with parameters depending analytically on (e, t) and satisfying (8.1).*

This graphon bears a superficial resemblance to the graphon in the F(1,1) phase. After all, both are bipodal, with c infinitesimally small, and with values involving e and $1 - e$. However, there are important differences.

- In the F(1,1) phase, c scales as $t - e^3$ and $\|g - e\|_2^2$ scales as c^1 . In the B(1,1) phase, c scales as $(e^3 - t)^{1/3}$ and $\|g - e\|_2^2$ scales as c^2 .
- In the F(1,1) phase, $d \approx 1 - e$ while b is close to e . Most of the “mass distribution” $(g(x, y) - e)^2 dx dy$ is supported on $(I_1 \times I_2) \cup (I_2 \times I_1)$, as that region has far more area than $I_1 \times I_1$. In the B(1,1) phase, it’s a that is approximately $1 - e$ while b and d are close to e . Most of the “mass” is from $I_1 \times I_1$.
- In the F(1,1) phase, the degree function takes on very different values (approximately $1 - e$ and e) on the two podes. In the B(1,1) phase, the degree function is nearly constant, with $g - e$ being nearly a rank one operator.

Proof The proof of Theorem 8.2 [45] involves some difficult technical analysis. However, the underlying ideas are relatively simple. Before getting into the details, let’s consider some considerations for constructing an optimal graphon and sketch the strategy of the proof. As usual, we write $g(x, y) = e + g_1(x, y)$. For a given size $\|g_1\|_2$, we want to minimize t and maximize $S(g)$.

As always, we have

$$t = e^3 + 3e \int_0^1 (d(x) - e)^2 dx + \text{tr}(g_1^3).$$

We minimize the quadratic term by insisting that g_1 have zero marginals. We minimize $\text{tr}(g_1^3)$ by making g_1 rank one, with a negative eigenvalue λ . In that case, $t = e^3 + \lambda^3$, so $\lambda = -\delta$, giving us

$$g_1(x) = -\delta v(x)v(y)$$

for some unit vector $v \in L^2([0, 1])$. The overall edge density is $e - \delta \left(\int_0^1 v(x) dx \right)^2 = e$, so we must have $\int_0^1 v(x) dx = 0$.

As usual, the change in entropy is

$$S(g) - H(e) = \iint g_1(x, y)^2 f(e, g_1(x, y)) dx dy,$$

where

$$f(e, e') = \frac{H(e') - H(e) - H'(e)(e' - e)}{(e' - e)^2}.$$

For fixed e , the function $f(e, e')$ takes on the maximum value of $\frac{H'(e)}{2e-1}$ at $e' = 1 - e$. We therefore want the “mass distribution” $g_1(x, y)^2 dx dy$ to be concentrated as much as possible around points where $g_1(x, y) \approx 1 - 2e$.

The two goals are not consistent, however. We can make g_1 have rank one, but we can't make $g_1(x, y)$ only take on the values $1 - 2e$ and 0, since then $\iint g_1(x, y) dx dy$ wouldn't be zero. The next best thing would be to make $g_1(x, y)$ equal $1 - 2e$ on a set of measure approximately $\delta^2/(2e - 1)^2$ and $\delta^2/(2e - 1)$ everywhere else. However, that wouldn't have rank one.

Even so, we can obtain an upper bound on $S(g)$ by pretending that both goals are met. Since $\delta^3 = e^3 - t \leq \|g_1\|_2^3$ and $S(g) - H(e) \leq f(e, 1 - e)\|g_1\|_2^2$, we must have

$$S(g) \leq H(e) + f(e, 1 - e)\delta^2. \quad (8.3)$$

We now exhibit a graphon that comes close to saturating the inequality (8.3). We take

$$g_0 = e - \delta v_0(x)v_0(y)$$

with

$$v_0(x) = \begin{cases} \sqrt{\frac{1-c}{c}} & x < c \\ -\sqrt{\frac{c}{1-c}} & x > c \end{cases}$$

and

$$c = \frac{\delta}{2e - 1 + \delta}.$$

For this graphon, $g_0 - e$ has rank one and all but $O(\delta^3)$ of the “mass” is where $g_0(x, y) = 1 - e$. Thus

$$S(g_0) = H(e) + \delta^2 f(e, 1 - e) + O(\delta^3).$$

Our strategy of proof is as follows:

1. Using the Method of Competing Inequalities, we show that any entropy-optimizing graphon must be close to g_0 in an integral sense. This is Lemma 8.3.

2. By studying the Euler-Lagrange equations for our variational problem, we show that the leading eigenvector $v_1(x)$ of $g - e$ is pointwise close to a step function taking on two values. This is Lemma 8.4. This then defines two natural podes.
3. We show that g can be approximated by a bipodal graphon g_b . This is essentially Lemma 8.5, although the bipodal graphon g_b may differ slightly from the graphon $e + \lambda v(x)v(y)$ constructed in that lemma.
4. We show that, if g isn't already bipodal, then g_b is a better graphon than g . Specifically, we can make small changes to the bipodal parameters of g_b to achieve the same edge and triangle densities as g , only with greater entropy. Since g is assumed optimal, this is a contradiction that proves that g is bipodal.
5. Working in the (4-dimensional) space of bipodal graphons, we show that the optimal graphon is unique and satisfies (8.1).

Lemma 8.3 (Preliminary estimates) *Let $g = e + g_1$ be an entropy maximizing graphon. Then*

- $g_1(x, y)$ is approximately rank one. Specifically, if we diagonalize g_1 as

$$g_1(x, y) = \sum_{i=1}^{\infty} \lambda_i v_i(x) v_i(y),$$

where $|\lambda_1| \geq |\lambda_2| \geq \dots$ and the functions $v_i(x)$ are orthonormal in $L^2([0, 1])$, and if we set

$$g_2(x, y) = \sum_{i=2}^{\infty} \lambda_i v_i(x) v_i(y),$$

then

$$\|g_2\|_2^2 = \sum_{i=2}^{\infty} \lambda_i^2 = O(\delta^3). \quad (8.4)$$

- The degree function is concentrated near e :

$$\int_0^1 (d(x) - e)^2 dx = O(\delta^4).$$

- $g_1(x, y)$ mostly takes values near zero and $1 - 2e$, in that

$$\iint \psi(g_1(x, y)) dx dy = O(\delta^3),$$

where

$$\psi(u) = \min(u^2, (u + 2e - 1)^2).$$

- $\lambda_1 v_1(x) v_1(y)$ also mostly takes values near zero and $1 - 2e$, in that

$$\iint \psi(\lambda_1 v_1(x) v_1(y)) dx dy = O(\delta^3).$$

Proof These estimates all follow from the Method of Competing Inequalities and comparison to the graphon g_0 . Since

$$S(g) \geq S(g_0) = H(e) + \delta^2 f(e, 1 - e) + O(\delta^3), \quad (8.5)$$

the total entropy cost that g pays for $d(x)$ not being constant, or for g_1 not being rank one, or for g_1 taking values not equal to $1 - 2e$ or 0, cannot be greater than $O(\delta^3)$. We take these one at a time.

We will make repeated use of the following identity for constants $a > 0$ and $x > -a^3$:

$$(a^3 + x)^{2/3} \leq a^2 + \frac{2x}{3a}. \quad (8.6)$$

The right hand side is the linear approximation to the left hand side near $x = 0$. The inequality comes from the fact that the second derivative of $(a^3 + x)^{2/3}$ is everywhere negative.

First we show that g_1 is almost rank one. Since $\sum_{j=1}^{\infty} \lambda_j^3 \leq -\delta^3$ and $\sum_{j=1}^{\infty} \lambda_j^2 = \delta^2 + O(\delta^3)$, we must have $\lambda_1 = -\delta + o(\delta)$ and $\sum_{j=2}^{\infty} \lambda_j^2 = o(\delta^2)$. But then λ_1^2 is especially close to δ^2 :

$$\begin{aligned} -\delta^3 &\geq \lambda_1^3 + \text{tr}(g_2^3) \\ \delta^3 &\leq (-\lambda_1)^3 + \text{tr}((-g_2)^3) \\ \delta^2 &\leq \lambda_1^2 - \frac{2 \text{tr}((-g_2)^3)}{3\lambda_1} \\ &\leq \lambda_1^2 + \frac{-2}{3\lambda_1} \|g_2\|_2^3 \\ &= \lambda_1^2 + o(\|g_2\|_2^2). \end{aligned}$$

Thus

$$\|g_1\|_2^2 = \delta^2 + \|g_2\|_2^2 + o(\|g_2\|_2^2).$$

Since $\|g_1\|_1^2$ is within $O(\delta^3)$ of δ^2 , $\|g_2\|_2^2$ must be $O(\delta^3)$, as claimed in (8.4).

Next we consider the concentration of degrees. From the expansion of $\text{tr}(g^3)$ we have

$$\begin{aligned} -\delta^3 = t - e^3 &= 3e \int_0^1 (d(x) - e)^2 dx + \text{tr}(g_1^3) \\ &\geq 3e \int_0^1 (d(x) - e)^2 dx - \|g_1\|_2^3. \end{aligned}$$

But then

$$\delta^3 \leq \|g_1\|_2^3 - 3e \int_0^1 (d(x) - e)^2 dx.$$

Equation (8.6) then says that

$$\delta^2 \leq \|g_1\|_2^2 - \frac{2e}{3\|g_1\|_2} \int_0^1 (d(x) - e)^2 dx.$$

Since $\|g_1\|_2^2 = \delta^2 + O(\delta^3)$, we must have

$$\int_0^1 (d(x) - e)^2 dx = O(\|g_1\|_2 \delta^3) = O(\delta^4),$$

as claimed.

We now consider closeness of g_1 to 0 and $1 - 2e$. The second derivative of the function $f(e, e')$ with respect to e' is negative at $e' = 1 - e$. (It happens to equal $2((e - \frac{1}{2})H''(e) + H'(e))(2e - 1)^3$, but that detail is not important.) This means that

$$f(e, g(x, y)) - f(e, 1 - e)$$

is bounded below by a constant times $(g(x, y) + e - 1)^2 = (g_1(x, y) + 2e - 1)^2$. Thus the entropy cost of not having $g(x, y)$ everywhere equal to e or $1 - e$ is bounded below by a multiple of

$$\iint (g_1(x, y) + 2e - 1)^2 g_1(x, y)^2 dx dy.$$

This integrand is approximately equal to $(2e - 1)^2 \psi(g_1(x, y))$ when $g_1(x, y)$ is close to 0 or $1 - 2e$ and is bounded below everywhere else. As such, the integral is bounded below by a multiple of $\iint \psi(g_1(x, y)) dx dy$. Since the entropy cost $\iint g_1(x, y)(f(e, g(x, y)) - f(e, 1 - e)) dx dy$ is $O(\delta^3)$, we must have

$$\iint \psi(g_1(x, y)) dx dy = O(\delta^3).$$

Finally, we bound $\iint \psi(\lambda_1 v_1(x) v_1(y)) dx dy$ in terms of $\iint \psi(g_1(x, y)) dx dy$ and $\|g_2\|_2^2$. At every point $(x, y) \in [0, 1]^2$,

$$\begin{aligned} \psi(\lambda_1 v_1(x) v_1(y)) &= \psi(g_1(x, y) - g_2(x, y)) \\ &= \min((g_1(x, y) - g_2(x, y))^2, (2e - 1 + g_1(x, y) - g_2(x, y))^2) \\ &\leq 2 \min(g_1(x, y)^2 + g_2(x, y)^2, (2e - 1 + g_1(x, y))^2 + g_2(x, y)^2) \\ &= 2(\psi(g_1(x, y)) + g_2(x, y)^2), \end{aligned}$$

where in the third line we used the fact that, for any two real numbers a and b ,

$(a - b)^2 \leq 2(a^2 + b^2)$. Integrating over the unit square we get

$$\begin{aligned} \iint \psi(\lambda_1 v_1(x) v_1(y)) &\leq 2 \iint \psi(g_1(x, y)) dx dy + 2\|g_2\|_2^2 \\ &= O(\delta^3) + O(\delta^3) = O(\delta^3). \end{aligned}$$

□

Step 2:

Having determined the approximate form that an entropy maximizing graphon must take, we now turn to the Euler-Lagrange equations for our variational problem. As usual, for each entropy maximizing graphon g there must exist constants α and β such that, at every point (x, y) ,

$$H'(g(x, y)) = \alpha + \beta G(x, y), \quad (8.7)$$

where

$$G(x, y) = \int_0^1 g(x, z)g(y, z) dz.$$

We then compute

$$G(x, y) = e^2 + e(d(x) + d(y) - 2e) + \sum_{j=1}^{\infty} \lambda_j^2 v_j(x) v_j(y).$$

We write this as $e^2 + \lambda_1(g(x, y) - e) + G_1(x, y)$, where

$$G_1(x, y) = e(d(x) + d(y) - 2e) + \sum_{j=2}^{\infty} (\lambda_j^2 - \lambda_1 \lambda_j) v_j(x) v_j(y).$$

Thanks to our degree estimates, and to the fact that $\lambda_j = O(\delta^{3/2})$ when $j > 1$, we have $|G_1| \ll |G_0|$ at most points where $g(x, y) \approx e$ and also at most points where $g(x, y) \approx 1 - e$. This implies that two of the three solutions to

$$H'(g) = \alpha + \beta(e^2 + \lambda_1(g - e)) \quad (8.8)$$

are close to e and $1 - e$. This in turn fixes the values of α and β , to within a small error:

$$\lambda_1 \beta = \frac{H'(e)}{e - \frac{1}{2}} + o(1), \quad \alpha = H'(e) - \beta e^2 + o(1).$$

The third solution to (8.8) is then close to $1/2$. We then rewrite equation (8.7) as

$$H'(g(x, y)) - (\alpha + \beta e^2) - \lambda_1 \beta (g(x, y) - e)$$

$$= \beta \left[e(d(x) + d(y) - 2e) + \sum_{j=2}^{\infty} (\lambda_j - \lambda_1) \lambda_j v_j(x) v_j(y) \right]. \quad (8.9)$$

Since $g(x, y)$ is approximately $2e - 1$ on a set of measure $\Theta(\delta^2)$, $\sqrt{-\lambda_1} v_1(x)$ must be close to $\pm\sqrt{2e - 1}$ on a set of measure $\Theta(\delta)$. If $\sqrt{-\lambda_1} v_1(x)$ is close to $+\sqrt{2e - 1}$ on a set of measure $\Theta(\delta)$, then $\sqrt{-\lambda_1} v_1(x)$ can only be close to $-\sqrt{2e - 1}$ on a set of measure $O(\delta^2)$, since otherwise the integral of $\psi(g(x, y))$ over the product of the two regions would not be $O(\delta^3)$. Likewise, if the set where $\sqrt{-\lambda_1} v_1(x)$ is close to $-\sqrt{2e - 1}$ has measure $\Theta(\delta)$, then the set where it is close to $+\sqrt{2e - 1}$ has measure $O(\delta^2)$. By flipping the overall sign of $v_1(x)$, we can assume that $\sqrt{-\lambda_1} v_1(x)$ is close to $\sqrt{2e - 1}$ on a larger set than the one on which it is close to $-\sqrt{2e - 1}$.

Lemma 8.4 *For any positive tolerance ϵ , and for all sufficiently small values of δ , $\sqrt{-\lambda_1} v_1(x)$ only takes values within ϵ of 0 or $\sqrt{2e - 1}$.*

Proof Let I_1 , I_2 and I_3 be the subsets of $[0, 1]$ on which $\sqrt{-\lambda_1} v_1(x)$ is within ϵ of $\sqrt{2e - 1}$, within ϵ of 0, and not within ϵ of either.

If $x \in I_3$ and $\sqrt{-\lambda_1} v_1(x)$ is not within ϵ of $\frac{1}{2}\sqrt{2e - 1}$, then for all $y \in I_1$, $e + \lambda_1 v_1(x) v_1(y)$ is not within $o(1)$ of e or $1 - e$ or $1/2$, so the left hand side of equation (8.9) is $\Theta(1)$. Since $\beta = \Theta(\delta^{-1})$, this implies that $G_1(x, y)$ is $\Theta(\delta)$ for all $y \in I_1$. The $\sum (\lambda_j - \lambda_1) \lambda_j v_j(x) v_j(y)$ terms cannot account for this, so either $d(x) - e$ must be $\Theta(\delta)$ or $d(y) - e$ must be $\Theta(\delta)$ for all y . In the first case, this bias would force $g(x, z)$ to be far from e (in a consistent direction!) for all $z \in I_2$, which it turn would make $d(x) - e$ be $\Theta(1)$, not $\Theta(\delta)$. In the second case, $d(x_1) + d(x_2) - 2e$ would be $\Theta(\delta)$ for all $(x_1, x_2) \in I_1 \times I_1$, so $g(x_1, x_2)$ would not be close to $1 - e$ there, contradicting the estimate $\iint \psi(g(x, y)) dx dy = O(\delta^3)$.

That is, the only values of $\sqrt{-\lambda_1} v_1(x)$ that are possible are (approximately) $\sqrt{2e - 1}$, 0, and $\frac{1}{2}\sqrt{2e - 1}$. The third possibility is not ruled out by equation (8.9), which only concerns pointwise variations in the value of $g(x, y)$. However, the variation of the entropy with respect to the size of I_3 is *not* zero. There is an entropy cost of approximately $2|I_3||I_1|(e - \frac{1}{2})^2(f(e, 1 - e) - f(e, 1/2))$ with no corresponding gain to first order in $|I_3|$. To maximize entropy, the size of this third node must be zero. $\square$

Without loss of generality, we can take $I_1 = [0, c]$ and $I_2 = [c, 1]$. Since $\int_0^1 v_1(x)^2 dx = 1$ and $\lambda_1 \approx -\delta$, we must have $c \approx \delta/(2e - 1)$. Since $\int_0^2 (d(x) - e)^2 dx = O(\delta^4)$, we must have $\int_0^1 v_1(x) dx = O(\delta^{3/2})$, so $v_1(x)$ must average out to around $-\sqrt{\delta/(2e - 1)}$ on I_2 . In other words, $v_1(x)$ is nearly the same as the function $v_0(x)$.

A heuristic digression:

The derivative of the left hand side of (8.9) with respect to $g(x, y)$ is

$$H''(g) - \lambda_1 \beta \approx H''(g) - \frac{H'(e)}{e - \frac{1}{2}}.$$

This is negative both near $g = e$ and near $g = 1 - e$. However, β is positive, λ_1 is negative, and $\lambda_1 < \lambda_j$, so the contribution to $\beta G_1(x, y)$ from each term in the sum over j is a positive multiple of $\lambda_j v_j(x) v_j(y)$. That is, the bigger $g(x, y)$ is, the bigger $G_1(x, y)$ is, and the *smaller* $g(x, y)$ needs to be on the left hand side.

This effect is particularly strong with regard to the degree function, as the coefficient is $\beta e = \Theta(\delta^{-1})$. Suppose that x_1 and x_2 are both in I_1 , or are both in I_2 , and that $d(x_1) > d(x_2)$. For each $y \in [0, 1]$, $g(x_1, y)$ and $g(x_2, y)$ are close to the same value (e or $1 - e$). The difference $d(x_1) - d(x_2)$ makes the right hand side of (8.9) greater at (x_1, y) than at (x_2, y) and so makes $g(x_1, y)$ less than $g(x_2, y)$. Integrating over y , we get that $d(x_1) < d(x_2)$, which is a contradiction. This strongly suggests that $d(x)$ must be constant on I_1 , and likewise constant on I_2 .

We can try to apply the same reasoning to each component $\lambda_j v_j(x) v_j(y)$ of $g(x, y)$ with $j > 1$. As long as $\lambda_j v_j(x) v_j(y)$ is pointwise small, the contribution to the left hand side of (8.9) has the opposite sign as the contribution to the right hand side, so λ_j should be zero. This suggests that $g - e$ must have rank 1 and that $v_1(x)$ must be constant on each pde. In other words, our graphon must be g_0 .

This subsection is called a “heuristic digression” for a reason. We were careful to say **only** that our arguments “(strongly) suggest” various results. While equation (8.9) is suggestive, it is a nonlinear coupled system of equations involving many quantities, namely the coefficients λ_j and the form of the functions $v_j(x)$.

In principle it should be possible to devise a bootstrap argument to derive better and better pointwise bounds on each $v_j(x)$ and bounds on the coefficients λ_j , achieving bipodality in the limit. However, that isn’t how we are going to complete the proof. Steps (3) and (4) of our proof sidestep these difficulties by finding a different way to approximate g with a bipodal graphon. The estimates are technical and are not very intuitive, but sometimes that’s the price of rigor.

Step 3:

So far, we have shown that g is well-approximated in L^2 by $e - \lambda_1 v_1(x) v_1(y)$ and that v_1 is pointwise close to being a step function taking two values. In

this step we show that we can obtain similar results when $v_1(x)$ is rounded to a function that only takes on two values.

Lemma 8.5 *There exists a constant λ and a function $v(x)$ such that*

1. v only take two values, one $\sqrt{(2e-1)/\delta} + O(\sqrt{\delta})$ and the other $O(\sqrt{\delta})$.
2. $\int_0^1 v(x) dx = 0$ and $\int_0^1 v(x)^2 dx = 0$.
3. $\lambda = -\delta + O(\delta^2)$,
4. $\|g(x, y) - e - \lambda v(x)v(y)\|_2^2 = O(\delta^3)$.
5. Let $I_1 = \{x | v(x) = \sqrt{(2e-1)/\delta} + O(\sqrt{\delta})\}$ and let I_2 be the complement of I_1 . For all $x \in I_1$, $\int_{I_1} (g(x, y) - (1-e))^2 dy \leq \int_{I_1} (g(x, y) - e)^2 dy + O(\delta^2)$.
6. For all $x \in I_2$, $\int_{I_1} (g(x, y) - e)^2 dy \leq \int_{I_1} (g(x, y) - (1-e))^2 dy + O(\delta^2)$.

Conditions 2–4 are nearly the same as the estimates that we already obtained with $v_1(x)$ and λ_1 . The fifth and sixth say that no points are assigned to the manifestly wrong node.

Proof We construct the function $v(x)$ and the constant λ by perturbing $v_1(x)$ and λ_1 . The first step is rounding. By Lemma 8.4, $v_1(x)$ only takes on values close to $\sqrt{(1-2e)/\lambda_1}$ or 0. Let I_1 and I_2 be the regions where $v_1(x)$ is close to $\sqrt{(1-2e)/\lambda_1}$ and 0, respectively. We define

$$\bar{v}(x) = \begin{cases} \sqrt{(1-2e)/\lambda_1} & \text{on } I_1, \\ 0 & \text{on } I_2. \end{cases}$$

Note that $\lambda_1 \bar{v}(x)\bar{v}(y)$ is $1-2e$ on $I_1 \times I_1$ and is zero elsewhere, and also that $\lambda_1 v_1(x)v_1(y)$ is closer to $1-2e$ than to 0 on $I_1 \times I_2$ and is closer to 0 elsewhere. As a result,

$$\|\lambda_1 \bar{v}(x)\bar{v}(y) - \lambda_1 v_1(x)v_1(y)\|_2^2 = \iint \psi(\lambda v_1(x)v_1(y)) dx dy = O(\delta^3).$$

Since

$$\|g(x, y) - e - \lambda_1 v_1(x)v_2(x)\|_2^2 = \sum_{j=2}^{\infty} \lambda_j^2 = O(\delta^3),$$

the triangle inequality implies that

$$\|g(x, y) - e - \lambda_1 \bar{v}(x)\bar{v}(y)\|_2^2 = O(\delta^3),$$

as desired.

Next we consider points x that have been misclassified, in the sense that the relative size of $\int_{I_1} (g(x, y) - e)^2 dy$ and $\int_{I_1} (g(x, y) - (1-e))^2 dy$ are not what you would expect from the values of $\bar{v}(x)$. These are points at which $\int_{I_1} (g(x, y) - e - \lambda_1 \bar{v}(x)\bar{v}(y))^2 dy$ is of order δ , and hence at which $\int_0^1 (g(x, y) -$

$e - \lambda_1 \bar{v}(x) \bar{v}(y))^2 dy$ is of order δ . However, we just showed that $\|g(x, y) - e - \lambda_1 \bar{v}(x) \bar{v}(y)\|_2^2 = O(\delta^3)$, so the total measure of these misplaced points is $O(\delta^2)$.

At each point $x \in I_1$ for which $\int_{I_1} (g(x, y) - e)^2 dy < \int_{I_1} (g(x, y) - (1 - e))^2 dy$, we redefine $\bar{v}(x)$ to be 0. Likewise, at each point $x \in I_2$ for which $\int_{I_1} (g(x, y) - e)^2 dy > \int_{I_1} (g(x, y) - (1 - e))^2 dy$, we redefine $\bar{v}(x)$ to be $\sqrt{(1 - 2e)/\lambda_1}$. In other words, we have slightly changed the sets I_1 and I_2 . In the process, we change the integrals of $\int_{I_1} (g(x, y) - (1 - e))^2 dy$ and $\int_{I_1} (g(x, y) - (1 - e))^2 dy$ by $O(\delta^3)$. As a result, with the redefined sets I_1 and I_2 , $\int_{I_1} (g(x, y) - (1 - e))^2 dy - \int_{I_1} (g(x, y) - (1 - e))^2 dy$ is within $O(\delta^2)$ of being negative when $x \in I_1$ and is within $O(\delta^2)$ of being positive when $x \in I_2$.

Next let $u(x)$ be the result of subtracting a constant from $\bar{v}(x)$ to make the integral over $[0, 1]$ equal to zero. Since the size of I_1 is $O(\delta)$, this constant is $O(\sqrt{\delta})$. The difference $\lambda_1 u(x) u(y) - \lambda_1 \bar{v}(x) \bar{v}(y)$ is then $O(\delta)$ on $I_1 \times I_1$, $I_1 \times I_2$ and $I_2 \times I_1$ and is $O(\delta^2)$ on $I_2 \times I_2$. Squaring and integrating, we get that

$$\|\lambda_1 u(x) u(y) - \lambda_1 \bar{v}(x) \bar{v}(y)\|_2^2 = O(\delta^3).$$

By the triangle inequality,

$$\|g(x, y) - e - \lambda_1 u(x) u(y)\|_2^2 = O(\delta^3).$$

Finally, we rescale $u(x)$ by a factor of $1 + O(\delta)$ to obtain a function $v(x)$ such that $\int_0^1 v(x)^2 dx = 1$. We similarly rescale λ_1 to obtain a constant λ such that $\lambda v(x) v(y) = \lambda_1 u(x) u(y)$. $\square$

Step 4:

We have shown that $g(x, y)$ can be approximated by a bipodal graphon whose podes I_1 and I_2 satisfy conditions 5 and 6 of Lemma 8.5. We now switch to a different bipodal graphon with the same podes. Let a , b and d be the average values of $g(x, y)$ on $I_1 \times I_1$, $I_2 \times I_2$, and $I_1 \times I_2$, respectively. Let

$$g_b(x, y) = \begin{cases} a & \text{on } I_1 \times I_1, \\ b & \text{on } I_2 \times I_2, \\ d & \text{on } (I_1 \times I_2) \cup (I_2 \times I_1), \end{cases}$$

let

$$r_1(x, y) = g_b(x, y) - (e + \lambda v(x) v(y)),$$

and let

$$\Delta g(x, y) = g(x, y) - g_b(x, y).$$

Note that $r_1(x, y)$ and $\Delta g(x, y)$ are orthogonal in $L^2([0, 1]^2)$, since $r_1(x, y)$

is constant on quadrants and $\Delta g(x, y)$ integrates to 0 on each quadrant. As a result,

$$\|r_1(x, y)\|_2^2 + \|\Delta g(x, y)\|_2^2 = \|g(x, y) - e - \lambda v(x)v(y)\|_2^2 = O(\delta^3),$$

so $\|r_1(x, y)\|_2^2$ and $\|\Delta g\|_2^2$ are both $O(\delta^3)$. From this and the bounds on $\iint \psi(g(x, y) - e) dx dy$ we conclude that

$$a = 1 - e + O(\sqrt{\delta}), \quad b = e + O(\delta^{3/2}), \quad \text{and} \quad d = e + O(\delta).$$

The edge density of g_b is the same as that of g . We now compare the entropy and triangle densities of g_b with those of g . Let $\Delta S = S(g) - S(g_b)$ and let $\Delta t = \tau(g) - \tau(g_b) = t - \tau(g_b)$. We are interested in the ratio $\Delta S/\Delta t$. Note that ΔS is always negative, as averaging a graphon over a region always increases the entropy.

It is possible to vary the bipodal parameters of g_b so as to change the entropy and triangle density, while holding the edge density fixed. Specifically, we increase the size of I_1 while holding a and d fixed, and adjusting b to keep the overall edge density constant. Since the values of a , b , c , and d are nearly the same as those of g_0 ,

$$\frac{dS}{dt} = \frac{dS/d\delta}{dt/d\delta} = -\frac{2f(e, 1-e)}{3\delta} + o(\delta^{-1}).$$

If $\Delta S + \frac{2\Delta t}{3\delta} > o(\Delta t/\delta)$, then a modified bipodal graphon will have more entropy, for the same triangle density, as g , which is a contradiction.

We will compare both ΔS and Δt to $\|\Delta g\|_2^2$, finding lower bounds for both $\Delta S/\|\Delta g\|_2^2$ and $\Delta t/\|\Delta g\|_2^2$. If both lower bounds were saturated, then $\Delta S/\Delta t$ would in fact be close to $-\frac{2f(e, 1-e)}{3\delta}$. However, we will see that saturating the first lower bound requires having Δg take a form that results in missing the second lower bound (by a factor of two). Since the two lower bounds cannot be simultaneously saturated (to within small tolerances), we will have achieved our contradiction.

Bound on S :

The change in entropy from g_b to g is a sum of terms, one for each quadrant. Since Δg integrates to 0 on each quadrant,

$$\begin{aligned} \Delta S = & \iint_{I_1 \times I_1} f(a, g(x, y)) \Delta g(x, y)^2 dx dy + \iint_{I_2 \times I_2} f(b, g(x, y)) \Delta g(x, y)^2 dx dy \\ & + 2 \iint_{I_1 \times I_2} f(d, g(x, y)) \Delta g(x, y)^2 dx dy. \end{aligned}$$

However,

$$f(a, g(x, y)) \leq f(a, 1-a) = f(1-e, e) + O(\delta^{1/2}) = f(e, 1-e) + O(\delta^{1/2}).$$

The integral over $I_1 \times I_1$ is at most $(f(e, 1-e) + o(1))$ times the contribution of $I_1 \times I_1$ to $\|\Delta g(x, y)\|_2^2$. Furthermore, this is only achieved when all but $o(\|\Delta g\|_2^2)$ of the “mass” $\Delta g(x, y)^2 dx dy$ is concentrated at points where $\Delta g(x, y) \approx 2e - 1$.

Similarly, $f(b, g)$ and $f(d, g)$ are also bounded by $f(e, 1-e) + o(1)$, so the contributions of $I_1 \times I_2$ and of $I_2 \times I_2$ to ΔS are bounded above by $f(e, 1-e) + o(1)$ times their contributions to $\|\Delta g\|_2^2$, with equality only when all but $o(\|\Delta g\|_2^2)$ of the mass is concentrated at points where $\Delta g(x, y) \approx 1 - 2e$.

The upshot is that

$$\Delta S \leq (f(e, 1-e) + o(1))\|\Delta g\|_2^2,$$

with equality if and only if all but an $o(1)$ fraction of the mass is concentrated at points where $g(x, y) \approx e$ and $g_b(x, y) \approx 1 - e$, or vice-versa.

Bound on t :

We must have

$$\iint \Delta g(x, y)v(x)v(y) dx dy = 0, \quad (8.10)$$

since $v(x)v(y)$ is constant on quadrants and Δg averages to zero on each quadrant. We define

$$w(x) = \int_0^1 \Delta g(x, y)v(y) dy.$$

Note that $v(x)$ and $w(x)$ are orthogonal in $L^2([0, 1])$, thanks to (8.10). We decompose Δg as

$$\Delta g(x, y) = v(x)w(y) + w(x)v(y) + r_2(x, y). \quad (8.11)$$

The remainder $r_2(x, y)$ then satisfies

$$\int_0^1 r_2(x, y)v(y) dy = \int_0^1 r_2(x, y)w(y) dy = 0.$$

That is, the decomposition (8.11) is an orthogonal decomposition, and

$$\|\Delta g\|_2^2 = 2\|w\|_2^2 + \|r_2(x, y)\|_2^2.$$

Now we compute Δt :

$$\begin{aligned} \Delta t &= \text{Tr}((g_b + \Delta g)^3) - \text{Tr}(g_b^3) \\ &= 3\text{Tr}(g_b^2 \Delta g) + 3\text{Tr}(g_b \Delta g^2) + \text{Tr}(\Delta g^3). \end{aligned}$$

The first term on the second line is zero, since g_b^2 is constant on quadrants and Δg integrates to zero on each quadrant. For the second term, we expand g_b as

$$g_b(x, y) = e + \lambda v(x)v(y) + r_1(x, y).$$

The contribution from e is positive-definite. The contribution from $\lambda v(x)v(y)$ is precisely $3\lambda\|w(x)\|_2^2$. The contribution from r_1 is $O(\delta^{3/2}\|\Delta g\|_2^2)$, as is the third term, since $\|r_1\|_2^2 + \|\Delta g\|_2^2 = O(\delta^3)$. The upshot is that

$$\begin{aligned}\Delta t &\geq 3\lambda\|w(x)\|_2^2 + O(\delta^{3/2}\|\Delta g\|_2^2) \\ \frac{\Delta t}{\|\Delta g\|_2^2} &\geq -3\delta\frac{\|w(x)\|_2^2}{\|\Delta g\|_2^2} + O(\delta^{3/2}).\end{aligned}$$

Note that if $\|r_2(x, y)\|_2^2$ is an infinitesimally small fraction of $\|\Delta g\|_2^2$, then $\|w(x)\|_2^2/\|\Delta g\|_2^2 \approx 1/2$ and $\Delta t \approx -3\delta\|\Delta g\|_2^2/2$. If we also had

$$\Delta S \approx f(e, 1-e)\|\Delta g\|_2^2,$$

then we would have $\Delta S \approx \frac{-2f(e, 1-e)\Delta t}{3\delta}$. However, that is the *only* way to get ΔS that big. In other words, the only way that $\Delta S/\Delta t$ can be competitive with $-2f(e, 1-e)/3\delta$ is if

- All but an $o(1)$ fraction of the “mass distribution” of Δg is from points in $I_1 \times I_1$ where $\Delta g(x, y) \approx 2e - 1$ or points elsewhere where $\Delta g(x, y) \approx 1 - 2e$.
- All but an $o(1)$ fraction of $\|\Delta g\|_2^2$ comes from $v(x)w(y) + w(x)v(y)$.
- For any $x \in I_1$, the measure of the set of points $y \in I_1$ with $\Delta g(x, y) \approx 2e - 1$ is bounded by (approximately) half the measure of I_1 . This is condition 5 in Lemma 8.5.
- For any $x \in I_2$, the measure of the set of points $y \in I_1$ with $\Delta g(x, y) \approx 1 - 2e$ is bounded by (approximately) half the measure of I_1 . This is condition 6.

Note the importance of the previously obscure properties 5 and 6 of Lemma 8.5. If you start with a bipodal graphon and make an infinitesimal change in the size of I_1 , say changing from $[0, c]$ to $[0, c + \Delta c]$ as in Figure 8.2, then $w(x)$ is close (in L^2) to a multiple of the characteristic function of $[c, c + \Delta c]$ (or $[c + \Delta c, c]$, if $\Delta c < 0$) and $\Delta g(x, y)$ is very close (in L^2) to $w(x)v(y) + v(x)w(y)$. However, this change violates the third bullet point if $\Delta c < 0$ and the fourth bullet point if $\Delta c > 0$.

Inconsistency:

We now show that these conditions are inconsistent. Suppose rather that all of the conditions are met.

In order for $\Delta g(x, y)$ to take on values close to $2e - 1$ on $I_1 \times I_1$, and for those values to dominate the mass of Δg , $w(x)$ must be close to $\sqrt{(2e - 1)\delta}$ on some $A_1 \subset I_1$ and must be much smaller on the rest of I_1 . Similarly, $w(x)$ must be close to $-\sqrt{(2e - 1)\delta}$ on some $A_2 \subset I_2$, and must be much smaller on the rest of I_2 . For each $x_1 \in A_1$ and each $y \in I_1$, we then have

$$\Delta g(x, y) \approx 2e - 1 + u(y)v(x) + r_2(x, y).$$

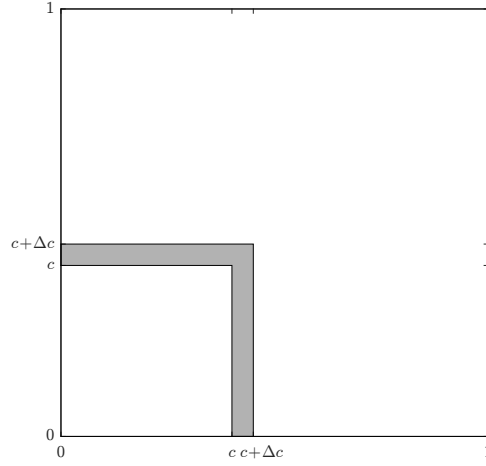


Figure 8.2 Changing the value of c results in a large change to the graphon in the shaded region.

However, $\Delta g_x(y)$ is only allowed to be close to $2e - 1$ on *half* of I_1 . Since the $u(y)v(x)$ term only makes Δg larger, $r_2(x, y)$ must have size approximately $1 - 2e$ on half the values of y , as seen in Figure 8.3. Likewise, for each $x_2 \in A_2$ and each $y \in I_1$, we have

$$\Delta g(x, y) \approx 1 - 2e + u(y)v(x) + r_2(x, y),$$

forcing $r_2(x, y)$ to be close to $2e - 1$ on half the values of $y \in I_1$. But then the contribution of $r_2(x, y)$ to $\|\Delta g\|_2^2$ on $I_1 \times I_1$ and $I_1 \times I_2$ is comparable (in fact at least equal to) the contribution of $v(x)w(y) + w(x)v(y)$.

As for $I_2 \times I_2$, $\int_{I_2} v(x)^2 dx$ is a factor of $O(\delta)$ smaller than $\int_{I_1} v(x)^2 dx$, so only an $O(\delta)$ fraction of the L^2 measure of $v(x)w(y) + w(x)v(y)$ is supported on $I_2 \times I_2$. Thus the total contribution of $r_2(x, y)$ to $\|\Delta g\|_2^2$ is comparable to that of $v(x)w(y) + w(x)v(y)$, contradicting the second condition.

We conclude that Δg must in fact be identically zero, in other words that g must already be bipodal.

Step 5:

With this result, we have reduced our problem from varying an arbitrary function $g(x, y)$ to varying four parameters (a, b, c, d) . Approximating the optimal graphon, showing that it is unique, and showing that it varies analytically with (e, t) then becomes a calculus problem. We refer the reader to [45] for

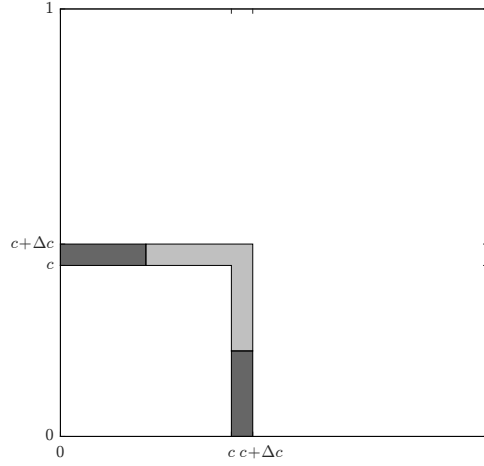


Figure 8.3 If $r_2(x, y)$ takes one sign for half the values of $y \in I_1$ and the other sign for the other half, then $\|r_2\|_2^2$ is at least as big as $\|v \otimes w + w \otimes v\|_2^2$.

details. The variational equations have a unique solution of the form

$$\begin{aligned} a &= 1 - e - \delta + O(\delta^2) \\ b &= e - \frac{\delta^2}{2e-1} + O(\delta^3) \\ c &= \frac{\delta}{2e-1} - \frac{2\delta^2}{2e-1} + O(\delta^3) \\ d &= e + \delta + \frac{\delta^2}{eH'(e)} \left(H''(e) - \left(e - \frac{1}{2} \right) H''(e) \right) + O(\delta^3). \end{aligned} \quad (8.12)$$

The peculiar δ^2 term in d can be understood from linearizing equation (8.9). Because of the difference between b and d at order δ , the left hand side of equation (8.9) differs by

$$\left(H''(e) - \frac{H'(e)}{e - \frac{1}{2}} \right) \delta + O(\delta^2)$$

between $I_1 \times I_2$ and $I_2 \times I_2$. This forces a difference of

$$\frac{\delta^2}{eH'(e)} \left(H''(e) - \frac{H'(e)}{e - \frac{1}{2}} \right) + O(\delta^3)$$

between the values of the degree function on the two nodes. Regardless of the value of a , this difference is equal to $d - (e + \delta)$ to leading order, so we see this difference in the $O(\delta^2)$ contribution to d .

□

8.3 The A(2,0) phase near $e = 1/2$

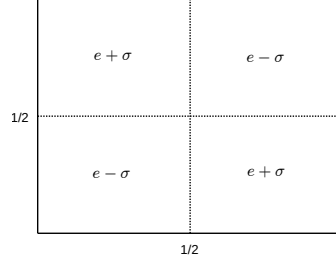


Figure 8.4 An A(2,0) graphon with edge density e and triangle density $e^3 - \sigma^3$, so $\sigma = (e^3 - t)^{1/3}$.

So far, we have proven the existence of phases near the boundary of the Razborov triangle and near the Erdős-Rényi curve. Our calculations have involved perturbing around the optimal graphons on the boundary, or of the Erdős-Rényi graphon, and estimating the resulting change in entropy and triangle number. For the F(1,1) and B(1,1) phases near the Erdős-Rényi curve, we have shown that the optimal graphon looks qualitatively different from the Erdős-Rényi graphon, with the values of $g(x, y)$ on one or more quadrants being close to $1 - e$, not e .

In this section we do something different. We study what happens when $e \approx 1/2$ and $t < e^3$ and show that the optimal graphon looks *just like* what we get when e is exactly $1/2$. Specifically, we show that the optimal graphon is symmetric bipodal, of the form shown in Figure 8.4, on an open set containing the line segment $e = 1/2$, $0 < t < 1/8$ [47].

Theorem 8.6 *For fixed $\sigma \in (0, \frac{1}{2})$ and for all sufficiently small δ (of either sign), the unique entropy maximizing graphon with edge density $e = \frac{1}{2} + \delta$ and triangle density $t = e^3 - \sigma^3$ is symmetric bipodal. Furthermore, the size of the allowed interval of δ 's varies continuously with σ .*

Note that the quantity $(e^3 - t)^{1/3}$ is denoted σ in this section. This is because the letter δ traditionally denotes a small parameter, while $(e^3 - t)^{1/3}$ can be as large as $1/2$. In this problem, the small parameter is the deviation of the edge density from $1/2$, so we take $\delta = e - \frac{1}{2}$.

Proof We use the Method of Competing Inequalities. As usual, we want $g - e$ to have rank 1, and we also want $(g - \frac{1}{2})^2$ to be constant. When $e = 1/2$, the two conditions are simultaneously satisfied by a symmetric bipodal graphon,

as in the proof of Theorem 3.7. When $e \neq 1/2$, the two conditions cannot be simultaneously satisfied, but we can get an upper bound on the entropy by pretending that they can. By measuring the difference between the upper bound and the actual entropy of a symmetric bipodal graphon, we bound the extent to which an entropy maximizing graphon can fail to meet these conditions.

More precisely, we diagonalize $g - e$ and write

$$g(x, y) = e + g_1(x, y) + g_2(x, y), \quad (8.13)$$

where

$$g_1(x) = \lambda_1 v_1(x) v_1(y)$$

and

$$g_2(x, y) = \sum_{j=2}^{\infty} \lambda_j v_j(x) v_j(y),$$

where the λ_j 's are the eigenvalues of $g - e$, viewed as an integral operator, and the $v_j(x)$'s are the normalized eigenvectors. Three key quantities are

$$\alpha^2 = \sum_{i=2}^{\infty} \lambda_i^2, \quad \beta^2 = \int_0^1 (v_1(x)^2 - 1)^2 dx, \quad \text{and} \quad \gamma = \int_0^1 v_1(x) dx. \quad (8.14)$$

A symmetric bipodal graphon is characterized by the conditions $\alpha = \beta = \gamma = 0$.

We first show that α , β and γ are all $O(\delta)$. Then we bound $\|g_2\|_{\infty}$ and $\|v_1(x)^2 - 1\|_{\infty}$. Using those bounds, we then show that having (α, β, γ) nonzero involves entropy costs that scale as $\alpha^2 + \beta^2 + \gamma^2$, while any entropy gains are higher order. We concluded that, for δ sufficiently small, the entropy-maximizing graphon is symmetric bipodal.

Step 1: An entropy bound

Lemma 8.7 *If g is a graphon with edge density $e = \frac{1}{2} + \delta$ and triangle density $e^3 - \sigma^3$, with $\sigma > 0$, then*

$$S(g) \leq H\left(\frac{1}{2} + \sqrt{\delta^2 + \sigma^2}\right). \quad (8.15)$$

Proof Let g be our arbitrary graphon, which we expand as in equation (8.13). For $i = 1, 2$, let $d_i(x) = \int_0^1 g_i(x, y) dy$ be the degree functions of g_1 and g_2 . Note that d_1 and d_2 are orthogonal in L^2 . A direct computation of the triangle density gives

$$\tau(g) = e^3 + \sum_{j=1}^{\infty} \lambda_j^3 + 3e \int_0^1 (d_1(x)^2 + d_2(x)^2) dx,$$

so

$$\sum_j \lambda_j^3 = - \left(\sigma^3 + 3e \int_0^1 (d_1(x)^2 + d_2(x)^2) dx \right).$$

The squared L^2 norm of $g_1 + g_2$ is

$$\sum_j \lambda_j^2 \geq \left| \sum_j \lambda_j^3 \right|^{2/3} \geq \sigma^2,$$

with equality if and only if $\lambda_1 = -\sigma$, $\lambda_2 = \lambda_3 = \dots = 0$, and $d_1(x) = 0$ for all x .

Next we maximize the entropy for a fixed $\|g_1 + g_2\|_2^2$, exactly as in the proof of Theorem 3.7. Expanding $H(u)$ in a power series in $u - \frac{1}{2}$ and integrating over $[0, 1]^2$ gives

$$S(g) = H\left(\frac{1}{2}\right) + \sum_{k=1}^{\infty} \frac{H^{(2k)}\left(\frac{1}{2}\right)}{(2k)!} \mu_{2k}, \quad (8.16)$$

where

$$\mu_{2k} = \iint (\delta + g_1(x, y) + g_2(x, y))^{2k} dx dy.$$

The second moment depends only on the size of g_1 and g_2 . Since $\iint g_1(x, y) + g_2(x, y) dx dy = 0$, there are no cross terms between δ and $g_1 + g_2$, leaving us with

$$\mu_2 = \delta^2 + \|g_1 + g_2\|_2^2 = \delta^2 + \sum_{j=1}^{\infty} \lambda_j^2.$$

Maximizing $S(g)$ is equivalent to minimizing all of the higher moments μ_{2k} with $k > 1$. This happens when $(\delta + g_1(x, y) + g_2(x, y))^2$ is constant, equaling μ_2 . In that case, g is everywhere equal to $\frac{1}{2} \pm \sqrt{\mu_2}$ and

$$S(g) = H\left(\frac{1}{2} + \sqrt{\delta^2 + \|g_1 + g_2\|_2^2}\right).$$

Since $\|g_1 + g_2\|_2^2 \geq \sigma^2$, and since $H(u)$ is a decreasing function of u for $u > 1/2$, we conclude that

$$S(g) \leq H\left(\frac{1}{2} + \sqrt{\delta^2 + \sigma^2}\right).$$

□

Corollary 8.8 *If g is an entropy-maximizing graphon, then $\int_0^1 v_1(x) dx$ is $O(\delta)$, while $\|g_2\|_2^2$, $\int_0^1 d_2(x)^2 dx$ and $\int_0^1 (v_1(x)^2 - 1)^2 dx$ are $O(\delta^2)$.*

Proof The symmetric bipodal graphon has entropy

$$\begin{aligned} \frac{1}{2} [H(e + \sigma) + H(e - \sigma)] &= \frac{1}{2} \left[H\left(\frac{1}{2} + \delta + \sigma\right) + H\left(\frac{1}{2} + \delta - \sigma\right) \right] \\ &= \frac{1}{2} \left[H\left(\frac{1}{2} + \sigma + \delta\right) + H\left(\frac{1}{2} + \sigma - \delta\right) \right] \\ &= H\left(\frac{1}{2} + \sigma\right) + \frac{\delta^2}{2} H''\left(\frac{1}{2} + \sigma\right) + O(\delta^4). \end{aligned}$$

By contrast, the upper bound (8.15) is

$$H\left(\frac{1}{2} + \sigma\right) + \frac{\delta^2}{2\sigma} H'\left(\frac{1}{2} + \sigma\right) + O(\delta^4).$$

Since the symmetric bipodal graphon comes within $O(\delta^2)$ of achieving the upper bound (8.15), the entropy-maximizing graphon must also come within $O(\delta^2)$ of that bound. In particular, $\|g_1 + g_2\|_2^2$ must be within $O(\delta^2)$ of σ^2 and the fourth moment μ_4 can be no more than $O(\delta^2)$ greater than $(\delta^2 + \sigma^2)^2 = \sigma^4 + O(\delta^2)$.

Now

$$\|g_1 + g_2\|_2^2 = \sum_j \lambda_j^2 \geq \left| \sum_j \lambda_j^3 \right|^{2/3} = \left(\sigma^3 + 3e \int_0^1 d_1(x)^2 + d_2(x)^2 dx \right)^{2/3}.$$

This can only be within $O(\delta^2)$ of σ^2 if $\int_0^1 d_1(x)^2 dx$ and $\int_0^1 d_2(x)^2 dx$ are both $O(\delta^2)$. However, $\int_0^1 d_1(x)^2 dx = \lambda_1^2 \left(\int_0^1 v_1(x) dx \right)^2$ and $\lambda_1 \approx -\sigma$, so $\int_0^1 v_1(x) dx$ must be $O(\delta)$.

We now turn to $\|g_2\|_2^2 = \sum_{j=2}^\infty \lambda_j^2$. Since

$$\lambda_1^3 = -(\sigma^3 + 3e \int_0^1 d_1(x)^2 + d_2(x)^2 dx - \sum_{j=2}^\infty \lambda_j^3),$$

and since $|\sum_{j=2}^\infty \lambda_j^3| \leq \|g_2\|_2^3$, $\lambda_1 \leq -\sigma + O(\|g_2\|_2^3)$. But then

$$\sum_{j=1}^\infty \lambda_j^2 \geq \sigma^2 + \|g_2\|_2^2 + O(\|g_2\|_2^3).$$

Since this must be within $O(\delta^2)$ of σ^2 , we must have $\|g_2\|_2^2 = O(\delta^2)$.

Finally, we consider the fourth moment μ_4 . The leading contribution is

$$\lambda_1^4 \left(\int_0^1 v_1(x)^4 dx \right)^2 = \lambda_1^4 \left(1 + \int_0^1 (v_1(x)^2 - 1)^2 dx \right)^2.$$

For this to be within $O(\delta^2)$ of σ^4 , $\int_0^1 (v_1(x)^2 - 1)^2 dx$ must be $O(\delta^2)$. $\square$

Step 2: Pointwise bounds

The upshot of step 1 is that g must be L^2 -close to a symmetric bipodal graphon, with λ_1 being close to $-\sigma$, with the sum of the other λ_j^2 being small, and with $v_1(x)$ being close to 1 on a set of measure approximately 1/2 and close to -1 on a set of measure approximately 1/2. We now upgrade those L^2 estimates into pointwise estimates:

Proposition 8.9 *If g is an entropy-maximizing graphon, then $\|g_2(x, y)\|_\infty$ is $o(1)$.*

Proposition 8.10 *If g is an entropy-maximizing graphon, then $\|v_1(x)^2 - 1\|_\infty$ is $o(1)$.*

We prove Propositions 8.9 and 8.10 with a series of lemmas. We begin by showing that g_1 and g_2 are pointwise bounded.

Lemma 8.11 *Let g be an entropy-maximizing graphon. For all $x, y \in [0, 1]$, the following bounds apply:*

$$|v_1(x)| \leq |\lambda_1|^{-1}, \quad |g_1(x, y)| \leq |\lambda_1|^{-1}, \quad |g_2(x, y)| \leq 1 + |\lambda_1|^{-1}.$$

Proof We use the fact that

$$0 \leq e + g_1(x, y) + g_2(x, y) \leq 1$$

for all (x, y) . The only way for $g_2(x, y)$ to be big and positive (resp. negative) is for $g_1(x, y)$ to be big and negative (resp. positive). This can only occur if $|v_1(x)|$ is large for some x .

Suppose that there is a point x_0 with $v_1(x_0) > 1/|\lambda_1|$. Let I_+ be the set of x for which $v_1(x) > 0.9|\lambda_1 v_1(x_0)|^{-1}$ and let I_- be the set of x for which $v_1(x) < -0.9|\lambda_1 v_1(x_0)|^{-1}$. We already know that the set of points with $v_1(x)$ close to ± 1 each have measure close to 1/2, since $\int_0^1 (v_1(x)^2 - 1)dx = O(\delta^2)$ and $\int_0^1 v_1(x)dx = O(\delta)$, so I_+ and I_- also each have measure close to 1/2.

Since λ_1 is negative, $g_1(x, y)$ is less than or equal to -0.9 for all $y \in I_+$ and is greater than or equal to 0.9 for all $y \in I_-$. Since e is close to $1/2$, $e + g_1(x_0, y)$ is close to or less than -1.4 when $y \in I_+$ (and in particular is less than -1.3) and is close to or greater than 1.4 (and in particular is greater than 1.3) when $y \in I_-$. As a result, $g_1(x_0, y)$ has magnitude at least 0.3 , and sign opposite to that of $g_2(x_0, y)$, for all $y \in I_+ \cup I_-$. This means that $g_1(x_0, y)g_2(x_0, y) < -0.27$ for all $y \in I_+ \cup I_-$.

When $y \notin I_- \cup I_+$, we have $|g_2(x_0, y)| \leq 0.9 < 1$, so $|g_2(x_0, y)| < 2$, so

$g_1(x_0, y)g_2(x_0, y) < 2$. Since $I_+ \cup I_-$ is a set of measure $m \approx 1$,

$$\int_0^1 g_1(x_0, y)g_2(x_0, y)dy \leq -0.27m + 2(1 - m) < 0.$$

However,

$$\int_0^1 g_1(x_0, y)g_2(x_0, y)dy = \sum_{j=2}^{\infty} \lambda_1 \lambda_j v_1(x_0) v_j(x_0) \int_0^1 v_1(y) v_j(y) dy = 0,$$

by the orthogonality of the functions $v_j(y)$ in $L^2([0, 1])$. This is a contradiction, so x_0 does not exist.

The same argument, with signs reversed, rules out the possibility that $v_1(x)$ is ever less than $-1/|\lambda_1|$. Since $|v_1(x)|$ is bounded by $|\lambda_1|^{-1}$, $|g_1(x, y)| = |\lambda_1 v_1(x) v_1(y)|$ is also bounded by $|\lambda_1|^{-1}$. Finally, we have that

$$-e - g_1(x, y) \leq g_2(x, y) \leq 1 - e - g_1(x, y).$$

Since e and $1-e$ are both less than 1, this implies that $|g_2(x, y)| < |g_1(x, y)| + 1 \leq |\lambda_1|^{-1} + 1$. $\square$

Lemma 8.11 is stated in terms of λ_1 , which of course depends on the graphon g . However, $\lambda_1 = -\sigma + o(1)$, so for small δ we can replace our bounds involving λ_1 with uniform bounds in terms of σ , at the cost of replacing the constant 1 with a slightly different number. For instance,

$$|v_1(x)| < 1.1\sigma^{-1}, \quad |g_1(x, y)| < 1.1\sigma^{-1}, \quad |g_2(x, y)| \leq 1 + 1.1\sigma^{-1}$$

whenever δ is sufficiently small. In practice, we do not need the specific bounds of Lemma 8.11. All we really need is for $v_1(x)$, $g_1(x, y)$ and $g_2(x, y)$ to be bounded.

We next turn to showing that $g_2(x, y)$ is not only bounded but small. Since $\|g_2\|_2^2$ is $O(\delta^2)$, the set of points where $|g_2(x, y)|$ is not small (say, larger than a fixed ϵ) has measure $O(\delta^2)$. We now establish a similar result for vertical strips.

Lemma 8.12 *Let g be an entropy-maximizing graphon. For any $\epsilon > 0$ and any $x \in [0, 1]$, the set of y -values for which $|g_2(x, y)| > \epsilon$ has measure $o(1)$.*

Proof Let $G(x, y) = \int_0^1 g(x, z)g(z, y) dz$. As an operator, this is the square of g . Expanding that square using $g(x, y) = e + g_1(x, y) + g_2(x, y)$, we let G_1 be the portion of G that comes from $e + g_1$, and let G_2 be the additional contributions that involve g_2 .

$$G_1(x, y) = \int_0^1 (e + g_1(x, z))(e + g_1(y, z))dz$$

$$\begin{aligned}
&= e^2 + \lambda_1^2 v_1(x) v_1(y) + e \left(\int_0^1 v_1(z) dz \right) (v_1(x) + v_1(y)). \\
&= e^2 + \lambda_1 g_1(x, y) + O(\delta), \\
&= \lambda_1 (e + g_1(x, y)) + e^2 - \lambda_1 e + O(\delta),
\end{aligned}$$

since $\int_0^1 v_1(z) dz = O(\delta)$ and $v_1(x)$ and $v_1(y)$ are bounded.

We next turn to G_2 . Since $\int_0^1 v_1(z) g_2(y, z) dz = 0$, there is no contribution from the product of g_1 and g_2 . We only have $e g_2$ and g_2^2 terms, specifically

$$G_2(x, y) = e(d_2(x) + d_2(y)) + \int_0^1 g_2(x, z) g_2(y, z) dz.$$

The function $d_2(y)$ has small L^2 -norm, and so must be $o(1)$ except on a set of small measure. (Since x is fixed, we cannot similarly argue that $d_2(x)$ is small.) Finally, since g_2 is bounded and has small L^2 norm, the integral $\int_0^1 g_2(x, z) g_2(y, z) dz$ for fixed x is small except for a set of y 's that has small measure. The result is an estimate

$$G_2(x, y) = e d_2(x) + o(1)$$

that is true for y in the complement of a set of measure $o(1)$, where that small set may depend on x .

Combining this with our estimate of G_1 , we have

$$G(x, y) = (e^2 - \lambda_1 e) + \lambda_1 g_1(x, y) + e d_2(x) + o(1)$$

for all but a small set of y 's.

Since g is assumed to maximize entropy subject to constraints on $\varepsilon(g)$ and $\tau(g)$, the functional derivative of $S(g)$ must be a linear combination of the functional derivatives of $\varepsilon(g)$ and $\tau(g)$. This yields the pointwise equations

$$H'(g(x, y)) = \Lambda_e + \Lambda_t G(x, y), \quad (8.17)$$

where Λ_e and Λ_t are Lagrange multipliers.¹

For all but a small set of y 's, equation (8.17) takes the form

$$H'(e + g_1(x, y) + g_2(x, y)) = \mu + \nu(e + g_1(x, y)) + \rho d_2(x) + o(1), \quad (8.18)$$

where

$$\mu = \Lambda_e + \Lambda_t(e^2 - \lambda_1 e), \quad \nu = \lambda_1 \Lambda_t, \quad \rho = e \Lambda_t.$$

For most values of (x, y) , $e + g_1(x, y)$ is close to $e \pm \sigma$ and $g_2(x, y)$ and $d_2(x)$

¹ In previous calculations, these multipliers were called α and β . Here, however, the first three Greek letters are already defined in (8.14), so we call them Λ_e and Λ_t .

are small. This fixes Λ_e and Λ_t , and therefore μ , ν , and ρ , to within a small error. Since $e = \frac{1}{2} + \delta$ and $H'(\frac{1}{2} - \sigma) = -H'(\frac{1}{2} + \sigma)$, we obtain

$$\begin{aligned} H'\left(\frac{1}{2} + \sigma\right) &= \Lambda_e + \Lambda_t \left(\frac{1}{4} - \sigma^2\right) + o(1) \\ -H'\left(\frac{1}{2} + \sigma\right) &= \Lambda_e + \Lambda_t \left(\frac{1}{4} + \sigma^2\right) + o(1), \end{aligned}$$

with solution

$$\begin{aligned} \Lambda_e &\approx -\frac{1}{4\sigma^2} H'\left(\frac{1}{2} + \sigma\right), & \Lambda_t &\approx \frac{1}{\sigma^2} H'\left(\frac{1}{2} + \sigma\right), \\ \mu &\approx -\frac{1}{2\sigma} H'\left(\frac{1}{2} + \sigma\right), & \nu &\approx \frac{1}{\sigma} H'\left(\frac{1}{2} + \sigma\right), \\ \rho &\approx -\frac{1}{2\sigma^2} H'\left(\frac{1}{2} + \sigma\right), \end{aligned}$$

where all of the approximations are “ $+o(1)$ ” as $\delta \rightarrow 0$.

From the explicit form of $H'(g) = \ln(1-g) - \ln(g)$, we see that there are only three roots to the equation $H'(g) = \mu + \nu g$, which are located near $g = e \pm \sigma$ and $g = e$. Our immediate goal is to show that $v_1(x)$ only takes values close to 0 and ± 1 , which implies that $g_1(x, y)$ is only close to e and $e \pm \sigma$.

Since $\int v_1(x) dx$ and $\int (v_1(x)^2 - 1)^2 dx$ are small, the function v_1 must be close to 1 on an interval (call it I_1) of measure close to $1/2$, must be close to -1 on an interval I_2 of measure close to $1/2$, may be close to 0 on a third interval I_3 of small measure, and may take on other values on a fourth interval I_4 of small measure.

Let x be an arbitrary point in $[0, 1]$, and let y_1 and y_2 be generic points in I_1 and I_2 . Equation (8.18) then determines $a = g_2(x, y_1)$ and $b = g_2(x, y_2)$ in terms of $d_2(x)$. What's more, g_2 takes values close to a on all of $\{x\} \times I_1$ (excepting those values of y where equation (8.18) doesn't apply), and takes values close to b on all of $\{x\} \times I_2$. We then compute

$$\int_0^1 g_2(x, y) v_1(y) dy \approx (a - b)/2.$$

However, this integral must be zero, since $v_1(y)$ is orthogonal to all of the functions $v_i(y)$ that make up $g_2(x, y)$. We conclude that $a \approx b$.

If $b \approx a$, then $d_2(x) = \int_0^1 g_2(x, y) dy$ also equals a and our equations at (x, y_2) and (x, y_1) become

$$H'\left(\frac{1}{2} + \sigma v_1(x) + a\right) \approx \mu + \nu \left(\frac{1}{2} + \sigma v_1(x)\right) + \rho a$$

$$H' \left(\frac{1}{2} - \sigma v_1(x) + a \right) \approx \mu + \nu \left(\frac{1}{2} - \sigma v_1(x) \right) + \rho a,$$

Adding these equations, and using the fact that $H'(\frac{1}{2} - \sigma v_1(x) + a) = -H'(\frac{1}{2} + \sigma v_1 - a)$, we get

$$H' \left(\frac{1}{2} + \sigma v_1(x) + a \right) - H' \left(\frac{1}{2} + \sigma v_1(x) - a \right) \approx 2\mu + \nu + 2\rho a \approx -\frac{a}{\sigma^2} H' \left(\frac{1}{2} + \sigma \right). \quad (8.19)$$

By the mean value theorem, the left hand side of equation (8.19) is $2aH''(u)$ for some u between $\frac{1}{2} + \sigma v_1(x) - a$ and $\frac{1}{2} + \sigma v_1(x) + a$. Regardless of the value of u , this is a negative multiple of a . However, the right hand side is a positive multiple of a , since $H'(\frac{1}{2} + \sigma) < 0$. Since a negative multiple of a equals a positive multiple, a must be (approximately) zero.

In particular, $g_2(x, y) \approx 0$ for all y such that equation (8.18) applies. That is, for fixed x , $g_2(x, y)$ is close to zero except on a set of y 's of small measure. $\square$

Proof of Proposition 8.9 By Lemma 8.11, $g_2(x, y)$ is bounded. By Lemma 8.12, for each x , $g_2(x, y)$ is small for all but a small set of y 's. Combining these results, we see that the degree function $d_2(x)$ is everywhere small, as is the integral $\int_0^1 g_2(x, z)g_2(y, z) dz$. That is, $G_2(x, y)$ is pointwise small. But then equation (8.18) applies everywhere, so $g_2(x, y)$ is small everywhere. $\square$

Proof of Proposition 8.10 In the notation of the proof of Lemma 8.12, we must show that the intervals I_3 and I_4 are empty, implying that $v_1(x)$ is everywhere close to ± 1 .

Since $G_2(x, y)$ is small for all (x, y) , we must have

$$H' \left(\frac{1}{2} + \sigma v_1(x) \right) = \mu + \nu \left(\frac{1}{2} + \sigma v_1(x) \right)$$

for all $y \in I_2$. However, the only solutions to this equation are (approximately) $\sigma v_1(x) = \pm\sigma$ or 0, implying that $v_1(x) \approx \pm 1$ or 0. In other words, $x \in I_1 \cup I_2 \cup I_3$ and I_4 is empty.

Showing that I_3 is empty requires a completely different argument, since equation (8.17) is indeed satisfied when $x \in I_3$. However, equation (8.17) only defines stationary points with respect to pointwise small changes in $g(x, y)$. We also have to consider infinitesimal changes in the boundary between I_1 , I_2 and I_3 .

So suppose that we increase the size of I_1 by an amount ϵ at the expense of I_3 . That is, we change the value of $v_1(x)$ from near 0 to near 1 on a set of small measure ϵ . To first order in ϵ , the change in entropy is $2\epsilon(H(\frac{1}{2} + \sigma) - H(\frac{1}{2}))$, since we are changing $g(x, y)$ from near zero to near $\frac{1}{2} \pm \sigma$ on a set of measure

$2\epsilon + O(\epsilon^2)$, and since $H(\frac{1}{2} - \sigma) = H(\frac{1}{2} + \sigma)$. The change in edge density is $-2\epsilon\lambda_1 \int_0^1 v_1(y)dy = O(\delta\epsilon)$. To leading order, the change in the triangle density is $3\lambda_1^3\epsilon$, since the λ_1^3 contribution to $\tau(g)$ is actually $\lambda_1^3(\int_0^1 v_1(x)^2 dx)^3$, which changes from λ_1^3 to $\lambda_1^3(1 + \epsilon)^3 \approx \lambda_1^3(1 + 3\epsilon)$.

The variational equations $dS = \Lambda_e d\epsilon + \Lambda_t d\tau/3$ then become

$$2 \left[H\left(\frac{1}{2} + \sigma\right) - H\left(\frac{1}{2}\right) \right] = \lambda_1^3 \Lambda_t = \sigma H'\left(\frac{1}{2} + \sigma\right). \quad (8.20)$$

We expand both sides of equation (8.20) as power series in σ . The left-hand side is

$$2 \sum_{k=1}^{\infty} \frac{H^{(2k)}(1/2)}{(2k)!} \sigma^{2k}.$$

The right-hand side is

$$\sum_{k=1}^{\infty} \frac{H^{(2k)}(1/2)}{(2k-1)!} \sigma^{2k}.$$

The coefficients agree when $k = 1$, but are strictly greater for the right-hand side when $k > 1$. Since all terms are strictly negative (insofar as all even derivatives of H are negative-definite), the right-hand side is strictly smaller than the left-hand side.

Since varying the size of I_3 does not satisfy the variational equation (8.20), we cannot be in the interior of our parameter space. Rather, the measure of I_3 must be zero. $\square$

Step 3: Cost-benefit analysis

In steps 1 and 2 we showed that $g_2(x, y)$ and $v_1(x)^2 - 1$ are pointwise small. In this section we show that they are zero, completing the proof of Theorem 8.6.

We use the expansion (8.16) and compare the moments μ_{2k} to those of the symmetric bipodal graphon. We will estimate costs (terms that increase μ_{2k}) and benefits (terms that decrease μ_{2k}). We will show that having α or β or γ nonzero comes with costs that go as α^2 , β^2 , and γ^2 , while the benefits are $o(\alpha^2 + \beta^2 + \gamma^2)$. When δ is sufficiently small, the costs exceed the benefits, so the symmetric bipodal graphon has more entropy than any graphon that isn't symmetric bipodal.

We first establish the costs. Our triangle density is

$$e^3 - \sigma^3 = t = e^3 + 3e \int_0^1 \left(d_1(x)^2 + d_2(x)^2 \right) dx + \lambda_1^3 + \sum_{j=2}^{\infty} \lambda_j^3.$$

Now

$$\int_0^1 d_1^2(x) dx = \lambda_1^2 \left(\int_0^1 v_1(x) dx \right)^2 = \lambda_1^2 \gamma^2,$$

while $\int_0^1 d_2(x)^2 dx > 0$ and

$$\left| \sum_{j=2}^{\infty} \lambda_j^3 \right| \leq \left(\sum_{j=2}^{\infty} \lambda_j^2 \right)^{3/2} = \alpha^3.$$

This implies that

$$\lambda_1^3 \leq -\sigma^3 - 3e\sigma^2\gamma^2 + O(\alpha^3),$$

so

$$\lambda_1^2 \geq \sigma^2 + 2e\sigma\gamma^2 + O((\alpha, \beta, \gamma)^3)$$

and

$$\mu_2 = \lambda_1^2 + \alpha^2 \geq \sigma^2 + \alpha^2 + 2e\sigma\gamma^2 + o(\alpha^2 + \beta^2 + \gamma^2).$$

That is, there are α^2 and γ^2 costs associated with μ_2 .

We next look at μ_4 . This contains a term

$$\iint g_1(x, y)^4 dx dy = \lambda_1^4 \left(\int_0^1 v_1(x)^4 dx \right)^2 = \lambda_1^4 (1 + \beta^2)^2 \geq \sigma^4 (1 + 2\beta^2).$$

That is, there is a cost proportional to β^2 . Having established costs proportional to α^2 , β^2 , and γ^2 , we just have to show that the benefits of having α, β, γ nonzero are smaller.

We are looking at even moments

$$\mu_{2k} = \iint (\delta + g_1 + g_2)^{2k} dx dy.$$

We expand out the power, getting terms proportional to a power of δ times a power of g_1 times a power of g_2 . We make repeated use of the following trick:

$$u^{2m} = 1 + u^{2m} - 1 = 1 + (u^2 - 1)p_m(u), \quad (8.21)$$

where

$$p_m(u) = 1 + u^2 + u^4 + \cdots + u^{2m-1}. \quad (8.22)$$

This means that

$$\begin{aligned} g_1(x, y)^{2m} &= \lambda_1^{2m} \left[1 + (v_1(x)^2 - 1)p_m(v_1(x)) \right] \left[1 + (v_1(y)^2 - 1)p_m(v_1(y)) \right] \\ &= \lambda_1^{2m} \left[1 + (v_1(x)^2 - 1)p_m(v_1(x)) + (v_1(y)^2 - 1)p_m(v_1(y)) \right] \end{aligned}$$

$$+ \left(v_1(x)^2 - 1 \right) \left(v_1(y)^2 - 1 \right) p_m(v_1(x)) p_m(v_1(y)) \Big] \quad (8.23)$$

and

$$\begin{aligned} g_1(x, y)^{2m+1} = & \lambda_1^{2m+1} \Big[v_1(y) \left(v_1(x) + \left(v_1(x)^3 - v_1(x) \right) p_m(v_1(x)) \right) \\ & + v_1(x) \left(v_1(y)^3 - v_1(y) \right) p_m(v_1(y)) \\ & + \left(v_1(x)^2 - 1 \right) \left(v_1(y)^2 - 1 \right) v_1(x) v_1(y) p_m(v_1(x)) p_m(v_1(y)) \Big]. \end{aligned} \quad (8.24)$$

We divide the terms obtained by expanding $\iint (\delta + g_1 + g_2)^{2k}$ into several classes.

1. Terms with three or more powers of g_2 . Since g_2 is pointwise small and g_1 is bounded, these are bounded by small multiples of $\iint g_2^2 = \alpha^2$. In other words, they are $o(\alpha^2)$.
2. Terms with two powers of g_2 , an even number of powers of δ and an even number of powers of g_1 . These are manifestly positive and represent costs, not benefits. Aside from the $\iint \delta^0 g_1^0 g_2^2 = \alpha^2$ contribution to μ_2 that we already considered, we do not keep track of these.
3. Terms with two powers of g_2 , an odd number of powers of δ and an odd number of powers of g_1 . The integrand is a positive power of δ times a bounded quantity times $g_2(x, y)^2$, making the integral $O(\delta \alpha^2)$.
4. Terms with one power of g_2 , an odd power of δ and an even power of g_1 . We expand these using equation (8.23) and consider each piece separately. First, we compute

$$\iint g_2(x, y) dx dy = - \iint g_1(x, y) dx dy = -\lambda_1 \left(\int_0^1 v_1(x) dx \right)^2 = -\lambda_1 \gamma^2.$$

Next, the L^2 norm of $v_1(x)^2 - 1$ is β and the L^2 norm of g_2 is α , so

$$\iint (v_1(x)^2 - 1) p_m(v_1(x)) g_2 dx dy = O(\alpha \beta).$$

The $(v_1(y)^2 - 1)$ piece is similar, while the $(v_1(x)^2 - 1)(v_1(y)^2 - 1)$ piece is $O(\alpha \beta^2)$. Thus $\iint g_1^{odd} g_2 dx dy = O(\alpha \beta) + O(\gamma^2)$, so $\iint \delta^{odd} g_1^{odd} g_2 dx dy = O(\delta \alpha \beta) + O(\delta \gamma^2) = o(\alpha^2 + \beta^2 + \gamma^2)$.

5. Terms with one power of g_2 , an even power of δ and an odd power of g_1 . We expand these using equation (8.24), noting that the first line in (8.24) has a single factor of $v_1(y)$ and the second line has a single factor of $v_1(x)$. However,

$$\int_0^1 v_1(y) g_2(x, y) dy = \int_0^1 v_1(x) g_2(x, y) dx = 0,$$

where x is arbitrary in the first integral and y is arbitrary in the second. Thus the first two lines contribute nothing and $\iint g_1(x, y)^{2m+1} g_2(x, y) dx dy$ is equal to λ_1^{2m+1} times

$$\iint \left(v_1(x)^2 - 1 \right) \left(v_1(y)^2 - 1 \right) \left(v_1(x) v_1(y) p_m(v_1(x)) p_m(v_1(y)) \right) g_2(x, y) dx dy.$$

The factor $\left(v_1(x)^2 - 1 \right) \left(v_1(y)^2 - 1 \right)$ has L^2 norm β^2 , the factor $g_2(x, y)$ has L^2 norm α , and the factor $v_1(x) v_1(y) p_m(v_1(x)) p_m(v_1(y))$ is bounded, so the integral is $O(\alpha\beta^2) = o(\alpha^2 + \beta^2 + \gamma^2)$.

6. Terms with no powers of g_2 , an even number of powers of δ and an even number of powers of g_1 . These are all positive and are at least as big as the corresponding terms for the symmetric bipodal graphon. We have already taken into account the costs associated with μ_2 and μ_4 . There are additional costs associated with higher moments, but they are not needed for this proof.
7. Finally, there are terms with no powers of g_2 , an odd number of powers of δ and an odd number of powers of g_1 . Note that

$$\iint g_1(x, y)^{2m+1} dx dy = \lambda_1^{2m+1} v_{2m+1}^2,$$

where

$$\begin{aligned} v_{2m+1} &= \int_0^1 v_1(x)^{2m+1} dx \\ &= \int_0^1 v_1(x) dx + \int_0^1 \left(v_1(x)^2 - 1 \right) v_1(x) p_m(v_1(x)) dx \\ &= \gamma + O(\beta), \end{aligned}$$

since $v_1(x)^2 - 1$ has L^2 -norm β and $v_1(x) p_m(v_1(x))$ is bounded. Squaring v_{2m+1} and multiplying by an odd power of δ gives $O(\delta\gamma^2) + O(\delta\beta^2) + O(\delta\beta\gamma)$, which is $o(\alpha^2 + \beta^2 + \gamma^2)$.

Putting everything together, we have identified costs proportional to α^2 , β^2 and γ^2 . Other costs only add to that total, so the total cost is at least a constant times $\alpha^2 + \beta^2 + \gamma^2$. All of the potential benefits are smaller, either involving three or more powers of α , β and γ , or δ times a quadratic function of α , β and γ , or the sup norm of g_2 times α^2 . When δ is sufficiently small, the costs outweigh the benefits, so the optimal graphon is symmetric bipodal.

□

8.4 Order parameters and symmetry

Now that we have proven the existence of the B(1,1) and A(2,0) phases, how do we know that these phases are actually different? How can we rule out the possibility, already realized in the edge-2star model (and similar to what happens in statistical mechanics with physical liquids and gases) that the two sides of a phase transition are actually linked by an analytic path that goes around a bifurcation point?

The answer is to construct an order parameter based on symmetry. The reduced density $\tilde{t}_2$ of 2-stars is given by

$$\tilde{t}_2 = \int_0^1 (d(x) - e)^2 dx.$$

This is exactly zero for symmetric bipodal graphons, as the degree function is the same on both podes. However, $\tilde{t}_2$ is nonzero in the B(1,1) phase, where $d(x) - e$ is given by a nonzero multiple of δ^2 (plus higher-order terms) on the small podes, and is $O(\delta^3)$ on the large podes. On any analytic path from one phase to the other, $\tilde{t}_2$ is zero on an open set, yet becomes nonzero elsewhere. As a result, $\tilde{t}_2$ cannot be an analytic function of whatever variable we are using to parametrize the path. There must be a point (or multiple points) where $\tilde{t}_2$ fails to be analytic, meaning that the underlying optimal graphon fails to be analytic, or maybe fails to be unique. Either way, we have a phase transition.

In Chapter 4, we constructed order parameters that measured the rank of the optimal graphon. Here both phases involve rank-2 graphons, so we use an order parameter $\tilde{t}_2$ that detects symmetry-breaking instead. The details are different but the game is qualitatively the same.

8.5 Just below Erdős-Rényi with $e < 1/2$

How far does the A(2,0) phase extend? We already know that the entropy maximizing graphon is unique and bipodal on an open set containing the interval $e = 1/2$, $0 < t < 1/8$ and the line segment $t = 0$, $0 \leq e \leq 1/2$. On the Erdős-Rényi curve $t = e^3$, the optimal (constant) graphon can be viewed as a special kind of symmetric bipodal graphon, one with $c = 1/2$ and $a = b = d = e$.

Question: Does the A(2,0) phase contain the entire region bounded by the Erdős-Rényi curve, the line $t = 0$, and the line $e = 1/2$?

For nearly a decade, the answer was thought to be “yes”. In this section, we show that it is “no”.

Theorem 8.13 ([47]) *If $e < e_0 := (3 - \sqrt{3})/6 \approx 0.2113$, and if $t < e^3$ is sufficiently close to e^3 , then the symmetric bipodal graphon is not entropy-maximizing.*

Proof We will construct an explicit family of graphons depending on parameters A , B and c , with A and B constant and $c \rightarrow 0$. We will then compute t and $S(g)$ in terms of these parameters and show that, for appropriate values of A and B and sufficiently small values of c , the entropy of this graphon is greater than that of a symmetric bipodal graphon with the same value of t . Note: This does not prove that our explicit graphon **is** entropy-maximizing, but it does prove that the symmetric bipodal graphon is **not** entropy-maximizing. For a numerical exploration of what the optimal graphon actually looks like, see [17].

We consider values of e and $t = e^3 - \delta^3$ where $e < e_0$ and δ is sufficiently small. The number e_0 is defined by the equation

$$3H'''(e_0)^2 = H''(e_0)H''''(e_0),$$

which simplifies to $6e_0^2 - 6e_0 + 1 = 0$, or $e_0 = (3 - \sqrt{3})/6 \approx 0.2113$. When $e < e_0$, $3H'''(e)^2$ is greater than $H''(e)H''''(e)$.

Let $A > B > 0$ and let

$$F(A, B) = \frac{H(e + A + B) + H(e - A + B) - 2H(e) - 2BH'(e)}{(A^3 - B^3)^{2/3}}.$$

We will eventually choose A and B to maximize $F(A, B)$. Pick a small number c and divide the interval $[0, 1]$ into three pieces:

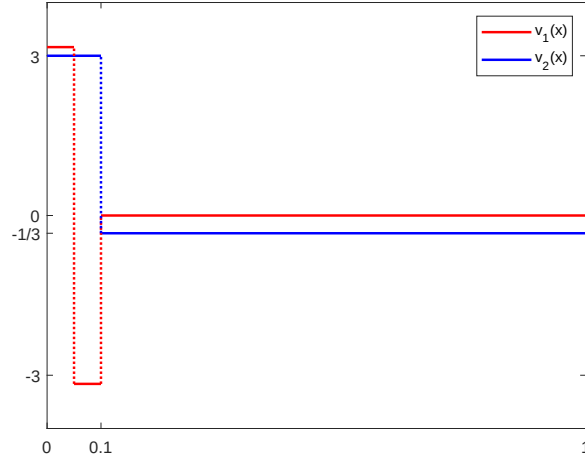
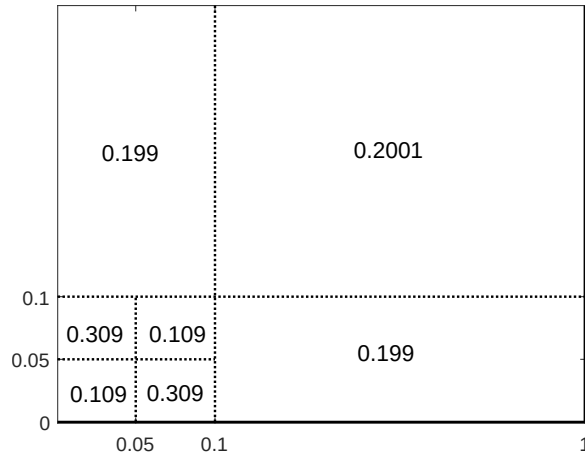
$$I_1 = [0, c/2], \quad I_2 = (c/2, c], \quad I_3 = (c, 1].$$

Consider the graphon

$$g(x, y) = \begin{cases} e - A + B(1 - c) & (x, y) \in I_1 \times I_1 \cup I_2 \times I_2 \\ e + A + B(1 - c) & (x, y) \in I_1 \times I_2 \cup I_2 \times I_1 \\ e - cB & (x, y) \in [(I_1 \cup I_2) \times I_3] \cup [I_3 \times (I_1 \cup I_2)] \\ e + \frac{c^2}{1-c}B & (x, y) \in I_3 \times I_3. \end{cases} \quad (8.25)$$

Equivalently, $g(x, y) = e - cAv_1(x)v_1(y) + cBv_2(x)v_2(y)$, where

$$v_1(x) = \begin{cases} c^{-1/2} & x \in I_1, \\ -c^{-1/2} & x \in I_2, \\ 0 & x \in I_3, \end{cases} \quad v_2(x) = \begin{cases} \sqrt{(1-c)/c} & x \in I_1 \cup I_2, \\ -\sqrt{c/(1-c)} & x \in I_3, \end{cases}$$

Figure 8.5 The functions $v_1(x)$ and $v_2(x)$ when $c = 0.1$.Figure 8.6 The graphon (8.25) when $e = 0.2$, $c = A = 0.1$ and $B = 0.01$. The axes are not drawn to scale.

as displayed in Figure 8.5. An example of such a graphon is displayed in Figure 8.6, in the particular case that $e = 0.2$, $c = A = 0.1$ and $B = 0.01$.

Since $\int_0^1 v_1(x) dx = \int_0^1 v_2(x) dx = 0$, the tripodal graphon (8.25) has constant degree function $d(x) = e$. Since the eigenvalues of $g - e$ are $-cA$ and cB , the triangle density is

$$\tau(g) = e^3 - c^3 A^3 + c^3 B^3.$$

Setting this equal to $e^3 - \delta^3$ gives

$$c = \delta(A^3 - B^3)^{-1/3}.$$

We now estimate the entropy

$$\begin{aligned} S(g) = & \frac{c^2}{2}H(e - A + (1 - c)B) + \frac{c^2}{2}H(e + A + (1 - c)B) \\ & + 2c(1 - c)H(e - cB) + (1 - c)^2H\left(e + \frac{c^2}{1 - c}B\right) \end{aligned}$$

to order c^2 , or equivalently to order δ^2 . The first two terms already are $O(c^2)$, so we can simply replace $e \pm A + (1 - c)B$ with $e \pm A + B$. For the remaining terms, we can use the linear approximation $H(u) \approx H(e) + H'(e)(u - e)$. The result is

$$\begin{aligned} S(g) = & H(e) + \frac{c^2}{2}\left(H(e - A + B) + H(e + A + B) - 2H(e) - 2BH'(e)\right) + O(c^3) \\ = & H(e) + \frac{1}{2}F(A, B)\delta^2 + O(\delta^3). \end{aligned}$$

For comparison, the symmetric bipodal graphon has entropy

$$H(e) + \frac{1}{2}H''(e)\delta^2 + O(\delta^4).$$

If we can find values of A and B such that $F(A, B) > H''(e)$, and if c (and therefore δ) is sufficiently small, then the tripodal graphon will have more entropy than the symmetric bipodal graphon.

To find such a pair (A, B) , let A be a small positive number and let

$$B = -\frac{H'''(e)}{2H''(e)}A^2.$$

Since $H'''(e) > 0 > H''(e)$, B is positive. Since $A^3 - B^3 = A^3 + O(A^6)$, $(A^3 - B^3)^{2/3} = A^2 + O(A^5)$. We compute the numerator of $F(A, B)$ to order A^4 by doing a 4-th order Taylor series expansion of $H(e - A + B)$ and $H(e + A + B)$ around e and keeping terms proportional to A^2 , A^4 , B , B^2 , and A^2B . (The expression is even in A , so we only get even powers of A .) The result is

$$\begin{aligned} F(A, B) = & \frac{(A^2 + B^2)H''(e) + A^2BH'''(e) + \frac{1}{12}A^4H''''(e) + O(A^6)}{(A^3 - B^3)^{2/3}} \\ = & \frac{A^2H''(e) + A^4\left(\frac{H''''(e)}{12} - \frac{(H'''(e))^2}{4H''(e)}\right) + O(A^6)}{A^2 + O(A^5)} \\ = & H''(e) + \left(\frac{H''''(e)}{12} - \frac{(H'''(e))^2}{4H''(e)}\right)A^2 + O(A^3). \end{aligned}$$

The coefficient of A^2 is positive for $e < e_0$, so $F(A, B) > H''(e)$ when A is small. $\square$

This proof suggests three questions:

1. Where did the functions $v_1(x)$ and $v_2(x)$ come from?
2. Why does the construction break down when $e > e_0$?
3. What actually happens when $e > e_0$?

As noted at the beginning of the chapter, it is useful to express the entropy of a graphon as

$$S(g) = H(e) + \sum_{k=2}^{\infty} \frac{H^{(k)}(e)}{k!} \mu_k,$$

where

$$\mu_k = \iint (g(x, y) - e)^k dx dy.$$

Since $e < 1/2$, all the odd derivatives are positive, while the even derivatives are negative, so we want to maximize the odd moments and minimize the even moments. If we diagonalize $g - e$ as

$$g(x, y) - e = \sum_{i=1}^{\infty} \lambda_i v_i(x) v_i(y),$$

then the second moment is simply

$$\mu_2 = \sum_{i=1}^{\infty} \lambda_i^2.$$

This is minimized when $\lambda_1 = -\delta$ and all the other eigenvalues are zero, in other words when $g - e$ has rank 1. Among such graphons, Theorem 8.1 states that the best graphon is symmetric bipodal.

To do better than symmetric bipodal, we must have a second nonzero eigenvalue and the eigenfunctions $v_1(x)$ and $v_2(x)$ must be chosen to make μ_3 as large as possible. In expanding μ_3 in terms of the eigenvalues and eigenfunctions of $g - e$, there is a term proportional to

$$\lambda_1^2 \lambda_2 \left(\int_0^1 v_1(x)^2 v_2(x) dx \right)^2.$$

Since we want to have $\int_0^1 v_2(x) dx = 0$, this is the same as

$$\lambda_1^2 \lambda_2 \left(\int_0^1 (v_1(x)^2 - 1) v_2(x) dx \right)^2.$$

For fixed λ_1 and λ_2 , this is maximized by maximizing the variance of $v_1(x)^2$ and by picking $v_2(x)$ proportional to $v_1(x)^2 - 1$. However, increasing the variance of $v_1(x)^2$ also results in increasing μ_4 .

The upshot is that this construction incurs

- An entropy cost from changes to μ_2 , relative to the symmetric bipodal graphon, that is proportional to $H''(e)\lambda_2^2$,
- A benefit from changes to μ_3 that is proportional to $H'''(e)\lambda_1^2\lambda_2$ times the variance of $v_1(x)^2$, and
- A cost from changes to μ_4 that is proportional to $H''''(e)\lambda_1^4$ times a combination of the variance and squared variance of $v_1(x)^2$.

For each choice of functions, this results in a quadratic function of λ_2/λ_1^2 . The discriminant of this function is proportional to $H''''(e)^2$ minus a multiple of $H''(e)H''''(e)$. The larger the variance of $v_1(x)^2$ is, the smaller the multiple, and the more it becomes possible to have a positive discriminant. In the limit as the variance goes to ∞ , the multiple becomes $1/3$ and the discriminant becomes positive for all $e < e_0$.

When $e > e_0$, the discriminant is negative regardless of the variance of $v_1(x)^2$. $H''''(e)$ just isn't big enough to allow the gains associated with changes to μ_3 to overcome the losses associated with changes to μ_2 and μ_4 .

In our construction, the function $v_1(x)$ was chosen to be a simple step function for which the variance of $v_1(x)^2$, namely $\frac{1-c}{c}$, diverged as $c \rightarrow 0$. The function $v_2(x)$ is $\sqrt{\frac{c}{1-c}}(v_1(x)^2 - 1)$.

There is a drawback, however, to using a function $v_1(x)$ that is pointwise very large. Our power series expansions only make sense when $\lambda_1 v_1(x)v_1(y)$ is everywhere much less than 1. This restricts the possible values of λ_1 and means that the superiority of a tripodal graphon over a symmetric bipodal graphon is only assured in a very small region just below the Erdős-Rényi curve. Indeed, numerical experiments [17] suggest that the entropy maximizing graphon is symmetric bipodal whenever δ is greater than a value δ_{max} that depends on e and is never greater than $e/8$. Since $\delta^3 < e^3/512$, it appears that the best graphon is symmetric bipodal whenever $t < \frac{511}{512}e^3$. The quantity δ_{max} decays as $(e_0 - e)^1$ as e approaches e_0 from below.

Which brings us to the third question. The evidence supports the following

Conjecture 8.14 *For all e between e_0 and $1/2$, and for all $\delta \in (0, e^{1/3})$, the unique entropy maximizing graphon with edge and triangle densities $(e, e^3 - \delta^3)$ is symmetric bipodal.*

In summary, the optimal graphon just below the Erdős-Rényi curve appears to be

- Bipodal without symmetry when $e > 1/2$. This is the B(1,1) phase whose existence we proved in Theorem 8.2.
- Symmetric bipodal when $e_0 \leq e \leq 1/2$. This is the A(2,0) phase whose optimality is not yet proven.
- Not bipodal when $e < e_0$, as proven in Theorem 8.13. We do not yet know the actual form of the optimal graphons. Numerics suggest multiple phases, one from $e = 0.0024$ to 0.2113 and at least one below 0.0024 , but this has not been proven.

The reader is invited to prove Conjecture 8.14 and to investigate the phase(s) below the Erdős-Rényi curve for $e < e_0$.

8.6 The edge-cycle model

So far we have studied the edge- k star model and the edge-triangle model. Here we consider the edge- H model where H is a cycle with an arbitrary number m of edges. The density of such an m -cycle is given by the functional

$$\tau_H(g) = \text{tr}(g^m),$$

and we refer to a typical value of the m -cycle density as t_H .

The problem for oversaturated graphs was already covered by Theorem 7.11. Since every vertex of an m -cycle has degree 2, the edge- H model for t_H just above e^m behaves much like the edge-2star model for t_2 just above e^2 .

Our first result is that the problem for undersaturated graphs only makes sense when m is odd.

Theorem 8.15 *Let g be any graphon with edge density e . If m is even, then*

$$\tau_H(g) \geq e^m.$$

If m is even and g is not the Erdős-Rényi graphon, then $\tau_H(g) > e^m$.

Proof Diagonalize $g(x, y)$, writing $g(x, y) = \sum_{i=1}^{\infty} \lambda_i v_i(x) v_i(y)$. Let $v_0(x) = 1$ for all $x \in [0, 1]$. Since $\iint g(x, y) v_0(x) v_0(y) dx dy = e$, one of the eigenvalues of g (call it λ_1) is at least e . But then

$$\tau_H(g) = \sum_{i=1}^{\infty} \lambda_i^m \geq \lambda_1^m \geq e^m.$$

If g is any graphon other than the Erdős-Rényi graphon, then

$$\sum_i \lambda_i^2 = \|g\|_2^2 = e^2 + \|g - e\|_2^2 > e^2,$$

so either $\lambda_1 > e$ or $\lambda_2 \neq 0$. Either way, $\tau_H(g)$ is strictly greater than e^m . $\square$

When m is odd, the edge- H model turns out to behave *very* similarly to the edge-triangle model, at least in the regions that we have considered so far. The only major difference is that our small parameter δ near Erdős-Rényi is $(e^m - t_H)^{1/m}$ rather than $(e^3 - t)^{1/3}$. We state some results and then present their proofs.

Theorem 8.16 *For all values of e , the maximum entropy at H -density $t_H = e^m - \delta^m$ is at least $H(e) + \frac{1}{2}H''(e)\delta^2 + O(\delta^4)$ and is at most $H(e) + f(e, 1-e)\delta^2$. In particular, the entropy cost of reducing the H -density from e^m to $e^m - \delta^m$ is $\Theta(\delta^2)$.*

There is a B(1,1)-like phase:

Theorem 8.17 *When $e > 1/2$ and δ is sufficiently small, the entropy maximizing graphon is unique and bipodal, satisfying the estimates (8.1).*

There is an A(2,0)-like phase:

Theorem 8.18 *When $t_H < e^m$ and e is sufficiently close to $1/2$, the unique entropy maximizing graphon is symmetric bipodal.*

The A(2,0)-like phase contains a neighborhood of the line segment $t_H = 0$, for $0 < e < 1/2$:

Theorem 8.19 *When $e < 1/2$ and t_H is sufficiently small, the unique entropy maximizing graphon is symmetric bipodal.*

The A(2,0)-like phase does not cover the entire undersaturated region with $e < 1/2$:

Theorem 8.20 *When $e < e_0 = (3 - \sqrt{3})/6$ and δ is sufficiently small, the symmetric bipodal graphon is not entropy maximizing.*

We prove (or at least sketch the proof of) each of these theorems in turn.

Proof of Theorem 8.16 The lower bound is easy, since a symmetric bipodal graphon with $a = b = e - \delta$, $c = 1/2$ and $d = e + \delta$ has $t_H = e^m - \delta^m$ and has entropy $\frac{1}{2}(H(e + \delta) + H(e - \delta)) = H(e) + \frac{1}{2}H''(e)\delta^2 + O(\delta^4)$.

For the upper bound, we already know that the entropy is bounded above by $H(e) + f(e, 1-e)\|g - e\|_2^2$. All that remains is proving that $\|g - e\|_2^2 \geq \delta^2$.

Let $\lambda_1, \lambda_2, \dots$ be the eigenvalues of g . (Note that we are diagonalizing g , not $g - e$.) Since $\iint v_0(x)v_0(y)g(x, y) dx dy = e$, $\lambda_1 \geq e$. Since

$$e^m - \delta^m = \sum_{i=1}^{\infty} \lambda_i^m,$$

we must have

$$\sum_{i=2}^{\infty} |\lambda_i|^m \geq \left| \sum_{i=2}^{\infty} \lambda_i^m \right| = |e^m - \delta^m - \lambda_1^m| \geq \delta^m,$$

which implies

$$\sum_{i=2}^{\infty} \lambda_i^2 \geq \delta^2.$$

But then

$$\begin{aligned} \|g - e\|_2^2 &= \|g\|_2^2 - e^2 \\ &= (\lambda_1^2 - e^2) + \sum_{i=2}^{\infty} \lambda_i^2 \\ &\geq 0 + \delta^2 = \delta^2. \end{aligned}$$

□

Since the entropy cost for decreasing t scales as δ^2 , it is reasonable to expect dB/dt_H to scale as δ^{2-m} and to diverge as $\delta \rightarrow 0$, where $B(e, t_H)$ is the Boltzmann entropy. Moreover, the bigger m gets, the more singular the derivative becomes.

For each fixed e , it does indeed appear that the entropy just below Erdős-Rényi can be expressed as a convergent power series in δ , justifying our taking derivatives term by term. (This is certainly true in the region covered by Theorem 8.17, and also in any places where the optimal graphon is symmetric bipodal.) However, we do not need to assume regularity of the entropy as a function of δ to conclude that there is a singularity at the Erdős-Rényi curve. Since $B(e, t) = H(e) + \Theta(\delta^2)$, the mean value theorem says that dB/dt_H at some point between $t_H = e^m - \delta^m$ and e^m is $\Theta(\delta^{2-m})$. This in turn implies that dB/dt diverges to ∞ in lim sup at least as δ^{2-m} . No matter what m is, there is *always* a phase transition as we cross from one side of the Erdős-Rényi curve to the other.

Sketch of proof of Theorem 8.17 The proof is similar to the proof of Theorem 8.2, with the following modifications. For more details, see [45].

- Cubes and cube roots are replaced throughout by m -th powers and m -th roots.
- When we diagonalize $g - e$ as $\sum_i \lambda_i v_i(x) v_i(y)$ and express t_H in terms the eigenvalues, we get

$$t_H = e^m + m e^{m-2} \int_0^1 (d(x) - e)^2 dx + \sum_i \lambda_i^m + \text{extra terms.}$$

The extra terms involve expressions like $\iint (d(x) - e)(g(x, y) - e)(d(y) - e) dx dy$ or $\iiint (d(x) - e)(g(x, y) - e)(g(y, z) - e)(d(z) - e) dx dy dz$ and products of such expressions. These terms are all $O\left(\|g - e\|_2 \int_0^1 (d(x) - e)^2 dx\right)$ and do not affect any of the estimates.

- The concentration of the degree function $d(x)$ near e is sharper than in the edge-triangle model. The preliminary estimate gives $\int_0^1 (d(x) - e)^2 dx = O(\delta^{m+1})$. The entropy determines $\|g_1\|_2^2$ to within $O(\delta^3)$ as before. This determines $\|g_1\|_2^m$ and therefore $\int_0^1 (d(x) - e)^2 dx$ to within $O(\delta^{m+1})$.
- The function $G(x, y)$ in step 2 is the kernel of the $m - 1$ st power of g , viewed as an integral operator. Equation (8.9) then becomes

$$\begin{aligned} H'(g(x, y)) - (\alpha + \beta e^{m-1}) - \lambda_1^{m-2} \beta (g(x, y) - e) = \\ \beta \left[e^{m-2} (d(x) + d(y) - 2e) + \text{smaller terms involving } d(x) \text{ and } d(y) \right. \\ \left. + \sum_{j=2}^{\infty} (\lambda_j^{m-2} - \lambda_1^{m-2}) \lambda_j v_j(x) v_j(y) \right]. \end{aligned}$$

- Linearizing equation (8.9) proceeds as before, leading to pointwise estimates of $v_1(x)$. However, β scales as δ^{2-m} , not as δ^{-1} . This means that a much smaller difference between the degree functions for the two nodes is needed to balance the difference between b and d on the left hand side of (8.9). As a result, the estimates (8.12) for a , b and c wind up holding, but the “peculiar” δ^2 contribution to d disappears. Instead, $d = e + \delta + O(\delta^{m-1})$.
- Steps 3, 4, and 5 are largely unchanged.

□

Sketch of proof of Theorem 8.18 This generalization of Theorem 8.6 requires less modification than the generalization of Theorem 8.2. The first step that is different is the proof of Lemma 8.12. The operator G is now the $(m-1)$ st power of g instead of the square. Defining G_1 to be the portion coming from $e + g_1$ and G_2 to be the rest, we obtain the expansions

$$\begin{aligned} G_1(x, y) &= e^{m-1} + \lambda_1^{m-1} v_1(x) v_1(y) + \text{terms involving } \int_0^1 v_1(x) dz \\ &= e^{m-1} + \lambda_1^{m-2} (e + g_1(x, y)) + O(\delta) \\ &= \lambda_1^{m-2} (e + g_1(x, y)) + e^{m-1} - \lambda_1^{m-2} e + O(\delta). \end{aligned}$$

and

$$\begin{aligned} G_2(x, y) &= e^{m-2} (d_2(x) + d_2(y)) + \text{smaller terms involving } d_2 \\ &\quad + g_2^{m-1}(x, y), \end{aligned}$$

where the last term is the kernel of the $m - 1$ st power of g_2 as an operator, not $g_2(x, y)^{m-1}$. The resulting arguments involving Lagrange multipliers are different in detail but the same in form.

Most of the cost-benefit analysis involves analyzing the entropy, not the triangle count, and so is unchanged. The key observation regarding the triangle count is that λ_i^3 contributes negligibly, so that (to lowest order) the $O(\alpha^2)$ cost of having eigenvalues beyond λ_1 is not offset by anything having to do with the triangle count. This remains the case with m -cycles, as the contribution of the i -th eigenvalue to t_H is λ_i^m , which is even smaller than λ_1^3 .

□

Sketch of Proof of Theorem 8.19 First we establish that an optimal graphon at $t_H = 0$ must be symmetric bipodal. This is because any graphon with a positive density of triangles also has a positive density of odd m -cycles, so any graphon with $\tau_H(g) = 0$ must also have $\tau(g) = 0$. But the unique maximum entropy for graphons with $\tau(g) = 0$ is symmetric bipodal, and that graphon has $\tau_H(g) = 0$, so it is the unique entropy maximizer among graphons with $\tau_H(g) = 0$.

Next we need to modify our pointwise variational equations and the definition of worth. As in the proof of Theorem 8.18, the operator G gets replaced by a multiple of the $(m-1)$ st power of the operator g . Similarly, $g(x, y)$ in the β term in the definition of $W(C)$ gets replaced by a multiple of the $(m-2)$ nd power of g .

The rest of the proof proceeds like the proof of Theorem 3.3. What t_H is small, an optimal graphon must be L^2 -close to a symmetric bipodal graphon. This gives good enough estimates of g^{m-2} and g^{m-1} to show that any worth-maximizing column must be L^2 -close to a column of a symmetric bipodal graphon. This divides the unit square into four rectangles on which the graphon is close to constant. Using the pointwise variational equations, we show that the fluctuation of g on each rectangle is bounded by a small multiple of itself, and so must be zero. Once we know that the optimal graphon is exactly bipodal, we do some calculations in the 4-dimensional space of bipodal graphons to show that symmetric graphons have more entropy than non-symmetric graphons. □

The role of symmetry in the edge- H model is similar to that in the edge-triangle model. The reduced 2star density $\tilde{t}_2$ is an order parameter that distinguishes between the B(1,1)-like and A(2,0)-like phases. The order parameters of Section 4.5 can distinguish between phases characterized by graphons of different rank, as are likely to appear near the bottom boundary of the space of possible (e, t_H) values.

Sketch of Proof of Theorem 8.20 This is the simplest modification of all. The

graphon (8.25) considered in the proof of Theorem 8.13 has m -cycle density $e^m + c^m(B^m - A^m)$, so $c = \delta(A^m - B^m)^{-1/m}$. We therefore change the denominator of $F(A, B)$ from $(A^3 - B^3)^{2/3}$ to $(A^m - B^m)^{2/m}$. Since B is eventually chosen to be proportional to A^2 , the denominator changes from $A^2 + O(A^5)$ to $A^2 + O(A^{2+m})$. The $O(A^5)$ term was already too small to affect anything, so this change has absolutely no impact on the calculation. The rest of the proof of Theorem 8.13 can be copied verbatim. $\square$

8.7 The edge- H model in general

Suppose that H is a fixed graph with V vertices and E edges. An Erdős-Rényi graphon g with edge density e will have $\tau_H(g) = e^E$. In this section, we consider the edge- H model for values of t_H slightly below e^E , especially when $e < 1/2$.

The results depend strongly on the cycles that are part of H . There are three possibilities.

1. H is bipartite, possibly with even cycles but with no odd cycles.
2. H has at least one odd cycle, but the smallest cycle is even.
3. H has one or more cycles, the smallest of which is odd.

In the first case, t_H is conjectured by Sidorenko [63] to be bounded below by e^E , so there is nothing to consider. (Sidorenko's conjecture is easy to prove when H is a tree or an even cycle, but it has not been proven in general.) In the second case, or in the case of counterexamples to Sidorenko's conjecture, we will show that there is a discontinuous phase transition at the Erdős-Rényi curve. In the third case, we will show that the model for t_H just below Erdős-Rényi behaves qualitatively like the edge- m -cycle model, where m is the size of the smallest cycle in H .

In both the second and third cases, it is useful to consider a graphon g that is close, in an L^2 sense, to Erdős-Rényi. That is, we write

$$g(x, y) = e + g_1(x, y) + g_2(x, y), \quad (8.26)$$

where

$$g_1(x, y) = d(x) + d(y) - 2e$$

and $g_2(x, y)$ has zero marginals:

$$\int_0^1 g_2(x, y) dy = 0 \text{ for all } x \in [0, 1].$$

To see that the decomposition (8.26) is possible, note that $e + g_1$ has the same degree function as g , and define g_2 to be $g - (e + g_1)$.

To understand the effect on $\tau_H(g)$ of having g_1 or g_2 nonzero, we expand the product

$$\tau_H(g) = \int \prod_{(i,j)} g(x_i, x_j) d^V x = \int \prod_{(i,j)} (e + g_1(x_i, x_j) + g_2(x_i, x_j)) d^V x \quad (8.27)$$

in terms of integrals over products of e , g_1 and g_2 .

Proposition 8.21 *In the expansion of equation 8.27, $\tau_H(g)$ receives*

- A contribution e^E from associating every edge with e ,
- Positive contributions of order $\Theta(\|g_1\|_2^2)$,
- Additional contributions of magnitude $O(\|g_1\|_2^2 \|g_2\|_2)$ and
- Additional contributions of magnitude $O(\|g_2\|_2^m)$, where m is the size of the smallest cycle in H .

Note that the third set of contributions is necessarily smaller than the second. In order to have $\tau_H(g) < e^E$, where E is the number of edges in H , the fourth set must be greater than the second, so $\|g_1\|_2$ must be $O(\|g_2\|_2^{m/2})$. While g_1 does not have to vanish, undersaturation mostly involves g_2 and the cycles of H .

Proof As noted in Lemma 7.10, the change in $\tau_H(g)$ relative to Erdős-Rényi is controlled, to within $O(\|g_1 + g_2\|_2^3)$, by the change in a positive linear combination of k star densities t_k . Since each k star density $\tilde{t}_k$ depends only on g_1 and is $\Theta(\|g_1\|_2^2)$, there is a positive $\Theta(\|g_1\|_2^2)$ contribution to $\tau(g)$.

Next we consider the effect of g_2 . For each term in our expansion of $\tau_H(g)$, let $H' \subset H$ be the set of edges to which we associate a factor of g_2 (as opposed to e or g_1). If H' is non-empty and does not contain a cycle, then it has at least two ends. Suppose that (i, j) is an edge, with j being an end. Since the marginals of $g_2(x_i, x_j)$ are zero, integrating over dx_j gives zero unless there is another edge (j, k) to which we associate g_1 . Moreover, a single edge cannot serve two different ends, since $\iint g_2(x_i, x_j) g_1(x_j, x_k) g_2(x_k, x_\ell) dx_j dx_k$ equals

$$\begin{aligned} & \iint g_2(x_i, x_j) (d(x_j) - e) g_2(x_k, x_\ell) dx_j dx_k \\ & + \iint g_2(x_i, x_j) (d(x_k) - e) g_2(x_k, x_\ell) dx_j dx_k. \end{aligned}$$

The first integral vanishes when we integrate over x_k , while the second vanishes when we integrate over x_j .

To get a nonzero contribution from an H' that does not contain a cycle,

we must have at least two edges associated with g_1 , so the contribution is $O(\|g_1\|_2^2 \|g_2\|_2)$. These contributions, which may be positive or negative, are dominated by the positive contributions associated with g_1 alone.

As for the terms that involve g_2 but not g_1 , the lowest order contributions are those where H' is the smallest possible cycle, namely an m -cycle. In that case, the contribution is a positive multiple of $\text{tr}(g_2^m) = \sum \lambda_i^m$, where $\{\lambda_i\}$ are the eigenvalues of g_2 . When m is even, this is necessarily positive. When m is odd, it may be positive or negative.

Finally, if H' contains a cycle (of length m or larger), and if there is at least one edge associated with g_1 , then this term contributes $O(\|g_1\|_2 \|g_2\|_2^m)$. Such terms can be absorbed into the $O(\|g_2\|^m)$ contributions of the previous paragraph. $\square$

Proposition 8.22 *Suppose that H contains an m -cycle and no smaller cycle. If m is even, then there is an $\epsilon > 0$ such that, if $0 < \|g_1 + g_2\|_2 < \epsilon$, then $\tau_H(g) > e^E$.*

Proof In the expansion of (8.27), we have

- The leading term e^E ,
- Positive terms of order $\Theta(\|g_1\|_2^2)$ from changes to the k star densities,
- Positive terms proportional to $\text{tr}(g_2^m) = \sum \lambda_i^m$ from m -cycles in H , and
- Corrections of order $\|g_1 + g_2\|_2^3$, $\|g_1\|_2 \|g_2\|_2^m$ and $\|g_2\|_2^{m+1}$.

When $g_1 + g_2$ is nonzero and small, the positive terms dominate the corrections and we have $\tau_H(g) > e^E$. $\square$

Theorem 8.23 *Suppose that $e < 1/2$, that H contains both even and odd cycles, and that the smallest cycle in H is even. Then the edge- H model has a discontinuous phase transition at the Erdős-Rényi curve. Specifically,*

- All values of t_H between 0 and e^E are possible, as are all values slightly above e^E .
- The limit of the Boltzmann entropy $B(e, t_H)$ as t_H approaches e^E from below is strictly less than $H(e)$, while the limit as t_H approaches e^E from above is equal to $H(e)$.

Proof Let m_e and m_o be the size of the smallest even and odd cycles in H . Consider the family of (symmetric) bipodal graphons with $a = b = e - \delta$, $c = 1/2$ and $d = e + \delta$, as δ ranges from 0 to e . For this family, the density of m_o -cycles is $e^{m_o} - \delta^{m_o}$, which goes to zero as $\delta \rightarrow e$. Since the density of H 's is bounded by the density of m_o -cycles, $\tau_H(g)$ must go from e^E to 0 as δ goes

from 0 to e . Since τ_H is a continuous function of δ , every value between e^E and 0 must be achievable.

Now consider any graphon g with $\varepsilon(g) = e$ and $\tau(g) < e^E$. By Proposition 8.22, we must have $\|g - e\|_2 \geq \epsilon$, so $S(g) \leq H(e) + f(e, 1 - e)\epsilon^2$. In particular, $B(e, t_H) \leq H(e) + f(e, 1 - e)\epsilon^2$ for all $t_H < e^E$.

Finally, consider a family of bipodal graphons with $a = e + \delta$, $b = e - \delta$, and $c = d = 1/2$. This family is **not** entropy-maximizing, but it does achieve values of t_H that are greater than e^E , with an entropy of

$$H(e) + \frac{1}{4}H''(e)\delta^2 + O(\delta^4) = H(e) - \Theta(t_H - e^E).$$

This entropy goes to $H(e)$ as t_H approaches e^E from above. Since the actual value of $B(e, t_H)$ is sandwiched between this quantity and $H(e)$, the limit of $B(e, t_H)$ as t_H approaches e^E from above is $H(e)$. $\square$

The theorem states that $B(e, t_H)$ jumps discontinuously as t_H crosses e^E , but perhaps a more important point is that the form of the optimal graphon jumps, implying that many other quantities, such as the densities of triangles and of other embedded subgraphs, also jump.

The assumption that $e < 1/2$ is important. If $e > 1/2$, then it may not be possible to decrease the density of m_o -cycles by enough to make up for the simultaneous increase in m_e -cycles. For instance, if H contains 1000 squares and one pentagon, and if e is close to 1, then it appears likely that the minimum value of t_H is e^E .

Finally, we consider what happens when there are no even cycles in H that are smaller than the smallest odd cycle.

Theorem 8.24 *Suppose that the smallest odd cycle in H is an m -cycle, and that there are no even cycles smaller than m . Then, for any $e \in (0, 1)$, all values of t_H in a neighborhood of e^E are possible, and*

$$B(e, t_H) = H(e) - \Theta\left((e^E - t_H)^{2/m}\right)$$

as t_H approaches e^E from below.

Proof The families of bipodal graphons considered in the proof of Theorem 8.23 show that it is possible both to increase and to decrease t_H near e^E .

In the case of the symmetric bipodal graphon, $e^E - t_H$ is a constant multiple of δ^m (with the multiple depending on e and on the number of m -cycles in H) plus $O(\delta^{m+1})$. The entropy of this graphon is then

$$H(e) + \frac{1}{2}H''(e)\delta^2 + O(\delta^4) = H(e) - \Theta\left((e^E - t_H)^{2/m}\right),$$

which gives us a lower bound for $B(e, t_H)$. For the upper bound, observe that $e^E - t_H = O(\|g_2\|_2^m) = O(\|g - e\|_2^m)$ and that

$$S(g) \leq H(e) + f(e, 1 - e)\|g - e\|_2^2.$$

Combining these observations, we get that

$$B(e, t_H) = H(e) - \Theta\left((e^E - t_H)^{2/m}\right)$$

when $t_H < e^E$. $\square$

In specific regimes, we can be more precise in carrying over our results for m -cycles to general H . When $e > 1/2$ and t_H is slightly less than e^E , the bipodal graphon of Theorem 8.17 maximizes the entropy to leading order in $e^E - t_H$. The proof of Theorem 8.2 was delicate, so this observation is not enough to prove that there is a $B(1,1)$ -like phase. However, we conjecture that there is:

Conjecture 8.25 *Suppose that the smallest odd cycle in H is an m -cycle, and that there are no even cycles smaller than m . When $e > 1/2$ and $e^E - t_H$ is sufficiently small, the entropy maximizing graphon is unique and bipodal, satisfying the estimates (8.1), where δ is a constant (depending of e) times $(e^E - t_H)^{1/m}$. That is, there is a $B(1,1)$ -like phase.*

At the other end, we have

Theorem 8.26 *Suppose that the smallest odd cycle in H is an m -cycle, and that there are no even cycles smaller than m . If $e < e_0 = (3 - \sqrt{3})/6$, and if t_H is sufficiently close to, but less than, e^E , then the entropy optimizing graphon is not symmetric bipodal.*

Sketch of proof We just need to compare the entropies of the symmetric bipodal graphon and the graphons considered in the proof of Theorem 8.13. In that comparison, we achieve a graphon that has more entropy, to leading order in δ , than the symmetric bipodal graphon. The argument is unchanged if we replace the m -cycle density with a multiple of the m -cycle density, or if t_H contains contributions that are higher-order in our parameters (λ_1 and λ_2 for one graphon, δ for the other). $\square$

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Appendix A

Numerical investigations

The numbered chapters in this book have all consisted of rigorous mathematics. Besides laying out the uniform random graph formalism for studying large constrained random graphs, we proved the existence of a number of phases in the edge-triangle model:

- The bipodal $F(1,1)$ phase, just above the Erdős-Rényi curve and also just below the upper boundary $t = e^{3/2}$,
- The bipodal $A(2,0)$ phase, near $e = 1/2$ and below the Erdős-Rényi curve and also just above line segment $t = 0$ for $0 < e < 1/2$,
- The bipodal $B(1,1)$ phase, just below the Erdős-Rényi curve when $e > 1/2$, and
- The $(n + 2)$ -podal $C(n,2)$ phase, just above the n -th scallop.

However, there are many questions that have defied rigorous proof, at least so far. We have not determined the extent within the phase space of these phases, the form of the remaining phases, or the nature of the transitions between them.

In this appendix, we take a more heuristic approach to those questions. In part, we describe the results of a series of illuminating numerical experiments. In part, we prove theorems with strong underlying assumptions. For instance, we might assume the existence of two phases and prove that any transition between them must be discontinuous. While neither approach belongs in a “prove everything from scratch” treatise, both provide insight on what is actually happening in the edge-triangle model.

A.1 The phase diagram

The most basic question concerns the structure of our phase diagram. In this section we explain the details of the conjectured phase diagram of the edge-

triangle model, repeated here as Figure A.1, the numerical evidence supporting this conjecture over most of the phase diagram, and a small region where the conjecture turned out to be incorrect.

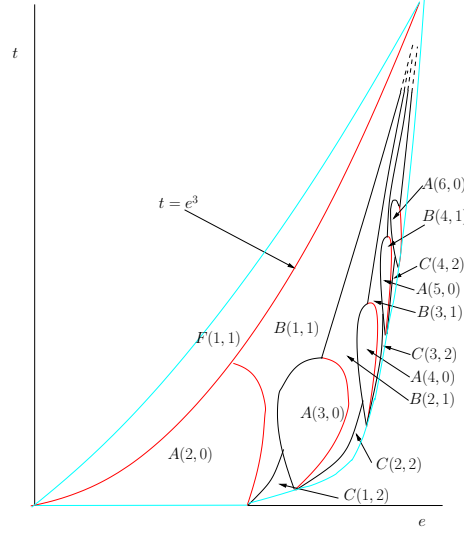


Figure A.1 Conjectured phase diagram of the edge-triangle model. Red curves indicate continuous phase transitions at which there is a unique optimal graphon. Black curves indicate discontinuous phase transitions at which there are two different graphons with the same entropy.

The numerical experiments of [51] (coauthored with K. Ren) suggested that

1. All entropy-maximizing graphons are multipodal.
2. There are three infinite families of phases for undersaturated graphs, the *A* family with $(n, 0)$ symmetry, the *B* family with $(n, 1)$ symmetry, and the *C* family with $(n, 2)$ symmetry. The set of all $A(2,0)$ graphons is described by two parameters that are completely determined by the edge and triangle densities. The space of $B(n,1)$ graphons is 5-dimensional when $n > 1$ and is 4-dimensional when $n = 1$. The space of $C(n,2)$ graphons is 6-dimensional when $n > 1$ and 5-dimensional when $n = 1$.
3. There is only one phase for supersaturated graphs, a phase that we call $F(1,1)$. While $F(1,1)$ has the same number of nodes as $B(1,1)$ and the same (lack of) symmetry, it is a distinct phase and is not an analytic continuation of $B(1,1)$.
4. Likewise, the $B(2,1)$ and $C(1,2)$ phases are distinct, despite having similar

decompositions into two podes of one type and a third pod of a different type.

5. Some phase transitions are continuous, featuring a unique entropy maximizer that is the limit of the structures of both phases. Other transitions are discontinuous, with two different entropy maximizers, each the limit of the optimizing graphons in one of the two phases.
6. The common boundary of two adjacent phases is a smooth curve. The common boundary of three or more phases is a single point.

A.1.1 Evidence for multipodality

The first remarkable discovery was that entropy maximizers seemed to always be multipodal. Of course every measurable function can be approximated, in an integral norm, by a piecewise constant function, but why should optimal graphons be *exactly* constant on rectangular regions?

In [51], we numerically sampled the entropy of bipodal, tripodal, . . . , up to 16-podal graphons with prescribed edge and triangle densities. If the optimal graphon was not multipodal, then it would be better approximated by a 16-podal graphon than by a smaller multipodal graphon. The maximum entropy would then be a strictly increasing function of the allowed podality, approaching the true maximum as the podality went to infinity.

Instead, in about 95% of the Razborov triangle, bipodal graphons did just as well as 16-podal graphons. In fact, the very best 16-podal graphons turned out to actually be bipodal, only with the two podes artificially divided into multiple pieces! In addition, there was a small region where the best graphon turned out to be tripodal, a still smaller region where the best graphon was 4-podal, and so on. The regions of relatively large podality tended to be near the “scallop”, especially for larger values of the edge density e .

What we observed experimentally is that the dominant structures on the boundary extend into the interior. The $F(1,1)$ phase is the extension of the clique structure at the top boundary. The $A(n,0)$ phases are extensions of the symmetric structures at the cusps and on the $t = 0$ part of the boundary. The $C(n,2)$ phases are extensions of the structures on the scallops. These observations about $F(1,1)$, $A(2,0)$ and $C(n,2)$ were eventually proved in 2024 [57], as presented in Chapter 4.

The $B(n,1)$ phases are somewhat different, as they only touch the boundary at the single point $(1, 1)$. They arise from symmetry breaking. One can perturb an $A(n,0)$ graphon to distinguish one pod from the others. Since the two parameters of an $A(n,0)$ graphon are completely determined by (e, t) , we can compute, as a function of (e, t) , the second derivative of the entropy with

respect to this perturbation. Where the second derivative is negative, the $A(n,0)$ graphon is stable. Where it becomes positive, the $A(n,0)$ graphon becomes unstable and transitions to a $B(n-1,1)$ graphon.

This transition, analogous to a bifurcation, occurs continuously. At the boundary, there is a unique optimal graphon with n -fold symmetry. As one moves into the $B(n-1,1)$ region, that symmetry is gradually broken. In [52], we assumed the existence of the $A(2,0)$ phase and computed the curve along which $A(2,0)$ becomes unstable. This is presumably the exact location of the $A(2,0) \leftrightarrow B(1,1)$ phase transition. However, this result must be viewed through the same heuristic lens as the rest of this appendix.

A.2 Discontinuous phase transitions

The conjectured phase diagram contains three kinds of continuous phase transitions:

- Between $F(1,1)$ and $A(2,0)$,
- Between $F(1,1)$ and $B(1,1)$, and
- Between $A(n,0)$ and $B(n-1,1)$.

The first two occur at the Erdős-Rényi curve. There, the unique entropy maximizing graphon $g(x, y) = e$ is the limit of a $B(1,1)$ or $F(1,1)$ graphon as the size of the small node goes to zero. It is also the limit of an $A(2,0)$ graphon as the values of the graphon on the four quadrants all approach e .

As noted above, the $A(n,0) \leftrightarrow B(n-1,1)$ transition involves symmetry breaking, with one node gradually becoming different or the same as the other $n-1$. At the boundary, the entropy maximizing graphon is unique and symmetric.

In the diagram, there are four other kinds of transitions between the A , B and C families of phases:

1. Between $B(n,1)$ and $C(n,2)$,
2. Between $B(n,1)$ and $A(n+2,0)$,
3. Between $B(n,1)$ and $B(n+1,1)$, and
4. Between $C(n,2)$ and $A(n+2,0)$.

Theorem A.1 *There are no continuous transitions between $B(n,1)$ and $C(n,2)$ or between $B(1,1)$ and $B(2,1)$, except possibly at a finite number of values of the edge density.*

Theorem A.2 *There are no continuous transitions between $A(n+2,0)$ and $B(n,1)$ for any n , or between $B(n,1)$ and $B(n+1,1)$ for $n > 1$.*

Note that Theorems A.1 and A.2 do not purport to prove the existence of (say) the A(4,0) and B(2,1) phases, nor do they prove that these two phases meet at a phase boundary. (In principle, there could be an as-yet-undiscovered third phase that separates them.) The theorems merely state that *if* these phases exist and meet at a common boundary, then the transition must be discontinuous.

The caveat in Theorem A.1 about finitely many values of the edge density is an artifact of the proof. Numerical investigations have not revealed any points of continuity when transitioning between B(n ,1) and C(n ,2) or between B(1,1) and B(2,1); we do not believe that any exist.

Proof of Theorem A.1 To illustrate the arguments, we will do the proof in three passes, first for B(1,1) $\leftrightarrow$ C(1,2), then for B(1,1) $\leftrightarrow$ B(2,1), and finally for B(n ,1) $\leftrightarrow$ C(n ,2) with $n > 1$.

There certainly are paths in $\widetilde{\mathcal{W}}$ that move continuously from B(1,1) graphons to C(1,2) graphons. At some point, one simply splits one of the podes in half. However, the entropy cost of splitting a pode continuously is extremely high, as we saw in Chapter 8. Decreasing the triangle count by Δt in this way comes with an entropy cost of order $(\Delta t)^{2/3}$.

Now suppose that we have such a path and that each graphon on the path is an entropy maximizer for its values of (e, t) , so that $dS = \alpha d\varepsilon + \frac{\beta}{3} d\tau$ for some Lagrange multipliers α and β . As we approach the transition point from the C(1,2) side, the Lagrange multiplier β must diverge. On the B(1,1) side, an infinite value of β corresponds to the so-called “natural boundary” of the B(1,1) phase, namely the point that minimizes $t = \tau(g)$ among all bipodal graphons with fixed edge density $e = \varepsilon(g)$. At this point, the graphon must take the form

$$g(x, y) = \begin{cases} p & x, y < c, \\ 1 & x < c < y \text{ or } y < c < x, \\ 0 & x, y > c, \end{cases} \quad (\text{A.1})$$

for some parameters p and c . The edge and triangle densities and entropy are then

$$\begin{aligned} e = \varepsilon(g) &= pc^2 + 2c(1 - c) \\ t = \tau(g) &= 3pc(1 - c)^2 + p^3c^3 \\ S(g) &= c^2H(p) \end{aligned}$$

Fixing e still leaves us with one degree of freedom, which we use to minimize t .

Since the entropy is given by an analytic function of (p, c) than is analytically independent of t and e , the values of (p, c) that minimize t do not maximize S (except possibly at coincidences at finitely many values of e). This means that,

at the values of (p, c) that minimize t , we can increase the entropy by δ while increasing t by $O(\delta^2)$. We will use this freedom to show that the τ -minimizing bipodal graphon (A.1) does not maximize entropy for fixed edge and triangle densities.

Starting with the graphon (A.1), vary p and c to hold the edge density fixed and to increase the entropy by δ , at a cost of increasing the triangle density by $O(\delta^2)$. Then decrease the triangle density by splitting the pde on $[0, c]$ in half, decreasing the value of $g(x, y)$ by $\mu = O(\delta^{2/3})$ on $[0, c/2] \times [0, c/2]$ and on $[c/2, c] \times [c/2, c]$ and increasing it by the same amount on $[0, c/2] \times [c/2, c]$ and $[c/2, c] \times [0, c/2]$. See Figure A.2. This brings $\tau(g)$ back down to t at an entropy cost of $O(\mu^2) = O(\delta^{4/3}) < \delta$. Since the entropy cost of splitting the pde by μ is smaller than the entropy gain of varying p and c , the original graphon was not entropy maximizing.

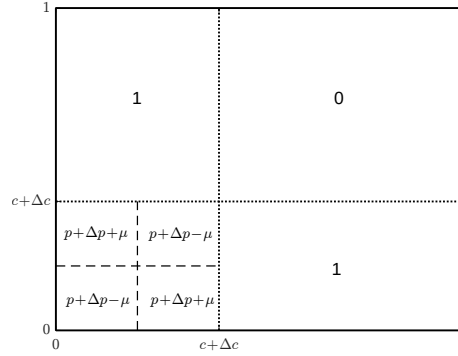


Figure A.2 An improvement to a triangle-minimizing $B(1,1)$ graphon. The quantities Δc and Δp are $O(\delta)$ while μ is $O(\delta^{2/3})$.

The argument for the $B(1,1) \leftrightarrow B(2,1)$ transition follows the same path. Any transition between the two phases would have to involve splitting one of the $B(1,1)$ podes in half. The divergence of β , the form of a putative optimal graphon on the boundary, and the moves needed to show that this graphon is not actually optimal are word-for-word identical.

The argument for the $B(n,1) \leftrightarrow C(n,2)$ transition is also almost identical. At the boundary point, the Lagrange multiplier β must diverge, forcing us to have a graphon that minimizes $\tau(g)$ among $B(n,1)$ graphons. The resulting graphon can take the value p on the odd pde $[0, c]$, but must take on the values 0 and 1 everywhere else. The specific formulas for e , t and S in terms of p and c are different than in $B(1,1)$, but the overall picture is the same. The values of p and c that minimize t are not the same as those that minimize S , except possibly

at finitely many values of the edge density. By varying p and c to increase the entropy by δ at a cost of increasing τ by $O(\delta^2)$, and by then splitting the odd podes in half by an amount $\mu = O(\delta^{2/3})$, we see that graphons at the natural boundary of the $B(n,1)$ phase are not entropy maximizers.

□

From the proof we see that the *actual* boundary between the $B(n,1)$ and $C(n,2)$ phases must occur along a curve that is strictly above the “natural boundary” of the $B(n,1)$ phase. At the actual boundary, there are two distinct entropy-maximizing graphons, one with $B(n,1)$ symmetry and one with $C(n,2)$ symmetry. Both have finite values of β , with β for the $B(n,1)$ graphon being larger than β for the $C(n,2)$ graphon.

Proof of Theorem A.2 Proving the first statement about the $B(n,1) \leftrightarrow A(n+2,0)$ transition does not require any considerations of entropy. Just of symmetry. There is simply no way to continuously change a $B(n,1)$ graphon, with $n+1$ podes, into an $A(n+2,0)$ graphon with $(n+2)$ identical podes.

The $B(n,1) \leftrightarrow B(n+1,1)$ transition is slightly subtler. There *is* a way to continuously change a $B(n,1)$ graphon into a $B(n+1,1)$ graphon, but only by first becoming an $A(n+1,0)$ graphon. In principle, a $B(n,1)$ graphon could become an $A(n+1,0)$ graphon by having the odd podes become indistinguishable from the others, while a $B(n+1,1)$ graphon could become an $A(n+1,0)$ graphon by having the odd podes shrink to zero size. However, the $B(n+1,1)$ phase does not touch the $A(n+1,0)$ phase! So on the boundary between the $B(n,1)$ and $B(n+1,1)$ phases, there are no points where the optimal graphon has $A(n+1,0)$ symmetry. As a result, there cannot be a unique entropy maximizing graphon on that boundary that is a limit both of $B(n,1)$ and $B(n+1,1)$ graphons. □

Finally, we consider transitions between $C(n,2)$ and $A(n+2,0)$. Numerical investigations indicate that these transitions are discontinuous. However, there is not yet any rigorous proof that continuous transitions do not occur.

A.3 Something is missing from the diagram

Unfortunately, the phase diagram of Figure A.1 is incomplete. As we saw in Theorem 8.13, the $A(2,0)$ phase does not extend up to the Erdős-Rényi curve when $e < e_0 \approx 0.2113$. The region just below the Erdős-Rényi curve for $e < e_0$ presumably contains at least one phase that is not bipodal. It might even admit an infinite family of phases. Preliminary numerical investigations [17] suggest that

- This region is extremely thin, only including values of $t \in (0.998e^3, e^3)$. It is especially thin as e approaches e_0 , shrinking as $\Theta((e_0 - e)^3)$.
- Most of the region is a tripodal phase, with the (unique) optimal graphon closely resembling the ansatz (8.25), with the values of A and B chosen to maximize the function $F(A, B)$.
- The transition between this tripodal phase and the $A(2,0)$ phase is discontinuous.
- Besides the tripodal phase, there is at least one more phase associated with values of e below 0.024. The nature of the optimal graphon in this corner is not known.
- We can neither demonstrate nor rule out the existence of an infinite family of phases accumulating near $(0, 0)$, much as the A , B and C families accumulate near $(1, 1)$.

We will discuss the possible implications of these phases on moderate deviations of Erdős-Rényi graphs in Appendix B.

This leaves plenty of open questions for future work. Some involve directly examining graphons with e small. Others are farther-reaching. Here is an open problem concerning graphons with $e > 1/2$.

The transition from $B(n,1)$ to $C(n,2)$ involves one node splitting in two, much as an Erdős-Rényi graphon becomes an $A(2,0)$ graphon. However, the transition from Erdős-Rényi to $A(2,0)$ is not direct when $e < e_0$, passing instead through a phase featuring tripodal graphons. Is it possible that the $B(n,1) \rightarrow C(n,2)$ transition also involves a quadrant of the optimal graphon similarly becoming tripodal before eventually becoming bipodal? We do not know the answer; we invite you to find out.

A.4 Spectral analysis of finite graphs

Next we consider the structure of *finite* random graphs with constrained edge and triangle densities, with n ranging from several hundred down to about 30. The leading eigenvalues and eigenvectors of the adjacency matrix are key. Since the adjacency matrix is (with high probability) close to the entropy-maximizing graphon, the eigenvalues of the adjacency matrix should be n times the eigenvalues of the graphon, plus fluctuations of order $\sqrt{n}$. When the eigenvalues of the graphon are nonzero, eigenvectors of the adjacency matrix should be close to the corresponding eigenvectors of the graphon, which of course are constant on nodes. If we track k different eigenfunctions, then the values of these functions map each vertex to a point in $\mathbb{R}^k$. The extent to which

the points cluster gives a geometric picture of the extent to which the graph can be called multipodal as well as identifying the population of vertices associated with each pde.

A.4.1 Probing the B(1,1) $\leftrightarrow$ A(3,0) transition at $n = 54$

For instance, consider a graph sampled from the A(3,0) phase. A general A(3,0) graphon takes the form

$$g(x, y) = \begin{cases} e - 2\delta & (x, y) \in [0, 1/3]^2 \cup [1/3, 2/3]^2 \cup [2/3, 1]^2, \\ e + \delta & \text{elsewhere} \end{cases}$$

and has rank 3. The nonzero eigenvalues are e (with multiplicity one) and $-\delta$ (with multiplicity two). The eigenfunction with eigenvalue e is constant and so does not help to distinguish vertices, so we study the eigenspace corresponding to the negative eigenvalue. This eigenspace is spanned by the two eigenfunctions

$$u_1(x) = \sqrt{\frac{3}{2}} \begin{cases} 1 & x \in [0, 1/3] \\ -1 & x \in [1/3, 2/3] \\ 0 & x \in [2/3, 1] \end{cases}$$

$$u_2(x) = \frac{\sqrt{2}}{2} \begin{cases} 1 & x \in [0, 2/3] \\ -2 & x \in [2/3, 1] \end{cases}$$

A typical random graph should then have an adjacency matrix with three eigenvalues of order n and the rest of order $\sqrt{n}$. We will ignore the positive eigenvalue $\lambda_0 \approx ne$ and concentrate on the two most negative eigenvalues λ_1 and λ_2 and the corresponding eigenvectors v and w . We should expect both λ_1 and λ_2 to be close to $-n\delta$ and for v and w to be approximated by linear combinations of u_1 and u_2 sampled at n points $x_1, \dots, x_n$. If we do a scatter plot of (v_i, w_i) as i ranges from 1 to n , we should expect the points to cluster around three locations that form an equilateral triangle.

Not surprisingly, that is exactly what happens. Figure A.3 shows the scatter plot for a fairly typical graph with $n = 54$ and $(e, t) = (0.67, 0.24)$. The clustering is sharpest when n is large, but is usually still clear when $n = 30$.

By contrast, Figure A.4 shows a typical graph with $n = 54$ and $(e, t) = (0.67, 0.26)$. This graph has a structure similar to a B(1,1) graphon. A bipodal graphon can only have one negative eigenvalue, so λ_2 is only of order $\sqrt{n}$, with an eigenvector w that is close to zero on one pde and is random on the other.

There are three key differences between A(3,0) and B(1,1) graphs:

1. In an A(3,0) graph the two largest negative eigenvalues λ_1 and λ_2 of the

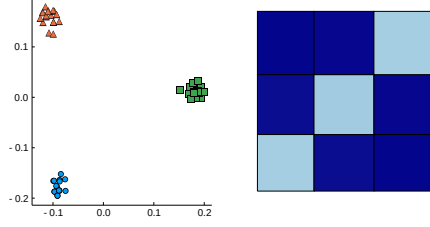


Figure A.3 Eigenvectors of a graph with $n = 54$ and $(e, t) = (0.67, 0.24)$. The three nodes are easily identified and the graph is described well by the tripodal graphon on the right.

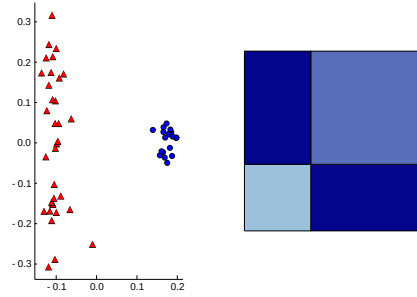


Figure A.4 Eigenvectors of a graph with $n = 54$ and $(e, t) = (0.67, 0.26)$. The graph is best described by the bipodal graphon on the right.

adjacency matrix have comparable size. In a $B(1,1)$ graph, λ_2 is much smaller (in magnitude) than λ_1 .

2. The spectral plot of an $A(3,0)$ graph has three clusters forming the vertices of an equilateral triangle, each corresponding to one of the three nodes. The spectral plot of a $B(1,1)$ graph has a cluster corresponding to one node and a vertical strip corresponding to the other.
3. The populations of the three clusters for the $A(3,0)$ graph are nearly equal. The populations for the two regions for the $B(1,1)$ graph are unequal, with the split depending on e . When $e = 0.67$, roughly 40% of the vertices are in one region and 60% are in the other.

In [44] (coauthored with J. Neeman), we used a Metropolis-Hastings Markov Chain Monte Carlo method to probe the transition between the $B(1,1)$ and $A(3,0)$ phases at $e = 0.67$. To generate graphs with edge density e and triangle

density t , we started with a graph with edge density e and arbitrary triangle density. We then moved edges from one randomly chosen pair of vertices to another randomly chosen pair, thereby keeping the edge density constant. These moves were allowed if they either decreased the number of triangles or resulted in a graph with triangle density t or less. This protocol converges to the uniform distribution on all graphs with triangle density t or less. However, the vast majority of such graphs actually have triangle density very close to t . So to sample graphs with edge density t , we merely had to sample from this ensemble and reject the occasional graph whose triangle density was significantly less than t .

The distribution of λ_2 for different values of t , all for $e = 0.67$ and $n = 54$, is shown in Figure A.5. When $t = 0.252$ or greater, most graphs have λ_2 between -6 and -8 , reflecting a B(1,1)-like structure. However, λ_2 is typically around -15 when t is 0.246 or less, reflecting an A(3,0)-like structure. That is, most graphs with triangle density $t = 0.250$ look like B(1,1), but there is a long tail extending in the A(3,0) direction. At $t = 0.248$ most graphs look like A(3,0), but there is a long tail in the B(1,1) direction. We can reasonably say that for $n = 54$ and $e = 0.67$, the transition between B(1,1) and A(3,0) occurs somewhere between $t = 0.248$ and $t = 0.250$.

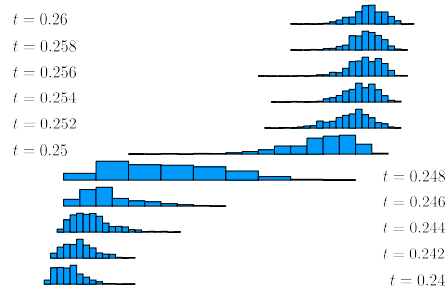


Figure A.5 Empirical distribution of λ_2 as a function of t for $n = 54$ and $e = 0.67$.

Note that the transition does not happen at a single sharply defined value of t . Discontinuous changes can only appear in the $n \rightarrow \infty$ limit. For any finite value of n , there is a transition region where both structures occur with non-negligible frequency. When $n = 54$, this transition region has size $\Delta t \approx 0.002$. It is an interesting open problem to determine how this size scales with n .

Not only does the thickness of the transition depend on n , but so does the location. When dealing with the $n \rightarrow \infty$ limit (in other words, with graphons), the transition for $e = 0.67$ occurs at $t = 0.2596$. However, when $n = 30$ it seems

to occur at a much lower value of t , around 0.24. As we have seen, it occurs between 0.248 and 0.250 when $n = 54$, and it occurs between 0.250 and 0.252 when $n = 60$. The smaller n is, the more easily the system manages to reduce the triangle count while maintaining a $B(1,1)$ -like structure.

More specifically, the $B(1,1)$ -like graphs tended to divide neatly into two podes. Each vertex in the first pod was connected to (nearly) every vertex in the second pod, as well as to some of the vertices in the first pod. There were almost no edges connecting vertices in the second pod. The overall picture was similar to that of Figure A.2, except that the first pod was not neatly split in half as in Figure A.2. Instead, the graph managed to somehow reduce the number of triangles in the first pod in ways that did not resemble splitting the pod.

These results suggest narrowing our focus to just the first pod. How can a finite graph that is nearly 1-podal reduce its triangle count short of transitioning to a bipodal phase? This is precisely the problem of moderate deviations of Erdős-Rényi graphs that will be discussed in Appendix B.

Appendix B

Moderate deviations

The concept of moderate deviations is very general and has generated countless papers. In this appendix, we are not going to attempt to cover all that. Instead, we will do something much more modest, reporting on some results about moderate deviations for undersaturated triangle and m -cycle counts in dense Erdős-Rényi $G(n, m)$ graphs, mostly from the papers [22] and [46].

B.1 What are moderate deviations of triangle counts?

Consider the ensemble $G(n, m)$ of all graphs with n vertices and m edges, where each graph is considered equally likely. Let $E = \binom{n}{2}$ be the number of possible edges and let $e = m/E$ be the edge density. The probability of any triple of vertices forming a triangle is

$$\binom{m}{3} / \binom{E}{3} = e^3 + O(n^{-2}),$$

so the expected number of triangles in the graph is

$$T_0 = \binom{n}{3} \binom{m}{3} / \binom{E}{3} = \frac{e^3 n^3}{6} + O(n^2).$$

The actual number T of triangles is a random variable with mean $\mu_T = T_0$ and variance σ_T^2 of order n^3 .¹

$O(n^{3/2})$ fluctuations in the triangle count, or equivalently $O(n^{-3/2})$ fluctuations in the triangle density, are called *small deviations* and are governed by

¹ In a $G(n, p)$ ensemble where each edge appears independently with fixed probability p , the variance of the triangle count scales as n^4 . However, most of the fluctuation in the triangle count is due to fluctuations in the number of edges. The *residual variance*, which measures fluctuations in the triangle count that are not “explained” by fluctuations in the edge count, scales as n^3 .

the Central Limit Theorem. At this scale, the probability distribution for the triangle count is well-approximated by a Gaussian of variance $\Theta(n^3)$, so the probability of raising or lowering the triangle count by ΔT or more is

$$\exp(-\Theta(n^{-3}(\Delta T)^2)).$$

Equivalently, the probability of raising or lowering the triangle density by Δt or more is

$$\exp(-\Theta(n^3(\Delta t)^2)).$$

$\Theta(n^3)$ fluctuations in the triangle count, or equivalently $\Theta(1)$ fluctuations in the triangle density, are called *large deviations*. These are governed by Chatterjee and Varadhan's Large Deviations Principle [15], or equivalently by the problem of finding an optimal graphon with triangle density $t = e^3 \pm \Delta t$. The probability of a fluctuation that raises or lowers the triangle density by Δt or more is governed by the Boltzmann entropy and goes as

$$\exp(n^2(B(e, e^3 \pm \Delta t) - H(e)) + o(n^2)),$$

with the sign depending on whether we are trying to increase or decrease the triangle count. As we have seen, this yields different formulas for increasing t ("upper tails") and for decreasing t ("lower tails"). Specifically, the probability for upper tails is $\exp(-\Theta(n^2\Delta t))$ and the probability for lower tails is $\exp(-\Theta(n^2(\Delta t)^{2/3}))$.

Fluctuations in the triangle density that are both $\Omega(n^{-3/2})$ and $o(1)$ are called *moderate deviations*. Such deviations in the triangle density are large compared to the standard deviation but small in absolute terms. Understanding moderate deviations for upper tails and lower tails are fundamentally different problems.

B.1.1 How big is big enough?

Back in Chapter 1, a motivating question for the edge-triangle model involved graphs with edge density $e = 0.6$ and triangle density $t = 0.215$ or 0.216 or 0.217 . We have seen that the entropy of the optimal graphon with edge density 0.6 and triangle density 0.216 is substantially higher than graphons with edge density 0.6 and triangle densities 0.215 or 0.217 . This means that among graphs with (say) $n = 10^9$ vertices and edge density 0.6 , there are many more graphs with triangle density 0.216 than with triangle density 0.215 or 0.217 . The difference in form between the optimal graphons at the three values also means that a typical graph with $t = 0.216$ looks different from one with $t = 0.215$ or 0.217 . This difference can be seen in the densities of other subgraphs such

as squares or tetrahedra as well as in the eigenvalues and eigenvectors of the adjacency matrix.

That description is fine for $n = 10^9$, which can reasonably be described as “close to ∞ ”. But what about smaller values of n ? Do optimal graphons still describe typical graphs when $n = 10^6$? What about $n = 1000$ or $n = 100$ or $n = 25$? How would the size needed to recognize the differences in structure change if we were comparing $t = 0.216$ to $t = 0.21599$ instead of to $t = 0.215$?

In general, there are three natural ways to think about scaling.

- We can fix Δt to a small value, like 10^{-3} or 10^{-6} or 10^{-9} , and vary n . When n is sufficiently large, Δt is a large deviation. When n is small, it is a small deviation. In between, it is a moderate deviation. At certain values of n we should see a qualitative change in the form of a typical graph. This is the “how big is big enough” problem.
- We can fix n to a large value, like 10^3 or 10^6 or 10^9 and vary Δt . As the name implies, large values of Δt are large deviations, small ones are small deviations, and intermediate ones are moderate deviations. As we saw in Appendix A in our exploration of the $B(1,1) \leftrightarrow A(3,0)$ phase transition, the form of a typical graph can change abruptly as t passes certain critical values, with the critical values of t depending on n .
- We can postulate a relation between n and Δt , such as “ $\Delta t = \Theta(n^{-\alpha})$ ” for some constant α , take a limit as $n \rightarrow \infty$, and compare results for different values of α . This is the setting for almost all of our theorems in this appendix. Precise mathematical statements almost always require taking a limit as $n \rightarrow \infty$.

B.2 Lower tails for triangles

We restrict our attention to lower tails, meaning fluctuations in the triangle density below e^3 . There are two reasons for this.

The first is theoretical. As we have seen, the phase structure in the edge-triangle model is far richer for undersaturated graphs than for supersaturated graphs. When t is slightly less than e^3 , we have the $B(1,1)$ phase when $e > 1/2$, it appears that we have the $A(2,0)$ phase when $e_0 < e < 1/2$, and we have something else entirely when $e < e_0$. When it comes to understanding moderate deviations as a limit of large deviations, lower tails are much more interesting than upper tails.

The second reason is far more prosaic. To date, there has been much less

progress on upper tails than on lower tails! When it comes to reporting about upper tails, there isn't very much to tell.

We follow the standard procedure of estimating the probability of getting a deviation of a certain size *or bigger*. Specifically, we fix an edge density e and work in the ensemble $G(n, m)$ of graphs on n vertices with m edges, where $m = e\binom{n}{2} + O(1)$. For a given $\Delta t > 0$ (which may depend on n in some way), we consider

$$P_\ell(n, \Delta t) := P\left(\frac{T}{\binom{n}{3}} < e^3 - \Delta t\right).$$

Since we are taking the uniform distribution on $G(n, m)$, this is the same as counting the number of graphs with n vertices, m edges, and fewer than $(e^3 - \Delta t)\binom{n}{3}$ triangles and dividing by $\binom{n}{m}$.

Here is what is known about lower tails of various sizes:

Theorem B.1 *If $\Delta t = O(n^{-3/2})$, then*

$$P_\ell(n, \Delta t) = \exp\left(-\frac{n^3(\Delta t)^2}{2e^3(1-e)^2(1+2e)}(1+o(1))\right).$$

Proof This is just the Central Limit Theorem, together with a computation of the variance of T . The calculation is not difficult, but neither is it necessary for our purposes. See [53]. $\square$

The best results for relatively small moderate deviations come from Goldschmidt et al [22]:

Theorem B.2 ([22]) *If $\Delta t = \Omega(n^{-3/2}) \cap o(n^{-1})$, then*

$$P_\ell(n, \Delta t) = \exp\left(-\Theta(n^3(\Delta t)^2)\right).$$

In other words, $o(n^{-1})$ moderate deviations have the same $\exp(-\Theta(n^3(\Delta t)^2))$ behavior as small deviations. This suggests that, if we vary Δt or n while holding the other fixed, the small deviation and moderate deviation regimes might well blend into one another without any abrupt change in the number of graphs or the form of a typical graph.

In [46], we studied relatively big moderate deviations.

Theorem B.3 ([46]) *If $\Delta t = \Omega(n^{-3/4}) \cap o(1)$, then*

$$P_\ell(n, \Delta t) = \exp\left(-\Theta(n^2(\Delta t)^{2/3})\right).$$

Sketch of proof One can view a graph as a random adjacency matrix. The proof in [46] involved applying random matrix theory to bound the eigenvalues of the difference between the adjacency matrix and its expectation. Combined

with technical estimates on the number of matrices with such eigenvalues, we obtained upper and lower bounds for $\ln(P_\ell)$, both of them scaling as $n^2(\Delta t)^{2/3}$. These technical estimates bear a strong resemblance to the calculations involving graphons (or more precisely, the difference between a graphon $g(x, y)$ and the constant e) that we saw in Chapter 8. $\square$

Finally, we have already established that large deviations are governed by our entropy functions. Since $B(e, t)$ is an increasing function of t for $t < e^3$, the probability of having a triangle density of t or less is controlled by the Boltzmann entropy $at(e, t)$, which in turn is the maximum of the Shannon entropy among all graphons with edge and triangle density (e, t) . That is, if $\Delta t = \Theta(1)$, then

$$P_\ell(n, \Delta t) = \exp \left(n^2 (B(e, e^3 - \Delta t) - H(e)) (1 + o(1)) \right).$$

Furthermore, we know that

$$B(e, e^3 - \Delta t) = H(e) - \Theta((\Delta t)^{2/3})$$

as $\Delta t \rightarrow 0$. Together, these imply that P_ℓ should behave as $\exp(-\Theta(n^2(\Delta t)^{2/3}))$ whenever Δt is small in absolute terms and n is large on a scale determined by Δt . That is, moderate deviations with $\Delta t = \Omega(n^{-3/4})$ behave like the $\Delta t \rightarrow 0$ limit of large variations, just as moderate deviations with $\Delta t = o(n^{-1})$ behave like the many-standard-deviations limit of small variations.

There is a gap between Theorems B.2 and B.3, with the first only going up to $\Delta t = o(n^{-1})$ and the second only going down to $\Omega(n^{-3/4})$. The formulas in the two theorems agree when $\Delta t = \Theta(n^{-3/4})$. We conjectured, and Alvarado, Dias and Griffiths proved, that this is where the system crosses over from behavior resembling small deviations to behavior resembling large deviations. Specifically:

Theorem B.4 ([4]) *For all values of Δt ,*

$$\begin{aligned} P_\ell(n, \Delta t) &= \exp \left(-\Theta(\min(n^3(\Delta t)^2, n^2(\Delta t)^{2/3})) \right) \\ &= \begin{cases} \exp(-\Theta(n^3(\Delta t)^2)) & \Delta t \in O(n^{-3/4}) \\ \exp(-\Theta(n^2(\Delta t)^{2/3})) & \Delta t \in \Omega(n^{-3/4}). \end{cases} \end{aligned}$$

B.3 Large e versus small e

Theorems B.2 and B.3 merely refer to $\ln(P_\ell)$ being bounded above and below by multiples of $n^3(\Delta t)^2$ or $n^2(\Delta t)^{2/3}$. When $e > 1/2$ it is possible to sharpen the estimates of Theorem B.3 considerably:

Theorem B.5 ([46]) *If $e \geq 1/2$ and $\Delta t = \Omega(n^{-3/4}) \cap o(1)$, then*

$$\ln(P_\ell(n, \Delta t)) = \frac{H'(e)}{2e-1} n^2 (\Delta t)^{2/3} (1 + o(1)).$$

The right hand side is exactly n^2 times the leading term in the expansion of $\Delta B := B(e, e^3 - \Delta t) - H(e)$ in Theorem 8.2.

Comments on proof Finding upper and lower bounds for $\ln(P_\ell)$ is similar to finding upper and lower bounds for $B(e, e^3 - \Delta t)$. This was the easy part of proving Theorem 8.2. We found upper bounds by minimizing $\|g - e\|_2$ for fixed Δt and by maximizing S for fixed $\|g - e\|$, while pretending that we could do both things with the same g . We found lower bounds by exhibiting an explicit graphon g . The upper and lower bounds agree to order $\Delta t^{2/3}$, giving us a precise leading-order approximation to $B(e, e^3 - \Delta t)$.

The proof of Theorem B.5 is similar. We find an upper bound for P_ℓ by combining a set of estimates that cannot actually be simultaneously saturated. We find a lower bound by constructing a class of graphs, similar to those generated by the optimal graphon in Theorem 8.2, and computing the rate of growth of this class. The upper and lower bounds agree to leading order, so we are done. $\square$

Things are trickier when $e < 1/2$. The upper bound that applies when $e \geq 1/2$ also applies when $e < 0$. Specifically,

$$\ln(P_\ell(n, \Delta t)) \leq \frac{H'(e)}{2e-1} n^2 (\Delta t)^{2/3} (1 + o(1)).$$

There are many classes of graphs that provide lower bounds proportional to $n^2 (\Delta t)^{2/3}$, but they do not have the same leading coefficient $\frac{H'(e)}{2e-1}$. This is related to the fact that we have not constructed any graphons with $e < 1/2$ for which $S(g) = H(e) + \frac{H'(e)}{2e-1} (\Delta t)^{2/3} + o((\Delta t)^{2/3})$. Indeed, we do not believe that any exist! Rather, the best known graphons when $e_0 < e < 1/2$ are symmetric bipodal, for which $S(g) = H(e) + \frac{1}{2} H''(e) (\Delta t)^{2/3} + O((\Delta t)^{4/3})$. One can improve on the coefficient $\frac{1}{2} H''(e)$ when $e < e_0$, but the best known constant is still less than $H'(e)/(2e-1)$.

The upshot is that there is a gap in our understanding, both for moderate deviations and for large deviations, when $e < 1/2$. We have upper and lower bounds on the change in entropy relative to Erdős-Rényi, quadratic in $\delta := (\Delta t)^{1/3}$, but the coefficients of the upper and lower bounds are not the same.

We conjecture that the two lacunae in our knowledge are in fact one, and that the behavior of moderate deviations with $\Delta t = \Omega(n^{-1})$ is always the same as the $\Delta t \rightarrow 0$ limit of large deviations.

Conjecture B.6 For all values of $e \in (0, 1)$ and for all $\Delta t \in o(1) \cap \Omega(n^{-1})$,

$$\ln(P_\ell(n, \Delta t)) = n^2(B(e, e^3 - \Delta t) - H(e))(1 + o(1)).$$

This conjecture underscores the importance of understanding the mystery phases discussed in Sections 8.5 and A.3. While the region supporting these phases is extremely thin, consisting of only a tiny range of t values for each $e < e_0 \approx 0.2113$, the Boltzmann energy function in these phases and the form of the optimal graphons are likely to control moderate deviations of lower tails for all $e < e_0$. Taking a further limit as $e \rightarrow 0$ is likely to be relevant to understanding moderate deviations of *sparse* Erdős-Rényi graphs.

B.4 Other m -cycles

As in Chapter 8, we can study edges and m -cycles, where m is an odd integer, rather than edges and triangles. As with Theorems 8.16–8.20, the estimates are similar to those in the edge-triangle model, but are in fact sharper. This yields the following analogue of Theorem B.4.

Theorem B.7 ([46]) Let $m > 3$ be an odd integer and let $\Delta t = \delta^m = \Omega(n^{-1}) \cap o(1)$. Let $P_m(n, \Delta t)$ be the probability that the density of m -cycles is less than $e^m - \Delta t$. Then

$$\ln(P_m(n, \Delta t)) = -\Theta(n^2 \delta^2).$$

Meanwhile, the analogue of Theorem B.2 is

Theorem B.8 ([22]) Let $m > 3$ be an odd integer and let $\Delta t = \delta^m = \Omega(n^{-3/2}) \cap o(n^{-1})$. Let $P_m(n, \Delta t)$ be the probability that the density of m -cycles is less than $e^m - \Delta t$. Then

$$\ln(P_m(n, \Delta t)) = -\Theta(n^3 \delta^{2m}) = -\Theta(n^3 (\Delta t)^2).$$

The crossover between $n^3 \delta^{2m}$ and $n^2 \delta^2$ is when Δt is $n^{-\frac{m}{2(m-1)}}$. However, the transition between Theorems B.7 and B.8 does not occur at that scale! Instead, it occurs at order n^{-1} , where there is a very sharp dropoff in the number of possible graphs. When Δt is of order $n^{-(1+\epsilon)}$, Theorem B.8 applies and $\ln(P_m)$ goes as $-n^{1+2\epsilon}$. When Δt is of the slightly larger order $n^{\epsilon-1}$, Theorem B.7 applies and $\ln(P_m)$ goes as $-n^{2-\frac{2}{m}+O(\epsilon)}$.

Appendix C

The functions $f(e, e')$ and $f_k(e, e')$

In equation (3.3) we introduced a function $f(e, e')$ that measured the entropy cost per change in g (as measured by the squared L^2 norm $\|g - e\|_2^2$) of moving away from a constant Erdős-Rényi graphon. In equation (6.2) we defined similar functions $f_k(e, e')$ that measure the entropy cost per change in the k star density. In this appendix we study those functions, reproducing results first found in [26]. The only difference is that, to avoid a sea of e 's, we will call the two variables u and v .

Our first function is defined as

$$f(u, v) = \frac{H(v) - H(u) - (v - u)H'(u)}{(v - u)^2}.$$

Our second family of functions, depending on a real parameter $k \geq 2$, is

$$f_k(u, v) = 2 \frac{H(v) - H(u) - (v - u)H'(u)}{v^k - u^k - k u^{k-1}(v - u)}.$$

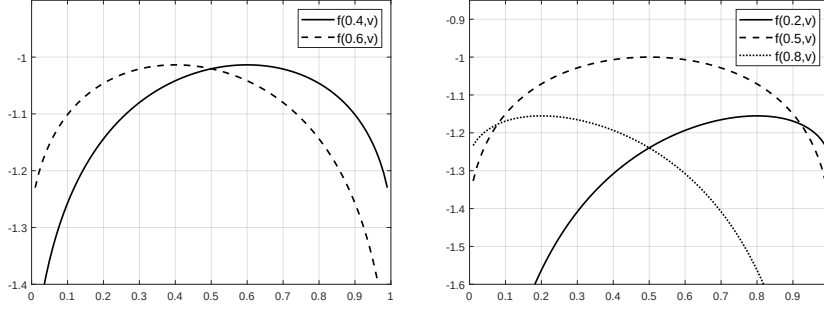
Our specific applications to graphs use f_k with integer values of k , but the function makes perfect sense for any real k . Note that $f_2(u, v) = 2f(u, v)$, so the first function is essentially a special case of the second.

Both $f(u, v)$ and $f_k(u, v)$ have removable singularities at $v = u$, which we fill by defining

$$f(u, u) = \frac{1}{2}H''(u), \quad f_k(u, u) = \frac{2H''(u)}{k(k-1)u^{k-2}}.$$

With this adjustment, both $f(u, v)$ and $f_k(u, v)$ become smooth (in fact analytic) functions of the variables u and v .

The first graph in Figure C.1 shows plots of $f(0.4, v)$ and $f(0.6, v)$ as functions of v . Note that $f(0.4, v)$ has a maximum at $v = 0.6$ and $f(0.6, v)$ has a maximum at $v = 0.4$. In the second graph, which shows $f(0.2, v)$, $f(0.5, v)$

Figure C.1 For fixed u , the maximum of $f(u, v)$ occurs at $v = 1 - u$

and $f(0.8, v)$, we continue to see the maximum of $f(u, v)$ occurring at $v = 1 - u$. This is not a coincidence!

Theorem C.1 For any fixed value of $u \in (0, 1)$, the maximum value of $f(u, v)$ is

$$f(u, 1 - u) = \begin{cases} \frac{H'(u)}{2u-1} & u \neq 1/2, \\ -1 & u = 1/2, \end{cases}$$

and is uniquely achieved at $v = 1 - u$.

Corollary C.2 For all values of u and v , $f(u, v) \leq -1$.

Proof of theorem We work separately with the numerator N and denominator D of f , defining

$$N = H(v) - H(u) - (v - u)H'(u) \quad \text{and} \quad D = (v - u)^2.$$

We write partial derivatives with respect to the first and second variable as ∂_1 and ∂_2 and compute

$$\partial_2 N = H'(v) - H'(u), \quad \partial_2 D = 2(v - u),$$

and

$$\partial_{22}^2 N = H''(v), \quad \partial_{22}^2 D = 2.$$

By the quotient rule,

$$\begin{aligned} \partial_2 f &= \frac{D \partial_2 N - N \partial_2 D}{D^2}, \\ \partial_{22}^2 f &= \frac{D \partial_{22}^2 N - N \partial_{22}^2 D}{D^2} - 2 \frac{\partial_2 D (D \partial_2 N - N \partial_2 D)}{D^3}. \end{aligned}$$

At a critical point, we must have

$$\begin{aligned} 0 &= D\partial_2 N - N\partial_2 D \\ &= (v-u)^2(H'(v) - H'(u)) - 2(v-u)(H(v) - H(u) - (v-u)H'(u)) \\ &= (v-u)^2(H'(u) + H'(v)) - 2(v-u)(H(v) - H(u)). \end{aligned}$$

Note that this expression is invariant under the interchange of u and v , so $\partial_2 f(u, v) = 0$ if and only if $\partial_2 f(v, u) = 0$.

The last line is obviously zero when $v = u$, and also when $v = 1 - u$, insofar as $H(v)$ is an even function of $v - \frac{1}{2}$ while $H'(v)$ is odd. This shows that $v = 1 - u$ is a critical point of f . We will show that it is the only critical point:

1. First, we will show that $v = u$ is not a critical point, except when $u = 1/2$. When $u = 1/2$, $\partial_2 f(1/2, v)$ has a simple root at $v = 1/2$ and no other roots.
2. Then we will consider what happens as we vary u away from $1/2$. Any critical points other than $v = 1 - u$ would either have to emerge from the boundary $v = 0$ or 1 , or from a double root where $\partial_2 f(u, v) = \partial_{22}^2 f(u, v) = 0$. We will rule out both possibilities.

When $v \approx u$, we expand the numerator in powers of $v - u$ and then divide by $(v - u)^2$:

$$\begin{aligned} N &= \frac{1}{2}H''(u)(v-u)^2 + \frac{1}{6}H'''(u)(v-u)^3 + \cdots, \\ f &= \frac{1}{2}H''(u) + \frac{1}{6}H'''(u)(v-u) + \cdots. \end{aligned}$$

There is a critical point at $v = u$ if and only if $H'''(u) = 0$. But

$$H'''(u) = \frac{1}{2} \left(\frac{1}{u^2} - \frac{1}{(1-u)^2} \right).$$

This vanishes at $u = 1/2$ and nowhere else. Furthermore, $H''''(1/2) \neq 0$, so $\partial_{22}^2 f(1/2, 1/2) \neq 0$, implying that $v = 1/2$ is a simple root of $\partial_2 f(1/2, v)$.

The Taylor series for $H(v)$ around $v = 1/2$ converges absolutely for all $v \in [0, 1]$, giving us an exact power series representation of $f(1/2, v)$ and its derivative

$$\begin{aligned} f(1/2, v) &= \sum_{k=1}^{\infty} \frac{H^{(2k)}(1/2)}{(2k)!} \left(v - \frac{1}{2}\right)^{2k-2} \\ \partial_2 f(1/2, v) &= \sum_{k=1}^{\infty} \frac{(2k-2)H^{(2k)}(1/2)}{(2k)!} \left(v - \frac{1}{2}\right)^{2k-3}. \end{aligned}$$

All of the even derivatives of $H(u)$ are negative at $u = 1/2$, so the only way for $\partial_2 f(1/2, v)$ to equal zero is for v to equal $1/2$. $\partial_2 f(1/2, v)$ has exactly one root, counted with multiplicity, namely $v = 1/2$.

We next consider the possibilities that a critical point can emerge elsewhere as we vary u away from $1/2$. Near the boundary ($v = 0$ or $v = 1$), $H'(v)$ diverges to $\pm\infty$ while $H(v)$ does not, so we cannot ever have $0 = (v - u)^2(H'(v) + H'(u)) - 2(v - u)(H(v) - H(u))$ when v is close to 0 or 1 and u is bounded away from 0 or 1.

The only remaining possibility is for a double root to emerge. To eliminate this possibility, we use a trick to express our equations using algebraic rather than transcendental functions. Note that

$$\partial_1 N(u, v) = -(v - u)H''(u) = \frac{v - u}{2u(1 - u)}.$$

Since $N(v, v) = 0$,

$$N(u, v) = \int_v^u \partial_1 N(x, v) dx = \int_v^u \frac{v - x}{2x(1 - x)} dx,$$

and likewise

$$D(u, v) = \int_v^u \partial_1 D(x, v) dx = \int_v^u 2(x - v) dx.$$

We can also take derivatives of these expressions with respect to v to get $\partial_2 N(u, v)$ and $\partial_2 D(u, v)$.

Now suppose that we have a double root of $\partial_2 f(u, v)$, in other words a point where $\partial_2 f(u, v) = \partial_{22}^2 f(u, v) = 0$. By the quotient rule, at such a point we must have

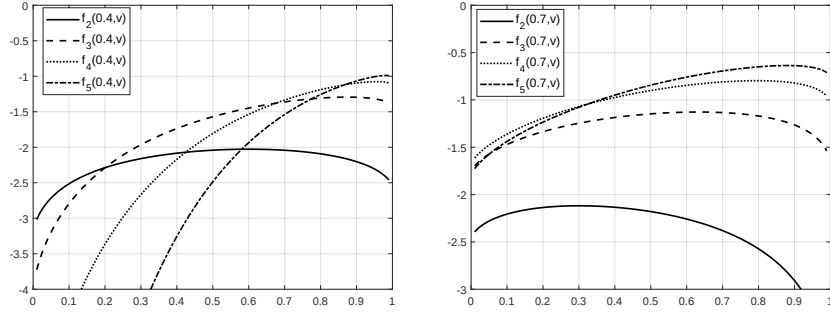
$$\frac{\partial_{22}^2 N}{\partial_{22}^2 D} = \frac{\partial_2 N}{\partial_2 D} = \frac{N}{D}.$$

But then $\partial_2 N \partial_{22}^2 D - \partial_2 D \partial_{22}^2 N = 0$. Expanding this using our trick gives

$$\begin{aligned} 0 &= \int_v^u \partial_{22}^2 D(u, v) \partial_{12}^2 N(x, v) - \partial_{22}^2 N(u, v) \partial_{12}^2 D(x, v) dx \\ &= \int_v^u \frac{v(1 - v) - x(1 - x)}{v(1 - v)x(1 - x)} dx. \end{aligned}$$

The numerator has one sign for all values of x between v and $1 - v$ and the opposite sign for x beyond $1 - v$. For the integral to equal zero, the integrand must take both signs, so u must be greater than $1 - v$ if $v < 1/2$ and must be less than $1 - v$ if $v > 1/2$. That is, we must have $|u - \frac{1}{2}| > |v - \frac{1}{2}|$.

However, the condition for being a critical point is symmetric in u and v , as is the preservation of this condition to leading order under changes in u and v . Repeating the argument with the roles of u and v reversed, we must have $|v - \frac{1}{2}| > |u - \frac{1}{2}|$. Contradiction. $\square$

Figure C.2 The functions $f_k(u, v)$ at $u = 0.4$ and $u = 0.7$

Proof of corollary For any u and v , we have shown that

$$f(u, v) \leq f(u, 1 - u) = \frac{H'(u)}{2u - 1}.$$

A power series expansion for $H'(u)$ around $u = 1/2$ gives

$$\frac{H'(u)}{2u - 1} = \frac{1}{2} \sum_{k=1}^{\infty} \frac{H^{(2k)}(1/2)}{(2k - 1)!} \left(u - \frac{1}{2}\right)^{2k-2} \leq \frac{H''(1/2)}{2} = -1,$$

since all of the terms with $k > 1$ contribute negatively to the sum. $\square$

Having understood the function $f(u, v)$ that appears in the edge-triangle model, we turn our attention to the family of functions $f_k(u, v)$ that appear in the edge- k -star model. See Figure C.2. The results are similar to those for $k = 2$, as is the proof.

Theorem C.3 *For any fixed value of $u \in (0, 1)$, the function $f_k(u, v)$ achieves its maximum value at a unique point $v = \zeta_k(u)$, where ζ_k is an involution of $[0, 1]$ with fixed point $(k - 1)/k$.*

Proof We need to do three things:

1. Show that the function $f(u, v)$, for fixed u , has exactly one critical point, which is the maximum. This allow us to define $\zeta_k(u)$ to be that critical point.
2. Show that v is a critical point of $f(u, v)$ if and only if u is a critical point of $f(v, u)$. This shows that ζ_k is an involution.
3. Show that there is a critical point at $v = u$ exactly when $u = (k - 1)/k$. This defines the fixed point of the involution.

When it comes to the calculations, it is convenient to do the third step first, then the second, and to save the first (and hardest) step for last.

As before, we study the numerator and denominator separately, only now

$$\begin{aligned} N_k(u, v) &= 2(H(v) - H(u) - (v - u)H'(u)) \\ D_k(u, v) &= v^k - u^k - ku^{k-1}(v - u). \end{aligned}$$

Note that the expressions

$$\begin{aligned} \partial_2 N_k(u, v) &= 2(H'(v) - H'(u)) \\ \partial_2 D_k(u, v) &= k(v^{k-1} - u^{k-1}) \end{aligned}$$

are odd under interchange of u and v . Furthermore, note that

$$\begin{aligned} (v - u)\partial_2 N_k(u, v) - N_k(u, v) &= 2[(v - u)(H'(u) - H'(v)) \\ &\quad - H(u) + H(v) + (v - u)H'(u)] \\ &= 2[H(v) - H(u) - (u - v)H'(v)] \\ &= N_k(v, u), \end{aligned}$$

and likewise

$$(v - u)\partial_2 D_k(u, v) - D_k(u, v) = D_k(v, u).$$

Both $N_k(u, v)$ and $D_k(u, v)$ admit simple power series expansions near $v = u$:

$$\begin{aligned} N_k(u, v) &= 2 \sum_{j=2}^{\infty} \frac{H^{(j)}(u)}{j!} (v - u)^j \\ D_k(u, v) &= \sum_{j=2}^k \binom{k}{j} (v - u)^j \end{aligned}$$

Dividing both by $(v - u)^2$, we obtain

$$f_k(u, v) = \frac{H''(u) + \frac{1}{3}H'''(u)(v - u) + \dots}{k(k-1)u^{k-2} + \frac{1}{3}k(k-1)(k-2)u^{k-3}(v - u) + \dots}$$

Setting the derivative with respect to v at $v = u$ equal to zero then gives

$$\begin{aligned} k(k-1)u^{k-2}H'''(u) &= k(k-1)(k-2)u^{k-3}H''(u) \\ -u \left(\frac{1}{u^2} - \frac{1}{(1-u)^2} \right) &= (k-2) \left(\frac{1}{u} + \frac{1}{1-u} \right) \\ -u \left(\frac{1}{u} - \frac{1}{1-u} \right) &= k-2 \\ \frac{u}{1-u} &= k-1 \\ u &= \frac{k-1}{k}. \end{aligned}$$

Now suppose that $\partial_2 f_k(u, v) = 0$. This implies that

$$\frac{N_k(u, v)}{D_k(u, v)} = \frac{\partial_2 N_k(u, v)}{\partial_2 D_k(u, v)}.$$

Call the common ratio r , so $N_k(u, v) = rD_k(u, v)$ and $\partial_2 N_k(u, v) = r\partial_2 D_k(u, v)$. But then

$$\begin{aligned} (v - u)\partial_2 N_k(u, v) - N_k(u, v) &= r(v - u)\partial_2 D_k(u, v) - D_k(u, v) \\ N_k(v, u) &= rD_k(v, u). \end{aligned}$$

Meanwhile,

$$\partial_2 N_k(v, u) = -\partial_2 N_k(u, v) = -r\partial_2 D_k(u, v) = r\partial_2 D_k(v, u).$$

But then

$$\frac{N_k(v, u)}{D_k(v, u)} = r = \frac{\partial_2 N_k(v, u)}{\partial_2 D_k(v, u)},$$

so u is a critical point of $f_k(v, u)$.

To show that there is a unique critical point of $f_k(u, v)$, we refer to Theorem C.1, which tells us that $f_2(u, v)$ has a unique critical point. We then consider what happens as we continuously increase k .

Supposing that there are values of k for which there are multiple critical points of $f_k(u, v)$ for the same value of u , we let k_0 be the infimum of those values. Taking limits, we obtain a point (u, v) that is a double root of $\partial_2 f_{k_0}(u, v)$. By the symmetry of the equations defining roots, this also implies that (v, u) is a double root of $\partial_2 f_{k_0}(v, u)$.

Now suppose that $\partial_2 f_{k_0}(u, v) = \partial_{22}^2 f_{k_0}(u, v) = 0$. Arguing as before, we obtain

$$\begin{aligned} 0 &= \int_v^u \partial_{22}^2 D_{k_0}(u, v) \partial_{12}^2 N_{k_0}(x, v) - \partial_{22}^2 N_{k_0}(u, v) \partial_{12}^2 D_{k_0}(x, v) dx \\ &= k_0(k_0 - 1) \int_v^u \frac{v^{k_0-1}(1-v) - x^{k_0-1}(1-x)}{v(1-v)x(1-x)} dx. \end{aligned}$$

For the integral to be zero, the integrand has to change sign, which means that there is a point x between v and u where $x^{k_0-1}(1-x) = v^{k_0-1}(1-v)$. But then $u^{k_0-1}(1-u)$ is strictly smaller than $v^{k_0-1}(1-v)$. However, the same argument applied to $f_{k_0}(v, u)$ shows that $v^{k_0-1}(1-v)$ is strictly smaller than $u^{k_0-1}(1-u)$. Contradiction.

□

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